Verification pack

Generated by explaneat export from the JSON records; order as pre-registered (`PRE` in `experiments/10seed/build_explanations_csv.py`).

#	Genome	Dataset	Test AUC	Struct.	Comp.	Recorded verdict
1	churn100-s2-c1	churn_onehot	0.8756	1	1	none
2	churn100-s2-c2	churn_onehot	0.8883	1	1	none
3	churn100-s4-c2	churn_onehot	0.8664	1	1	none
4	churn100-s4-c3	churn_onehot	0.8671	1	1	none
5	churn100-s7-c1	churn_onehot	0.9007	1	1	none
6	heart-s1-c1	Heart Disease (binary, prepared)	0.861	1	1	none
7	heart-s2-c1	Heart Disease (binary, prepared)	0.8966	1	1	none
8	heart-s2-c2	Heart Disease (binary, prepared)	0.9009	1	1	none
9	heart-s2-c3	Heart Disease (binary, prepared)	0.8987	1	1	none
10	heart-s4-c1	Heart Disease (binary, prepared)	0.8858	1	1	none
11	heart-s5-c1	Heart Disease (binary, prepared)	0.8869	1	1	none
12	heart-s5-c2	Heart Disease (binary, prepared)	0.8922	1	1	none
13	heart-s5-c3	Heart Disease (binary, prepared)	0.9084	1	1	none

For each genome: a verification checklist (recorded vs live checks, what to look at, verdict fields and the command that records the verdict), then its explanation certificate.

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1. churn100-s2-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	6.66e-16	6.66e-16	6.66e-16	agree
Closed-form max abs error (all leaves)	0.03121	0.03121	0.03121	agree
Numeric-function max abs error	n/a	2.66e-15	2.66e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	churn_logit	churn_logit	agree
Leaf / composition counts	8 / 2	n/a	8 / 2	agree
Operation counts	17 splits, 2 identity	add identity node 2, annotate 10, consolidate node 3, rename node 71, split node 17	add identity node 2, annotate 10, consolidate node 3, rename node 71, split node 17	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `state_offsets` (0.0312), then `linear_backbone` (0.0152). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `csc_usage_gate`: *"Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.098/0.092/0.099/0.147/0.333/0.645/0.923;"* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `voicemail_chain`: *"All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 2 hidden-node split(s), which decide how a unit's roles are divided: `2627`→`2627_a`, `2627_b`; `9555`→`9555_a`, `9555_b`. Do the copies really play the different roles the annotations give them?
- 3 consolidation(s): copies merged back after a split. Check that the merged copies carry the same role.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): split_node cannot re-split a consolidated copy ('Cannot re-split -71_bde: expected 3 outputs, got 5'), so the hidden splits (2627, 9555) had to come before the input splits (undo_operation from seq 71, then redo). The Bash sandbox refused shell loops/variables around uv, so tool calls were driven from a scratch Python script that runs experiments/species_leads/xn.py per call (all calls logged).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `churn100-s2-c1-plain-v5.pdf` in this genome's folder):

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up seven parts: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 13% at 249 minutes. [C057] An international plan takes the

chance from about 1% to about 13%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 5% at 32 messages. [C058, C067, C059] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C062, C063, C064, C070] Through it, the state moves the chance from about 1% in West Virginia to about 4% in California and in Washington, and to under 1% in Hawaii, Vermont and Colorado (41 to 80 training customers per state). [C060, C061, C066, C068, C071] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C111] Through it, daytime minutes lower the chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way: for a typical customer, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] This part reads the voice-mail plan and a few states: through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Through this part, evening minutes add nothing until about 239 minutes, then lower the chance steadily; it acts on 885 of 4000 training customers, 165 of whom left, and for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C135, C136] This part reads only the state and moves the chance by less than half a point: up in Arizona and Ohio, down in Nevada. [C138, C139, C140] This part reads the state and daytime calls: being in Tennessee raises a typical customer's chance from about 1% to about 2%, and daytime calls barely change it. [C144, C145] This small part moves the chance by less than a tenth of a point, up in Arkansas and Utah. [C150, C153]

Worth checking: Worth checking: the state moves the chance by several points (for example California and Washington about 6% and 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C016] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C012, C005] Worth checking: the many-service-calls part is switched off for customers in West Virginia, the state of the typical customer. [C130] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C057, C112, C064, C115, C062, C114]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C056	structure: made of linear_backbone, service_call_nonlinear, intl_state_unit, eve_minutes_hinge, voicemail_chain, state_offsets, day_calls_state42; added		log-odds n/a, OR n/a	4000	train	yes	
C057	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 13%	log-odds 5.19, OR 179; range across patients 5.19 to 5.19	4000	train; test agrees (AUC gain 0.188 [0.143, 0.234], n=1000)	yes	clearly
C058	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2; range across patients 2.58 to 2.58	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C059	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C060	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.41, OR 4.10; range across patients 1.41 to 1.41	122, 41	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C061	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.23, OR 3.41; range across patients 1.23 to 1.23	122, 72	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C062	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.76, OR 5.79; range across patients 1.76 to 1.76	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.30, OR 3.66; range across patients 1.30 to 1.30	4000	train; test agrees (AUC gain 0.020 [0.009, 0.032], n=1000)	yes	slightly
C064	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.14, OR 3.12; range across patients 1.14 to 1.14	4000	train; test agrees (AUC gain 0.017 [0.007, 0.027], n=1000)	yes	slightly
C066	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -2.18, OR 0.11; range across patients -2.18 to -2.18	122, 73	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C067	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.04, OR 0.13; range across patients -2.04 to -2.04	2939, 1061	train; test agrees (AUC gain 0.018 [0.004, 0.034], n=1000)	yes	slightly
C068	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C070	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.78 to 0.78	4000	train; test agrees (AUC gain 0.006 [0.0005, 0.012], n=1000)	yes	slightly
C071	direction: state: West Virginia to Colorado (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.39, OR 0.25; range across patients -1.39 to -1.39	122, 78	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C103	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 154 minutes; 100% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 17%; total evening charge at or below 17.12 US dollars: 2% to 10%	log-odds difference 1.53, OR 4.60	1998, 2002	train; test agrees (AUC gain 0.009 [-0.003, 0.020], n=1000); group difference on test 1.53 [1.35, 1.69]	yes	clearly
C107	threshold: total daytime minutes: fold at about 170 minutes (stops) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C107 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	38% to 1% (typical patient with a fold)	fold 170 minutes (IQR 153 to 191); past it log-odds -4.88, OR 0.008; range across patients -7.17 to -3.26	3999, 1839	train; test agrees (AUC gain 0.062 [0.035, 0.095], n=1000)	yes	clearly
C108	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 73% of patients same; fallback; risk saturates near 0%*	3% to 11% (typical patient with a fold)	fold 3.02 calls (IQR 2.14 to 3.02); past it log-odds 1.84, OR 6.28; range across patients 0.60 to 3.83	2668, 577	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	clearly
C111	structure: made of csc_usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C125	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -0.64, OR 0.53; range across patients -0.64 to -0.64	2939, 1061	train; test unclear (AUC gain 0.002 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C126	direction: state: West Virginia to state other than Alabama, North Carolina, West Virginia (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 2%	log-odds 0.35, OR 1.42; range across patients 0.35 to 0.35	122, 3699	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C135	threshold: total evening minutes: fold at about 239 minutes (starts) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1% (typical patient with a fold)	fold 239 minutes (IQR 239 to 239); past it log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	4000, 885	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C136	subgroup: who this part acts on: 885 of 4000 patients	5% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.22; log-odds -0.35, OR 0.70; range across patients -0.77 to -0.05	885, 3115	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C138	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Arizona (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3508, 73	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C139	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Ohio (through this part)	1% to 1%	log-odds 0.21, OR 1.24; range across patients 0.21 to 0.21	3508, 87	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C140	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Nevada (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.26, OR 0.77; range across patients -0.26 to -0.26	3508, 77	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C144	direction: state: state other than Tennessee to Tennessee (through this part) *shape: categorical; highest to lowest: Tennessee, state other than Tennessee*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.38 to 0.39	3932, 68	train; test disagrees (AUC gain -0.0006 [-0.001, -7e-05], n=1000)	yes	slightly; on the training data only
C145	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	4000	train; test unclear (AUC gain 4e-05 [-0.0002, 0.0003], n=1000)	yes	slightly; on the training data only
C150	direction: state: state other than Arkansas, New Mexico, Utah to Arkansas (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.05, OR 1.06; range across patients 0.03 to 0.08	3761, 78	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only
C153	direction: state: state other than Arkansas, New Mexico, Utah to Utah (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.009, OR 1.01; range across patients 0.00 to 0.03	3761, 89	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict churn100-s2-c1 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/churn100-s2-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict churn100-s2-c1 --verified --minutes N --changes "..." --notes "..."`

1 Explanation certificate: churn100-s2-c1

This section is the explanation certificate of the churn100-s2-c1 network. It is rendered from the canonical JSON record `churn100-s2-c1.json` (schema 1.4.0, SHA-256 `e92cc46d5657fbeb`…), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Drafter's summary. logit P(churn) = -1.05 + a sparse linear backbone over 40 inputs (day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71,

Dataset	churn_onehot
Experiment	churn100-s2, seed 2
Genome	`f188e353-beb3-4b23-9e19-b64aba8cb5ba` (NEAT key 24636, generation 99)
Explanation	`6eb705c3-d493-4a02-ab0a-6e01b8c5d696` (103 operations)
Train AUC / accuracy	0.9009 / 0.9008 (n = 4000)
Test AUC / accuracy	0.8756 / 0.901 (n = 1000)
Output activation	sigmoid
Phenotype	94 nodes (71 inputs, 22 hidden, 1 output), 95 enabled edges
Explained model state	116 nodes, 100 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 3, split_node 17
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 6.66 \times 10^{-16}$ on the training split
Annotations	10 (8 leaf, 2 composition)
Structural coverage	1 (115 of 115 candidate nodes; 24 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`e92cc46d5657fbeb...`

Table 1: Explanation certificate summary: churn100-s2-c1.

voicemail messages +0.62, ...) + a service-call x low-day-usage gate + a 4+-calls hinge + an evening-minutes hinge + exactly linear voicemail and state-offset units, all additive in the logit. The gate is a low-day-minutes hinge whose kink moves up with each customer-service call (median 147 day minutes at 0 calls, 220 at 4, 276 at 6), so with the direct day-minutes edge the logit is V-shaped in day minutes and frequent callers with low usage churn. Service calls act only through hinges. Train AUC 0.901, accuracy 0.901.

What resisted explanation. Nothing structural. Hidden nodes 2627 and 9555 (the always-on service-call encoding) were split so the near-dead state:2/state:44 unit 252 could be its own leaf; 161 and 3861 are identically 0 on train (dead). The gate under-predicts churn for frequent callers with low day usage (CSC>=4 and day minutes <160: 113 rows, model 0.66 vs empirical 0.88) and over-predicts it for low-usage customers with few calls (CSC<4, day minutes <120: 476 rows, 0.08 vs 0.05). closed_form_max_abs_err 0.031 is 3-significant-figure rounding on state one-hots (standardised values up to 8); the numeric function is exact.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up seven parts: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 13% at 249 minutes. [C057] An international plan takes the chance from about 1% to about 13%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 5% at 32 messages. [C058, C067, C059] A higher international charge, more evening minutes, a higher evening charge and more

night-time minutes also raise it steadily, each by one to two points. [C062, C063, C064, C070] Through it, the state moves the chance from about 1% in West Virginia to about 4% in California and in Washington, and to under 1% in Hawaii, Vermont and Colorado (41 to 80 training customers per state). [C060, C061, C066, C068, C071] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C111] Through it, daytime minutes lower the chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way: for a typical customer, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] This part reads the voice-mail plan and a few states: through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Through this part, evening minutes add nothing until about 239 minutes, then lower the chance steadily; it acts on 885 of 4000 training customers, 165 of whom left, and for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C135, C136] This part reads only the state and moves the chance by less than half a point: up in Arizona and Ohio, down in Nevada. [C138, C139, C140] This part reads the state and daytime calls: being in Tennessee raises a typical customer's chance from about 1% to about 2%, and daytime calls barely change it. [C144, C145] This small part moves the chance by less than a tenth of a point, up in Arkansas and Utah. [C150, C153]

The whole model. The model adds up seven parts into one score: fixed-weight inputs, service calls, a voice-mail discount, heavy evening use, state units, a daytime-calls unit and a small state unit. [C056] *Evidence.* It is made up of these seven parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C056] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 13%, and a voice-mail plan lowers it to under 1%. [C001, C012] More customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C008] The model ignores the daytime charge, evening calls, international calls and the night-time charge. [C052, C053, C054, C055]

Fixed-weight inputs. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C057, C058, C059, C067] *Evidence.* Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 13% at 249 minutes, and an international plan from about 1% to about 13%; a voice-mail plan lowers the chance to under 1%. [C057, C058, C067] The state moves it from about 1% in West Virginia to about 4% in California and Washington, and to under 1% in Hawaii and Vermont. [C060, C061, C066, C068] Its formula reproduces the part exactly on all 4000 training customers. [C096]

Service calls. This part raises the chance of leaving with calls to customer service, from 4 calls upwards and in an uneven way, and more so for customers with low daytime use; it adds up two parts. [C111, C108, C107] *Evidence.* It is made of two parts that are added: calls with low usage, and many service calls. [C111] Through it, daytime minutes lower a typical customer's chance only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), in an uneven way, from about 38% at 91 minutes to about 1% at 352 minutes. [C107] Customer-service calls add to it from 4 calls upwards, in an uneven way: for a typical such customer, from about 3% at 0 calls to about 11% at 4 calls. [C108] With an evening charge above 17.12 US dollars, the chance is lowest at about 154 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 17% at 249 minutes, against about 2% to about 10% at or below that charge. [C103]

Calls with low usage. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 170 minutes (between 147 and 216 minutes for different customers), and calls raise it, both in an uneven way. [C119, C113] *Evidence.* It acts on 3121 of 4000 training customers, 402 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C123] Through it, a typical customer's chance falls in an uneven way from about 38% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 170 minutes (between 147 and 216 minutes for different customers). [C119] A higher international charge lowers it only while it stays below about 2.95 US dollars (between 2.95 and 4.34 US dollars for different customers), in an uneven way, from about 3% to about 1%. [C121] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 3% at 3 calls. [C113] Its formula reproduces the part exactly on all 4000 training customers. [C124]

Many service calls. This part raises the chance of leaving with calls to customer service, adding nothing until about 3 calls, then steadily, except in West Virginia. [C131, C130] *Evidence.* It acts on 306 of 4000 training customers, 154 of whom left; for the typical customer it acts on, the chance is about 16% without it and about 26% with it. [C132] It is switched off in West Virginia: for the typical customer it acts on, being in any other state takes the chance from about 16% to about 26%. [C130] Its formula reproduces the part exactly on all 4000 training customers. [C133]

Voice-mail discount. This part gives customers with a voice-mail plan a small discount on the chance of leaving. [C125] *Evidence.* Through it, a voice-mail plan lowers a typical customer's chance by less than a point, and customers in states other than Alabama, North Carolina and West Virginia get about 2% instead of about 1%. [C125, C126] Its formula reproduces the part exactly on all 4000 training customers. [C129]

Heavy evening use. This part lowers the chance of leaving a little for customers with heavy evening use: evening minutes add nothing until about 239 minutes, then lower it steadily. [C134] *Evidence.* It acts on 885 of 4000 training customers, 165 of whom left; for the typical customer it acts on, the chance is about 5% without it and about 4% with it. [C136] Through it, evening minutes change nothing until about 239 minutes, then lower a typical customer's chance steadily, from about 1% at 138 minutes to under 1% at 266 minutes. [C134] Its formula reproduces the part exactly on all 4000 training customers. [C137]

State units. This part adds small state offsets: up in Arizona and Ohio, down in Nevada. [C138, C139, C140] *Evidence.* Through it, a typical customer's chance moves by less than half a point: up in Arizona and Ohio, down in Nevada, with Missouri and Massachusetts almost unchanged. [C138, C139, C140, C141, C142] Its formula reproduces the part exactly on all 4000 training customers. [C143]

Daytime calls and Tennessee. This part gives customers in Tennessee a small rise in the chance of leaving; daytime calls barely change it. [C144, C145] *Evidence.* Through it, being in Tennessee takes a typical customer's chance from about 1% to about 2%, and going from 75 to 125 daytime calls barely changes it. [C144, C145] Its formula reproduces the part exactly on all 4000 training customers. [C149]

Small state unit. This small part barely changes the chance of leaving: it adds less than a tenth of a point in Arkansas and Utah. [C150, C153] *Evidence.* Through it, the chance moves by less than a tenth of a point in Arkansas and Utah, a higher international charge lowers it steadily by even less, and customer-service calls barely move it. [C150, C153, C151, C152] Its formula reproduces the part exactly on all 4000 training customers. [C161]

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example California and Washington about 6% and 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C016] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C012, C005] Worth checking: the many-service-calls part is switched off for customers in West Virginia, the state of the typical customer. [C130] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C057, C112, C064, C115, C062, C114]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2; range across patients 2.58 to 2.58	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	1% to 13%	log-odds 1.55, OR 4.72; range across patients -0.77 to 3.15	4000	train; test agrees (AUC gain 0.092 [0.059, 0.128], n=1000)	no	clearly
C003	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 6%	log-odds 1.84, OR 6.30; range across patients 1.76 to 1.76	122, 41	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	on the training data only
C004	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 5%	log-odds 1.66, OR 5.24; range across patients 1.58 to 1.58	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 3%	log-odds 1.17, OR 3.22; range across patients 1.09 to 1.09	122, 75	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C007	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 3%	log-odds 1.04, OR 2.83; range across patients 0.96 to 0.96	122, 82	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C008	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C009	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.90, OR 2.45; range across patients 0.82 to 0.82	122, 87	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C010	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.88, OR 2.41; range across patients 0.80 to 0.81	122, 68	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C011	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.67, OR 1.95; range across patients 0.29 to 0.88	4000	train; test unclear (AUC gain 0.005 [-0.001, 0.012], n=1000)	yes	slightly; on the training data only
C012	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.68, OR 0.07; range across patients -2.68 to -2.68	2939, 1061	train; test agrees (AUC gain 0.033 [0.015, 0.053], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.74, OR 2.10; range across patients 0.67 to 0.67	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C014	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: several pieces; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98	4000	train; test agrees (AUC gain 0.009 [0.002, 0.016], n=1000)	yes	slightly
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.62 to 0.62	122, 84	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -1.75, OR 0.17; range across patients -1.82 to -1.82	122, 73	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 13.46 US dollars; 75% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds -0.05, OR 0.95; range across patients -1.93 to 1.14	4000	train; test agrees (AUC gain 0.012 [0.002, 0.024], n=1000)	no	slightly
C018	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.78 to 0.78	4000	train; test agrees (AUC gain 0.006 [0.0005, 0.012], n=1000)	yes	slightly
C019	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -1.40, OR 0.25; range across patients -1.47 to -1.47	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.63, OR 1.87; range across patients 0.55 to 0.55	122, 73	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C021	direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to <1%	log-odds -0.96, OR 0.38; range across patients -1.04 to -1.04	122, 78	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.65; range across patients 0.42 to 0.46	122, 89	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.64; range across patients 0.42 to 0.42	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to state other than Alaska, Alabama, Hawaii, Massachusetts, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, New Mexico, Nevada, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Tennessee, Utah, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.50, OR 1.64; range across patients 0.42 to 0.42	122, 1587	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.49, OR 1.64; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C026	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.47, OR 1.59; range across patients 0.37 to 0.43	122, 78	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C027	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.40, OR 1.49; range across patients 0.33 to 0.33	122, 74	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C028	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 2%	log-odds 0.40, OR 1.49; range across patients 0.32 to 0.32	122, 68	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C029	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.40, OR 0.67; range across patients -0.47 to -0.47	122, 56	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C030	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.39, OR 0.68; range across patients -0.46 to -0.46	122, 56	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	4000	train; test disagrees (AUC gain -0.002 [-0.004, 0.0007], n=1000)	yes	slightly; on the training data only
C032	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.29 to 0.29	122, 79	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C033	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	4000	train; test disagrees (AUC gain -0.003 [-0.006, -8e-05], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.28, OR 0.76; range across patients -0.35 to -0.35	122, 91	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C035	direction: state: West Virginia to Wyoming (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.27, OR 0.77; range across patients -0.34 to -0.34	122, 89	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C036	direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.20 to -0.20	122, 86	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.24, OR 1.27; range across patients 0.16 to 0.16	122, 77	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.15, OR 0.86; range across patients -0.15 to -0.15	2016, 1024	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C039	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds -0.07, OR 0.93; range across patients -0.14 to -0.14	122, 80	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C040	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.13 to -0.13	2016, 960	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C041	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.48, OR 1.62; range across patients -0.37 to 1.32	122, 66	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	no	slightly; on the training data only
C042	direction: state: West Virginia to New Hampshire (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.05 to 0.05	122, 74	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C043	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	4000	train; test disagrees (AUC gain -0.0002 [-0.0007, 0.0003], n=1000)	yes	slightly; on the training data only
C044	direction: state: West Virginia to Alabama (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.04, OR 1.04; range across patients -0.04 to -0.04	122, 105	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	no	slightly; on the training data only
C045	direction: state: West Virginia to Nebraska (whole model) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 1%	log-odds 0.09, OR 1.10; range across patients 0.02 to 0.02	122, 72	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	slightly; on the training data only
C046	direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	4000	train; test disagrees (AUC gain -0.0001 [-0.0004, 0.0001], n=1000)	yes	slightly; on the training data only
C047	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C048	magnitude: state: West Virginia to California, typical patient *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Ohio, Tennessee …*	1% to 6%	log-odds 1.76, OR 5.84	122, 41	train; test disagrees (AUC gain -0.016 [-0.030, -0.004], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C050	magnitude: customer-service calls: 0 calls to 3 calls, typical patient *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 0.93, OR 2.52	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C051	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 87% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.88, OR 2.41	4000	train; test unclear (AUC gain 0.005 [-0.001, 0.012], n=1000)	yes	slightly; on the training data only
C052	null: model ignores total daytime charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C053	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C054	null: model ignores total international calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C055	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C056	structure: made of linear_backbone, service_call_nonlinear, intl_state_unit, eve_minutes_hinge, voicemail_chain, state_offsets, day_calls_state42; added		log-odds n/a, OR n/a	4000	train	yes	
C057	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 13%	log-odds 5.19, OR 179; range across patients 5.19 to 5.19	4000	train; test agrees (AUC gain 0.188 [0.143, 0.234], n=1000)	yes	clearly
C058	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 13%	log-odds 2.58, OR 13.2; range across patients 2.58 to 2.58	3620, 380	train; test agrees (AUC gain 0.060 [0.035, 0.086], n=1000)	yes	clearly
C059	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 5%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	4000	train; test unclear (AUC gain 0.008 [-0.002, 0.019], n=1000)	yes	slightly; on the training data only
C060	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.41, OR 4.10; range across patients 1.41 to 1.41	122, 41	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 4%	log-odds 1.23, OR 3.41; range across patients 1.23 to 1.23	122, 72	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C062	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.76, OR 5.79; range across patients 1.76 to 1.76	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.30, OR 3.66; range across patients 1.30 to 1.30	4000	train; test agrees (AUC gain 0.020 [0.009, 0.032], n=1000)	yes	slightly
C064	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.14, OR 3.12; range across patients 1.14 to 1.14	4000	train; test agrees (AUC gain 0.017 [0.007, 0.027], n=1000)	yes	slightly
C065	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.74, OR 2.10; range across patients 0.74 to 0.74	122, 75	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C066	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -2.18, OR 0.11; range across patients -2.18 to -2.18	122, 73	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C067	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.04, OR 0.13; range across patients -2.04 to -2.04	2939, 1061	train; test agrees (AUC gain 0.018 [0.004, 0.034], n=1000)	yes	slightly
C068	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 82	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C070	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.78 to 0.78	4000	train; test agrees (AUC gain 0.006 [0.0005, 0.012], n=1000)	yes	slightly
C071	direction: state: West Virginia to Colorado (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -1.39, OR 0.25; range across patients -1.39 to -1.39	122, 78	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -0.83, OR 0.44; range across patients -0.83 to -0.83	122, 56	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to <1%	log-odds -0.82, OR 0.44; range across patients -0.82 to -0.82	122, 56	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C074	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.73, OR 0.48; range across patients -0.73 to -0.73	122, 66	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C075	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.70, OR 0.49; range across patients -0.70 to -0.70	122, 91	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C076	direction: state: West Virginia to Wyoming (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 89	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C077	direction: state: West Virginia to Wisconsin (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.55, OR 0.57; range across patients -0.55 to -0.55	122, 86	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C078	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.46, OR 0.63; range across patients -0.46 to -0.46	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C079	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 2%	log-odds 0.31, OR 1.37; range across patients 0.31 to 0.31	122, 80	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C080	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	4000	train; test disagrees (AUC gain -0.002 [-0.004, 0.0007], n=1000)	yes	slightly; on the training data only
C081	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.32, OR 1.38; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.003 [-0.006, -8e-06], n=1000)	yes	slightly; on the training data only
C082	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.27, OR 1.31; range across patients 0.27 to 0.27	122, 84	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C083	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 74	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C084	direction: state: West Virginia to Ohio (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 87	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: West Virginia to Nebraska (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.33, OR 0.72; range across patients -0.33 to -0.33	122, 72	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C086	direction: state: West Virginia to New Hampshire (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.30, OR 0.74; range across patients -0.30 to -0.30	122, 74	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C087	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.15, OR 0.86; range across patients -0.15 to -0.15	2016, 1024	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C088	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 510, area code 408*	1% to 1%	log-odds -0.13, OR 0.88; range across patients -0.13 to -0.13	2016, 960	train; test disagrees (AUC gain -0.002 [-0.003, -0.0001], n=1000)	yes	slightly; on the training data only
C089	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.10, OR 0.91; range across patients -0.10 to -0.10	122, 73	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C090	direction: state: West Virginia to state other than Alaska, Hawaii, Arkansas, Maryland, Missouri, Montana, North Carolina, North Dakota, Nebraska, Arizona, New Hampshire, Ohio, Oklahoma, Oregon, Pennsylvania, California, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado, Wyoming, Connecticut (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07	122, 2084	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C091	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.07, OR 0.94; range across patients -0.07 to -0.07	122, 79	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C092	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	4000	train; test disagrees (AUC gain -0.0002 [-0.0007, 0.0003], n=1000)	yes	slightly; on the training data only
C093	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.03, OR 0.97; range across patients -0.03 to -0.03	122, 68	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C094	direction: total night-time calls: 74 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	4000	train; test disagrees (AUC gain -0.0001 [-0.0004, 0.0001], n=1000)	yes	slightly; on the training data only
C095	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: California, Washington, Montana, Maryland, Connecticut, Oregon …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 78	train; test disagrees (AUC gain -0.012 [-0.025, -0.0006], n=1000)	yes	slightly; on the training data only
C096	structure: its formula reproduces it on 4000 training patients (largest difference 3.55271e-15)		log-odds 4e-15, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C107 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	log-odds -3.63, OR 0.03; range across patients -5.95 to -2.04	4000	train; test agrees (AUC gain 0.062 [0.035, 0.095], n=1000)	yes	clearly
C098	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 84% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.097 [0.060, 0.139], n=1000)	yes	slightly
C099	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -1.46 to -0.88	4000	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C100	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	log-odds -1.19, OR 0.30; range across patients -3.06 to 0.00	4000	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C101	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 2%	log-odds 0.86, OR 2.36; range across patients 0.00 to 1.70	122, 66	train; test disagrees (AUC gain -0.002 [-0.004, -0.0001], n=1000)	yes	slightly; on the training data only
C102	direction: state: West Virginia to state other than South Dakota, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, West Virginia, state other than South Dakota, West Virginia*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.00 to 0.00	122, 3812	train; test disagrees (AUC gain -0.002 [-0.004, -0.0001], n=1000)	yes	slightly; on the training data only
C103	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 154 minutes; 100% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 17%; total evening charge at or below 17.12 US dollars: 2% to 10%	log-odds difference 1.53, OR 4.60	1998, 2002	train; test agrees (AUC gain 0.009 [-0.003, 0.020], n=1000); group difference on test 1.53 [1.35, 1.69]	yes	clearly
C104	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus West Virginia; stronger in West Virginia *shape: dip at 160 minutes; 98% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; West Virginia: 1% to 13%	log-odds difference -1.09, OR 0.34	66, 122	train; test agrees (AUC gain -0.002 [-0.004, -0.0001], n=1000); group difference on test -1.05 [-1.92, -0.17]	yes	clearly
C105	interaction: effect of customer-service calls (0 calls to 3 calls) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge at or below 17.12 US dollars *shape: accelerating; 89% of patients same; fallback; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 2%; total evening charge at or below 17.12 US dollars: 1% to 5%	log-odds difference -0.81, OR 0.45	1998, 2002	train; test agrees (AUC gain 0.009 [-0.003, 0.020], n=1000); group difference on test -0.91 [-1.04, -0.76]	yes	slightly
C106	interaction: effect of total	South Dakota:	log-odds difference	66, 122	train; test unclear	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C109 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.62); past it log-odds -1.42, OR 0.24; range across patients -2.21 to -1.11	3851, 2752	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C110	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C110 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 14.6 to 18.7); past it log-odds -2.12, OR 0.12; range across patients -3.60 to 0.00	71, 34	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C111	structure: made of csc_usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C112	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 170 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C119 (median fold 170 minutes, typical patient 160 minutes, 10th-90th percentile 147 minutes to 216 minutes, wide)*	15% to 1%	log-odds -3.63, OR 0.03; range across patients -5.95 to -2.04	4000	train; test agrees (AUC gain 0.062 [0.035, 0.095], n=1000)	yes	clearly
C113	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 80% of patients same; fallback; risk saturates near 0%*	1% to 3%	log-odds 1.19, OR 3.30; range across patients 0.00 to 2.96	4000	train; test agrees (AUC gain 0.061 [0.035, 0.091], n=1000)	yes	slightly
C114	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 84% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -1.46 to -0.88	4000	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C115	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.11 US dollars; 65% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 13.46 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to 1%	log-odds -1.19, OR 0.30; range across patients -3.06 to 0.00	4000	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C116	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	1% to 2%	log-odds 0.78, OR 2.19; range across patients 0.00 to 1.70	3934, 66	train; test disagrees (AUC gain -0.0007 [-0.002, 0.0001], n=1000)	yes	slightly; on the training data only
C117	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: South Dakota minus state other than South Dakota; stronger in state other than South Dakota *shape: dip at 160 minutes; 100% of patients same; verified; risk saturates near 0%*	South Dakota: 5% to 9%; state other than South Dakota: 1% to 13%	log-odds difference -1.26, OR 0.28	66, 3934	train; test agrees (AUC gain -0.0007 [-0.002, 0.0001], n=1000); group difference on test -1.38 [-2.09, -0.66]	yes	clearly
C118	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: South Dakota minus state other than South Dakota; stronger in South Dakota *shape: dip at 18.52 US*	South Dakota: 4% to 1%; state other than South Dakota: 1% to 2%	log-odds difference -0.90, OR 0.41	66, 3934	train; test unclear (AUC gain -0.0007 [-0.002, 0.0001], n=1000); group difference on test -0.60 [-1.23, 0.05]	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C121 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.34 US dollars, wide)*	3% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.62); past it log-odds -1.42, OR 0.24; range across patients -2.21 to -1.11	3851, 2752	train; test unclear (AUC gain 0.006 [-0.001, 0.014], n=1000)	yes	slightly; on the training data only
C122	threshold: total evening charge: fold at about 18.11 US dollars (stops) for customers in Colorado *shape: hinge at 18.11 US dollars; 75% of patients same; fallback; risk saturates near 0%; one point with C122 (median fold 18.11 US dollars, typical patient 17.01 US dollars, 10th-90th percentile 7.97 US dollars to 21.09 US dollars, wide)*	2% to <1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 14.6 to 18.7); past it log-odds -2.12, OR 0.12; range across patients -3.60 to 0.00	71, 34	train; test unclear (AUC gain 0.009 [-0.003, 0.020], n=1000)	yes	slightly; on the training data only
C123	subgroup: who this part acts on: 3121 of 4000 patients	<1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.78; log-odds 2.00, OR 7.39; range across patients 0.18 to 5.02	3121, 879	train; test agrees (AUC gain 0.141 [0.093, 0.190], n=1000)	yes	slightly
C124	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C125	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -0.64, OR 0.53; range across patients -0.64 to -0.64	2939, 1061	train; test unclear (AUC gain 0.002 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C126	direction: state: West Virginia to state other than Alabama, North Carolina, West Virginia (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 2%	log-odds 0.35, OR 1.42; range across patients 0.35 to 0.35	122, 3699	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C127	direction: state: West Virginia to Alabama (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	log-odds -0.10, OR 0.90; range across patients -0.10 to -0.10	122, 105	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C128	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: state other than Alabama, North Carolina, West Virginia, North Carolina, West Virginia, Alabama*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 74	train; test disagrees (AUC gain -0.0005 [-0.002, 0.001], n=1000)	yes	slightly; on the training data only
C129	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C130	direction: state: West Virginia to state other than West Virginia (through this part) *shape: categorical; highest to lowest: state other than West Virginia, West Virginia*	16% to 26% (typical patient this part acts on)	log-odds 0.08, OR 1.08; range across patients 0.00 to 0.00	122, 3878	train; test disagrees (AUC gain -0.001 [-0.003, 0.0002], n=1000)	yes	on the training data only
C131	threshold: customer-service calls: fold at about 4 calls (starts) *shape: hinge at 3 calls; 100% of patients same; verified; risk saturates near 0%; one point with C131 (median fold 4 calls, typical patient 3 calls, 10th-90th percentile 3 calls to 3 calls)*	1% to 1% (typical patient with a fold)	fold 3.02 calls (IQR 3.02 to 3.02); past it log-odds 0.58, OR 1.79; range across patients 0.60 to 0.60	4000, 306	train; test unclear (AUC gain 0.002 [-0.001, 0.006], n=1000)	yes	slightly; on the training data only
C132	subgroup: who this part acts on: 306 of 4000 patients	16% to 26% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 0.95, OR 2.58; range across patients 0.60 to 1.83	306, 3694	train; test unclear (AUC gain 0.001 [-0.002, 0.005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C133	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C134	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1%	log-odds -0.32, OR 0.73; range across patients -0.32 to -0.32	4000	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C135	threshold: total evening minutes: fold at about 239 minutes (starts) *shape: hinge at 239 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C135 (median fold 239 minutes, typical patient 236 minutes, 10th-90th percentile 239 minutes to 239 minutes)*	1% to 1% (typical patient with a fold)	fold 239 minutes (IQR 239 to 239); past it log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	4000, 885	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C136	subgroup: who this part acts on: 885 of 4000 patients	5% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.22; log-odds -0.35, OR 0.70; range across patients -0.77 to -0.05	885, 3115	train; test disagrees (AUC gain -0.004 [-0.009, 0.0002], n=1000)	yes	slightly; on the training data only
C137	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C138	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Arizona (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3508, 73	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C139	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Ohio (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds 0.21, OR 1.24; range across patients 0.21 to 0.21	3508, 87	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C140	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Nevada (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.26, OR 0.77; range across patients -0.26 to -0.26	3508, 77	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C141	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Missouri (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.04, OR 0.97; range across patients -0.04 to -0.04	3508, 80	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C142	direction: state: state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming to Massachusetts (through this part) *shape: categorical; highest to lowest: Arizona, Ohio, Wyoming, state other than Massachusetts, Missouri, Arizona, Nevada, Ohio, Wyoming, Massachusetts, Missouri …*	1% to 1%	log-odds -0.001, OR 1.00; range across patients -0.001 to -0.001	3508, 86	train; test disagrees (AUC gain -0.0003 [-0.001, 0.001], n=1000)	yes	slightly; on the training data only
C143	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C145	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	4000	train; test unclear (AUC gain 4e-05 [-0.0002, 0.0003], n=1000)	yes	slightly; on the training data only
C146	interaction: effect of total daytime calls (75 calls to 125 calls) by state: Tennessee minus state other than Tennessee; stronger in state other than Tennessee *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Tennessee: 2% to 3%; state other than Tennessee: 1% to 1%	log-odds difference -0.003, OR 1.00	68, 3932	train; test agrees (AUC gain -0.0006 [-0.001, -7e-05], n=1000); group difference on test -0.003 [-0.003, -0.003]	no	slightly
C147	threshold: total daytime calls: fold at about 114 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 113 calls (IQR 113 to 113); past it log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	3932, 2926	train; test unclear (AUC gain 4e-05 [-0.0002, 0.0003], n=1000)	no	slightly; on the training data only
C148	subgroup: who this part acts on: 2994 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.75; log-odds 0.01, OR 1.01; range across patients 0.001 to 0.01	2994, 1006	train; test disagrees (AUC gain -0.0005 [-0.001, 9e-06], n=1000)	no	slightly; on the training data only
C149	structure: its formula reproduces it on 4000 training patients (largest difference 5.55112e-17)		log-odds 6e-17, OR n/a	4000	train	yes	
C150	direction: state: state other than Arkansas, New Mexico, Utah to Arkansas (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.05, OR 1.06; range across patients 0.03 to 0.08	3761, 78	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only
C151	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.001, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000)	yes	slightly; on the training data only
C152	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	yes	slightly; on the training data only
C153	direction: state: state other than Arkansas, New Mexico, Utah to Utah (through this part) *shape: categorical; highest to lowest: Arkansas, Utah, New Mexico, state other than Arkansas, New Mexico, Utah*	1% to 1%	log-odds 0.009, OR 1.01; range across patients 0.00 to 0.03	3761, 89	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000)	yes	slightly; on the training data only
C154	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in Arkansas *shape: accelerating; 82% of patients same; fallback; risk saturates near 0%*	Arkansas: 2% to 4%; state other than Arkansas, New Mexico, Utah: 1% to 3%	log-odds difference 0.28, OR 1.32	78, 3761	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000); group difference on test -0.27 [-0.85, 0.37]	no	slightly; on the training data only
C155	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Arkansas minus state other than Arkansas, New Mexico, Utah; stronger in state other than Arkansas, New Mexico, Utah *shape: accelerating; 91% of patients same; verified; risk saturates near 0%*	Arkansas: 1% to 3%; state other than Arkansas, New Mexico, Utah: 1% to 2%	log-odds difference -0.07, OR 0.93	78, 3761	train; test disagrees (AUC gain -5e-05 [-0.0002, 0.0001], n=1000); group difference on test 0.01 [-0.12, 0.14]	no	slightly; on the training data only
C156	interaction: effect of customer-service calls (0 calls to 3 calls) by total international charge: total international charge above 2.78 US dollars minus total international charge at or below 2.78 US dollars; stronger in total international charge at or below 2.78 US dollars *shape: accelerating; 78% of patients same; fallback; risk saturates near 0%*	total international charge above 2.78 US dollars: 2% to 3%; total international charge at or below 2.78 US dollars: 1% to 3%	log-odds difference -0.06, OR 0.94	1999, 2001	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000); group difference on test -0.12 [-0.26, 0.03]	no	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C157	threshold: customer-service calls: fold at about 5 calls (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	2% to 2% (typical patient with a fold)	fold 4.49 calls (IQR 3.72 to 5.40); past it log-odds -0.01, OR 0.99; range across patients -0.02 to -0.004	66, 63	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	no	slightly; on the training data only
C158	threshold: total international charge: fold at about 0.43 US dollars (stops) *shape: flat; 97% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 0.43 US dollars (IQR 0.34 to 0.43); past it log-odds -0.002, OR 1.00; range across patients 0.00 to 0.00	3890, 153	train; test unclear (AUC gain 2e-05 [-6e-05, 0.0001], n=1000)	no	slightly; on the training data only
C159	subgroup: who this part acts on: 153 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.04; log-odds 0.04, OR 1.04; range across patients 0.005 to 0.07	153, 3847	train; test disagrees (AUC gain -2e-05 [-0.0002, 0.0002], n=1000)	no	slightly; on the training data only
C160	null: this part barely changes predictions (switched off)		max change 0.020 (max |change in predicted risk|)	4000	train	no	
C161	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	4000	train	yes	
C162	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C163	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C164	boundary: boundary		log-odds 1.00, OR n/a	4000	train	yes	

Not supported by the model.

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (17% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C041: direction claim about state in the whole model: not supported by the model (45% of patients have the opposite sign (allowed: 10%)).
- C044: direction claim about state in the whole model: not supported by the model (92% of patients have the opposite sign (allowed: 10%)).
- C146: interaction claim about total daytime calls and state in part ann_97: not supported by the model (the group difference is below 0.1 log-odds).
- C147: threshold claim about total daytime calls in part ann_97: not supported by the model (the fold changes the curve by only 0.014 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C148: subgroup claim about state and total daytime calls in part ann_97: not supported by the model (its effect is negligible).
- C154: interaction claim about number of calls to customer service and state in part ann_92: not supported by the model (the part's own counterfactual difference has the opposite sign).
- C155: interaction claim about total international charge and state in part ann_92: not supported by the model (the group difference is below 0.1 log-odds).
- C156: interaction claim about number of calls to customer service and total international charge in part ann_92: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C157: threshold claim about number of calls to customer service in part ann_92: not supported by the model (the fold changes the curve by only 0.015 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C158: threshold claim about total international charge in part ann_92: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C159: subgroup claim about state and total international charge in part ann_92: not supported by the model (its effect is negligible).
- C160: null claim about state and total international charge in part ann_92: not supported by the model (predictions move by up to 0.020).

- C162: boundary claim about total daytime minutes and number of calls to customer service in part ann_101: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C163: boundary claim about total daytime minutes and number of calls to customer service in part ann_91: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (24):** `-10` (state:17), `-11` (state:18), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-19` (state:25), `-26` (state:31), `-29` (state:34), `-3` (state:10), `-34` (state:39), `-36` (state:40), `-39` (state:43), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-51` (state:9), `-6` (state:13), `-61` (totalDayCharge), `-63` (totalEveCalls), `-67` (totalNightCharge), `-69` (totalIntlCalls), `-7` (state:14), `-8` (state:15), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)
- ▷ A2: intl_state_unit (leaf, section 1.6.2)
- ▷ A3: eve_minutes_hinge (leaf, section 1.6.3)
- ▷ A4: voicemail_chain (leaf, section 1.6.4)
- ▷ A5: state_offsets (leaf, section 1.6.5)
- ▷ A6: day_calls_state42 (leaf, section 1.6.6)
- ▷ A7: linear_backbone (leaf, section 1.6.7)
- ▷ A8: service_call_nonlinear (composition, section 1.6.8)
  - ▷ A9: csc_usage_gate (leaf, section 1.6.9)
  - ▷ A10: csc_hinge (leaf, section 1.6.10)

1.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s2-c1-fig-full.tex`.

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 6.66×10^{-16} (tolerance 1.00×10^{-9}).

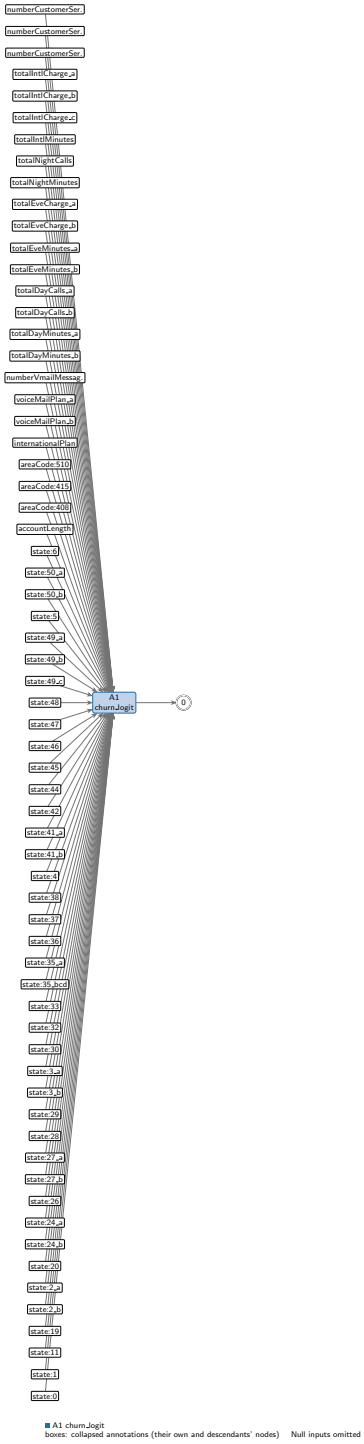


Figure 1: churn100-s2-c1: collapsed view at hierarchy level 0 (fully recomposed).

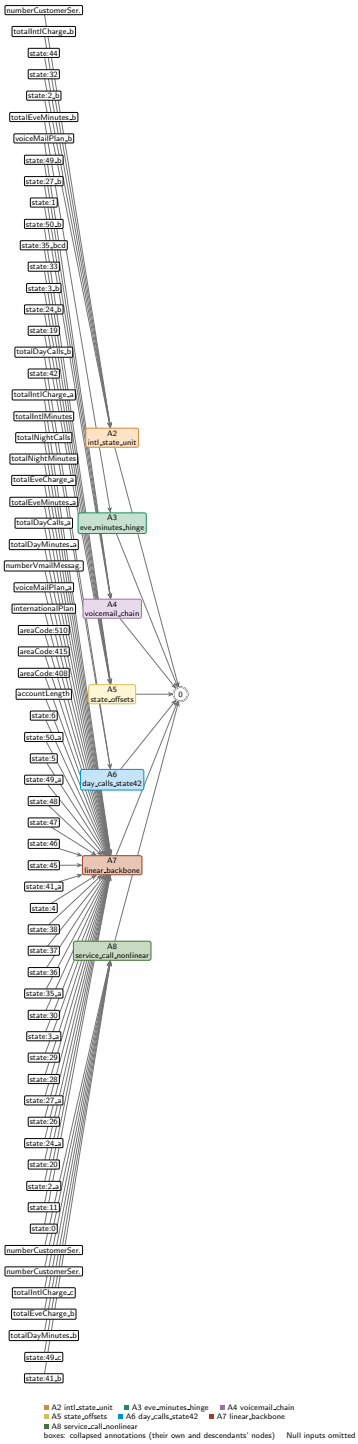


Figure 2: churn100-s2-c1: collapsed view at hierarchy level 1.

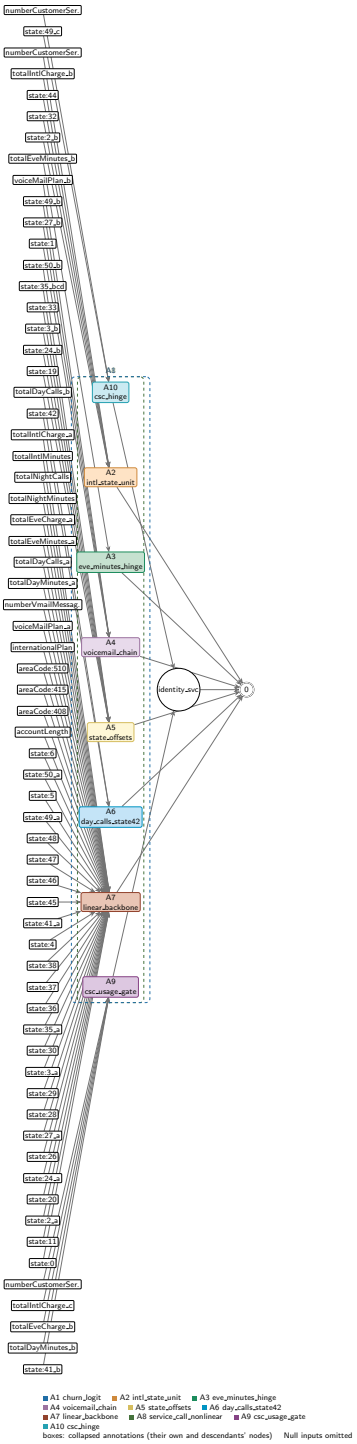


Figure 3: churn100-s2-c1: collapsed view at hierarchy level 2.

- #71: split 2627 into 2627_a, 2627_b.
- #72: split 9555 into 9555_a, 9555_b.
- #73: split -71 into -71_a, -71_b, -71_c, -71_d, -71_e, -71_f, -71_g, -71_h.
- #74: split -70 into -70_a, -70_b, -70_c.
- #75: split -64 into -64_a, -64_b.
- #76: split -62 into -62_a, -62_b.
- #77: split -60 into -60_a, -60_b.
- #78: split -59 into -59_a, -59_b.
- #79: split -57 into -57_a, -57_b.
- #80: split -47 into -47_a, -47_b.
- #81: split -45 into -45_a, -45_b, -45_c.
- #82: split -37 into -37_a, -37_b.
- #83: split -30 into -30_a, -30_b, -30_c, -30_d.
- #84: split -24 into -24_a, -24_b.
- #85: split -21 into -21_a, -21_b.
- #86: split -18 into -18_a, -18_b.
- #87: split -13 into -13_a, -13_b.
- #88: consolidate -71_a, -71_b, -71_c, -71_f, -71_h.
- #89: consolidate -71_e, -71_g.
- #90: consolidate -30_b, -30_c, -30_d.
- #98: identity node identity_lin before 0 (40 connection(s)).
- #100: identity node identity_svc before 0 (3 connection(s)).

1.6 Annotations

1.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6), A7 (section 1.6.7), A8 (section 1.6.8).

- **Entry nodes:** -69 (totalIntlCalls), -68 (totalIntlMinutes), -67 (totalNightCharge), -66 (totalNightCalls), -65 (totalNightMinutes), -63 (totalEveCalls), -61 (totalDayCharge), -58 (numberVmailMessages), -56 (internationalPlan), -55 (areaCode:510), -54 (areaCode:415), -53 (areaCode:408), -52 (accountLength), -51 (state:9), -50 (state:8), -49 (state:7), -48 (state:6), -46 (state:5), -44 (state:48), -43 (state:47), -42 (state:46), -41 (state:45), -40 (state:44), -39 (state:43), -38 (state:42), -36 (state:40), -35 (state:4), -34 (state:39), -33 (state:38), -32 (state:37), -31 (state:36), -29 (state:34), -28 (state:33), -27 (state:32), -26 (state:31), -25 (state:30), -23 (state:29), -22 (state:28), -20 (state:26), -19 (state:25), -17 (state:23), -16 (state:22), -15 (state:21), -14 (state:20), -12 (state:19), -11 (state:18), -10 (state:17), -9 (state:16), -8 (state:15), -7 (state:14), -6 (state:13), -5 (state:12), -4 (state:11), -3 (state:10), -2 (state:1), -1 (state:0), -71_d (numberCustomerServiceCalls_d), -70_a (totalIntlCharge_a), -70_b (totalIntlCharge_b), -70_c (totalIntlCharge_c), ... (29 more; see the JSON record).

- **Exit nodes:** 0.
- **Internal nodes:** none.
- **Edges (weight):** 252→0 (0.0205), 476→0 (-0.55), 1133→0 (-0.95), 3861→0 (-0.15), 4051→0 (-0.994), 8896→0 (0.0326), 10249→0 (0.04), 11427→0 (0.0502), 13827→0 (0.432), 14942→0 (0.0416), identity_lin→0 (1), identity_svc→0 (1).
- **With its descendants:** 116 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 115 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.05 + linear_backbone + service_call_nonlinear (identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208) - 0.95*1133 - 0.99*4051 (voicemail_chain) - 0.55*476 (eve_minutes_hinge) + state_offsets + 0.02*252 + 0.05*11427, purely additive in the logit; the sigmoid is the only coupling. 24 inputs are unconnected (total intl calls, night charge, evening calls, day charge and 20 state indicators) and have zero effect. Train AUC 0.901, accuracy 0.901. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.837 (an L2 logistic regression refit on all 71 inputs gets 0.843); restoring csc_usage_gate alone gives 0.898, csc_hinge alone 0.851, eve_minutes_hinge alone 0.834, the others 0.837 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11/46/5 offsets of -2.24/-1.89/-1.46 logit on 73-80 customers (empirical -1.35/-0.91/-0.51); state:4 +1.35 (41 rows, empirical +0.80); state:47 +1.16 (72 rows, +0.92); state:49 -0.42 offset, down to -2.87 for 12 customers with 4+ calls (turns off the 4+-calls hinge); state:35 +0.40 (empirical +0.06).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.05 + 1.0\,\text{linear_backbone}\big(x_{state:0}, x_{state:11}, \\
&\quad x_{state:20}, x_{state:26}, x_{state:28}, x_{state:29}, x_{state:30}, x_{state:36}, \\
&\quad x_{state:37}, x_{state:38}, x_{state:4}, x_{state:45}, x_{state:46}, x_{state:47}, \\
&\quad x_{state:48}, x_{state:5}, x_{state:6}, x_{accountLength}, x_{areaCode:408}, x_{areaCode:415}, \\
&\quad x_{areaCode:510}, x_{internationalPlan}, x_{numberVmailMessages}, x_{totalNightMinutes}, \\
&\quad x_{totalNightCalls}, x_{totalIntlMinutes}, x_{totalIntlCharge_a}, x_{totalEveCharge_a}, \\
&\quad x_{totalEveMinutes_a}, x_{totalDayCalls_a}, x_{totalDayMinutes_a}, x_{voiceMailPlan_a}, \\
&\quad x_{state:50_a}, x_{state:49_a}, x_{state:41_a}, x_{state:35_a}, x_{state:3_a}, \\
&\quad x_{state:27_a}, x_{state:24_a}, x_{state:2_a}\big) \\
&\quad + 1.0\,\text{service_call_nonlinear}\big(x_{numberCustomerServiceCalls_abcfh}, \\
&\quad x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}, \\
&\quad x_{numberCustomerServiceCalls_d}, x_{state:49_c}\big) \\
&\quad + 0.04\,\text{state_offsets_1}\big(x_{state:19}, x_{state:3_b}, x_{state:35_bcd}, \\
&\quad x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0416\,\text{state_offsets_2}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0326\,\text{state_offsets_0}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad + 0.0205\,\text{intl_state_unit}\big(x_{numberCustomerServiceCalls_eg}, \\
&\quad x_{state:32}, x_{state:2_b}, x_{state:44}, x_{totalIntlCharge_b}\big) \\
&\quad + 0.0502\,\text{day_calls_state42}\big(x_{state:42}, x_{totalDayCalls_b}\big) \\
&\quad + 0.432\,\text{state_offsets_3}\big(x_{state:19}, x_{state:3_b}, \\
&\quad x_{state:35_bcd}, x_{state:33}, x_{state:24_b}, x_{state:50_b}\big) \\
&\quad - 0.55\,\text{eve_minutes_hinge}\big(x_{totalEveMinutes_b}\big)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.15\,\text{state_offsets_4}\big(x_{state:19}, x_{state:3_b}, x_{state:35_bcd},\\
&x_{state:33}, x_{state:24_b}, x_{state:50_b}\big)\\
&- 0.95\,\text{voicemail_chain_0}\big(x_{voiceMailPlan_b}, x_{state:1},\\
&x_{state:27_b}, x_{state:49_b}\big)\\
&- 0.994\,\text{voicemail_chain_1}\big(x_{voiceMailPlan_b}, x_{state:1},\\
&x_{state:27_b}, x_{state:49_b}\big)
\end{aligned}
$$

The fully expanded form is in the JSON record. where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00582, and the exact function by at most 6.66×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.05 + linear_backbone + service_call_nonlinear (1.60*807 + 1.81*1264 + 1.48*7208) - 0.95*1133 - 0.99*4051 (voicemail_chain) - 0.55*476 (eve_minutes_hinge) + state_offsets + 0.02*252 + 0.05*11427, purely additive in the logit; the sigmoid is the only coupling. 24 inputs are unconnected (total intl calls, night charge, evening calls, day charge and 20 state indicators) and h

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1.6.2 A2: intl_state_unit

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-71_eg` (numberCustomerServiceCalls_eg), `-27` (state:32), `-13_b` (state:2_b), `-40` (state:44), `-70_b` (totalIntlCharge_b).
- **Exit nodes:** `252`.
- **Internal nodes:** `9555_a`, `2627_a`, `161`.
- **Edges (weight):** `-27`→`161` (-0.0113), `-40`→`252` (0.437), `161`→`252` (0.791), `2627_a`→`252` (-0.378), `9555_a`→`2627_a` (0.325), `-70_b`→`252` (-0.869), `-13_b`→`252` (0.76), `-71_eg`→`2627_a` (0.278), `-71_eg`→`9555_a` (0.855).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

Hypothesis. Near-dead unit 252 = relu(-2.22 + 0.76 state:2 + 0.44 state:44 - 0.87 intlCharge + 0.79*161 - 0.38 g(csc)), with 161 = relu(-0.14 - 0.011 state:32) identically 0 on train (dead). 252 is active on 153 of 4000 training rows (78 in state:2, 55 in state:44, mostly with low international charge); its logit contribution is at most +0.10 (weight 0.02). Hypothesis: a negligible state-specific low-international-charge offset. Evidence: mean ablation and linear replacement leave train AUC at 0.9009; restoring it onto the all-linear model changes AUC by -0.0001.

Closed form:

$$
\begin{aligned}
y_0 = \text{relu}\big(&-2.22 + 0.76 x_{state:2_b}\\
&+ 0.791\,\text{relu}\big(-0.14 - 0.0113 x_{state:32}\big) + 0.437 x_{state:44}\\
&- 0.378\,\text{relu}\big(1.01 + 0.325\,\text{relu}\big(-0.0966\\
&+ 0.855 x_{numberCustomerServiceCalls_eg}\big)\\
&+ 0.278 x_{numberCustomerServiceCalls_eg}\big) - 0.869 x_{totalIntlCharge_b}\big)
\end{aligned}
$$

where y_0 is node `252`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00409, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Near-dead unit 252 = relu(-2.22 + 0.76 state:2 + 0.44 state:44 - 0.87 intlCharge + 0.79*161 - 0.38 g(csc)), with 161 = relu(-0.14 - 0.011 state:32) identically 0 on train (dead). 252 is active on 153 of 4000 training rows (78 in state:2, 55 in state:44, mostly with low international charge); its logit contribution is at most +0.10 (weight 0.02). Hypothesis: a negligible state-specific low-internat

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: eve_minutes_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-62_b` (totalEveMinutes_b).
- **Exit nodes:** `476`.
- **Internal nodes:** none.
- **Edges (weight):** `-62_b`→`476` (1.07).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 476 = relu(1.07 eveMinutes - 0.81), logit weight -0.55. Hypothesis: above about 239 evening minutes (885 of 4000 training customers) this hinge flattens the direct evening-minutes slope (+0.51 per SD direct, -0.59 per SD through the hinge), so evening minutes stop adding risk above about 239 minutes (evening charge, +0.45 per SD, still adds). Evidence: line plot of 476 on train from 0 to 2.7; mean ablation AUC 0.9002, linear replacement 0.9005; restoring it alone onto the all-linear model lowers AUC (0.834 vs 0.837): a small fit-to-train detail.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.805 + 1.07 x_{totalEveMinutes_b}\big)$$

where y_0 is node `476`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00493, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 476 = relu(1.07 eveMinutes - 0.81), logit weight -0.55. Hypothesis: above about 239 evening minutes (885 of 4000 training customers) this hinge flattens the direct evening-minutes slope (+0.51 per SD direct, -0.59 per SD through the hinge), so evening minutes stop adding risk above about 239 minutes (evening charge, +0.45 per SD, still adds). Evidence: line plot of 476 on train from 0 to 2.7; mean

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1.6.4 A4: voicemail_chain

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-57_b` (voiceMailPlan_b), `-2` (state:1), `-21_b` (state:27_b), `-45_b` (state:49_b).
- **Exit nodes:** `1133`, `4051`.
- **Internal nodes:** `5577`.

- **Edges (weight):** `-2`→`5577` (0.255), `1133`→`4051` (1.15), `5577`→`1133` (0.137), `-57_b`→`5577` (0.991), `-45_b`→`4051` (0.0611), `-21_b`→`1133` (0.0189).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 5577 = relu(0.99 + 0.99 voiceMailPlan + 0.26 state:1), 1133 = relu(0.68 + 0.14*5577 + 0.02 state:27), 4051 = relu(2.09 + 1.15*1133 + 0.06 state:49). All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0). Hypothesis: a linear voicemail-plan term (net logit -0.95*1133 - 0.99*4051 = about -0.28 per SD of voicemail plan on top of the direct -0.90) plus small state:1/27/49 offsets. Evidence: exact linear fit; linear replacement AUC 0.9009 (unchanged).

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.68 + 0.137\operatorname{relu}\big(0.991 \\
&\qquad + 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) + 0.0189x_{state:27_b}\big) \\
y_1 &= \operatorname{relu}\big(2.09 + 1.15\operatorname{relu}\big(0.68 \\
&\qquad + 0.137\operatorname{relu}\big(0.991 + 0.255x_{state:1} + 0.991x_{voiceMailPlan_b}\big) \\
&\qquad + 0.0189x_{state:27_b}\big) + 0.0611x_{state:49_b}\big)
\end{aligned}
$$

where y_0 is node `1133`; y_1 is node `4051`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.01, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 5577 = relu(0.99 + 0.99 voiceMailPlan + 0.26 state:1), 1133 = relu(0.68 + 0.14*5577 + 0.02 state:27), 4051 = relu(2.09 + 1.15*1133 + 0.06 state:49). All three are always active on train (minimum pre-activations 0.35, 0.73, 2.91), so the chain is exactly affine in its binary inputs (least-squares R2 = 1.0). Hypothesis: a linear voicemail-plan term (net logit -0.95*1133 - 0.99*4051 = about -0.29 per

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-12` (state:19), `-24_b` (state:3_b), `-30_bcd` (state:35_bcd), `-28` (state:33), `-18_b` (state:24_b), `-47_b` (state:50_b).
- **Exit nodes:** `8896`, `10249`, `14942`, `13827`, `3861`.
- **Internal nodes:** `11951`, `6730`.
- **Edges (weight):** `-12`→`8896` (-1), `-28`→`6730` (-0.418), `11951`→`6730` (-9.39×10^{-4}), `6730`→`13827` (1.01), `-47_b`→`3861` (-0.771), `-24_b`→`10249` (0.998), `-18_b`→`6730` (-0.0113), `-30_bcd`→`6730` (0.00244), `-30_bcd`→`11951` (1), `-30_bcd`→`14942` (1).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

Hypothesis. Units 8896 (state:19), 10249 (state:3), 11951 and 14942 (state:35), 6730 -> 13827 (state:33, state:24, state:35), 3861 (state:50, identically 0 on train). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest

offsets here are state:33 about -0.25 (switches off 6730 -> 13827), state:35 +0.21 (14942), state:3 +0.30 (10249). Evidence: linear replacement AUC 0.9009 (unchanged); mean ablation 0.9006.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.106 - 1.0 x_{state:19}\big) \\
y_1 &= \mathrm{relu}\big(0.172 + 0.998 x_{state:3_b}\big) \\
y_2 &= \mathrm{relu}\big(-1.74 + 1.0 x_{state:35_bcd}\big) \\
y_3 &= \mathrm{relu}\big(-0.737 + 1.01\,\mathrm{relu}\big(1.26 \\
&\qquad + 0.00244 x_{state:35_bcd} - 0.0113 x_{state:24_b} - 0.000939\,\mathrm{relu}\big(-1.09 \\
&\qquad + 1.0 x_{state:35_bcd}\big) - 0.418 x_{state:33}\big)\big) \\
y_4 &= \mathrm{relu}\big(-0.431 - 0.771 x_{state:50_b}\big)
\end{aligned}
$$

where y_0 is node `8896`; y_1 is node `10249`; y_2 is node `14942`; y_3 is node `13827`; y_4 is node `3861`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0312, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Units 8896 (state:19), 10249 (state:3), 11951 and 14942 (state:35), 6730 -> 13827 (state:33, state:24, state:35), 3861 (state:50, identically 0 on train). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets h

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.6 A6: day_calls_state42

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-38` (state:42), `-60_b` (totalDayCalls_b).
- **Exit nodes:** `11427`.
- **Internal nodes:** none.
- **Edges (weight):** `-38`→`11427` (0.995), `-60_b`→`11427` (-0.105).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 11427 = relu(0.20 + 1.00 state:42 - 0.11 dayCalls), logit weight +0.05. Hypothesis: a tiny hinge that is on for up to about 110 day calls and switches off above about 120 (on for 100% of rows below 110 day calls, 44% at 110-119, 2% at 120+), plus a state:42 offset; contribution at most +0.40 logit. Evidence: linear fit R2 0.999; mean ablation and linear replacement leave AUC at 0.9009.

Closed form:

$$
y_0 = \mathrm{relu}\big(0.198 + 0.995 x_{state:42} - 0.105 x_{totalDayCalls_b}\big)
$$

where y_0 is node `11427`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0035, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).

- EMPIRICAL, `ablation_study`; data split: train. 11427 = relu(0.20 + 1.00 state:42 - 0.11 dayCalls), logit weight +0.05. Hypothesis: a tiny hinge that is on for up to about 110 day calls and switches off above about 120 (on for 100% of rows below 110 day calls, 44% at 110-119, 2% at 120+), plus a state:42 offset; contribution at most +0.40 logit. Evidence: linear fit R2 0.999; mean ablation and linear replacement leave AUC at 0.9009.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.7 A7: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-1` (state:0), `-4` (state:11), `-14` (state:20), `-20` (state:26), `-22` (state:28), `-23` (state:29), `-25` (state:30), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-35` (state:4), `-41` (state:45), `-42` (state:46), `-43` (state:47), `-44` (state:48), `-46` (state:5), `-48` (state:6), `-52` (accountLength), `-53` (areaCode:408), `-54` (areaCode:415), `-55` (areaCode:510), `-56` (internationalPlan), `-58` (numberVmailMessages), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-68` (totalIntlMinutes), `-70_a` (totalIntlCharge_a), `-64_a` (totalEveCharge_a), `-62_a` (totalEveMinutes_a), `-60_a` (totalDayCalls_a), `-59_a` (totalDayMinutes_a), `-57_a` (voiceMailPlan_a), `-47_a` (state:50_a), `-45_a` (state:49_a), `-37_a` (state:41_a), `-30_a` (state:35_a), `-24_a` (state:3_a), `-21_a` (state:27_a), `-18_a` (state:24_a), `-13_a` (state:2_a).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`identity_lin` (-0.104), `-4`→`identity_lin` (-0.3), `-14`→`identity_lin` (0.0772), `-20`→`identity_lin` (0.0915), `-22`→`identity_lin` (-0.0124), `-23`→`identity_lin` (-0.0533), `-25`→`identity_lin` (-0.0496), `-31`→`identity_lin` (-0.0184), `-32`→`identity_lin` (0.0293), `-33`→`identity_lin` (-0.105), `-35`→`identity_lin` (0.135), `-41`→`identity_lin` (-0.115), `-42`→`identity_lin` (-0.265), `-43`→`identity_lin` (0.154), `-44`→`identity_lin` (-0.0899), `-46`→`identity_lin` (-0.201), `-48`→`identity_lin` (0.0346), `-52`→`identity_lin` (0.138), `-53`→`identity_lin` (-0.0272), `-54`→`identity_lin` (0.0454), `-55`→`identity_lin` (-0.0161), `-56`→`identity_lin` (0.756), `-58`→`identity_lin` (0.623), `-65`→`identity_lin` (0.307), `-66`→`identity_lin` (-0.00711), `-68`→`identity_lin` (-0.0175), `-70_a`→`identity_lin` (0.705), `-64_a`→`identity_lin` (0.45), `-62_a`→`identity_lin` (0.511), `-60_a`→`identity_lin` (0.127), `-59_a`→`identity_lin` (2.04), `-57_a`→`identity_lin` (-0.899), `-47_a`→`identity_lin` (-0.112), `-45_a`→`identity_lin` (-0.0115), `-37_a`→`identity_lin` (-0.101), `-30_a`→`identity_lin` (0.0276), `-24_a`→`identity_lin` (-0.0222), `-21_a`→`identity_lin` (0.0267), `-18_a`→`identity_lin` (-0.074), `-13_a`→`identity_lin` (-0.0117).
- **Covers** (coverage, own and as part of the hierarchy): 40 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (identity). Exactly linear in 40 inputs: day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, evening minutes +0.51, evening charge +0.45, night minutes +0.31, state:11 -0.30, state:46 -0.27, state:5 -0.20, state:47 +0.15, account length +0.14, state:4 +0.14, day calls +0.13, and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes and voicemail/international plan dominate. Evidence: closed form is exact (identity); sensitivities match the weights.

Closed form:

$$\begin{aligned} y_0 = {} & 0.154x_{state:47} + 0.135x_{state:4} + 0.623x_{numberVmailMessages} \\ & + 0.756x_{internationalPlan} + 0.0267x_{state:27_a} + 0.127x_{totalDayCalls_a} \\ & + 2.04x_{totalDayMinutes_a} + 0.0346x_{state:6} + 0.307x_{totalNightMinutes} \\ & + 0.0772x_{state:20} + 0.0915x_{state:26} + 0.45x_{totalEveCharge_a} \end{aligned}$$

$$
\begin{aligned}
&+ 0.511x_{totalEveMinutes_a} + 0.138x_{accountLength} + 0.705x_{totalIntlCharge_a} \\
&+ 0.0454x_{areaCode:415} + 0.0276x_{state:35_a} + 0.0293x_{state:37} \\
&- 0.074x_{state:24_a} - 0.0272x_{areaCode:408} - 0.0115x_{state:49_a} \\
&- 0.0496x_{state:30} - 0.265x_{state:46} - 0.0184x_{state:36} - 0.0899x_{state:48} \\
&- 0.0124x_{state:28} - 0.201x_{state:5} - 0.112x_{state:50_a} - 0.0161x_{areaCode:510} \\
&- 0.0175x_{totalIntlMinutes} - 0.3x_{state:11} - 0.0223x_{state:3_a} \\
&- 0.0117x_{state:2_a} - 0.101x_{state:41_a} - 0.104x_{state:0} - 0.105x_{state:38} \\
&- 0.0533x_{state:29} - 0.899x_{voiceMailPlan_a} - 0.00711x_{totalNightCalls} \\
&- 0.115x_{state:45}
\end{aligned}
$$

where y_0 is node `identity_lin`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0152, and the exact function by at most 2.66×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `empirical_viz`; data split: train. Direct input-to-output edges collected through identity_lin (identity). Exactly linear in 40 inputs: day minutes +2.04 per SD, voicemail plan -0.90, international plan +0.76, international charge +0.71, voicemail messages +0.62, evening minutes +0.51, evening charge +0.45, night minutes +0.31, state:11 -0.30, state:46 -0.27, state:5 -0.20, state:47 +0.15, account length +0.14, state:4 +0.14, day c

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.8 A8: service_call_nonlinear

Kind: Composition annotation (depth 1); parent A1 (section 1.6.1); children A9 (section 1.6.9), A10 (section 1.6.10).

- **Entry nodes:** `-71_abcfh` (numberCustomerServiceCalls_abcfh), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-59_b` (totalDayMinutes_b), `-70_c` (totalIntlCharge_c), `-71_d` (numberCustomerServiceCalls_d), `-45_c` (state:49_c).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `807`→`identity_svc` (1.6), `1264`→`identity_svc` (1.8), `7208`→`identity_svc` (1.48).
- **With its descendants:** 16 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 15 candidate nodes.

Hypothesis. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.60*807 + 1.81*1264 + 1.48*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.736, linear replacement of both 0.834 (all-linear model 0.837); the gate alone restores 0.898 of 0.901.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 = 1.8\,\mathrm{csc_usage_gate_1}\big(&x_{numberCustomerServiceCalls_abcfh}, \\
&x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 1.6\,\mathrm{csc_usage_gate_0}\big(x_{numberCustomerServiceCalls_abcfh},\\
&x_{totalEveCharge_b}, x_{state:41_b}, x_{totalDayMinutes_b}, x_{totalIntlCharge_c}\big)\\
&+ 1.48\,\mathrm{csc_hinge}\big(x_{numberCustomerServiceCalls_d}, x_{state:49_c}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = 1.8\,\mathrm{relu}\big(&0.0759 + 0.363\,\mathrm{relu}\big(-1.29\\
&+ 0.622\,\mathrm{relu}\big(0.416 + 0.154x_{state:41_b}\\
&+ 0.342\,\mathrm{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big)\\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\,\mathrm{relu}\big(1.01\\
&+ 0.325\,\mathrm{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big)\\
&- 0.326x_{totalIntlCharge_c}\big) + 1.6\,\mathrm{relu}\big(-1.29\\
&+ 0.622\,\mathrm{relu}\big(0.416 + 0.154x_{state:41_b}\\
&+ 0.342\,\mathrm{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big)\\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\,\mathrm{relu}\big(1.01\\
&+ 0.325\,\mathrm{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big)\\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big)\\
&+ 1.48\,\mathrm{relu}\big(-0.673 + 0.544\,\mathrm{relu}\big(0.00784\\
&+ 1.0x_{numberCustomerServiceCalls_d}\big) - 0.359x_{state:49_c}\big)
\end{aligned}
$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0274, and the exact function by at most 3.55×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_usage_gate and csc_hinge: their exits enter the logit additively (+1.60*807 + 1.81*1264 + 1.48*7208); they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating b

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.9 A9: csc_usage_gate

Kind: Leaf annotation (depth 2); parent A8 (section 1.6.8).

- **Entry nodes:** `-71_abcfh` (numberCustomerServiceCalls_abcfh), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-59_b` (totalDayMinutes_b), `-70_c` (totalIntlCharge_c).
- **Exit nodes:** `807`, `1264`.
- **Internal nodes:** `9555_b`, `2627_b`, `12926`, `2796`.
- **Edges (weight):** `2796`→`807` (0.622), `807`→`1264` (0.363), `12926`→`2796` (0.342), `2627_b`→`807` (0.675), `9555_b`→`2627_b` (0.325), `-70_c`→`1264` (-0.326), `-64_b`→`2796` (-1), `-59_b`→`807` (-1.4), `-37_b`→`2796` (0.154), `-71_abcfh`→`807` (0.0405), `-71_abcfh`→`2796` (0.36), `-71_abcfh`→`12926` (0.987), `-71_abcfh`→`2627_b` (0.278), `-71_abcfh`→`9555_b` (0.855).

- **Covers** (coverage, own and as part of the hierarchy): 9 candidate nodes.

Hypothesis. Service-call x low-day-usage gate. 807 = relu(-1.29 - 1.40 dayMin + 0.04 csc + 0.675 g(csc) + 0.622 relu(0.42 - 1.00 eveCharge + 0.36 csc + 0.34 relu(0.99 csc - 0.65) + 0.15 state:41)), with g(csc) = relu(1.01 + 0.28 csc + 0.33 relu(0.86 csc - 0.10)) always active (convex in csc, kink near 2 calls); 1264 = relu(0.08 + 0.36*807 - 0.33 intlCharge). The logit gets +1.60*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 159 at 1, 174 at 2, 198 at 3, 220 at 4, 244 at 5, 276 at 6); lower evening charge also raises the kink. Below the kink the gate's slope is -3.15 logit per SD of day minutes, which more than cancels the direct +2.04, so the logit is V-shaped in day minutes (net -1.12 per SD below the kink). 1264 adds a second hinge on low international charge that is active on 1281 rows where 807 is off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.885, model mean 0.655; CSC<4 and day minutes < 120: 476 customers, empirical 0.046, model 0.084. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.098/0.092/0.099/0.147/0.333/0.645/0.923; share of rows with 807 on rises 0.32 -> 0.96. Sensitivity: 807 mostly day minutes (0.64), service calls (0.26), evening charge (0.22). Ablation: mean-ablation AUC 0.764 (full 0.901); least-squares linear replacement 0.848 (R2 0.70/0.75); restored alone onto an all-linearised model 0.898 (all-linear 0.837). This is the model's main nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.29 + 0.622\operatorname{relu}\big(0.416 \\
&+ 0.154x_{state:41_b} + 0.342\operatorname{relu}\big(-0.649 \\
&+ 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0759 + 0.363\operatorname{relu}\big(-1.29 \\
&+ 0.622\operatorname{relu}\big(0.416 + 0.154x_{state:41_b} \\
&+ 0.342\operatorname{relu}\big(-0.649 + 0.987x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.36x_{numberCustomerServiceCalls_abcfh} - 1.0x_{totalEveCharge_b}\big) \\
&+ 0.0405x_{numberCustomerServiceCalls_abcfh} + 0.675\operatorname{relu}\big(1.01 \\
&+ 0.325\operatorname{relu}\big(-0.0966 + 0.855x_{numberCustomerServiceCalls_abcfh}\big) \\
&+ 0.278x_{numberCustomerServiceCalls_abcfh}\big) - 1.4x_{totalDayMinutes_b}\big) \\
&- 0.326x_{totalIntlCharge_c}\big)
\end{aligned}
$$

where y_0 is node `807`; y_1 is node `1264`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0119, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Service-call x low-day-usage gate. 807 = relu(-1.29 - 1.40 dayMin + 0.04 csc + 0.675 g(csc) + 0.622 relu(0.42 - 1.00 eveCharge + 0.36 csc + 0.34 relu(0.99 csc - 0.65) + 0.15 state:41)), with g(csc) = relu(1.01 + 0.28 csc + 0.33 relu(0.86 csc - 0.10)) always active (convex in csc, kink near 2 calls); 1264 = relu(0.08 + 0.36*807 - 0.33 intlCharge). The logit gets +1.60*807 + 1.81*1264. Hypothesis: 8

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.10 A10: csc_hinge

Kind: Leaf annotation (depth 2); parent A8 (section 1.6.8).

- **Entry nodes:** `-71_d` (numberCustomerServiceCalls_d), `-45_c` (state:49_c).
- **Exit nodes:** `7208`.
- **Internal nodes:** `17032`.
- **Edges (weight):** `17032`→`7208` (0.544), `-71_d`→`17032` (1), `-45_c`→`7208` (-0.359).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 7208 = relu(-0.67 + 0.54 relu(csc + 0.008) - 0.36 state:49), logit weight +1.48. Hypothesis: a hinge on customer-service calls that switches on from 4 calls upward (kink at 3.17 calls) and adds about +0.61 logit per extra call (contribution 0.60 at 4 calls, 1.22 at 5, 1.83 at 6, 2.45 at 7); state:49 switches it off (12 state:49 customers with 4+ calls, 5 of them churned). Active on 306 of 318 customers with 4+ calls. Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.66); mean ablation AUC 0.8996, linear replacement 0.8986; restored alone onto the all-linear model 0.851 (all-linear 0.837), but redundant once the gate is present (gate alone 0.898).

Closed form:

$$y_0 = \mathrm{relu}\big(-0.673 + 0.544\,\mathrm{relu}\big(0.00784 + 1.0x_{numberCustomerServiceCalls_d}\big) - 0.359x_{state:49_c}\big)$$

where y_0 is node `7208`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00181, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 7208 = relu(-0.67 + 0.54 relu(csc + 0.008) - 0.36 state:49), logit weight +1.48. Hypothesis: a hinge on customer-service calls that switches on from 4 calls upward (kink at 3.17 calls) and adds about +0.61 logit per extra call (contribution 0.60 at 4 calls, 1.22 at 5, 1.83 at 6, 2.45 at 7); state:49 switches it off (12 state:49 customers with 4+ calls, 5 of them churned). Active on 306 of 318 cust

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

2. churn100-s2-c2: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	5.55e-16	5.55e-16	5.55e-16	agree
Closed-form max abs error (all leaves)	0.04264	0.04264	0.04264	agree
Numeric-function max abs error	n/a	2.66e-15	2.66e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	churn_logit	churn_logit	agree
Leaf / composition counts	7 / 2	n/a	7 / 2	agree
Operation counts	13 splits, 2 identity	add identity node 2, annotate 9, consolidate node 3, rename node 71, split node 13	add identity node 2, annotate 9, consolidate node 3, rename node 71, split node 13	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `state_offsets` (0.0426), then `linear_backbone` (0.0165). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `csc_usage_gate`: *"Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.087/0.078/0.080/0.113/0.246/0.503/0.820;"* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `area_code_units`: *"161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 13 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- 3 consolidation(s): copies merged back after a split. Check that the merged copies carry the same role.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): None new. Following the sibling agent's note, all splits were input splits before consolidation (no hidden split was needed). The Bash sandbox refused shell loops/variables/heredocs around uv, so tool calls were driven from a scratch Python script that runs experiments/species_leads/xn.py per call (all calls logged).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `churn100-s2-c2-plain-v5.pdf` in this genome's folder):

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up six parts: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 11% at 249 minutes. [C053] An international plan takes the chance from about 1% to about 14%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily,

from about 1% at 0 to about 4% at 32 messages. [C052, C061, C056] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C057, C059, C060, C065] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] This part reads the voice-mail plan and one state: through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] This part reads only the state: it raises a typical customer's chance from about 1% to about 2% in Washington and in Montana, and by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] This part reads the area code and barely changes the predictions (by at most half a point). [C137] This small part reads the international charge and being in Utah, and barely changes the predictions (by at most 0.2 points). [C144]

Worth checking: Worth checking: the state moves the chance by several points (for example Washington about 9% and California about 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C012] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C010, C005] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C053, C104, C060, C107, C057, C105]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C051	structure: made of linear_backbone, service_call_nonlinear, voicemail_chain, intl_state44_unit, area_code_units, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C052	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5; range across patients 2.60 to 2.60	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C053	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 11%	log-odds 4.70, OR 110; range across patients 4.70 to 4.70	4000	train; test agrees (AUC gain 0.157 [0.117, 0.200], n=1000)	yes	clearly
C054	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.47, OR 4.33; range across patients 1.47 to 1.47	122, 72	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C055	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.42, OR 4.15; range across patients 1.42 to 1.42	122, 41	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91; range across patients 1.36 to 1.36	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C057	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.73, OR 5.63; range across patients 1.73 to 1.73	4000	train; test agrees (AUC gain 0.020 [0.008, 0.032], n=1000)	yes	slightly
C059	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C060	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.87, OR 2.40; range across patients 0.87 to 0.87	4000	train; test agrees (AUC gain 0.007 [0.0006, 0.014], n=1000)	yes	slightly
C061	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.98, OR 0.14; range across patients -1.98 to -1.98	2939, 1061	train; test disagrees (AUC gain -0.0003 [-0.015, 0.015], n=1000)	yes	slightly; on the training data only
C062	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to <1%	log-odds -1.86, OR 0.16; range across patients -1.86 to -1.86	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C065	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.70 to 0.70	4000	train; test unclear (AUC gain 0.004 [-0.0008, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C095	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in total evening charge above 17.12 US dollars *shape: dip at 148 minutes; 98% of patients same; verified; risk saturates near 0%*	total evening charge above 17.12 US dollars: 1% to 13%; total evening charge at or below 17.12 US dollars: 2% to 9%	log-odds difference 1.47, OR 4.36	1998, 2002	train; test agrees (AUC gain 0.008 [-0.003, 0.019], n=1000); group difference on test 1.49 [1.33, 1.64]	yes	clearly
C098	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	fold 166 minutes (IQR 151 to 188); past it log-odds -4.16, OR 0.02; range across patients -6.25 to -2.68	3999, 1743	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C099	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 93% of patients same; verified*	3% to 8% (typical patient with a fold)	fold 3.19 calls (IQR 2.36 to 3.19); past it log-odds 1.47, OR 4.35; range across patients 0.52 to 3.17	2700, 528	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	clearly
C103	structure: made of csc_usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C119	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.29, OR 0.28; range across patients -1.29 to -1.29	2939, 1061	train; test disagrees (AUC gain -0.005 [-0.015, 0.005], n=1000)	yes	slightly; on the training data only
C120	direction: state: state other than North Carolina to North Carolina (through this part) *shape: categorical; highest to lowest: state other than North Carolina, North Carolina*	1% to 1%	log-odds -0.42, OR 0.66; range across patients -0.42 to -0.42	3926, 74	train; test agrees (AUC gain 0.0004 [0.0001, 0.0008], n=1000)	yes	slightly
C125	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.76, OR 2.14; range across patients 0.76 to 0.76	3386, 72	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C126	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	3386, 75	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C127	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	3386, 81	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C128	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Ohio (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota,*	1% to 2%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3386, 87	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict churn100-s2-c2 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/churn100-s2-c2.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict churn100-s2-c2 --verified --minutes N --changes "..." --notes "..."`

2 Explanation certificate: churn100-s2-c2

This section is the explanation certificate of the churn100-s2-c2 network. It is rendered from the canonical JSON record `churn100-s2-c2.json` (schema 1.4.0, SHA-256 `3994ace43ba74682...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 2 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s2, seed 2
Genome	`e5394267-5807-44ae-8629-5e157bcfb43a` (NEAT key 24564, generation 99)
Explanation	`4792056e-ee3e-4fde-a1bb-4c4d809e2d61` (98 operations)
Train AUC / accuracy	0.8944 / 0.896 (n = 4000)
Test AUC / accuracy	0.8883 / 0.9 (n = 1000)
Output activation	sigmoid
Phenotype	97 nodes (71 inputs, 25 hidden, 1 output), 89 enabled edges
Explained model state	114 nodes, 91 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 3, split_node 13
Function preservation	max $\|\Delta$ output$\| = 5.55 \times 10^{-16}$ on the training split
Annotations	9 (7 leaf, 2 composition)
Structural coverage	1 (113 of 113 candidate nodes; 30 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`3994ace43ba74682...`

Table 2: Explanation certificate summary: churn100-s2-c2.

Drafter's summary. logit P(churn) = -1.15 + a sparse linear backbone over 33 inputs (day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, ...) + a service-call x low-day-usage gate + a 4+-calls hinge + exactly linear voicemail, state-offset and area-code units, all additive in the logit. The gate is a low-day-minutes hinge whose kink moves up with each customer-service call (median

147 day minutes at 0 calls, 215 at 4, 270 at 6), lowered by high evening charge and switched off in state:0, so with the direct day-minutes edge the logit is V-shaped in day minutes and frequent callers with low usage churn. Service calls act only through hinges. Train AUC 0.894, accuracy 0.896.

What resisted explanation. Nothing structural: no hidden node needed splitting (13 input splits, 3 consolidations). Node 14653 (area code 408) is never active on train (dead); 252 (state:44 / very low international charge) is active on 100 rows with at most +0.03 logit. The gate under-predicts churn for frequent callers with low day usage (CSC>=4 and day minutes <160: 113 rows, model 0.52 vs empirical 0.88). The voicemail chain is exactly linear, and mean-ablating it raises train AUC (0.8944 -> 0.8979): its extra voicemail-plan penalty slightly hurts ranking. closed_form_max_abs_err 0.043 (state_offsets) is 3-significant-figure rounding on state one-hots (standardised values up to 8); the numeric function is exact (2.7e-15).

2.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up six parts: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 11% at 249 minutes. [C053] An international plan takes the chance from about 1% to about 14%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 4% at 32 messages. [C052, C061, C056] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C057, C059, C060, C065] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] This part reads the voice-mail plan and one state: through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] This part reads only the state: it raises a typical customer's chance from about 1% to about 2% in Washington and in Montana, and by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] This part reads the area code and barely changes the predictions (by at most half a point). [C137] This small part reads the international charge and being in Utah, and barely changes the predictions (by at most 0.2 points). [C144]

The whole model. The model adds up six parts into one score: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] *Evidence.* It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C051] Changing one input at a time across the whole model, an

international plan takes a typical customer's chance of leaving from about 1% to about 14%, and a voice-mail plan lowers it to under 1%. [C001, C010] In Washington the chance is about 9%, against about 1% in West Virginia. [C003] More customer-service calls raise it slowly at first, then faster, from about 1% at 0 calls to about 2% at 3 calls. [C011] The model ignores the daytime charge, evening calls, international minutes, night-time calls and the night-time charge. [C046, C047, C048, C049, C050]

Fixed-weight inputs. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C053, C052, C056, C061] *Evidence.* Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 11% at 249 minutes, and an international plan from about 1% to about 14%; a voice-mail plan lowers the chance to under 1%. [C053, C052, C061] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] Its formula reproduces the part exactly on all 4000 training customers. [C085]

Service calls. This part raises the chance of leaving with calls to customer service, from 4 calls upwards, slowly at first, then faster, and more so for customers with low daytime use, in an uneven way; it adds up two parts. [C103, C099, C098] *Evidence.* It is made of two parts that are added: calls with low usage, and many service calls. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095]

Calls with low usage. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), and calls raise it, both in an uneven way. [C112, C106] *Evidence.* It acts on 3063 of 4000 training customers, 393 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C117] Through it, a typical customer's chance falls in an uneven way from about 28% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 166 minutes (between 145 and 213 minutes for different customers). [C112] A higher international charge lowers it steadily up to about 2.95 US dollars (between 2.95 and 4.10 US dollars for different customers), then no further: for a typical such customer, from about 4% to about 1%. [C115] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 2% at 3 calls. [C106] Its formula reproduces the part exactly on all 4000 training customers. [C118]

Many service calls. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C122] *Evidence.* It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 7% without it and about 12% with it. [C123] Through it, the chance rises slowly at first, then faster: going from 0 to 4 calls moves a typical such customer's chance from about 1% to about 2%. [C122] Its formula reproduces the part exactly on all 4000 training customers. [C124]

Voice-mail discount. This part gives customers with a voice-mail plan a further discount on the chance of leaving. [C119] *Evidence.* Through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] Its formula reproduces the part exactly on all 4000 training customers. [C121]

State units. This part reads only the state and raises the chance of leaving in a few states, most in Washington and Montana. [C125, C126] *Evidence.* Through it, a typical customer's chance goes from about 1% to about 2% in Washington and in Montana, and up by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] In Arizona, Delaware, North Dakota and Nevada it lowers the chance by less than a tenth of a point. [C129, C130, C131, C132] Its formula reproduces the part exactly on all 4000 training customers. [C133]

Area-code unit. This part reads the area code and barely changes the predictions. [C137] *Evidence.* Through it, a typical customer's chance stays at about 1% in area codes 408 and 510, as in area code 415. [C134, C135] Switching it off changes the predictions by at most half a point. [C137] Its formula reproduces the part exactly on all 4000 training customers. [C138]

Small Utah unit. This small part reads the international charge and being in Utah, and barely changes the predictions. [C144] *Evidence.* Through it, the international charge barely changes the chance, which stays at about 1% for the typical customer it acts on. [C139] Switching it off changes the predictions by at most 0.2 points. [C144] Its formula reproduces the part exactly on all 4000 training customers. [C145]

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example Washington about 9% and California about 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C012] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C010, C005] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C053, C104, C060, C107, C057, C105]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5; range across patients 2.60 to 2.60	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 154 minutes; 99% of patients same; verified; risk saturates near 0%*	1% to 11%	log-odds 1.64, OR 5.16; range across patients -0.45 to 3.10	4000	train; test agrees (AUC gain 0.081 [0.053, 0.112], n=1000)	no	clearly
C003	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 9%	log-odds 1.66, OR 5.24; range across patients 0.76 to 2.22	122, 72	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	on the training data only
C004	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 5%	log-odds 1.42, OR 4.15; range across patients 1.42 to 1.42	122, 41	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91; range across patients 1.36 to 1.36	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 4%	log-odds 1.17, OR 3.23; range across patients 1.17 to 1.17	122, 75	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C007	direction: state: West Virginia to Texas (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 3%	log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96	122, 95	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C008	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C009	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.61, OR 1.85; range across patients 0.15 to 0.79	4000	train; test unclear (AUC gain 0.006 [-4e-05, 0.012], n=1000)	yes	slightly; on the training data only
C010	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -3.26, OR 0.04; range across patients -3.26 to -3.26	2939, 1061	train; test agrees (AUC gain 0.024 [0.0005, 0.047], n=1000)	yes	slightly
C011	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	slightly
C012	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to <1%	log-odds -1.86, OR 0.16; range across patients -1.86 to -1.86	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 66	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C014	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.60, OR 1.83; range across patients 0.60 to 0.60	122, 82	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C015	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.70 to 0.70	4000	train; test unclear (AUC gain 0.004 [-0.0008, 0.008], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	122, 77	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 12.96 US dollars; 71% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds -0.20, OR 0.82; range across patients -1.79 to 0.87	4000	train; test unclear (AUC gain 0.005 [-0.005, 0.015], n=1000)	no	slightly; on the training data only
C018	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.47, OR 1.59; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C019	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.42, OR 1.52; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	122, 91	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C021	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	122, 87	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Minnesota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	122, 100	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -1.01, OR 0.36; range across patients -2.44 to -0.52	122, 56	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 84	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.38, OR 0.68; range across patients -0.38 to -0.38	4000	train; test unclear (AUC gain 0.0005 [-0.002, 0.003], n=1000)	yes	slightly; on the training data only
C026	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.06, OR 1.07; range across patients -0.51 to 0.78	122, 66	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	no	slightly; on the training data only
C027	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C028	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.17, OR 0.85; range across patients -0.17 to -0.17	122, 80	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C029	direction: state: West Virginia to Washington DC (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.12, OR 0.89; range across patients -0.12 to -0.12	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C030	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.08, OR 1.09; range across patients 0.08 to 0.09	122, 89	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C031	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Kentucky, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Ohio, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Utah, Virginia, Washington, West Virginia, Washington DC, Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.08 to 0.08	122, 1947	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C032	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.08, OR 0.92; range across patients -0.08 to -0.08	2016, 960	train; test disagrees (AUC gain -0.0003 [-0.001, 0.0004], n=1000)	yes	slightly; on the training data only
C033	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.08, OR 0.93; range across patients -0.08 to -0.08	122, 68	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C035	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 79	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C036	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.05, OR 0.95; range across patients -0.05 to -0.05	122, 74	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: West Virginia to Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.04, OR 1.04; range across patients 0.04 to 0.04	122, 81	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1024	train; test disagrees (AUC gain -0.0003 [-0.001, 0.0004], n=1000)	yes	slightly; on the training data only
C039	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 82	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C040	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C042	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 9%	log-odds 2.08, OR 8.00	122, 72	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	on the training data only
C043	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C044	magnitude: total evening minutes: 138 minutes to 266 minutes, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C045	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.79, OR 2.20	4000	train; test unclear (AUC gain 0.006 [-4e-05, 0.012], n=1000)	yes	slightly; on the training data only
C046	null: model ignores total daytime charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C047	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C048	null: model ignores total international minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C050	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C051	structure: made of linear_backbone, service_call_nonlinear, voicemail_chain, intl_state44_unit, area_code_units, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C052	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5; range across patients 2.60 to 2.60	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C053	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 11%	log-odds 4.70, OR 110; range across patients 4.70 to 4.70	4000	train; test agrees (AUC gain 0.157 [0.117, 0.200], n=1000)	yes	clearly
C054	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.47, OR 4.33; range across patients 1.47 to 1.47	122, 72	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C055	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.42, OR 4.15; range across patients 1.42 to 1.42	122, 41	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91; range across patients 1.36 to 1.36	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C057	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.73, OR 5.63; range across patients 1.73 to 1.73	4000	train; test agrees (AUC gain 0.020 [0.008, 0.032], n=1000)	yes	slightly
C058	direction: state: West Virginia to Texas (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 3%	log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96	122, 95	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C059	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C060	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.87, OR 2.40; range across patients 0.87 to 0.87	4000	train; test agrees (AUC gain 0.007 [0.0006, 0.014], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.98, OR 0.14; range across patients -1.98 to -1.98	2939, 1061	train; test disagrees (AUC gain -0.0003 [-0.015, 0.015], n=1000)	yes	slightly; on the training data only
C062	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to <1%	log-odds -1.86, OR 0.16; range across patients -1.86 to -1.86	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C063	direction: state: West Virginia to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 66	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C064	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.60, OR 1.83; range across patients 0.60 to 0.60	122, 82	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C065	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.70 to 0.70	4000	train; test unclear (AUC gain 0.004 [-0.0008, 0.008], n=1000)	yes	slightly; on the training data only
C066	direction: state: West Virginia to Nevada (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 77	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C067	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.46, OR 1.59; range across patients 0.46 to 0.46	122, 75	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C068	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.42, OR 1.52; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	122, 91	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C070	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.37, OR 1.45; range across patients 0.37 to 0.37	122, 74	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C071	direction: state: West Virginia to Minnesota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	122, 100	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 1%	log-odds -0.52, OR 0.59; range across patients -0.52 to -0.52	122, 56	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.51, OR 0.60; range across patients -0.51 to -0.51	122, 66	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C074	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 84	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C075	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.38, OR 0.68; range across patients -0.38 to -0.38	4000	train; test unclear (AUC gain 0.0005 [-0.002, 0.003], n=1000)	yes	slightly; on the training data only
C076	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C077	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.17, OR 0.85; range across patients -0.17 to -0.17	122, 80	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C078	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.11, OR 1.12; range across patients 0.11 to 0.11	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C079	direction: state: West Virginia to Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.12, OR 0.89; range across patients -0.12 to -0.12	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C080	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Virginia, Washington, West Virginia, Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.08 to 0.08	122, 2285	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C081	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 79	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C082	direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code other than area code 510, area code 510*	1% to 1%	log-odds -0.06, OR 0.94; range across patients -0.06 to -0.06	3040, 960	train; test disagrees (AUC gain -6e-05 [-0.0006, 0.0005], n=1000)	yes	slightly; on the training data only
C083	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.06, OR 0.94; range across patients -0.06 to -0.06	122, 68	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C084	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 82	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	structure: its formula reproduces it on 4000 training patients (largest difference 5.32907e-15)		log-odds 5e-15, OR n/a	4000	train	yes	
C086	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	log-odds -3.06, OR 0.05; range across patients -5.15 to -1.60	4000	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C087	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	log-odds -1.11, OR 0.33; range across patients -1.57 to -0.94	4000	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C088	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	slightly
C089	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.08 US dollars; 67% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 12.96 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -2.67 to 0.00	4000	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C090	direction: state: state other than Alaska, South Dakota, Washington to Washington (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds -0.57, OR 0.57; range across patients -1.46 to 0.00	3806, 72	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C091	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds 0.57, OR 1.77; range across patients 0.00 to 1.29	3806, 66	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C092	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds -0.50, OR 0.61; range across patients -1.92 to 0.00	3806, 56	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C093	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C094	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Washington minus state other than Alaska, South Dakota, Washington; stronger in Washington *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Washington: 4% to 18%; state other than Alaska, South Dakota, Washington: 1% to 2%	log-odds difference 1.14, OR 3.11	72, 3806	train; test agrees (AUC gain 0.0004 [-0.0005, 0.001], n=1000); group difference on test 1.15 [1.14, 1.17]	yes	clearly
C095	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in	total evening charge above 17.12 US dollars: 1% to 13%; total evening charge at or below 17.12 US dollars: 2% to	log-odds difference 1.47, OR 4.36	1998, 2002	train; test agrees (AUC gain 0.008 [-0.003, 0.019], n=1000); group difference on test 1.49 [1.33, 1.64]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: Alaska minus state other than Alaska, South Dakota, Washington; stronger in Alaska *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Alaska: <1% to 1%; state other than Alaska, South Dakota, Washington: 1% to 2%	log-odds difference 1.08, OR 2.95	56, 3806	train; test agrees (AUC gain 0.0004 [-0.0005, 0.001], n=1000); group difference on test 1.14 [1.07, 1.21]	yes	slightly
C098	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	fold 166 minutes (IQR 151 to 188); past it log-odds -4.16, OR 0.02; range across patients -6.25 to -2.68	3999, 1743	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C099	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 93% of patients same; verified*	3% to 8% (typical patient with a fold)	fold 3.19 calls (IQR 2.36 to 3.19); past it log-odds 1.47, OR 4.35; range across patients 0.52 to 3.17	2700, 528	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	clearly
C100	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 13.7 to 18.8); past it log-odds -1.95, OR 0.14; range across patients -3.27 to 0.00	71, 35	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C101	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.47); past it log-odds -1.47, OR 0.23; range across patients -2.17 to -1.19	3886, 2729	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C102	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C102 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	fold 75.0 calls (IQR 68.2 to 96.7); past it log-odds 0.31, OR 1.37; range across patients 0.02 to 0.45	12, 9 (small n)	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; the model has little data on this; on the training data only
C103	structure: made of csc_usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C104	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	log-odds -3.06, OR 0.05; range across patients -5.15 to -1.60	4000	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C105	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	log-odds -1.11, OR 0.33; range across patients -1.57 to -0.94	4000	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C106	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 77% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.062 [0.036, 0.092], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds 0.57, OR 1.77; range across patients 0.00 to 1.29	3806, 66	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C110	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1% (typical patient this part acts on)	log-odds -0.50, OR 0.61; range across patients -1.92 to 0.00	3806, 56	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C111	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C112	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	fold 166 minutes (IQR 151 to 188); past it log-odds -4.16, OR 0.02; range across patients -6.25 to -2.68	3999, 1743	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C113	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C113 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 13.7 to 18.8); past it log-odds -1.95, OR 0.14; range across patients -3.27 to 0.00	71, 35	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C114	threshold: customer-service calls: fold at about 4 calls (starts) for customers in Wisconsin *shape: accelerating; 61% of patients same; fallback; risk saturates near 0%*	2% to 5% (typical patient with a fold (customers in Wisconsin))	fold 3.08 calls (IQR 2.33 to 4.11); past it log-odds 1.11, OR 3.04; range across patients 0.00 to 2.88	59, 12 (small n)	train; test agrees (AUC gain 0.062 [0.036, 0.092], n=1000)	yes	slightly; the model has little data on this
C115	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.47); past it log-odds -1.47, OR 0.23; range across patients -2.17 to -1.19	3886, 2729	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C116	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C116 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	fold 75.0 calls (IQR 68.2 to 96.7); past it log-odds 0.31, OR 1.37; range across patients 0.02 to 0.45	12, 9 (small n)	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; the model has little data on this; on the training data only
C117	subgroup: who this part acts on: 3063 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.77; log-odds 1.77, OR 5.84; range across patients 0.17 to 4.40	3063, 937	train; test agrees (AUC gain 0.134 [0.089, 0.181], n=1000)	yes	slightly
C118	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		log-odds 1e-15, OR n/a	4000	train	yes	
C119	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.29, OR 0.28; range across patients -1.29 to -1.29	2939, 1061	train; test disagrees (AUC gain -0.005 [-0.015, 0.005], n=1000)	yes	slightly; on the training data only
C120	direction: state: state other than North Carolina to North Carolina (through this part) *shape: categorical; highest*	1% to 1%	log-odds -0.42, OR 0.66; range across patients -0.42 to -0.42	3926, 74	train; test agrees (AUC gain 0.0004 [0.0001, 0.0008], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C122	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	1% to 2% (typical patient with a fold)	fold 3.19 calls (IQR 3.19 to 3.19); past it log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	4000, 318	train; test agrees (AUC gain 0.006 [0.0008, 0.012], n=1000)	yes	slightly
C123	subgroup: who this part acts on: 318 of 4000 patients	7% to 12% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 0.89, OR 2.44; range across patients 0.52 to 1.82	318, 3682	train; test agrees (AUC gain 0.006 [0.0008, 0.012], n=1000)	yes	slightly
C124	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C125	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.76, OR 2.14; range across patients 0.76 to 0.76	3386, 72	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C126	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	3386, 75	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C127	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	3386, 81	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C128	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Ohio (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3386, 87	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C129	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Arizona (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	log-odds -0.05, OR 0.96; range across patients -0.05 to -0.05	3386, 73	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C130	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Delaware (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	3386, 81	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C133	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C134	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1024	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	yes	slightly; on the training data only
C135	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 960	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	yes	slightly; on the training data only
C136	subgroup: who this part acts on: 2016 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.50; log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	no	slightly; on the training data only
C137	null: this part barely changes predictions (switched off)		max change 0.005 (max |change in predicted risk|)	4000	train	yes	
C138	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C139	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	yes	slightly; on the training data only
C140	direction: state: state other than Utah to Utah (through this part) *shape: categorical; highest to lowest: Utah, state other than Utah*	1% to 1%	log-odds 0.005, OR 1.01; range across patients 0.00 to 0.01	3911, 89	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	yes	slightly; on the training data only
C141	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Utah minus state other than Utah; stronger in Utah *shape: accelerating; 96% of patients same; verified; risk saturates near 0%*	Utah: 1% to 3%; state other than Utah: 1% to 2%	log-odds difference 0.003, OR 1.00	89, 3911	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000); group difference on test 0.03 [-0.09, 0.12]	no	slightly; on the training data only
C142	threshold: total international charge: fold at about 0.77 US dollars (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 0.77 US dollars (IQR 0.77 to 0.77); past it log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000, 100	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	no	slightly; on the training data only
C143	subgroup: who this part acts on: 100 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.03; log-odds 0.007, OR 1.01; range across patients 0.001 to 0.01	100, 3900	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	no	slightly; on the training data only
C144	null: this part barely changes predictions (switched off)		max change 0.002 (max |change in predicted risk|)	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C145	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C146	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C147	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	

Not supported by the model.

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (14% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C026: direction claim about state in the whole model: not supported by the model (53% of patients have the opposite sign (allowed: 10%)).

- C136: subgroup claim about area code in part ann_94: not supported by the model (its effect is negligible).
- C141: interaction claim about total international charge and state in part ann_93: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C142: threshold claim about total international charge in part ann_93: not supported by the model (the fold changes the curve by only 0.037 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C143: subgroup claim about state and total international charge in part ann_93: not supported by the model (its effect is negligible).
- C146: boundary claim about number of calls to customer service and total daytime minutes in part ann_96: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C147: boundary claim about total daytime minutes and number of calls to customer service in part ann_90: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

2.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (30):** `-11` (state:18), `-13` (state:2), `-15` (state:21), `-16` (state:22), `-19` (state:25), `-2` (state:1), `-23` (state:29), `-25` (state:30), `-26` (state:31), `-27` (state:32), `-29` (state:34), `-3` (state:10), `-33` (state:38), `-34` (state:39), `-38` (state:42), `-42` (state:46), `-44` (state:48), `-46` (state:5), `-47` (state:50), `-48` (state:6), `-5` (state:12), `-51` (state:9), `-6` (state:13), `-61` (totalDayCharge), `-63` (totalEveCalls), `-66` (totalNightCalls), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-7` (state:14), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

2.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, ...) are used in the figures.

▷ A1: churn_logit (composition, section 2.6.1)
- ▷ A2: linear_backbone (leaf, section 2.6.2)
- ▷ A3: voicemail_chain (leaf, section 2.6.3)
- ▷ A4: intl_state44_unit (leaf, section 2.6.4)
- ▷ A5: area_code_units (leaf, section 2.6.5)
- ▷ A6: state_offsets (leaf, section 2.6.6)
- ▷ A7: service_call_nonlinear (composition, section 2.6.7)
  - ▷ A8: csc_usage_gate (leaf, section 2.6.8)
  - ▷ A9: csc_hinge (leaf, section 2.6.9)

2.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s2-c2-fig-full.tex`.

2.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 5.55×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`, `-71_d`, `-71_e`.
- #72: split `-70` into `-70_a`, `-70_b`, `-70_c`.
- #73: split `-64` into `-64_a`, `-64_b`.
- #74: split `-59` into `-59_a`, `-59_b`.
- #75: split `-57` into `-57_a`, `-57_b`, `-57_c`, `-57_d`.
- #76: split `-43` into `-43_a`, `-43_b`, `-43_c`.
- #77: split `-37` into `-37_a`, `-37_b`.
- #78: split `-28` into `-28_a`, `-28_b`.
- #79: split `-24` into `-24_a`, `-24_b`.
- #80: split `-22` into `-22_a`, `-22_b`.
- #81: split `-21` into `-21_a`, `-21_b`.
- #82: split `-20` into `-20_a`, `-20_b`, `-20_c`.
- #83: split `-1` into `-1_a`, `-1_b`.
- #84: consolidate `-71_a`, `-71_b`, `-71_d`, `-71_e`.
- #85: consolidate `-57_b`, `-57_c`, `-57_d`.
- #86: consolidate `-20_b`, `-20_c`.
- #87: identity node `identity_lin` before `0` (33 connection(s)).
- #88: identity node `identity_svc` before `0` (3 connection(s)).

2.6 Annotations

2.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 2.6.2), A3 (section 2.6.3), A4 (section 2.6.4), A5 (section 2.6.5), A6 (section 2.6.6), A7 (section 2.6.7).

- **Entry nodes:** `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-60` (totalDayCalls), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-42` (state:46), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-36` (state:40), `-35` (state:4), `-34` (state:39), `-33` (state:38), `-32` (state:37), `-31` (state:36),

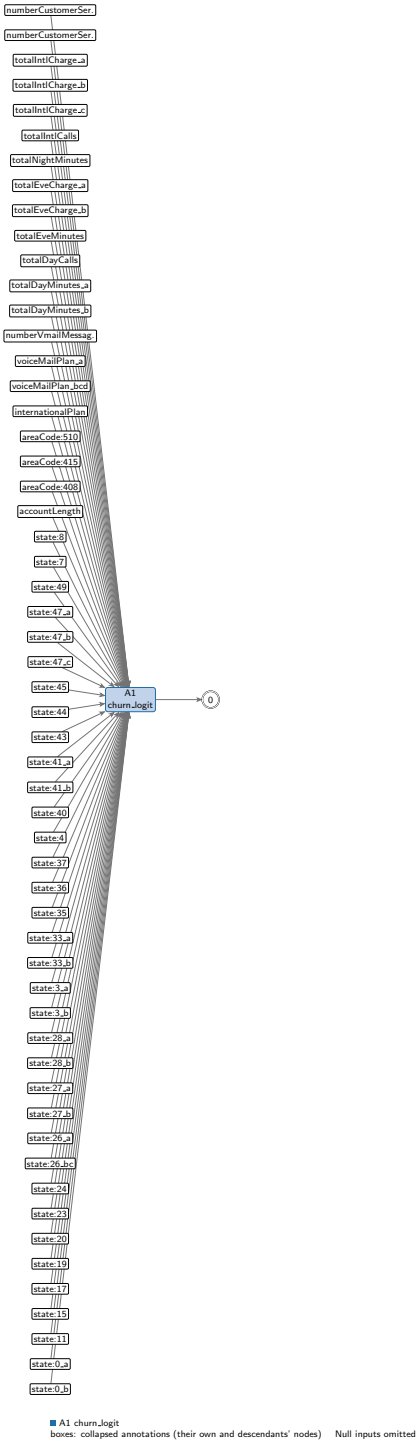


Figure 4: churn100-s2-c2: collapsed view at hierarchy level 0 (fully recomposed).

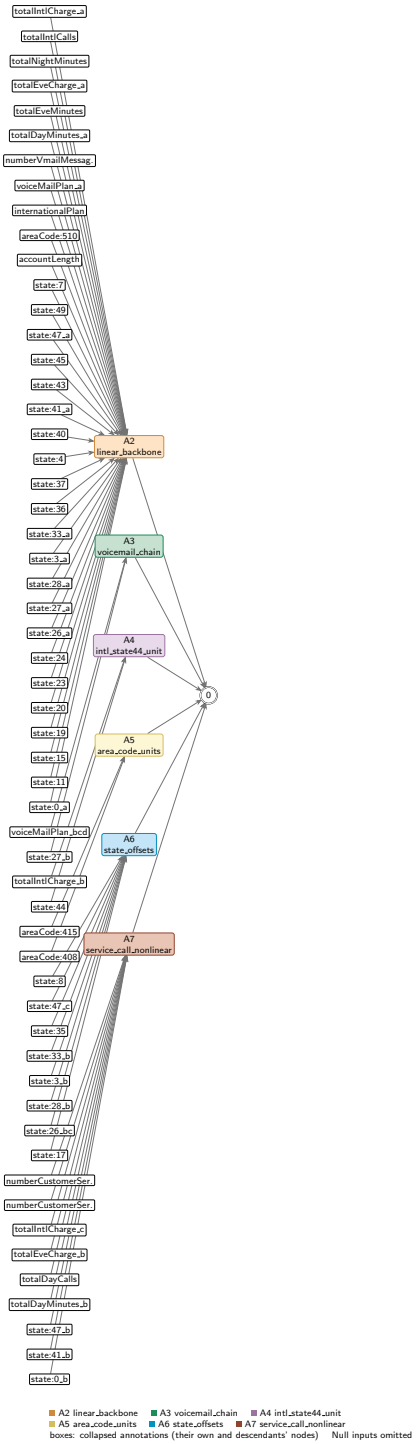


Figure 5: churn100-s2-c2: collapsed view at hierarchy level 1.

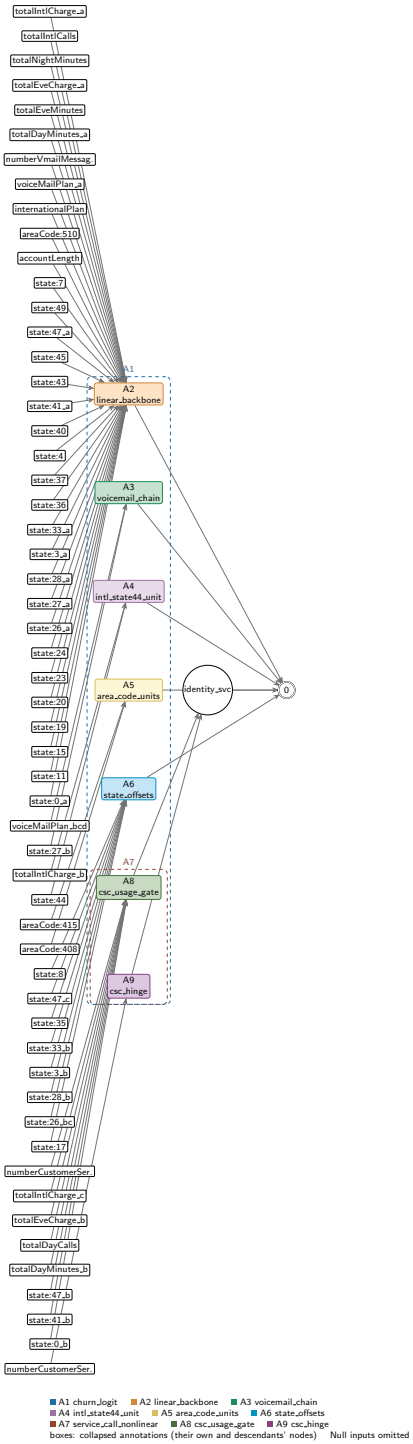


Figure 6: churn100-s2-c2: collapsed view at hierarchy level 2.

-30 (state:35), -29 (state:34), -27 (state:32), -26 (state:31), -25 (state:30), -23 (state:29), -19 (state:25), -18 (state:24), -17 (state:23), -16 (state:22), -15 (state:21), -14 (state:20), -13 (state:2), -12 (state:19), -11 (state:18), -10 (state:17), -9 (state:16), -8 (state:15), -7 (state:14), -6 (state:13), -5 (state:12), -4 (state:11), -3 (state:10), -2 (state:1), -71_c (numberCustomerServiceCalls_c), -70_a (totalIntlCharge_a), ... (26 more; see the JSON record).

- **Exit nodes:** 0.
- **Internal nodes:** none.
- **Edges (weight):** 161→0 (0.0489), 252→0 (0.00702), 1133→0 (-0.891), 4051→0 (-0.999), 7647→0 (-0.00656), 8418→0 (-0.00263), 8896→0 (-0.00632), 10332→0 (0.115), 13551→0 (0.0461), 13725→0 (0.0479), 14653→0 (-0.254), 16344→0 (0.0539), 16902→0 (0.0548), identity_lin→0 (1), identity_svc→0 (1).
- **With its descendants:** 114 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 113 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.15 + linear_backbone + service_call_nonlinear (identity_svc) - 0.89*1133 - 1.00*4051 (voicemail_chain) + state_offsets + 0.05*161 - 0.25*14653 (area codes, 14653 dead) + 0.007*252, purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, night charge, night calls, evening calls, day charge and 25 state indicators) and have zero effect. Train AUC 0.8944, accuracy 0.896. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8314 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring csc_usage_gate alone gives 0.8917, csc_hinge alone 0.8462, the others 0.8314 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11 -1.93 logit (73 rows, empirical -1.35); state:47 +1.47 mean (72 rows, empirical +0.92; direct +1.39, +0.76 via 10332, but it switches off 1264, down to -1.38 for some); state:4 +1.34 (41 rows, +0.80); state:0 -1.13 mean, down to -5.0 (56 rows, -0.76; switches off 2796 in the gate); state:26 +1.09 (75 rows, +0.50; via direct + state offsets); state:43 +0.88 (95 rows, +0.48); state:45 -0.71 (91 rows, -0.85).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.15 + 1.0\,\text{linear_backbone}\big(x_{state:19}, x_{state:20}, \\
&\quad x_{state:23}, x_{state:24}, x_{state:0_a}, x_{state:26_a}, x_{state:27_a}, x_{state:28_a}, \\
&\quad x_{state:3_a}, x_{state:33_a}, x_{state:36}, x_{state:37}, x_{state:4}, x_{state:40}, \\
&\quad x_{state:41_a}, x_{state:43}, x_{state:11}, x_{state:45}, x_{state:47_a}, x_{state:49}, \\
&\quad x_{state:7}, x_{accountLength}, x_{areaCode:510}, x_{internationalPlan}, x_{voiceMailPlan_a}, \\
&\quad x_{numberVmailMessages}, x_{totalDayMinutes_a}, x_{totalEveMinutes}, x_{totalEveCharge_a}, \\
&\quad x_{totalNightMinutes}, x_{totalIntlCalls}, x_{totalIntlCharge_a}, x_{state:15}\big) \\
&\quad + 1.0\,\text{service_call_nonlinear}\big(x_{numberCustomerServiceCalls_abde}, \\
&\quad x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b}, \\
&\quad x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}, \\
&\quad x_{numberCustomerServiceCalls_c}\big) \\
&\quad + 0.0479\,\text{state_offsets_5}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big) \\
&\quad + 0.0489\,\text{area_code_units_0}\big(x_{areaCode:415}, x_{areaCode:408}\big) \\
&\quad + 0.0539\,\text{state_offsets_6}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big) \\
&\quad + 0.0461\,\text{state_offsets_4}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.0548\,\text{state_offsets_7}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&+ 0.00702\,\text{intl_state44_unit}\big(x_{state:44}, x_{totalIntlCharge_b}\big)\\
&+ 0.115\,\text{state_offsets_3}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.891\,\text{voicemail_chain_0}\big(x_{voiceMailPlan_bcd}, x_{state:27_b}\big)\\
&- 0.00632\,\text{state_offsets_2}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.254\,\text{area_code_units_1}\big(x_{areaCode:415}, x_{areaCode:408}\big)\\
&- 0.00263\,\text{state_offsets_1}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.00656\,\text{state_offsets_0}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.999\,\text{voicemail_chain_1}\big(x_{voiceMailPlan_bcd}, x_{state:27_b}\big)
\end{aligned}
$$

The fully expanded form is in the JSON record. where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00617, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.15 + linear_backbone + service_call_nonlinear (identity_svc) - 0.89*1133 - 1.00*4051 (voicemail_chain) + state_offsets + 0.05*161 - 0.25*14653 (area codes, 14653 dead) + 0.007*252, purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, night charge, night calls, evening calls, day charge and 25 state indicators) a

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 2.6.1).

- **Entry nodes:** `-12` (state:19), `-14` (state:20), `-17` (state:23), `-18` (state:24), `-1_a` (state:0_a), `-20_a` (state:26_a), `-21_a` (state:27_a), `-22_a` (state:28_a), `-24_a` (state:3_a), `-28_a` (state:33_a), `-31` (state:36), `-32` (state:37), `-35` (state:4), `-36` (state:40), `-37_a` (state:41_a), `-39` (state:43), `-4` (state:11), `-41` (state:45), `-43_a` (state:47_a), `-45` (state:49), `-49` (state:7), `-52` (accountLength), `-55` (areaCode:510), `-56` (internationalPlan), `-57_a` (voiceMailPlan_a), `-58` (numberVmailMessages), `-59_a` (totalDayMinutes_a), `-62` (totalEveMinutes), `-64_a` (totalEveCharge_a), `-65` (totalNightMinutes), `-69` (totalIntlCalls), `-70_a` (totalIntlCharge_a), `-8` (state:15).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-4`→`identity_lin` (-0.259), `-8`→`identity_lin` (-0.0136), `-12`→`identity_lin` (0.0494), `-14`→`identity_lin` (0.0744), `-17`→`identity_lin` (0.0442), `-18`→`identity_lin` (-0.0346), `-31`→`identity_lin` (-0.00299), `-32`→`identity_lin` (0.0331), `-35`→`identity_lin` (0.135), `-36`→`identity_lin` (0.0673), `-39`→`identity_lin` (0.133), `-41`→`identity_lin` (-0.106), `-45`→`identity_lin` (-0.0136), `-49`→`identity_lin` (-0.0269), `-52`→`identity_lin` (0.126), `-55`→`identity_lin` (-0.0249), `-56`→`identity_lin` (0.763), `-58`→`identity_lin` (0.575), `-62`→`identity_`

lin (0.497), -65→identity_lin (0.274), -69→identity_lin (-0.159), -70_a→identity_lin (0.694), -64_a→identity_lin (0.346), -59_a→identity_lin (1.84), -57_a→identity_lin (-0.872), -43_a→identity_lin (0.184), -37_a→identity_lin (-0.0745), -28_a→identity_lin (0.0622), -24_a→identity_lin (0.00424), -22_a→identity_lin (-0.0176), -21_a→identity_lin (0.0391), -20_a→identity_lin (0.0518), -1_a→identity_lin (-0.0703).

- **Covers** (coverage, own and as part of the hierarchy): 33 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 33 inputs: day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, evening charge +0.35, night minutes +0.27, state:11 -0.26 (-1.93 logit for its 73 customers), state:47 +0.18 (+1.39), state:4 +0.14 (+1.34), state:43 +0.13 (+0.88), account length +0.13, international calls -0.16, state:45 -0.11 (-0.71), and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes, voicemail plan and international plan dominate. Evidence: closed form exact (identity); train sensitivities equal the weights (day minutes 1.84, voicemail plan 0.87, international plan 0.76).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.0518x_{state:26_a} + 1.84x_{totalDayMinutes_a} + 0.0673x_{state:40} \\
& + 0.346x_{totalEveCharge_a} + 0.0442x_{state:23} + 0.497x_{totalEveMinutes} \\
& + 0.0622x_{state:33_a} + 0.126x_{accountLength} + 0.0331x_{state:37} + 0.133x_{state:43} \\
& + 0.135x_{state:4} + 0.00424x_{state:3_a} + 0.273x_{totalNightMinutes} \\
& + 0.575x_{numberVmailMessages} + 0.0744x_{state:20} + 0.0391x_{state:27_a} \\
& + 0.694x_{totalIntlCharge_a} + 0.184x_{state:47_a} + 0.763x_{internationalPlan} \\
& + 0.0494x_{state:19} - 0.0703x_{state:0_a} - 0.159x_{totalIntlCalls} \\
& - 0.00299x_{state:36} - 0.0136x_{state:49} - 0.0745x_{state:41_a} - 0.0346x_{state:24} \\
& - 0.0249x_{areaCode:510} - 0.106x_{state:45} - 0.0136x_{state:15} - 0.259x_{state:11} \\
& - 0.0176x_{state:28_a} - 0.0269x_{state:7} - 0.872x_{voiceMailPlan_a}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0165, and the exact function by at most 2.66×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 33 inputs: day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, evening charge +0.35, night minutes +0.27, state:11 -0.26 (-1.93 logit for its 73 customers), state:47 +0.18 (+1.39), state:4 +0.14 (+1.34),

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.3 A3: voicemail_chain

Kind: Leaf annotation (depth 1); parent A1 (section 2.6.1).

- **Entry nodes:** -57_bcd (voiceMailPlan_bcd), -21_b (state:27_b).
- **Exit nodes:** 1133, 4051.
- **Internal nodes:** 5577, 13852, 14955.
- **Edges (weight):** 1133→4051 (0.942), 5577→1133 (0.191), 13852→1133 (0.221), 14955→1133 (0.0312), -21_b→14955 (0.993), -57_bcd→1133 (0.111), -57_bcd→5577 (0.962), -57_bcd→13852 (1.01).

- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 5577 = relu(0.99 + 0.96 vmPlan), 13852 = relu(-1.52 + 1.01 vmPlan) (on exactly for the 26.5% with a voicemail plan), 14955 = relu(0.51 + 0.99 state:27), 1133 = relu(0.72 + 0.11 vmPlan + 0.19*5577 + 0.22*13852 + 0.03*14955), 4051 = relu(2.09 + 0.94*1133). All but 13852 are always active, and voicemail plan and state:27 are binary, so the chain is exactly affine in them (least-squares R2 = 1.0). Hypothesis: a further voicemail-plan penalty (chain logit -4.76 with a plan vs -3.47 without, i.e. -1.29 on top of the direct -1.97 for having a plan) plus a small state:27 offset. Evidence: exact linear fit; linear replacement AUC 0.8944 (unchanged). Mean ablation RAISES train AUC to 0.8979: the chain's extra voicemail penalty slightly hurts ranking.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&0.723 + 0.111 x_{voiceMailPlan_bcd} \\
&+ 0.191 \operatorname{relu}\big(0.994 + 0.962 x_{voiceMailPlan_bcd}\big) \\
&+ 0.221 \operatorname{relu}\big(-1.52 + 1.01 x_{voiceMailPlan_bcd}\big) \\
&+ 0.0312 \operatorname{relu}\big(0.513 + 0.993 x_{state:27_b}\big)\big) \\
y_1 = \operatorname{relu}\big(&2.09 + 0.942 \operatorname{relu}\big(0.723 \\
&+ 0.111 x_{voiceMailPlan_bcd} + 0.191 \operatorname{relu}\big(0.994 \\
&+ 0.962 x_{voiceMailPlan_bcd}\big) + 0.221 \operatorname{relu}\big(-1.52 \\
&+ 1.01 x_{voiceMailPlan_bcd}\big) + 0.0312 \operatorname{relu}\big(0.513 \\
&+ 0.993 x_{state:27_b}\big)\big)\big)
\end{aligned}
$$

where y_0 is node `1133`; y_1 is node `4051`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00153, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 5577 = relu(0.99 + 0.96 vmPlan), 13852 = relu(-1.52 + 1.01 vmPlan) (on exactly for the 26.5% with a voicemail plan), 14955 = relu(0.51 + 0.99 state:27), 1133 = relu(0.72 + 0.11 vmPlan + 0.19*5577 + 0.22*13852 + 0.03*14955), 4051 = relu(2.09 + 0.94*1133). All but 13852 are always active, and voicemail plan and state:27 are binary, so the chain is exactly affine in them (least-squares R2 = 1.0). Hyp

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.4 A4: intl_state44_unit

Kind: Leaf annotation (depth 1); parent A1 (section 2.6.1).

- **Entry nodes:** `-40` (state:44), `-70_b` (totalIntlCharge_b).
- **Exit nodes:** `252`.
- **Internal nodes:** none.
- **Edges (weight):** `-40`→`252` (0.432), `-70_b`→`252` (-0.851).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 252 = relu(-2.22 + 0.43 state:44 - 0.85 intlCharge), logit weight +0.007. Active on 100 of 4000 training rows (71 in state:44, the rest with very low international charge); logit contribution at most +0.03. Hypothesis: a negligible state:44 / low-international-charge offset. Evidence: sensitivity 0.01/0.02; mean ablation and linear replacement leave train AUC at 0.8944.

Closed form:

$$y_0 = \mathrm{relu}\big(-2.22 + 0.432 x_{state:44} - 0.851 x_{totalIntlCharge_b}\big)$$

where y_0 is node `252`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.008, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 252 = relu(-2.22 + 0.43 state:44 - 0.85 intlCharge), logit weight +0.007. Active on 100 of 4000 training rows (71 in state:44, the rest with very low international charge); logit contribution at most +0.03. Hypothesis: a negligible state:44 / low-international-charge offset. Evidence: sensitivity 0.01/0.02; mean ablation and linear replacement leave train AUC at 0.8944.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.5 A5: area_code_units

Kind: Leaf annotation (depth 1); parent A1 (section 2.6.1).

- **Entry nodes:** `-54` (areaCode:415), `-53` (areaCode:408).
- **Exit nodes:** `161`, `14653`.
- **Internal nodes:** none.
- **Edges (weight):** `-54`→`161` (0.577), `-53`→`14653` (-0.535).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415). 14653 = relu(-1.07 - 0.54 areaCode:408) is never active on train (dead). Hypothesis: area code barely matters (direct area code 510 term -0.02 per SD in the backbone). Evidence: sensitivity 0.29 for 161 on its unit scale, 0 for 14653; linear R2 1.0; mean ablation AUC 0.8945.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.137 + 0.577 x_{areaCode:415}\big)$$
$$y_1 = \mathrm{relu}\big(-1.07 - 0.535 x_{areaCode:408}\big)$$

where y_0 is node `161`; y_1 is node `14653`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 3.38×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415). 14653 = relu(-1.07 - 0.54 areaCode:408) is never active on train (dead). Hypothesis: area code barely matters (direct area code 510 term -0.02 per SD in the backbone). Evidence: sensitivity 0.29 for 161 on its unit scale, 0 for 14653; linear R2 1.0; m

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.6 A6: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 2.6.1).

- **Entry nodes:** `-50` (state:8), `-22_b` (state:28_b), `-24_b` (state:3_b), `-43_c` (state:47_c), `-30` (state:35), `-20_bc` (state:26_bc), `-10` (state:17), `-28_b` (state:33_b).
- **Exit nodes:** `7647`, `8418`, `8896`, `10332`, `13551`, `13725`, `16344`, `16902`.
- **Internal nodes:** `4967`, `11261`.
- **Edges (weight):** `-50`→`4967` (0.979), `-10`→`11261` (0.99), `-30`→`13551` (0.998), `4967`→`7647` (0.984), `11261`→`16902` (1), `-43_c`→`10332` (0.894), `-28_b`→`11261` (-0.0231), `-24_b`→`8896` (1.01), `-22_b`→`8418` (0.987), `-20_bc`→`13725` (0.991), `-20_bc`→`16344` (0.997).
- **Covers** (coverage, own and as part of the hierarchy): 10 candidate nodes.

Hypothesis. Units 4967 -> 7647 (state:8), 8418 (state:28), 8896 (state:3), 10332 (state:47), 13551 (state:35), 13725 and 16344 (state:26), 11261 -> 16902 (state:17, state:33). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets here: state:47 +0.76 (10332, adding to its direct +1.39), state:26 +0.71 (13725 + 16344, adding to its direct +0.38), state:17 +0.39 (16902), state:35 +0.30 (13551); state:8/28/3 under 0.05. Evidence: linear replacement AUC 0.8944 (unchanged); mean ablation 0.8935.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-1.83 + 0.984\,\mathrm{relu}\big(1.76 \\
&\qquad + 0.979 x_{state:8}\big)\big) \\
y_1 &= \mathrm{relu}\big(0.806 + 0.987 x_{state:28_b}\big) \\
y_2 &= \mathrm{relu}\big(-0.144 + 1.01 x_{state:3_b}\big) \\
y_3 &= \mathrm{relu}\big(-0.0137 + 0.894 x_{state:47_c}\big) \\
y_4 &= \mathrm{relu}\big(-0.23 + 0.998 x_{state:35}\big) \\
y_5 &= \mathrm{relu}\big(-0.597 + 0.991 x_{state:26_bc}\big) \\
y_6 &= \mathrm{relu}\big(1.04 + 0.997 x_{state:26_bc}\big) \\
y_7 &= \mathrm{relu}\big(0.0457 + 1.0\,\mathrm{relu}\big(1.42 \\
&\qquad + 0.99 x_{state:17} - 0.0231 x_{state:33_b}\big)\big)
\end{aligned}
$$

where y_0 is node `7647`; y_1 is node `8418`; y_2 is node `8896`; y_3 is node `10332`; y_4 is node `13551`; y_5 is node `13725`; y_6 is node `16344`; y_7 is node `16902`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0426, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Units 4967 -> 7647 (state:8), 8418 (state:28), 8896 (state:3), 10332 (state:47), 13551 (state:35), 13725 and 16344 (state:26), 11261 -> 16902 (state:17, state:33). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.7 A7: service_call_nonlinear

Kind: Composition annotation (depth 1); parent A1 (section 2.6.1); children A8 (section 2.6.8), A9 (section 2.6.9).

- **Entry nodes:** `-71_abde` (numberCustomerServiceCalls_abde), `-59_b` (totalDayMinutes_b), `-60` (totalDayCalls), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-1_b` (state:0_b), `-70_c` (totalIntlCharge_c), `-43_b` (state:47_b), `-71_c` (numberCustomerServiceCalls_c).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `807`→`identity_svc` (1.59), `1264`→`identity_svc` (1.81), `7208`→`identity_svc` (1.52).
- **With its descendants:** 17 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 16 candidate nodes.

Hypothesis. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.59*807 + 1.81*1264 + 1.52*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.7406, linear replacement of both 0.8313 (all-linear model 0.8314); the gate alone restores 0.8917 of 0.8944; sensitivity of identity_svc on train: day minutes 1.18, state:0 0.61, state:47 0.50, service calls 0.46+0.07, international charge 0.44, evening charge 0.43.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 = {} & 1.81\,\mathrm{csc_usage_gate_1}\big(x_{numberCustomerServiceCalls_abde},\\
& x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b},\\
& x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}\big)\\
& + 1.52\,\mathrm{csc_hinge}\big(x_{numberCustomerServiceCalls_c}\big)\\
& + 1.59\,\mathrm{csc_usage_gate_0}\big(x_{numberCustomerServiceCalls_abde},\\
& x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b},\\
& x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = {} & 1.81\,\mathrm{relu}\big(0.0266 + 0.329\,\mathrm{relu}\big(-1.3\\
& + 0.584\,\mathrm{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde}\\
& + 0.359\,\mathrm{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big)\\
& + 0.608\,\mathrm{relu}\big(0.951 + 0.362\,\mathrm{relu}\big(-0.199\\
& + 0.917x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls}\\
& - 1.3x_{totalDayMinutes_b}\big) - 0.349x_{totalIntlCharge_c}\\
& - 0.397x_{state:47_b}\big) + 1.52\,\mathrm{relu}\big(-0.695\\
& + 0.559x_{numberCustomerServiceCalls_c}\big) + 1.59\,\mathrm{relu}\big(-1.3\\
& + 0.584\,\mathrm{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde}\\
& + 0.359\,\mathrm{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big)\\
& + 0.608\,\mathrm{relu}\big(0.951 + 0.362\,\mathrm{relu}\big(-0.199
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.917 x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235 x_{numberCustomerServiceCalls_abde}\big) + 0.0575 x_{totalDayCalls} \\
&- 1.3 x_{totalDayMinutes_b}\big)
\end{aligned}
$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0374, and the exact function by at most 3.55×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.59*807 + 1.81*1264 + 1.52*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.8 A8: csc_usage_gate

Kind: Leaf annotation (depth 2); parent A7 (section 2.6.7).

- **Entry nodes:** `-71_abde` (numberCustomerServiceCalls_abde), `-59_b` (totalDayMinutes_b), `-60` (totalDayCalls), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-1_b` (state:0_b), `-70_c` (totalIntlCharge_c), `-43_b` (state:47_b).
- **Exit nodes:** `807`, `1264`.
- **Internal nodes:** `9555`, `2627`, `15509`, `2796`.
- **Edges (weight):** `-60`→`807` (0.0575), `2627`→`807` (0.608), `2796`→`807` (0.584), `807`→`1264` (0.329), `9555`→`2627` (0.362), `15509`→`2796` (0.359), `-70_c`→`1264` (-0.349), `-64_b`→`2796` (-0.996), `-59_b`→`807` (-1.3), `-43_b`→`1264` (-0.397), `-37_b`→`2796` (0.129), `-1_b`→`2796` (-1.41), `-71_abde`→`2627` (0.235), `-71_abde`→`2796` (0.314), `-71_abde`→`9555` (0.917), `-71_abde`→`15509` (0.99).
- **Covers** (coverage, own and as part of the hierarchy): 12 candidate nodes.

Hypothesis. Service-call x low-day-usage gate. 807 = relu(-1.30 - 1.30 dayMin + 0.06 dayCalls + 0.61 g1(csc) + 0.58 g2), with g1 = relu(0.95 + 0.24 csc + 0.36 relu(0.92 csc - 0.20)) always active (convex in calls) and g2 = 2796 = relu(0.38 + 0.31 csc + 0.36 relu(0.99 csc - 0.20) - 1.00 eveCharge - 1.41 state:0 + 0.13 state:41); 1264 = relu(0.03 + 0.33*807 - 0.35 intlCharge - 0.40 state:47). The logit gets +1.59*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 157 at 1, 170 at 2, 194 at 3, 215 at 4, 237 at 5, 270 at 6); high evening charge lowers the kink (2796 is on for 98% of customers in the lowest evening-charge band, 22% in the highest) and state:0 switches 2796 off (all 56 customers). Below the kink the gate's slope is about -2.84 logit per SD of day minutes, more than cancelling the direct +1.84, so the logit is V-shaped in day minutes. 1264 passes 807 on and adds a low-international-charge hinge (on for 1319 rows where 807 is off); state:47 switches 1264 off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.88, model mean 0.52 (under-predicted); CSC<4 and day minutes < 120: 476 customers, empirical 0.05, model 0.06. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.087/0.078/0.080/0.113/0.246/0.503/0.820; share with 807 on rises 0.31 -> 0.93. Sensitivity (807): day minutes 0.57, state:0 0.29, service calls 0.22, evening charge 0.21. Ablation: mean-ablation AUC 0.7690 (full 0.8944); least-squares linear replacement

0.8461 (R2 0.68/0.74); restored alone onto an all-linearised model 0.8917 (all-linear 0.8314). This is the model's main nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.3 + 0.584\operatorname{relu}\big(0.382 \\
&+ 0.314x_{numberCustomerServiceCalls_abde} + 0.359\operatorname{relu}\big(-0.2 \\
&+ 0.99x_{numberCustomerServiceCalls_abde}\big) + 0.129x_{state:41_b} \\
&- 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0266 + 0.329\operatorname{relu}\big(-1.3 \\
&+ 0.584\operatorname{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde} \\
&+ 0.359\operatorname{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) - 0.349x_{totalIntlCharge_c} \\
&- 0.397x_{state:47_b}\big)
\end{aligned}
$$

where y_0 is node `807`; y_1 is node `1264`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0112, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Service-call x low-day-usage gate. 807 = relu(-1.30 - 1.30 dayMin + 0.06 dayCalls + 0.61 g1(csc) + 0.58 g2), with g1 = relu(0.95 + 0.24 csc + 0.36 relu(0.92 csc - 0.20)) always active (convex in calls) and g2 = 2796 = relu(0.38 + 0.31 csc + 0.36 relu(0.99 csc - 0.20) - 1.00 eveCharge - 1.41 state:0 + 0.13 state:41); 1264 = relu(0.03 + 0.33*807 - 0.35 intlCharge - 0.40 state:47). The logit gets +1.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.6.9 A9: csc_hinge

Kind: Leaf annotation (depth 2); parent A7 (section 2.6.7).

- **Entry nodes:** `-71_c` (numberCustomerServiceCalls_c).
- **Exit nodes:** `7208`.
- **Internal nodes:** none.
- **Edges (weight):** `-71_c`→`7208` (0.559).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 7208 = relu(-0.70 + 0.56 csc), logit weight +1.52. Hypothesis: a hinge on customer-service calls that is off for 0-3 calls and switches on from 4 calls (kink at 3.2 calls), adding about +0.65 logit per extra call (contribution +0.52 at 4 calls, +1.17 at 5, +1.82 at 6). Active on exactly the 318 customers with 4+ calls (empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.67); mean ablation AUC 0.8901, linear replacement 0.8916; restored alone onto the all-linear model 0.8462 (all-linear 0.8314), but largely redundant once the gate is present (gate alone 0.8917).

Closed form:

$$y_0 = \mathrm{relu}\big(-0.695 + 0.559 x_{numberCustomerServiceCalls_c}\big)$$

where y_0 is node `7208`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00224, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 7208 = relu(-0.70 + 0.56 csc), logit weight +1.52. Hypothesis: a hinge on customer-service calls that is off for 0-3 calls and switches on from 4 calls (kink at 3.2 calls), adding about +0.65 logit per extra call (contribution +0.52 at 4 calls, +1.17 at 5, +1.82 at 6). Active on exactly the 318 customers with 4+ calls (empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot o

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

2.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

3. churn100-s4-c2: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	4.44e-16	4.44e-16	4.44e-16	agree
Closed-form max abs error (all leaves)	0.01996	0.01996	0.01996	agree
Numeric-function max abs error	n/a	1.78e-15	1.78e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	churn_logit	churn_logit	agree
Leaf / composition counts	7 / 2	n/a	7 / 2	agree
Operation counts	5 splits, 2 identity	add identity node 2, annotate 9, consolidate node 1, rename node 71, split node 5	add identity node 2, annotate 9, consolidate node 1, rename node 71, split node 5	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `vmail_state18_unit` (0.02), then `evening_hinge` (0.0165). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `csc_high_hinge`: *"Active on exactly the 8% of customers with 4+ calls (318 rows;"* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `state_offsets`: *"Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 5 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- 1 consolidation(s): copies merged back after a split. Check that the merged copies carry the same role.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): None new. Hidden-node splits were not needed; input splits were done before the one consolidation. Display name of the consolidated copy -71_acd is None (cosmetic). Tool calls driven from scratch Python scripts that run experiments/species_leads/xn.py per call (all logged), because the Bash sandbox refuses shell variables/loops around uv.

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `churn100-s4-c2-plain-v5.pdf` in this genome's folder):

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 3%; it adds up six parts: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] A higher evening charge and more international

minutes also raise it steadily, each from about 2% to about 4% across their usual range. [C055, C056] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] This part adds up two parts: a rise from 3 customer-service calls upwards and a second rise from 4 calls upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] This part reads only the state: it raises a typical customer's chance from about 3% to about 22% in Rhode Island, and by less than a point in Indiana. [C100, C101] This part reads evening minutes: nothing until about 217 minutes, then steadily, but a typical customer's chance stays at about 3%. [C103, C104] This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions (by at most one point). [C111]

Worth checking: Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to under 1% while the state units raise it to about 22%, so overall it stays at about 3%. [C053, C100, C033] Worth checking: the state moves the chance by several points (Washington about 11%, New Jersey about 7%, Hawaii under 1%), based on 72 to 122 training customers per state. [C002, C005, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C006, C003] Worth checking: voice-mail messages raise the chance through the fixed-weight inputs but lower it through heavy daytime use. [C049, C080]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C046	structure: made of linear_backbone, service_calls, day_usage_hinges, evening_hinge, vmail_state18_unit, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C047	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7; range across patients 2.82 to 2.82	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C048	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.48, OR 4.40; range across patients 1.48 to 1.48	122, 72	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	on the training data only
C049	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	3% to 10%	log-odds 1.47, OR 4.35; range across patients 1.47 to 1.47	4000	train; test agrees (AUC gain 0.016 [0.003, 0.029], n=1000)	yes	clearly
C050	direction: state: West Virginia to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 7%	log-odds 1.09, OR 2.97; range across patients 1.09 to 1.09	122, 90	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C051	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08; range across patients -2.54 to -2.54	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C052	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	122, 73	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C053	direction: state: West Virginia to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.36, OR 0.09; range across patients -2.36 to -2.36	122, 82	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C055	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.88, OR 2.42; range across patients 0.88 to 0.88	4000	train; test agrees (AUC gain 0.013 [0.004, 0.022], n=1000)	yes	slightly
C056	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.85, OR 2.35; range across patients 0.85 to 0.85	4000	train; test unclear (AUC gain 0.0008 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C077	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	4000	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	slightly
C078	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 18% (typical patient with a fold)	fold 2.80 calls (IQR 2.80 to 2.80); past it log-odds 2.07, OR 7.90; range across patients 2.07 to 2.07	4000, 848	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C079	structure: made of csc_moderate_chain, csc_high_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C080	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: hinge at 11 messages; 64% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	12% to 6% (typical patient this part acts on)	log-odds -0.26, OR 0.77; range across patients -1.08 to 0.00	4000	train; test agrees (AUC gain 0.006 [0.003, 0.009], n=1000)	yes	clearly
C082	direction: state: state other than Washington to Washington (through this part) *shape: categorical; highest to lowest: state other than Washington, Washington*	12% to 7% (typical patient this part acts on)	log-odds -0.21, OR 0.81; range across patients -0.84 to 0.00	3928, 72	train; test agrees (AUC gain 0.0004 [2e-05, 0.0008], n=1000)	yes	slightly
C088	threshold: total daytime minutes: fold at about 221 minutes (starts) *shape: hinge at 221 minutes; 48% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 19% (typical patient with a fold)	fold 221 minutes (IQR 213 to 221); past it log-odds 1.98, OR 7.25; range across patients 1.48 to 2.79	3995, 1099	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C100	direction: state: state other than Indiana, Rhode Island to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 22%	log-odds 2.36, OR 10.6; range across patients 2.36 to 2.36	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	on the training data only
C101	direction: state: state other than Indiana, Rhode Island to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 3%	log-odds 0.27, OR 1.31; range across patients 0.27 to 0.27	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	slightly; on the training data only
C103	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3%	log-odds 0.20, OR 1.22; range across patients 0.20 to 0.20	4000	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C104	threshold: total evening minutes: fold at about 217 minutes (starts) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3% (typical patient with a fold)	fold 217 minutes (IQR 217 to 217); past it log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000, 1467	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C111	null: this part barely changes predictions (switched off)		max change 0.01 (max |change in predicted risk|)	4000	train	yes	

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict churn100-s4-c2 --verified --minutes N --notes "..."`.

Verdict.

Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/churn100-s4-c2.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`

```
uv run python -m explaneat export record-verdict churn100-s4-c2 --verified --minutes N --changes
"..." --notes "..."
```


3 Explanation certificate: churn100-s4-c2

This section is the explanation certificate of the churn100-s4-c2 network. It is rendered from the canonical JSON record `churn100-s4-c2.json` (schema 1.4.0, SHA-256 `36f617b704b0542d...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 3 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s4, seed 4
Genome	`4206744f-b5ce-4488-bc53-a39aa08fe972` (NEAT key 24427, generation 99)
Explanation	`33d6db5d-fbdf-41c8-a597-fc997715b92f` (88 operations)
Train AUC / accuracy	0.8908 / 0.888 (n = 4000)
Test AUC / accuracy	0.8664 / 0.898 (n = 1000)
Output activation	sigmoid
Phenotype	85 nodes (71 inputs, 13 hidden, 1 output), 58 enabled edges
Explained model state	93 nodes, 60 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 1, split_node 5
Function preservation	max $\|\Delta \text{output}\| = 4.44 \times 10^{-16}$ on the training split
Annotations	9 (7 leaf, 2 composition)
Structural coverage	1 (92 of 92 candidate nodes; 37 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`36f617b704b0542d...`

Table 3: Explanation certificate summary: churn100-s4-c2.

Drafter's summary. logit P(churn) = -1.30 + a sparse linear backbone over 29 inputs (voicemail plan -1.12 per SD, international plan +0.83, voicemail messages +0.62, evening charge +0.35, international minutes +0.34, state offsets) + a customer-service-calls threshold (two hinges: nothing for 0-2 calls, then about +2.4 logit per call from 4) + two day-usage hinges (nothing below about 210 day minutes, then steeply up, +4.3 logit above 280 minutes; the 3118 hinge only for customers without voicemail messages, the 10131 hinge off in state:47) + a small evening-minutes hinge and exact state:39/state:15 offsets, all additive in the logit. Day minutes and service calls have no linear term. Train AUC 0.891, accuracy 0.888. Compared with the run's cluster-1 lead (churn100-s4, key

24573), the same two thresholds (service calls, high day usage) carry the nonlinearity, but here day usage is gated by voicemail messages and state:47 rather than combined with evening minutes.

What resisted explanation. Nothing structural: no hidden node needed splitting (5 input splits, 1 consolidation of the customer-service-calls copies). 6089 (state:18 / 21+ voicemail messages) has logit weight 0.005 (at most +0.05): negligible. The state:39 pair (1295 -> 7872) is a constant -2.33 logit switched off for state:39, almost cancelling state:39's direct -2.61 (net -0.25 vs empirical -0.93). The service-call threshold is placed right (3 -> 4 calls) but the model under-predicts at 4 calls (0.33 vs 0.44) and is over-confident from 5 calls (0.75-0.99 vs 0.55-0.68; 119 customers). state:18 gets +0.41 logit against an empirical -0.48 (65 rows). closed_form_max_abs_err 0.020 (vmail_state18_unit) is 3-significant-figure rounding; numeric error 1.8e-15.

3.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 3%; it adds up six parts: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] A higher evening charge and more international minutes also raise it steadily, each from about 2% to about 4% across their usual range. [C055, C056] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] This part adds up two parts: a rise from 3 customer-service calls upwards and a second rise from 4 calls upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] This part reads only the state: it raises a typical customer's chance from about 3% to about 22% in Rhode Island, and by less than a point in Indiana. [C100, C101] This part reads evening minutes: nothing until about 217 minutes, then steadily, but a typical customer's chance stays at about 3%. [C103, C104] This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions (by at most one point). [C111]

The whole model. The model adds up six parts into one score: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] *Evidence.* It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C046] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C001, C006] More daytime minutes raise it in an uneven way, from about 3% at 112 minutes to about 8% at 249 minutes, flattening out near the bottom

because the predicted chance cannot go much lower. [C004] In Washington the chance is about 11%, against about 3% in West Virginia. [C002] The model ignores evening calls, international calls, the international charge, night-time calls, the night-time charge and night-time minutes. [C040, C041, C042, C043, C044, C045]

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan, voice-mail messages, the evening charge and international minutes raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C047, C049, C055, C056, C051] *Evidence.* Through it, an international plan takes a typical customer from about 3% to about 31%, and a voice-mail plan lowers the chance to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] Its formula reproduces the part exactly on all 4000 training customers. [C076]

Service calls. This part raises the chance of leaving with calls to customer service from 3 calls upwards, slowly at first, then faster; it adds up two parts. [C079, C078, C077] *Evidence.* It is made of two parts that are added: calls from 3 upwards and calls from 4 upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]

Calls from 4 upwards. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C093] *Evidence.* It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 8% without it and about 18% with it. [C094] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 6%. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C095]

Calls from 3 upwards. This part raises the chance of leaving from 3 customer-service calls upwards, slowly at first, then faster. [C097, C096] *Evidence.* It acts on 848 of 4000 training customers, 215 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 4% with it. [C098] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 8%. [C097] Its formula reproduces the part exactly on all 4000 training customers. [C099]

Heavy daytime use. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way, and voice-mail messages and being in Washington lower it. [C088, C080, C082] *Evidence.* This part acts on 1099 of 4000 training customers, 291 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 12% with it. [C091] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] Its formula reproduces the part exactly on all 4000 training customers. [C092]

State units. This part reads only the state and raises the chance of leaving in Rhode Island, and a little in Indiana. [C100, C101] *Evidence.* Through it, a typical customer's chance goes from about

3% to about 22% in Rhode Island, and up by less than a point in Indiana. [C100, C101] Its formula reproduces the part exactly on all 4000 training customers. [C102]

Evening unit. This part reads evening minutes and barely changes the chance of leaving: nothing until about 217 minutes, then steadily. [C103, C104] *Evidence.* Through it, evening minutes change nothing until about 217 minutes, then raise the chance steadily, but a typical customer's chance stays at about 3% from 138 to 266 minutes. [C103] Its formula reproduces the part exactly on all 4000 training customers. [C105]

Small voice-mail unit. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions. [C111] *Evidence.* Through it, a typical customer's chance stays at about 3% in Louisiana as elsewhere, and voice-mail messages barely change it (about 3% at 0 and at 32 messages). [C106, C107] Switching it off changes the predictions by at most one point. [C111] Its formula reproduces the part exactly on all 4000 training customers. [C112]

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to under 1% while the state units raise it to about 22%, so overall it stays at about 3%. [C053, C100, C033] Worth checking: the state moves the chance by several points (Washington about 11%, New Jersey about 7%, Hawaii under 1%), based on 72 to 122 training customers per state. [C002, C005, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C006, C003] Worth checking: voice-mail messages raise the chance through the fixed-weight inputs but lower it through heavy daytime use. [C049, C080]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7; range across patients 2.82 to 2.82	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.27, OR 3.58; range across patients 0.64 to 1.48	122, 72	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	on the training data only
C003	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	log-odds 1.21, OR 3.35; range across patients 0.39 to 1.47	4000	train; test agrees (AUC gain 0.017 [0.005, 0.031], n=1000)	yes	clearly
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	log-odds 1.13, OR 3.10; range across patients 0.83 to 1.79	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C005	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 7%	log-odds 1.09, OR 2.97; range across patients 1.09 to 1.09	122, 90	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C006	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08; range across patients -2.54 to -2.54	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C007	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	122, 73	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C008	direction: state: West Virginia to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.67, OR 1.95; range across patients 0.67 to 0.67	122, 65	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C009	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.66, OR 1.94; range across patients 0.66 to 0.66	122, 80	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C010	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.88, OR 2.42; range across patients 0.88 to 0.88	4000	train; test agrees (AUC gain 0.013 [0.004, 0.022], n=1000)	yes	slightly
C011	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.85, OR 2.35; range across patients 0.85 to 0.85	4000	train; test unclear (AUC gain 0.0008 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C012	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	log-odds -1.68, OR 0.19; range across patients -1.68 to -1.68	122, 80	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	122, 68	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C014	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.57, OR 1.77; range across patients 0.57 to 0.57	122, 86	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 84	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	122, 82	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C017	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.47, OR 1.61; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C018	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 91	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C019	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	122, 66	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 79	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C021	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.33, OR 1.40; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.0008], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40	122, 60	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.37, OR 0.69; range across patients -0.37 to -0.37	122, 56	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 68	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: West Virginia to state other than Georgia, Hawaii, Indiana, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 2286	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C026	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C027	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.22, OR 1.25; range across patients 0.22 to 0.22	122, 73	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C028	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: hinge at 218 minutes; 100% of patients same; verified; risk saturates near 0%; matches the fold at 217 minutes*	3% to 3%	log-odds 0.20, OR 1.22; range across patients 0.20 to 0.20	4000	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	4000	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	slightly
C030	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.14, OR 1.15; range across patients 0.14 to 0.14	122, 78	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C031	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0004, 5e-05], n=1000)	yes	slightly; on the training data only
C032	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 42.20 US dollars; 77% of patients same; fallback; risk saturates near 0%; matches the fold at 39.34 US dollars*	3% to 3%	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.94	4000	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; on the training data only
C033	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.004, OR 1.00; range across patients 0.004 to 0.004	122, 82	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds -0.0006, OR 1.00; range across patients -0.0006 to -0.0006	122, 78	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C035	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C036	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.48, OR 4.40	122, 72	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	log-odds 1.47, OR 4.37	4000	train; test agrees (AUC gain 0.017 [0.005, 0.031], n=1000)	yes	clearly
C038	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	log-odds 1.15, OR 3.17	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C039	magnitude: voice-mail plan: no voice-mail plan to a voice-mail plan, typical patient *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C040	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C041	null: model ignores total international calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C042	null: model ignores total international charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C043	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C044	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C045	null: model ignores total night-time minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C046	structure: made of linear_backbone, service_calls, day_usage_hinges, evening_hinge, vmail_state18_unit, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C047	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7; range across patients 2.82 to 2.82	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C048	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.48, OR 4.40; range across patients 1.48 to 1.48	122, 72	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	3% to 10%	log-odds 1.47, OR 4.35; range across patients 1.47 to 1.47	4000	train; test agrees (AUC gain 0.016 [0.003, 0.029], n=1000)	yes	clearly
C050	direction: state: West Virginia to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 7%	log-odds 1.09, OR 2.97; range across patients 1.09 to 1.09	122, 90	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C051	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08; range across patients -2.54 to -2.54	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C052	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	122, 73	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C053	direction: state: West Virginia to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.36, OR 0.09; range across patients -2.36 to -2.36	122, 82	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C054	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.66, OR 1.94; range across patients 0.66 to 0.66	122, 80	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C055	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.88, OR 2.42; range across patients 0.88 to 0.88	4000	train; test agrees (AUC gain 0.013 [0.004, 0.022], n=1000)	yes	slightly
C056	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.85, OR 2.35; range across patients 0.85 to 0.85	4000	train; test unclear (AUC gain 0.0008 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C057	direction: state: West Virginia to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.63, OR 1.87; range across patients 0.63 to 0.63	122, 65	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C058	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -1.68, OR 0.19; range across patients -1.68 to -1.68	122, 80	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C059	direction: state: West Virginia to Tennessee (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	122, 68	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C060	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.57, OR 1.77; range across patients 0.57 to 0.57	122, 86	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 84	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C062	direction: state: West Virginia to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.47, OR 1.61; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C063	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 91	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C064	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	122, 66	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C065	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 79	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C066	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.33, OR 1.40; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.0008], n=1000)	yes	slightly; on the training data only
C067	direction: state: West Virginia to Georgia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40	122, 60	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C068	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.37, OR 0.69; range across patients -0.37 to -0.37	122, 56	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 68	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C070	direction: state: West Virginia to state other than Georgia, Hawaii, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 2368	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C071	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.22, OR 1.25; range across patients 0.22 to 0.22	122, 73	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.14, OR 1.15; range across patients 0.14 to 0.14	122, 78	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C074	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0004, 5e-05], n=1000)	yes	slightly; on the training data only
C075	direction: state: West Virginia to Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds -0.0006, OR 1.00; range across patients -0.0006 to -0.0006	122, 78	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C076	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		log-odds 1e-15, OR n/a	4000	train	yes	
C077	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	4000	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	slightly
C078	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 18% (typical patient with a fold)	fold 2.80 calls (IQR 2.80 to 2.80); past it log-odds 2.07, OR 7.90; range across patients 2.07 to 2.07	4000, 848	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	clearly
C079	structure: made of csc_moderate_chain, csc_high_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C080	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: hinge at 11 messages; 64% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	12% to 6% (typical patient this part acts on)	log-odds -0.26, OR 0.77; range across patients -1.08 to 0.00	4000	train; test agrees (AUC gain 0.006 [0.003, 0.009], n=1000)	yes	clearly
C081	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 221 minutes; 50% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 8%	log-odds 1.13, OR 3.10; range across patients 0.83 to 1.79	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C082	direction: state: state other than Washington to Washington (through this part) *shape: categorical; highest to lowest: state other than Washington, Washington*	12% to 7% (typical patient this part acts on)	log-odds -0.21, OR 0.81; range across patients -0.84 to 0.00	3928, 72	train; test agrees (AUC gain 0.0004 [2e-05, 0.0008], n=1000)	yes	slightly
C083	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 39.34 US dollars; 77% of patients same; fallback; risk saturates near 0%; one point with C090 (median fold 39.34 US dollars, typical patient 42.20 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	3% to 3%	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.94	4000	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; on the training data only
C084	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge above 30.69 US dollars *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 6%	log-odds difference 0.54, OR 1.71	1997, 2003	train; test agrees (AUC gain 0.005 [-0.0004, 0.012], n=1000); group difference on test 0.53 [0.49, 0.57]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: Washington minus state other than Washington; stronger in state other than Washington *shape: hinge at 223 minutes; 51% of patients same; fallback; risk saturates near 0%*	Washington: 11% to 14%; state other than Washington: 3% to 8%	log-odds difference -0.75, OR 0.47	72, 3928	train; test agrees (AUC gain 0.0004 [2e-05, 0.0008], n=1000); group difference on test -0.80 [-0.92, -0.65]	yes	slightly
C086	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge at or below 30.69 US dollars *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 11%	log-odds difference -0.53, OR 0.59	1997, 2003	train; test agrees (AUC gain 0.005 [-0.0004, 0.012], n=1000); group difference on test -0.55 [-0.61, -0.49]	yes	slightly
C087	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime minutes: total daytime minutes above 180 minutes minus total daytime minutes at or below 180 minutes; stronger in total daytime minutes at or below 180 minutes *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime minutes above 180 minutes: 3% to 11%; total daytime minutes at or below 180 minutes: 3% to 11%	log-odds difference -0.53, OR 0.59	1997, 2003	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000); group difference on test -0.55 [-0.61, -0.49]	yes	slightly
C088	threshold: total daytime minutes: fold at about 221 minutes (starts) *shape: hinge at 221 minutes; 48% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 19% (typical patient with a fold)	fold 221 minutes (IQR 213 to 221); past it log-odds 1.98, OR 7.25; range across patients 1.48 to 2.79	3995, 1099	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C089	threshold: voice-mail messages: fold at about 11 messages (stops) for customers in New Jersey *shape: hinge at 11 messages; 74% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	24% to 13% (typical patient with a fold (customers in New Jersey))	fold 10.8 messages (IQR 6.77 to 14.9); past it log-odds -0.93, OR 0.40; range across patients -1.97 to -0.27	36, 26	train; test agrees (AUC gain 0.006 [0.003, 0.009], n=1000)	yes	clearly
C090	threshold: total daytime charge: fold at about 39.34 US dollars (starts) for customers in North Carolina *shape: hinge at 39.34 US dollars; 88% of patients same; fallback; one point with C090 (median fold 39.34 US dollars, typical patient 39.00 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	4% to 5% (typical patient with a fold (customers in North Carolina))	fold 39.3 US dollars (IQR 34.2 to 45.8); past it log-odds 0.51, OR 1.67; range across patients 0.00 to 1.28	50, 17 (small n)	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; the model has little data on this; on the training data only
C091	subgroup: who this part acts on: 1099 of 4000 patients	3% to 12% (typical patient this part acts on (risk without the part, then with it))	share 0.27; log-odds 1.49, OR 4.42; range across patients 0.12 to 3.47	1099, 2901	train; test agrees (AUC gain 0.080 [0.048, 0.112], n=1000)	yes	clearly
C092	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C093	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 6% (typical patient with a fold)	fold 3.21 calls (IQR 3.21 to 3.21); past it log-odds 0.91, OR 2.49; range across patients 0.91 to 0.91	4000, 318	train; test agrees (AUC gain 0.023 [0.012, 0.035], n=1000)	yes	slightly
C094	subgroup: who this part acts on: 318 of 4000 patients	8% to 18% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 1.71, OR 5.51; range across patients 0.91 to 3.72	318, 3682	train; test agrees (AUC gain 0.023 [0.012, 0.035], n=1000)	yes	clearly
C095	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.0, OR n/a	4000	train	yes	
C096	direction: customer-service calls: 0	3% to 3%	log-odds 0.19, OR 1.21; range across patients	4000	train; test agrees (AUC gain 0.028	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 8% (typical patient with a fold)	fold 2.80 calls (IQR 2.80 to 2.80); past it log-odds 1.16, OR 3.18; range across patients 1.16 to 1.16	4000, 848	train; test agrees (AUC gain 0.028 [0.014, 0.042], n=1000)	yes	clearly
C098	subgroup: who this part acts on: 848 of 4000 patients	3% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.21; log-odds 0.76, OR 2.13; range across patients 0.19 to 2.12	848, 3152	train; test agrees (AUC gain 0.028 [0.014, 0.042], n=1000)	yes	slightly
C099	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C100	direction: state: state other than Indiana, Rhode Island to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 22%	log-odds 2.36, OR 10.6; range across patients 2.36 to 2.36	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	on the training data only
C101	direction: state: state other than Indiana, Rhode Island to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 3%	log-odds 0.27, OR 1.31; range across patients 0.27 to 0.27	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	slightly; on the training data only
C102	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C103	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3%	log-odds 0.20, OR 1.22; range across patients 0.20 to 0.20	4000	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C104	threshold: total evening minutes: fold at about 217 minutes (starts) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3% (typical patient with a fold)	fold 217 minutes (IQR 217 to 217); past it log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000, 1467	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C105	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C106	direction: state: state other than Louisiana to Louisiana (through this part) *shape: categorical; highest to lowest: Louisiana, state other than Louisiana*	3% to 3%	log-odds 0.04, OR 1.04; range across patients 0.04 to 0.05	3935, 65	train; test disagrees (AUC gain -6e-05 [-0.0002, 5e-05], n=1000)	yes	slightly; on the training data only
C107	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.004, OR 1.00; range across patients 0.003 to 0.003	4000	train; test unclear (AUC gain 2e-05 [-9e-05, 0.0001], n=1000)	yes	slightly; on the training data only
C108	interaction: effect of voice-mail messages (0 messages to 32 messages) by state: Louisiana minus state other than Louisiana; stronger in Louisiana *shape: steady; 78% of patients same; fallback*	Louisiana: 5% to 19%; state other than Louisiana: 3% to 11%	log-odds difference 0.14, OR 1.15	65, 3935	train; test unclear (AUC gain -6e-05 [-0.0002, 5e-05], n=1000); group difference on test 0.11 [-0.15, 0.29]	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	threshold: voice-mail messages: fold at about 22 messages (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3% (typical patient with a fold)	fold 21.0 messages (IQR 21.0 to 21.0); past it log-odds 0.005, OR 1.00; range across patients 0.005 to 0.005	3935, 876	train; test unclear (AUC gain 2e-05 [-9e-05, 0.0001], n=1000)	no	slightly; on the training data only
C110	subgroup: who this part acts on: 941 of 4000 patients	2% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.24; log-odds 0.006, OR 1.01; range across patients 0.0009 to 0.007	941, 3059	train; test disagrees (AUC gain -5e-05 [-0.0002, 9e-05], n=1000)	no	slightly; on the training data only
C111	null: this part barely changes predictions (switched off)		max change 0.01 (max |change in predicted risk|)	4000	train	yes	
C112	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C113	boundary: boundary		log-odds 0.00, OR n/a	4000	train	no	

Not supported by the model.

- C109: threshold claim about number of voice-mail messages in part ann_84: not supported by the model (the fold changes the curve by only 0.007 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C110: subgroup claim about state and number of voice-mail messages in part ann_84: not supported by the model (its effect is negligible).
- C113: boundary claim about total daytime minutes and number of voice-mail messages in part ann_82: not supported by the model (the boundary rule matches 0.0 of training rows (needs 0.99)).

3.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (37):** `-1` (state:0), `-14` (state:20), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-19` (state:25), `-2` (state:1), `-20` (state:26), `-21` (state:27), `-23` (state:29), `-25` (state:30), `-27` (state:32), `-28` (state:33), `-29` (state:34), `-30` (state:35), `-35` (state:4), `-36` (state:40), `-39` (state:43), `-40` (state:44), `-44` (state:48), `-46` (state:5), `-47` (state:50), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-53` (areaCode:408), `-55` (areaCode:510), `-6` (state:13), `-63` (totalEveCalls), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-67` (totalNightCharge), `-69` (totalIntlCalls), `-7` (state:14), `-70` (totalIntlCharge), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

3.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 3.6.1)

  ▷ A2: linear_backbone (leaf, section 3.6.2)

- ▷ A3: day_usage_hinges (leaf, section 3.6.3)
- ▷ A4: evening_hinge (leaf, section 3.6.4)
- ▷ A5: vmail_state18_unit (leaf, section 3.6.5)
- ▷ A6: state_offsets (leaf, section 3.6.6)
- ▷ A7: service_calls (composition, section 3.6.7)
  - ▷ A8: csc_moderate_chain (leaf, section 3.6.8)
  - ▷ A9: csc_high_hinge (leaf, section 3.6.9)

3.4 Network figures

3.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 4.44×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`, `-71_d`.
- #72: split `-58` into `-58_a`, `-58_b`, `-58_c`.
- #73: split `-43` into `-43_a`, `-43_b`.
- #74: split `-34` into `-34_a`, `-34_b`.
- #75: split `-11` into `-11_a`, `-11_b`.
- #76: consolidate `-71_a`, `-71_c`, `-71_d`.
- #77: identity node `identity_lin` before `0` (29 connection(s)).
- #78: identity node `identity_svc` before `0` (3 connection(s)).

3.6 Annotations

3.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 3.6.2), A3 (section 3.6.3), A4 (section 3.6.4), A5 (section 3.6.5), A6 (section 3.6.6), A7 (section 3.6.7).

- **Entry nodes:** `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-64` (totalEveCharge), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-60` (totalDayCalls), `-59` (totalDayMinutes), `-57` (voiceMailPlan), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-42` (state:46), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-37` (state:41), `-36` (state:40), `-35` (state:4), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-28` (state:33), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-23` (state:29), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), … (17 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `1295`→`0` (-1.01), `3118`→`0` (1.53), `6089`→`0` (0.00514), `7872`→`0` (-1.21), `9954`→`0` (0.0405), `10131`→`0` (1.91), `15524`→`0` (0.201), `identity_lin`→`0` (1), `identity_svc`→`0` (1).

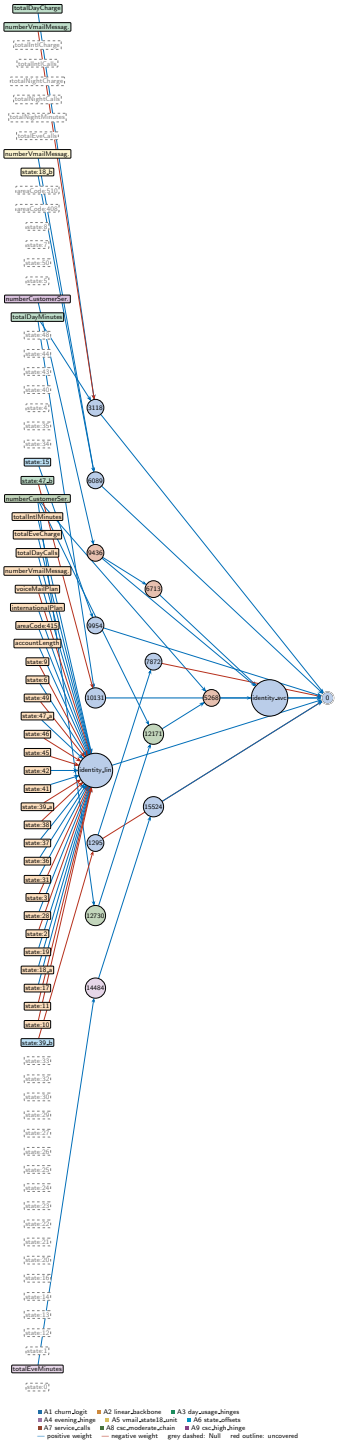


Figure 7: churn100-s4-c2: full view of the explained model state.

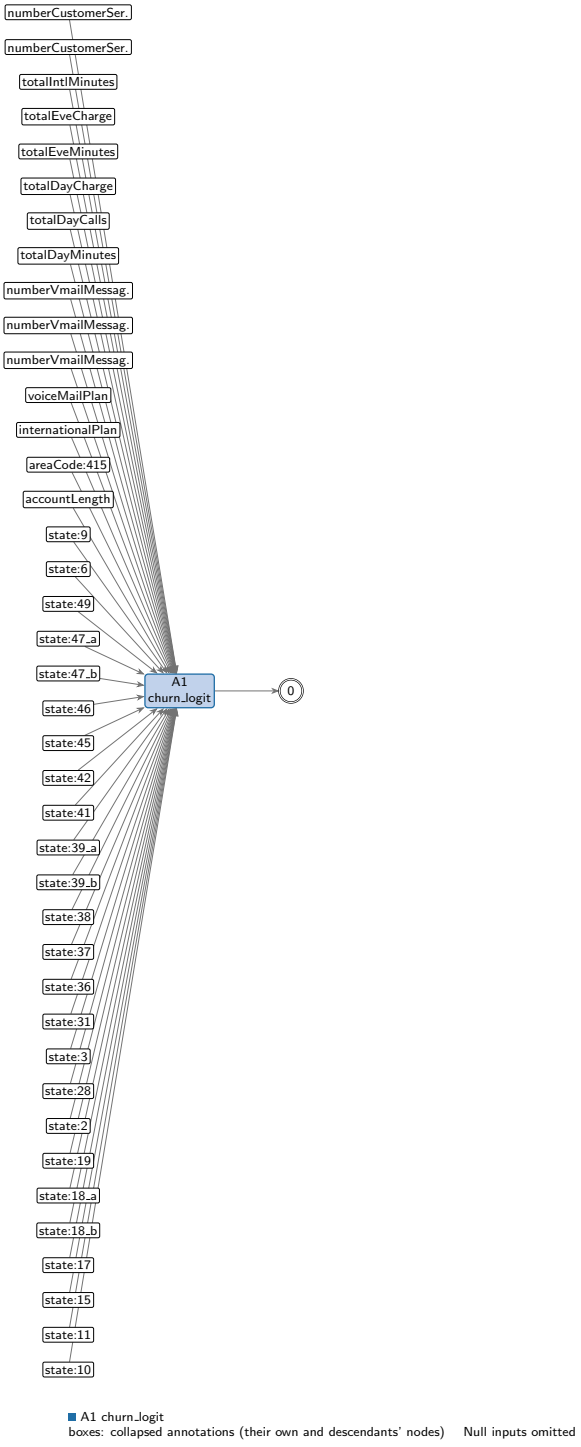


Figure 8: churn100-s4-c2: collapsed view at hierarchy level 0 (fully recomposed).

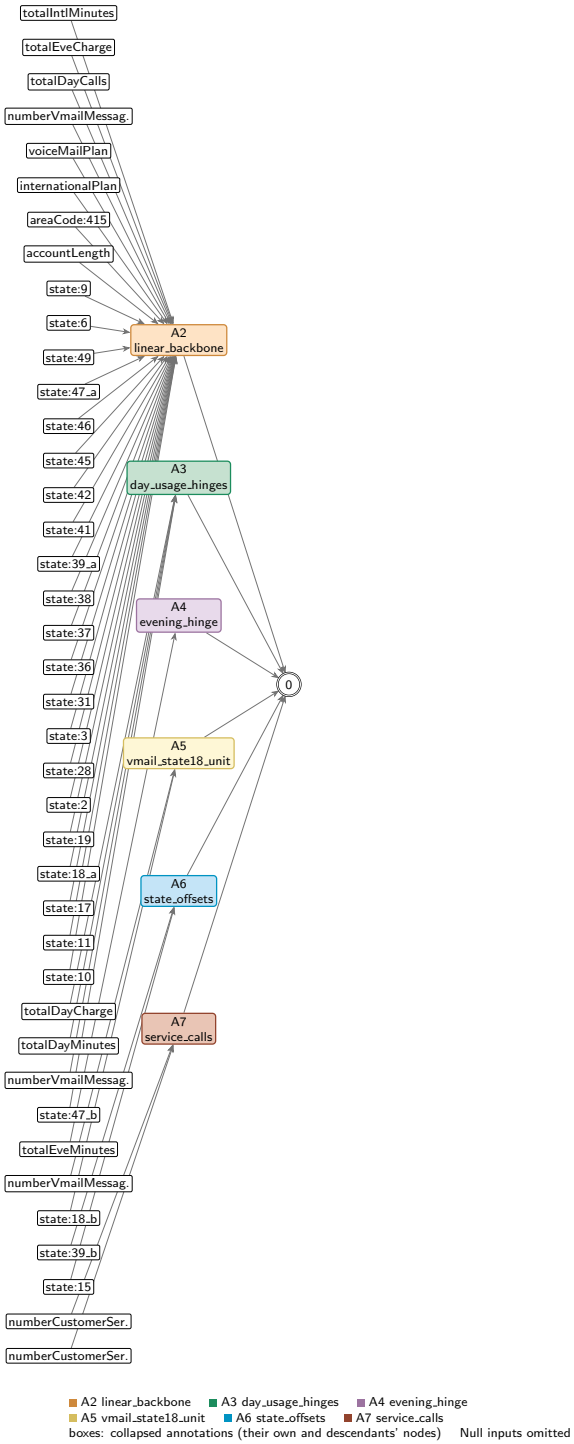


Figure 9: churn100-s4-c2: collapsed view at hierarchy level 1.

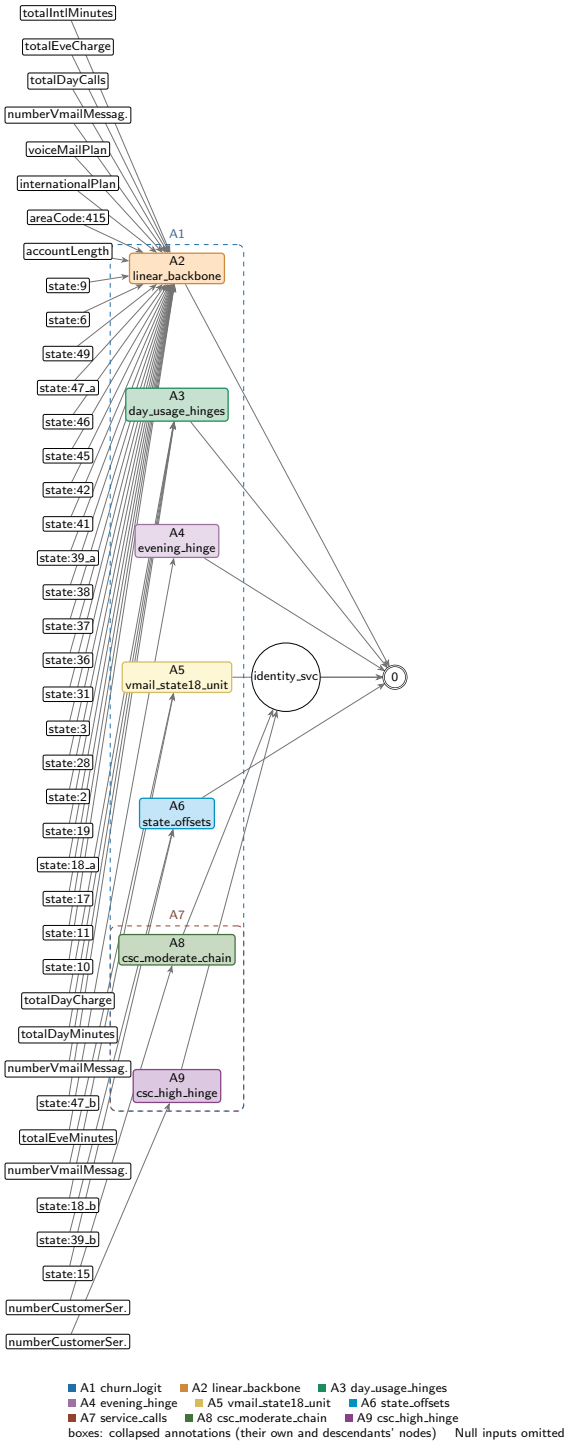


Figure 10: churn100-s4-c2: collapsed view at hierarchy level 2.

- **With its descendants:** 93 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 92 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/ calls, evening calls, area codes 408/510 and 29 state indicators) and have zero effect. Train AUC 0.8908, accuracy 0.888. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8333 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_ usage_hinges alone gives 0.8707, service_calls alone 0.8659, both 0.8907; evening, state and voicemail units restore nothing. So the gain over linear (+0.058) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.75 logit (73 rows, empirical -1.35); state:46 -1.94 (80 rows, -0.91); state:45 -0.94 (91, -0.85); state:47 +0.90 mean (72 rows, empirical +0.92; direct +1.23, but it switches off 10131 so heavy day users lose up to 2.75); state:31 +0.83 (90, +0.67); state:18 +0.41 (65 rows, empirical -0.48: opposite sign); state:39 -0.25 net (82, -0.93: direct -2.61 almost cancelled by the state units).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -1.3 + 1.0\,\text{linear_backbone}\big(x_{state:17}, x_{state:18_a},\\
&\quad x_{state:19}, x_{state:2}, x_{state:28}, x_{state:3}, x_{state:31}, x_{state:10},\\
&\quad x_{state:36}, x_{state:37}, x_{state:38}, x_{state:39_a}, x_{state:41}, x_{state:42},\\
&\quad x_{state:11}, x_{state:45}, x_{state:46}, x_{state:47_a}, x_{state:49}, x_{state:6},\\
&\quad x_{state:9}, x_{accountLength}, x_{areaCode:415}, x_{internationalPlan}, x_{voiceMailPlan},\\
&\quad x_{numberVmailMessages_a}, x_{totalDayCalls}, x_{totalEveCharge}, x_{totalIntlMinutes}\big)\\
&\quad + 1.0\,\text{service_calls}\big(x_{numberCustomerServiceCalls_acd},\\
&\quad x_{numberCustomerServiceCalls_b}\big)\\
&\quad + 0.201\,\text{evening_hinge}\big(x_{totalEveMinutes}\big)\\
&\quad + 1.53\,\text{day_usage_hinges_0}\big(x_{state:47_b}, x_{numberVmailMessages_b},\\
&\quad x_{totalDayCharge}, x_{totalDayMinutes}\big)\\
&\quad + 0.00514\,\text{vmail_state18_unit}\big(x_{state:18_b},\\
&\quad x_{numberVmailMessages_c}\big) + 0.0405\,\text{state_offsets_2}\big(x_{state:15},\\
&\quad x_{state:39_b}\big) + 1.91\,\text{day_usage_hinges_1}\big(x_{state:47_b},\\
&\quad x_{numberVmailMessages_b}, x_{totalDayCharge}, x_{totalDayMinutes}\big)\\
&\quad - 1.01\,\text{state_offsets_0}\big(x_{state:15}, x_{state:39_b}\big)\\
&\quad - 1.21\,\text{state_offsets_1}\big(x_{state:15}, x_{state:39_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -1.3 + 0.00719x_{areaCode:415} + 0.201\,\text{relu}\big(0.714\\
&\quad + 1.02\,\text{relu}\big(-0.33 + 1.01x_{totalEveMinutes}\big)\big)\\
&\quad + 0.106x_{accountLength} + 0.0569x_{state:6} + 0.163x_{state:47_a}\\
&\quad + 1.53\,\text{relu}\big(-0.856 + 0.382x_{totalDayCharge} + 0.547x_{totalDayMinutes}\\
&\quad - 0.65x_{numberVmailMessages_b}\big) + 1.76\,\text{relu}\big(-0.582\\
&\quad + 0.462x_{numberCustomerServiceCalls_b}\big) + 0.133x_{totalDayCalls}\\
&\quad + 0.00514\,\text{relu}\big(-0.668 + 1.14x_{state:18_b}\\
&\quad + 0.828x_{numberVmailMessages_c}\big) + 0.337x_{totalIntlMinutes} + 0.044x_{state:42}\\
&\quad + 1.66\,\text{relu}\big(-0.557 + 0.358\,\text{relu}\big(-0.443
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.719\,\mathrm{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\
&+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\
&+ 0.254 x_{numberCustomerServiceCalls_acd}\big) + 0.007 x_{state:36} + 0.00781 x_{state:41} \\
&+ 0.621 x_{numberVmailMessages_a} + 0.0397 x_{state:37} \\
&+ 0.0405\,\mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big) + 0.35 x_{totalEveCharge} \\
&+ 0.0453 x_{state:19} + 0.047 x_{state:18_a} + 1.54\,\mathrm{relu}\big(-0.127 \\
&+ 1.43\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \\
&+ 0.826 x_{internationalPlan} + 0.000897 x_{state:28} + 1.91\,\mathrm{relu}\big(-0.721 \\
&+ 0.837 x_{totalDayMinutes} - 0.672 x_{state:47_b}\big) + 0.0307 x_{state:17} \\
&+ 0.124 x_{state:31} - 0.37 x_{state:39_a} - 0.0355 x_{state:9} - 0.0158 x_{state:2} \\
&- 0.00422 x_{state:3} - 0.271 x_{state:46} - 1.12 x_{voiceMailPlan} - 0.141 x_{state:45} \\
&- 0.044 x_{state:49} - 1.01\,\mathrm{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
&- 0.0737 x_{state:38} - 1.21\,\mathrm{relu}\big(0.0138 \\
&+ 1.02\,\mathrm{relu}\big(0.868 - 1.27 x_{state:39_b}\big)\big) - 0.0797 x_{state:10} \\
&- 0.368 x_{state:11}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00219, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/calls, evening calls, ar

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 3.6.1).

- **Entry nodes:** `-10` (state:17), `-11_a` (state:18_a), `-12` (state:19), `-13` (state:2), `-22` (state:28), `-24` (state:3), `-26` (state:31), `-3` (state:10), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-34_a` (state:39_a), `-37` (state:41), `-38` (state:42), `-4` (state:11), `-41` (state:45), `-42` (state:46), `-43_a` (state:47_a), `-45` (state:49), `-48` (state:6), `-51` (state:9), `-52` (accountLength), `-54` (areaCode:415), `-56` (internationalPlan), `-57` (voiceMailPlan), `-58_a` (numberVmailMessages_a), `-60` (totalDayCalls), `-64` (totalEveCharge), `-68` (totalIntlMinutes).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-3`→`identity_lin` (-0.0797), `-4`→`identity_lin` (-0.368), `-10`→`identity_lin` (0.0307), `-12`→`identity_lin` (0.0453), `-13`→`identity_lin` (-0.0158), `-22`→`identity_lin` (8.97×10^{-4}), `-24`→`identity_lin` (-0.00422), `-26`→`identity_lin` (0.124), `-31`→`identity_lin` (0.007), `-32`→`identity_lin` (0.0397), `-33`→`identity_lin` (-0.0737), `-37`→`identity_lin` (0.00781), `-38`→`identity_lin` (0.044), `-41`→`identity_lin` (-0.141), `-42`→`identity_lin` (-0.271), `-45`→`identity_lin` (-0.044), `-48`→`identity_lin` (0.0569), `-51`→`identity_lin` (-0.0355), `-52`→`identity_lin` (0.106), `-54`→`identity_lin` (0.00719), `-56`→`identity_lin` (0.826), `-57`→`identity_lin` (-1.12), `-60`→`identity_lin` (0.133), `-64`→`identity_lin` (0.35),

-68→identity_lin (0.337), -58_a→identity_lin (0.621), -43_a→identity_lin (0.163), -34_a→identity_lin (-0.37), -11_a→identity_lin (0.047).

- **Covers** (coverage, own and as part of the hierarchy): 29 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state offsets: state:11 -0.37 (-2.75 logit for its 73 customers), state:39 -0.37 (-2.61, but the state_offsets units give back +2.36), state:46 -0.27 (-1.94), state:47 +0.16 (+1.23), state:31 +0.12 (+0.83), state:45 -0.14 (-0.94), then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.12, voicemail messages 0.62, international plan 0.83).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.00719x_{areaCode:415} + 0.106x_{accountLength} + 0.0569x_{state:6} \\
& + 0.163x_{state:47_a} + 0.133x_{totalDayCalls} + 0.337x_{totalIntlMinutes} \\
& + 0.044x_{state:42} + 0.007x_{state:36} + 0.00781x_{state:41} \\
& + 0.621x_{numberVmailMessages_a} + 0.0397x_{state:37} + 0.35x_{totalEveCharge} \\
& + 0.0453x_{state:19} + 0.047x_{state:18_a} + 0.826x_{internationalPlan} \\
& + 0.000897x_{state:28} + 0.0307x_{state:17} + 0.124x_{state:31} - 0.37x_{state:39_a} \\
& - 0.0355x_{state:9} - 0.0158x_{state:2} - 0.00422x_{state:3} - 0.271x_{state:46} \\
& - 1.12x_{voiceMailPlan} - 0.141x_{state:45} - 0.044x_{state:49} - 0.0737x_{state:38} \\
& - 0.0797x_{state:10} - 0.368x_{state:11}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00544, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state o

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.3 A3: day_usage_hinges

Kind: Leaf annotation (depth 1); parent A1 (section 3.6.1).

- **Entry nodes:** -43_b (state:47_b), -58_b (numberVmailMessages_b), -61 (totalDayCharge), -59 (totalDayMinutes).
- **Exit nodes:** 3118, 10131.
- **Internal nodes:** none.
- **Edges (weight):** -59→3118 (0.546), -61→3118 (0.382), -59→10131 (0.837), -58_b→3118 (-0.65), -43_b→10131 (-0.672).

- **Covers** (coverage, own and as part of the hierarchy): 4 candidate nodes.

Hypothesis. 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customers with no voicemail messages (its kink moves up about 25 minutes per 10 messages: 267 at 20, 292 at 30), so it is on for only 2% of message users; 10131 switches on at about 221 day minutes for everyone except state:47 (off for all 72 state:47 customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.62 at 220-240, +1.61 at 240-260, +2.59 at 260-280, +4.26 above 280; churn by band 0.06 (200-220), 0.14, 0.26, 0.46, 0.67 vs model 0.08, 0.15, 0.28, 0.43, 0.68 (well calibrated). Sensitivity: 3118 vmail messages 0.15, day minutes 0.12, day charge 0.09; 10131 day minutes 0.18, state:47 0.14. Mean ablation AUC 0.8107 (the largest single drop); linear replacement 0.8657 (R2 0.46/0.45); restored alone onto all-linear 0.8707 (all-linear 0.8333).

Closed form:

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.856 + 0.382 x_{totalDayCharge} + 0.547 x_{totalDayMinutes} \\
&- 0.65 x_{numberVmailMessages_b}\big) \\
y_1 = \mathrm{relu}\big(&-0.721 + 0.837 x_{totalDayMinutes} \\
&- 0.672 x_{state:47_b}\big)
\end{aligned}
$$

where y_0 is node `3118`; y_1 is node `10131`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 8.05×10^{-4}, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customer

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.4 A4: evening_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 3.6.1).

- **Entry nodes:** `-62` (totalEveMinutes).
- **Exit nodes:** `15524`.
- **Internal nodes:** `14484`.
- **Edges (weight):** `-62`→`14484` (1.01), `14484`→`15524` (1.02).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation AUC 0.8907, linear replacement 0.8907 (R2 0.62): negligible for ranking.

Closed form:

$$y_0 = \mathrm{relu}\big(0.714 + 1.02\,\mathrm{relu}\big(-0.33 + 1.01 x_{totalEveMinutes}\big)\big)$$

where y_0 is node `15524`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0165, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.5 A5: vmail_state18_unit

Kind: Leaf annotation (depth 1); parent A1 (section 3.6.1).

- **Entry nodes:** `-11_b` (state:18_b), `-58_c` (numberVmailMessages_c).
- **Exit nodes:** `6089`.
- **Internal nodes:** none.
- **Edges (weight):** `-58_c`→`6089` (0.828), `-11_b`→`6089` (1.14).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.668 + 1.14 x_{state:18_b} + 0.828 x_{numberVmailMessages_c}\big)$$

where y_0 is node `6089`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.02, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.6 A6: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 3.6.1).

- **Entry nodes:** `-8` (state:15), `-34_b` (state:39_b).
- **Exit nodes:** `1295`, `7872`, `9954`.
- **Internal nodes:** none.
- **Edges (weight):** `-8`→`9954` (1.01), `1295`→`7872` (1.02), `-34_b`→`1295` (-1.27).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling state:39's direct -2.61 (net -0.25, against an empirical log-odds offset of -0.93 for its 82 customers); 9954 adds +0.27 for state:15. Evidence: sensitivity 1.24/1.27 for state:39, 0.02 for state:15; linear replacement AUC 0.8908 (unchanged); mean ablation 0.8898.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
y_1 &= \mathrm{relu}\big(0.0138 + 1.02\,\mathrm{relu}\big(0.868 \\
&\qquad - 1.27 x_{state:39_b}\big)\big) \\
y_2 &= \mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big)
\end{aligned}
$$

where y_0 is node `1295`; y_1 is node `7872`; y_2 is node `9954`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00698, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.7 A7: service_calls

Kind: Composition annotation (depth 1); parent A1 (section 3.6.1); children A8 (section 3.6.8), A9 (section 3.6.9).

- **Entry nodes:** `-71_acd` (numberCustomerServiceCalls_acd), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `5268`→`identity_svc` (1.66), `9436`→`identity_svc` (1.76), `6713`→`identity_svc` (1.54).

- **With its descendants:** 8 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 7 candidate nodes.

Hypothesis. Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at 4, +4.44 at 5, +6.81 at 6, +9.18 at 7). Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.118/0.108/0.105/0.115/0.327/0.749/0.956/0.986: the threshold is right, but the model under-predicts at 4 calls and is over-confident from 5 calls (116 customers). Mean-ablating both children gives AUC 0.7976, linear replacement 0.8711; restoring service_calls alone onto the all-linear model gives 0.8659 (all-linear 0.8333).

Closed form (children as named functions):

$$\begin{aligned} y_0 = {} & 1.76\,\mathrm{csc_high_hinge_0}\big(x_{numberCustomerServiceCalls_b}\big) \\ & + 1.66\,\mathrm{csc_moderate_chain}\big(x_{numberCustomerServiceCalls_acd}\big) \\ & + 1.54\,\mathrm{csc_high_hinge_1}\big(x_{numberCustomerServiceCalls_b}\big) \end{aligned}$$

Expanded over the children:

$$\begin{aligned} y_0 = {} & 1.76\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\ & + 1.66\,\mathrm{relu}\big(-0.557 + 0.358\,\mathrm{relu}\big(-0.443 \\ & + 0.719\,\mathrm{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ & + 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ & + 0.254 x_{numberCustomerServiceCalls_acd}\big) + 1.54\,\mathrm{relu}\big(-0.127 \\ & + 1.43\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \end{aligned}$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0106, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.8 A8: csc_moderate_chain

Kind: Leaf annotation (depth 2); parent A7 (section 3.6.7).

- **Entry nodes:** `-71_acd` (numberCustomerServiceCalls_acd).
- **Exit nodes:** `5268`.
- **Internal nodes:** `12730`, `12171`.
- **Edges (weight):** `12171`→`5268` (0.358), `12730`→`12171` (0.719), `-71_acd`→`5268` (0.254), `-71_acd`→`12171` (0.845), `-71_acd`→`12730` (0.801).

- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): line plot of 5268 against calls rises from 0 to 3.6 (correlation 0.74); 5268 active on 21% of customers (all with 3+ calls). Mean ablation AUC 0.8733 (full 0.8908); linear replacement 0.8852 (R2 0.55); restored alone onto the all-linearised model 0.8565 (all-linear 0.8333).

Closed form:

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.557 + 0.358\operatorname{relu}\big(-0.443 \\ &+ 0.719\operatorname{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.254 x_{numberCustomerServiceCalls_acd}\big) \end{aligned}$$

where y_0 is node `5268`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00332, and the exact function by at most 1.11×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): li

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.6.9 A9: csc_high_hinge

Kind: Leaf annotation (depth 2); parent A7 (section 3.6.7).

- **Entry nodes:** `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `9436`, `6713`.
- **Internal nodes:** none.
- **Edges (weight):** `9436`→`6713` (1.43), `-71_b`→`9436` (0.462).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot correlation 0.66; mean ablation AUC 0.8761; linear replacement 0.8857 (R2 0.44/0.41); restored alone onto all-linear 0.8571.

Closed form:

$$\begin{aligned} y_0 &= \operatorname{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\ y_1 &= \operatorname{relu}\big(-0.127 + 1.43\operatorname{relu}\big(-0.582 \\ &\qquad + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \end{aligned}$$

where y_0 is node `9436`; y_1 is node `6713`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0119, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls t

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

3.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

4. churn100-s4-c3: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	4.44e-16	4.44e-16	4.44e-16	agree
Closed-form max abs error (all leaves)	0.0277	0.0277	0.0277	agree
Numeric-function max abs error	n/a	1.78e-15	1.78e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	churn_logit	churn_logit	agree
Leaf / composition counts	5 / 2	n/a	5 / 2	agree
Operation counts	8 splits, 2 identity	add identity node 2, annotate 7, consolidate node 1, rename node 71, split node 8	add identity node 2, annotate 7, consolidate node 1, rename node 71, split node 8	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `state_offsets` (0.0277). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `day_usage_hinges`: *"churn by band 0.06/0.14/0.26/0.46/0.58/0.79 against model 0.04/0.08/0.15/0.24/0.37/0.64 (right shape, under-predicted, as is the whole model)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `state_offsets`: *"net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 5786 (state:26/33/34: +0.38 for state:26 and state:34, -0.68 for state:33 against its direct +1.05), 1390 (state:25 +0.39),..."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 8 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- 1 consolidation(s): copies merged back after a split. Check that the merged copies carry the same role.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): None new. Hidden-node splits were not needed; input splits were done before the one consolidation. The line_plot for csc_early_hinge sweeps only its first entry (state:29), so it is flat at 0; sensitivity and local forward passes were used instead. Display name of the consolidated copy -71_ac is None (cosmetic). Tool calls driven from scratch Python scripts that run experiments/species_leads/xn.py per call (all logged), because the Bash sandbox refuses shell variables around uv.

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `churn100-s4-c3-plain-v5.pdf` in this genome's folder):

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up four parts: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it

to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] This part adds up two parts: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108]

Worth checking: Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to below 1% while the state units raise it to about 12%, so overall it is about 1%. [C061, C104, C025] Worth checking: for customers in Nevada the fixed-weight inputs raise the chance to about 4% while the state units lower it to below 1%, so overall it is about 2%. [C058, C108, C018] Worth checking: the state moves the chance by several points (Washington about 5%, Maine about 3%, Hawaii below 1%), based on 72 to 86 training customers per state. [C004, C006, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C008, C002] Worth checking: the early rise with customer-service calls does not happen for customers in Nebraska. [C093]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C054	structure: made of linear_backbone, service_calls, day_usage_hinges, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C055	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6; range across patients 2.74 to 2.74	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00; range across patients 1.61 to 1.61	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C057	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Washington (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 5%	log-odds 1.35, OR 3.86; range across patients 1.35 to 1.35	2456, 72	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C058	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 4%	log-odds 1.05, OR 2.85; range across patients 1.05 to 1.05	2456, 77	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C059	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.01, OR 2.73; range across patients 1.01 to 1.01	4000	train; test agrees (AUC gain 0.014 [0.003, 0.026], n=1000)	yes	slightly
C061	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.85, OR 0.06; range across patients -2.85 to -2.85	2456, 82	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C062	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	2456, 73	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C063	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail*	1% to <1%	log-odds -2.38, OR 0.09; range across patients -2.38 to -2.38	2939, 1061	train; test agrees (AUC gain 0.038 [0.017, 0.061], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C089	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 10% (typical patient with a fold)	fold 2.73 calls (IQR 2.73 to 2.73); past it log-odds 2.06, OR 7.84; range across patients 2.08 to 2.08	4000, 831	train; test agrees (AUC gain 0.106 [0.065, 0.145], n=1000)	yes	clearly
C090	structure: made of csc_late_chain, csc_early_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C099	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: <1% to <1%; no voice-mail plan: 1% to 6%	log-odds difference -1.11, OR 0.33	1061, 2939	train; test agrees (AUC gain 0.007 [0.003, 0.011], n=1000); group difference on test -1.11 [-1.11, -1.11]	yes	slightly
C100	threshold: total daytime minutes: fold at about 209 minutes (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 16% (typical patient with a fold)	fold 209 minutes (IQR 209 to 224); past it log-odds 2.24, OR 9.41; range across patients 1.15 to 2.63	4000, 1074	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	clearly
C102	subgroup: who this part acts on: 1074 of 4000 patients	1% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.27; log-odds 1.28, OR 3.61; range across patients 0.12 to 3.02	1074, 2926	train; test agrees (AUC gain 0.080 [0.049, 0.113], n=1000)	yes	slightly
C104	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 12%	log-odds 2.30, OR 9.97; range across patients 2.30 to 2.30	3039, 82	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	clearly
C105	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 3%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	3039, 80	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C106	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Tennessee (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.54, OR 1.72; range across patients 0.54 to 0.54	3039, 68	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C107	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Virginia (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 91	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C108	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict churn100-s4-c3 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/churn100-s4-c3.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict churn100-s4-c3 --verified --minutes N --changes "..." --notes "..."`

4 Explanation certificate: churn100-s4-c3

This section is the explanation certificate of the churn100-s4-c3 network. It is rendered from the canonical JSON record `churn100-s4-c3.json` (schema 1.4.0, SHA-256 `99f3bce0c6f3eff7...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 4 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s4, seed 4
Genome	`27f540ac-6acb-4d58-9972-cd71fae31cf0` (NEAT key 24445, generation 99)
Explanation	`ffdb207f-1e0f-4d2e-a330-f47b1ca75c4a` (89 operations)
Train AUC / accuracy	0.8884 / 0.8795 (n = 4000)
Test AUC / accuracy	0.8671 / 0.875 (n = 1000)
Output activation	sigmoid
Phenotype	93 nodes (71 inputs, 21 hidden, 1 output), 76 enabled edges
Explained model state	103 nodes, 78 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 1, split_node 8
Function preservation	max $\|\Delta$ output$\|$ = 4.44×10^{-16} on the training split
Annotations	7 (5 leaf, 2 composition)
Structural coverage	1 (102 of 102 candidate nodes; 30 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`99f3bce0c6f3eff7...`

Table 4: Explanation certificate summary: churn100-s4-c3.

Drafter's summary. logit P(churn) = -1.35 + a sparse linear backbone over 28 inputs (voicemail plan -1.05 per SD, international plan +0.80, voicemail messages +0.68, evening charge +0.40, state offsets) + a customer-service-calls threshold (an early hinge from 3 calls, off in state:29, plus a late chain from 5 calls gated by state:4/state:5: nothing for 0-2 calls, then about +2.3 logit per call from 4) + three day-usage hinges (nothing below about 200 day minutes, then rising in stages to

+4.7 logit above 300 minutes; the first hinge starts about 54 minutes later with a voicemail plan) + exact single-state offsets, all additive in the logit. Day minutes and service calls have no linear term. Train AUC 0.888, accuracy 0.880. Compared with the run's cluster-1 lead (churn100-s4, key 24573), the same two thresholds (service calls, high day usage) carry the nonlinearity, but here day usage is gated by the voicemail plan rather than combined with evening minutes, there is no evening term beyond the linear charge, and many more single-state units.

What resisted explanation. Nothing structural: no hidden node needed splitting (8 input splits, 1 consolidation of two customer-service-calls copies). 11 of the 21 hidden nodes are single-state offsets (exactly affine in one-hot indicators) and 5533 is never active on train (contributes exactly 0), so the nonlinearity sits in 8 nodes: the service-call hinges and the day-usage hinges. Near-cancelling state pairs: state:39 direct -2.85 vs unit +2.30 (net -0.55, empirical -0.93), state:46 -2.22 vs +0.77, state:33 +1.05 vs -0.68. The model is badly calibrated in level: mean prediction 0.082 against a base rate of 0.1415 (0-2 calls: 0.05-0.06 vs 0.11; 4 calls 0.21 vs 0.44), and over-confident from 6 calls (0.89-0.98 vs 0.55-0.68, 42 customers). state:18 gets +0.36 logit against an empirical -0.48 (65 rows); state:4 switches off the late service-call chain although its 41 customers churn more (empirical +0.80). closed_form_max_abs_err 0.028 (state_offsets) is 3-significant-figure rounding; numeric error 1.8e-15.

4.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up four parts: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] This part adds up two parts: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] This part acts on 1074 of 4000 training customers, 290 of whom left; for the

typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108]

The whole model. The model adds up four parts into one score: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] *Evidence.* It is made up of these four parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C054] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C001, C008] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C002] More daytime minutes raise it in an uneven way, from about 1% at 112 minutes to about 6% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C003] The model ignores evening calls, evening minutes, international calls, the international charge, international minutes, night-time calls, the night-time charge and night-time minutes. [C046, C047, C048, C049, C050, C051, C052, C053]

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan, voice-mail messages and the evening charge raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C055, C056, C059, C063] *Evidence.* Through it, an international plan takes a typical customer from about 1% to about 18%, and a voice-mail plan lowers the chance to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] Its formula reproduces the part exactly on all 4000 training customers. [C083]

Service calls. This part raises the chance of leaving with calls to customer service, slowly at first, then faster from about 3 calls upwards; it adds up two parts. [C090, C089] *Evidence.* It is made of two parts that are added: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]

Early rise in calls. This part raises the chance of leaving from about 3 customer-service calls upwards, in an uneven way. [C094] *Evidence.* It acts on 830 of 4000 training customers, 213 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 2% with it. [C095] Through it, calls add to the chance only from about 3 calls upwards, in an uneven way: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C094] Up to 3 calls it raises the chance slowly at first, then faster, but only outside Nebraska: there from about 1% to about 2%, while in Nebraska it stays at about 1%, flattening out near the bottom because the predicted chance cannot go much lower. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C096]

Late rise in calls. This part raises the chance of leaving for a small group of customers, more in Colorado and less in California. [C122, C117, C118] *Evidence.* It acts on 134 of 4000 training customers, 74 of whom left; for the typical customer it acts on, the chance is about 39% without it and about 75% with it. [C122] For the typical customer it acts on, the chance through it is about 75% in Colorado and about 39% in California, against about 52% in other states. [C117, C118] Its formula reproduces the part exactly on all 4000 training customers. [C123]

Heavy daytime use. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 209 minutes, in an uneven way, and much less with a voice-mail plan. [C100, C099] *Evidence.* This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] A voice-mail plan lowers it: for the typical customer it acts on, from about 4% to about 2%. [C098] Its formula reproduces the part exactly on all 4000 training customers. [C103]

State units. This part reads only the state: it raises the chance of leaving in Rhode Island, Vermont and Tennessee, and lowers it in Virginia and Nevada. [C104, C105, C106, C107, C108] *Evidence.* This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108] Its formula reproduces the part exactly on all 4000 training customers. [C116]

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to below 1% while the state units raise it to about 12%, so overall it is about 1%. [C061, C104, C025] Worth checking: for customers in Nevada the fixed-weight inputs raise the chance to about 4% while the state units lower it to below 1%, so overall it is about 2%. [C058, C108, C018] Worth checking: the state moves the chance by several points (Washington about 5%, Maine about 3%, Hawaii below 1%), based on 72 to 86 training customers per state. [C004, C006, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C008, C002] Worth checking: the early rise with customer-service calls does not happen for customers in Nebraska. [C093]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6; range across patients 2.74 to 2.74	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C002	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00; range across patients 1.61 to 1.61	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C003	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.30, OR 3.67; range across patients 0.49 to 1.59	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C004	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	log-odds 1.35, OR 3.86; range across patients 1.35 to 1.35	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C005	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.01, OR 2.73; range across patients 1.01 to 1.01	4000	train; test agrees (AUC gain 0.014 [0.003, 0.026], n=1000)	yes	slightly
C006	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Maine (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 3%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	1693, 86	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C007	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	1693, 73	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C008	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.62, OR 0.07; range across patients -3.50 to -2.38	2939, 1061	train; test agrees (AUC gain 0.049 [0.028, 0.072], n=1000)	yes	slightly
C009	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts,	1% to <1%	log-odds -1.46, OR 0.23; range across patients -1.46 to -1.46	1693, 80	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Illinois (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	1693, 68	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C014	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	1693, 56	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C015	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	1693, 91	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C016	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.64, OR 0.53; range across patients -0.64 to -0.64	1693, 60	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C017	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Mississippi (whole model) *shape: categorical; highest to lowest: Washington*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	1693, 77	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.55, OR 0.58; range across patients -0.55 to -0.55	1693, 82	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C026	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	log-odds 0.35, OR 1.42; range across patients 0.35 to 0.35	1693, 86	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C027	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nebraska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.81, OR 0.44; range across patients -0.83 to -0.53	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C028	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Florida (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.44, OR 0.64; range across patients -0.44 to -0.44	1693, 78	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.30, OR 1.36; range across patients 0.31 to 0.31	4000	train; test agrees (AUC gain 0.106 [0.065, 0.145], n=1000)	yes	slightly
C030	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.001], n=1000)	yes	slightly; on the training data only
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.001], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Arizona (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	1693, 73	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1984	train; test disagrees (AUC gain -7e-05 [-0.0004, 0.0003], n=1000)	yes	slightly; on the training data only
C039	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to California (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds -0.03, OR 0.97; range across patients 0.00 to 0.00	1693, 41	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C040	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	log-odds 0.11, OR 1.11; range across patients 0.00 to 0.22	1693, 78	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C042	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C043	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.59, OR 4.92	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C044	magnitude: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode	1% to 5%	log-odds 1.35, OR 3.86	1693, 72	train; test disagrees (AUC gain -0.013 [-0.028, -0.0007], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	null: model ignores total international charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C050	null: model ignores total international minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C051	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C052	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C053	null: model ignores total night-time minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C054	structure: made of linear_backbone, service_calls, day_usage_hinges, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C055	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	log-odds 2.74, OR 15.6; range across patients 2.74 to 2.74	3620, 380	train; test agrees (AUC gain 0.066 [0.035, 0.098], n=1000)	yes	clearly
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	log-odds 1.61, OR 5.00; range across patients 1.61 to 1.61	4000	train; test agrees (AUC gain 0.014 [0.0002, 0.029], n=1000)	yes	clearly
C057	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Washington (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 5%	log-odds 1.35, OR 3.86; range across patients 1.35 to 1.35	2456, 72	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C058	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 4%	log-odds 1.05, OR 2.85; range across patients 1.05 to 1.05	2456, 77	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C059	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.01, OR 2.73; range across patients 1.01 to 1.01	4000	train; test agrees (AUC gain 0.014 [0.003, 0.026], n=1000)	yes	slightly
C060	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Maine (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 3%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	2456, 86	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.85, OR 0.06; range across patients -2.85 to -2.85	2456, 82	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C062	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	2456, 73	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C063	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -2.38, OR 0.09; range across patients -2.38 to -2.38	2939, 1061	train; test agrees (AUC gain 0.038 [0.017, 0.061], n=1000)	yes	slightly
C064	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	log-odds -2.22, OR 0.11; range across patients -2.22 to -2.22	2456, 80	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C065	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	log-odds 0.54, OR 1.71; range across patients 0.54 to 0.54	2456, 66	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C066	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Alaska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.97, OR 0.38; range across patients -0.97 to -0.97	2456, 56	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C067	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada,	1% to 1%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	2456, 68	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nebraska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.53, OR 0.59; range across patients -0.53 to -0.53	2456, 72	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C074	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.001], n=1000)	yes	slightly; on the training data only
C075	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.001], n=1000)	yes	slightly; on the training data only
C076	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oregon (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23	2456, 84	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C077	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Florida (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.30, OR 0.74; range across patients -0.30 to -0.30	2456, 78	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C078	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds -0.13, OR 0.87; range across patients -0.13 to -0.13	2456, 74	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C079	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	log-odds 0.06, OR 1.07; range across patients 0.06 to 0.06	2456, 68	train; test disagrees (AUC gain -0.013 [-0.030, 0.002], n=1000)	yes	slightly; on the training data only
C080	direction: state: state	1% to 1%	log-odds 0.03, OR 1.03;	2456,	train; test disagrees	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: state other than Nebraska, California, Colorado to Nebraska (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds -0.29, OR 0.75; range across patients -0.31 to 0.00	3809, 72	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C086	direction: state: state other than Nebraska, California, Colorado to California (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds -0.03, OR 0.97; range across patients 0.00 to 0.00	3809, 41	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C087	direction: state: state other than Nebraska, California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	log-odds 0.11, OR 1.11; range across patients 0.00 to 0.22	3809, 78	train; test unclear (AUC gain 0.0001 [-0.0007, 0.002], n=1000)	yes	slightly; on the training data only
C088	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska, California, Colorado; stronger in state other than Nebraska, California, Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska, California, Colorado: 1% to 2%	log-odds difference -0.31, OR 0.74	72, 3809	train; test agrees (AUC gain 0.0001 [-0.0007, 0.002], n=1000); group difference on test -0.31 [-0.31, -0.31]	yes	slightly
C089	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 10% (typical patient with a fold)	fold 2.73 calls (IQR 2.73 to 2.73); past it log-odds 2.06, OR 7.84; range across patients 2.08 to 2.08	4000, 831	train; test agrees (AUC gain 0.106 [0.065, 0.145], n=1000)	yes	clearly
C090	structure: made of csc_late_chain, csc_early_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C091	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.30, OR 1.35; range across patients 0.31 to 0.31	4000	train; test agrees (AUC gain 0.082 [0.050, 0.114], n=1000)	yes	slightly
C092	direction: state: state other than Nebraska to Nebraska (through this part) *shape: categorical; highest to lowest: state other than Nebraska, Nebraska*	2% to 1% (typical patient this part acts on)	log-odds -0.29, OR 0.75; range across patients -0.31 to 0.00	3928, 72	train; test unclear (AUC gain 8e-05 [-0.0002, 0.0003], n=1000)	yes	slightly; on the training data only
C093	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska; stronger in state other than Nebraska *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska: 1% to 2%	log-odds difference -0.31, OR 0.73	72, 3928	train; test agrees (AUC gain 8e-05 [-0.0002, 0.0003], n=1000); group difference on test -0.31 [-0.31, -0.31]	yes	slightly
C094	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 10% (typical patient with a fold)	fold 2.73 calls (IQR 2.73 to 2.73); past it log-odds 2.08, OR 8.00; range across patients 2.08 to 2.08	3928, 830	train; test agrees (AUC gain 0.082 [0.050, 0.114], n=1000)	yes	clearly
C095	subgroup: who this part acts on: 830 of 4000 patients	1% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.21; log-odds 1.36, OR 3.88; range across patients 0.31 to 3.85	830, 3170	train; test agrees (AUC gain 0.083 [0.051, 0.116], n=1000)	yes	slightly
C096	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	log-odds 1.30, OR 3.67; range across patients 0.49 to 1.59	4000	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	slightly
C098	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	4% to 2% (typical patient this part acts on)	log-odds -0.24, OR 0.79; range across patients -1.12 to 0.00	2939, 1061	train; test agrees (AUC gain 0.007 [0.003, 0.011], n=1000)	yes	slightly
C099	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: <1% to <1%; no voice-mail plan: 1% to 6%	log-odds difference -1.11, OR 0.33	1061, 2939	train; test agrees (AUC gain 0.007 [0.003, 0.011], n=1000); group difference on test -1.11 [-1.11, -1.11]	yes	slightly
C100	threshold: total daytime minutes: fold at about 209 minutes (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 16% (typical patient with a fold)	fold 209 minutes (IQR 209 to 224); past it log-odds 2.24, OR 9.41; range across patients 1.15 to 2.63	4000, 1074	train; test agrees (AUC gain 0.074 [0.044, 0.105], n=1000)	yes	clearly
C101	threshold: total daytime charge: fold at about 51.68 US dollars (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 51.7 US dollars (IQR 51.7 to 51.7); past it log-odds 0.00, OR 1.00; range across patients 0.00 to 0.00	4000, 41	train; test unclear (AUC gain 3e-05 [-3e-05, 0.0001], n=1000)	no	slightly; on the training data only
C102	subgroup: who this part acts on: 1074 of 4000 patients	1% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.27; log-odds 1.28, OR 3.61; range across patients 0.12 to 3.02	1074, 2926	train; test agrees (AUC gain 0.080 [0.049, 0.113], n=1000)	yes	slightly
C103	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C104	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 12%	log-odds 2.30, OR 9.97; range across patients 2.30 to 2.30	3039, 82	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	clearly
C105	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 3%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	3039, 80	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C106	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Tennessee (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.54, OR 1.72; range across patients 0.54 to 0.54	3039, 68	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C107	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Virginia (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 91	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C108	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia,	1% to 1%	log-odds -0.66, OR 0.52; range across patients -0.66 to -0.66	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Mississippi (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C110	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Kansas (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 77	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C111	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Montana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 75	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C112	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to New York (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	3039, 93	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C113	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	log-odds 0.37, OR 1.45; range across patients 0.37 to 0.37	3039, 82	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C114	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Delaware (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.20, OR 0.82; range across patients -0.20 to -0.20	3039, 81	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C115	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Florida (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	log-odds -0.14, OR 0.87; range across patients -0.14 to -0.14	3039, 78	train; test agrees (AUC gain 0.015 [0.003, 0.029], n=1000)	yes	slightly
C116	structure: its formula reproduces it on 4000 training patients (largest difference 1.11022e-16)		log-odds 1e-16, OR n/a	4000	train	yes	
C117	direction: state: state other than California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Colorado, state*	52% to 75% (typical patient this part acts on)	log-odds 0.1[illegible]R 1.11; range across patients 0.00 to 0.22	3881, 78	train; test unclear (AUC gain 2e-05 [-0.0009, 0.001], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	threshold: customer-service calls: fold at about 5 calls (starts) *shape: hinge at 4 calls; 98% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 4.21 calls (IQR 4.21 to 4.21); past it log-odds 0.02, OR 1.02; range across patients 0.00 to 0.00	3959, 134	train; test unclear (AUC gain 0.0003 [-0.0007, 0.002], n=1000)	no	slightly; on the training data only
C122	subgroup: who this part acts on: 134 of 4000 patients	39% to 75% (typical patient this part acts on (risk without the part, then with it))	share 0.03; log-odds 0.85, OR 2.35; range across patients 0.53 to 1.88	134, 3866	train; test unclear (AUC gain 0.0003 [-0.0008, 0.002], n=1000)	yes	on the training data only
C123	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	

Not supported by the model.

- C101: threshold claim about total daytime charge in part ann_85: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C121: threshold claim about number of calls to customer service in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients).

4.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (30):** `-10` (state:17), `-13` (state:2), `-14` (state:20), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-2` (state:1), `-25` (state:30), `-26` (state:31), `-27` (state:32), `-30` (state:35), `-37` (state:41), `-39` (state:43), `-44` (state:48), `-45` (state:49), `-47` (state:50), `-48` (state:6), `-49` (state:7), `-5` (state:12), `-53` (areaCode:408), `-55` (areaCode:510), `-6` (state:13), `-62` (totalEveMinutes), `-63` (totalEveCalls), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-69` (totalIntlCalls), `-70` (totalIntlCharge).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

4.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 4.6.1)
  ▷ A2: linear_backbone (leaf, section 4.6.2)
  ▷ A3: day_usage_hinges (leaf, section 4.6.3)
  ▷ A4: state_offsets (leaf, section 4.6.4)
  ▷ A5: service_calls (composition, section 4.6.5)
    ▷ A6: csc_late_chain (leaf, section 4.6.6)
    ▷ A7: csc_early_hinge (leaf, section 4.6.7)

4.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s4-c3-fig-full.tex`.

4.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 4.44×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`.
- #72: split `-60` into `-60_a`, `-60_b`.
- #73: split `-57` into `-57_a`, `-57_b`.
- #74: split `-51` into `-51_a`, `-51_b`.
- #75: split `-42` into `-42_a`, `-42_b`.
- #76: split `-34` into `-34_a`, `-34_b`.
- #77: split `-28` into `-28_a`, `-28_b`.
- #78: split `-23` into `-23_a`, `-23_b`.
- #79: consolidate `-71_a`, `-71_c`.
- #80: identity node `identity_lin` before `0` (28 connection(s)).
- #81: identity node `identity_svc` before `0` (3 connection(s)).

4.6 Annotations

4.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 4.6.2), A3 (section 4.6.3), A4 (section 4.6.4), A5 (section 4.6.5).

- **Entry nodes:** `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-64` (totalEveCharge), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-59` (totalDayMinutes), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-43` (state:47), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-37` (state:41), `-36` (state:40), `-35` (state:4), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-11` (state:18), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), `-6` (state:13), `-5` (state:12), `-4` (state:11), ... (19 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `1295`→`0` (-1.11), `1390`→`0` (-1.17), `3118`→`0` (1.48), `4000`→`0` (1.5), `5533`→`0` (1.98), `5786`→`0` (-0.557), `7872`→`0` (-1.25), `9954`→`0` (0.056), `10131`→`0` (1.89), `11361`→`0` (0.0653), `12690`→`0` (-0.0283), `13068`→`0` (-0.0199), `15362`→`0` (0.0582), `15490`→`0` (0.0266), `15904`→`0` (0.0449), `17102`→`0` (-0.13), `identity_lin`→`0` (1), `identity_svc`→`0` (1).

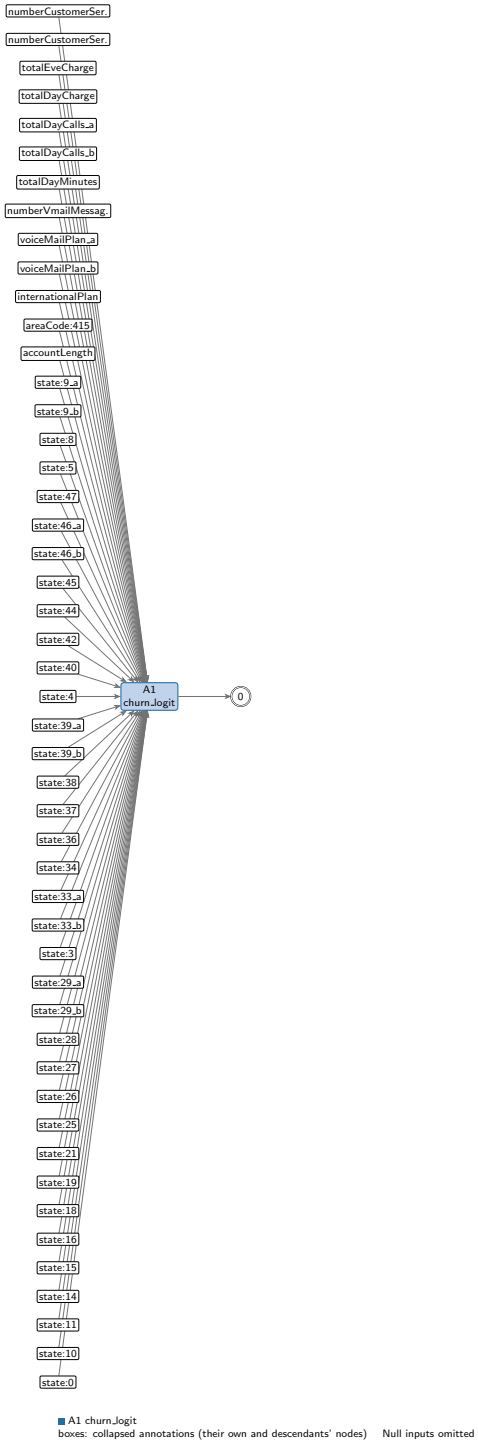


Figure 11: churn100-s4-c3: collapsed view at hierarchy level 0 (fully recomposed).

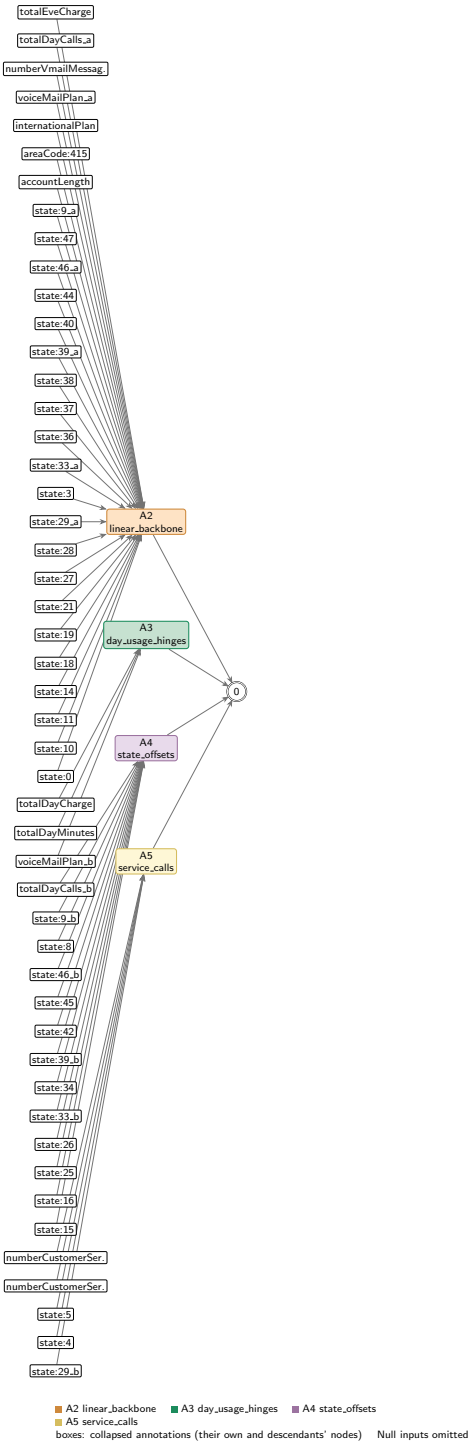


Figure 12: churn100-s4-c3: collapsed view at hierarchy level 1.

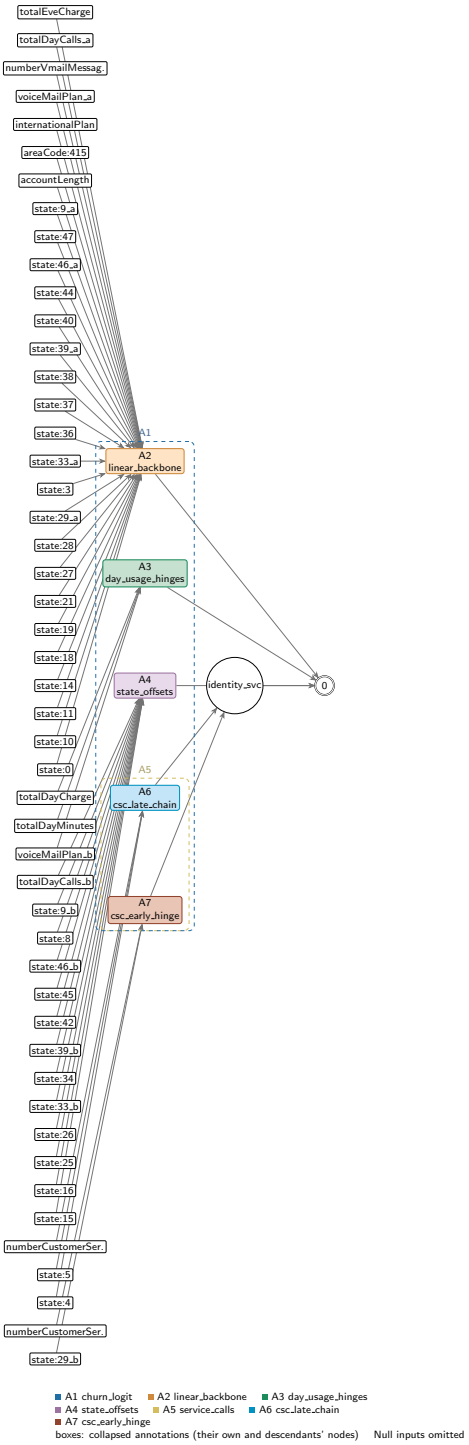


Figure 13: churn100-s4-c3: collapsed view at hierarchy level 2.

- **With its descendants:** 103 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 102 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night minutes/charge/calls, evening minutes and calls, area codes 408/510 and 20 state indicators) and have zero effect. Train AUC 0.8884, accuracy 0.8795; the mean predicted churn is 0.082 against a base rate of 0.1415 (the model ranks well but under-predicts). Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8352 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8713, service_calls alone 0.8657, both 0.8884; the state units restore nothing (exactly linear). So the gain over linear (+0.053) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.50 logit (73 rows, empirical -1.35); state:46 -1.46 net (80 rows, -0.91); state:47 +1.35 (72 rows, +0.92); state:0 -0.97 (56, -0.76); state:14 -0.94 (68, -0.53); state:29 -0.53 direct plus switching off the early service-call hinge (mean -0.78, down to -6.15 for many-call customers; 72 rows, -0.79); state:39 -0.55 net (82, -0.93: direct -2.85 mostly cancelled by the state units); state:33 +0.38 net (direct +1.05, unit -0.68); state:18 +0.36 (65 rows, empirical -0.48: opposite sign); state:4 switches off the late chain (41 rows, empirical +0.80).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.35 + 1.0\,\mathrm{linear_backbone}\big(x_{state:0}, x_{state:18}, \\
&\quad x_{state:19}, x_{state:21}, x_{state:27}, x_{state:28}, x_{state:29_a}, x_{state:3}, \\
&\quad x_{state:33_a}, x_{state:10}, x_{state:36}, x_{state:37}, x_{state:38}, x_{state:39_a}, \\
&\quad x_{state:40}, x_{state:11}, x_{state:44}, x_{state:46_a}, x_{state:47}, x_{state:9_a}, \\
&\quad x_{accountLength}, x_{areaCode:415}, x_{internationalPlan}, x_{voiceMailPlan_a}, \\
&\quad x_{numberVmailMessages}, x_{totalDayCalls_a}, x_{totalEveCharge}, x_{state:14}\big) \\
&\quad + 1.0\,\mathrm{service_calls}\big(x_{numberCustomerServiceCalls_ac}, x_{state:5}, \\
&\quad x_{state:4}, x_{state:29_b}, x_{numberCustomerServiceCalls_b}\big) \\
&\quad + 1.48\,\mathrm{day_usage_hinges_0}\big(x_{voiceMailPlan_b}, x_{totalDayMinutes}, \\
&\quad x_{totalDayCharge}\big) + 1.5\,\mathrm{day_usage_hinges_1}\big(x_{voiceMailPlan_b}, \\
&\quad x_{totalDayMinutes}, x_{totalDayCharge}\big) \\
&\quad + 0.0266\,\mathrm{state_offsets_8}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&\quad + 0.056\,\mathrm{state_offsets_3}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&\quad + 1.89\,\mathrm{day_usage_hinges_2}\big(x_{voiceMailPlan_b}, x_{totalDayMinutes}, \\
&\quad x_{totalDayCharge}\big) + 0.0653\,\mathrm{state_offsets_4}\big(x_{totalDayCalls_b}, \\
&\quad x_{state:33_b}, x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, \\
&\quad x_{state:9_b}, x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&\quad + 0.0449\,\mathrm{state_offsets_9}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&\quad x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&\quad + 0.0582\,\mathrm{state_offsets_7}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&\quad x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b},
\end{aligned}
$$

$$
\begin{aligned}
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&+ 1.98\,\text{state_offsets_12}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.0199\,\text{state_offsets_6}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.17\,\text{state_offsets_2}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.557\,\text{state_offsets_11}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.25\,\text{state_offsets_1}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.13\,\text{state_offsets_10}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 1.11\,\text{state_offsets_0}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big) \\
&- 0.0283\,\text{state_offsets_5}\big(x_{totalDayCalls_b}, x_{state:33_b}, \\
&x_{state:34}, x_{state:26}, x_{state:45}, x_{state:42}, x_{state:16}, x_{state:9_b}, \\
&x_{state:8}, x_{state:46_b}, x_{state:15}, x_{state:25}, x_{state:39_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= -1.35 + 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
&\quad + 1.48\,\text{relu}\big(-0.836 + 0.365x_{totalDayCharge}\big) \\
&\quad + 1.5\,\text{relu}\big(-0.565 + 0.7x_{totalDayMinutes}\big) \\
&\quad + 0.0266\,\text{relu}\big(0.361 + 0.984x_{state:42}\big) \\
&\quad + 0.056\,\text{relu}\big(-0.38 + 1.01x_{state:15}\big) \\
&\quad + 1.88\,\text{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\
&\quad - 1.12x_{state:29_b}\big) + 1.89\,\text{relu}\big(-0.625 \\
&\quad + 0.789x_{totalDayMinutes} - 0.347x_{voiceMailPlan_b}\big) \\
&\quad + 0.0653\,\text{relu}\big(1.36 + 1.65x_{state:46_b}\big) + 0.0685x_{state:40} \\
&\quad + 0.00461x_{state:36} + 0.0449\,\text{relu}\big(-0.0356 \\
&\quad + 0.993\,\text{relu}\big(0.361 + 0.984x_{state:42}\big)\big) + 0.18x_{state:47} \\
&\quad + 0.398x_{totalEveCharge} + 0.103x_{state:21} + 1.71\,\text{relu}\big(-0.557 \\
&\quad + 0.256\,\text{relu}\big(0.476 + 0.74x_{numberCustomerServiceCalls_ac}\big) \\
&\quad + 0.327x_{numberCustomerServiceCalls_ac} - 1.39\,\text{relu}\big(0.219 \\
&\quad + 0.414x_{state:4} - 1.86x_{state:5}\big)\big) + 0.126x_{totalDayCalls_a} \\
&\quad + 0.00838x_{state:28} + 0.00889x_{areaCode:415} + 0.0459x_{state:18} \\
&\quad + 0.805x_{internationalPlan} + 0.0513x_{state:19} + 0.106x_{accountLength} \\
&\quad + 0.0534x_{state:44} + 1.73\,\text{relu}\big(-0.0961
\end{aligned}
$$

$$
\begin{aligned}
&+ 1.49\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\
&- 1.12x_{state:29_b}\big)\big) + 0.0582\,\mathrm{relu}\big(-0.581 \\
&+ 0.999x_{state:16}\big) + 1.98\,\mathrm{relu}\big(-1.54 \\
&+ 0.655\,\mathrm{relu}\big(0.383 + 0.163x_{state:33_b} - 1.5x_{state:26} \\
&- 0.732x_{state:34}\big) - 0.00543x_{totalDayCalls_b}\big) \\
&- 0.0199\,\mathrm{relu}\big(-0.372 + 1.02x_{state:9_b}\big) \\
&- 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 1.17\,\mathrm{relu}\big(0.252 \\
&- 0.574x_{state:25}\big) - 0.404x_{state:39_a} - 0.557\,\mathrm{relu}\big(0.383 \\
&+ 0.163x_{state:33_b} - 1.5x_{state:26} - 0.732x_{state:34}\big) \\
&- 1.25\,\mathrm{relu}\big(0.0571 + 1.02\,\mathrm{relu}\big(0.78 \\
&- 1.28x_{state:39_b}\big)\big) - 0.311x_{state:46_a} - 0.0781x_{state:10} \\
&- 0.0904x_{state:38} - 0.00327x_{state:3} - 0.114x_{state:0} \\
&- 0.13\,\mathrm{relu}\big(-1.42 + 0.995x_{state:45}\big) \\
&- 1.11\,\mathrm{relu}\big(0.78 - 1.28x_{state:39_b}\big) - 0.07x_{state:29_a} \\
&- 0.0181x_{state:27} - 0.0419x_{state:9_a} - 0.0283\,\mathrm{relu}\big(-0.0337 \\
&+ 1.0x_{state:8}\big) - 0.122x_{state:14}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00303, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night m

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 4.6.1).

- **Entry nodes:** `-1` (state:0), `-11` (state:18), `-12` (state:19), `-15` (state:21), `-21` (state:27), `-22` (state:28), `-23_a` (state:29_a), `-24` (state:3), `-28_a` (state:33_a), `-3` (state:10), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-34_a` (state:39_a), `-36` (state:40), `-4` (state:11), `-40` (state:44), `-42_a` (state:46_a), `-43` (state:47), `-51_a` (state:9_a), `-52` (accountLength), `-54` (areaCode:415), `-56` (internationalPlan), `-57_a` (voiceMailPlan_a), `-58` (numberVmailMessages), `-60_a` (totalDayCalls_a), `-64` (totalEveCharge), `-7` (state:14).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`identity_lin` (-0.114), `-3`→`identity_lin` (-0.0781), `-4`→`identity_lin` (-0.335), `-7`→`identity_lin` (-0.122), `-11`→`identity_lin` (0.0459), `-12`→`identity_lin` (0.0513), `-15`→`identity_lin` (0.103), `-21`→`identity_lin` (-0.0181), `-22`→`identity_lin` (0.00838), `-24`→`identity_lin` (-0.00327), `-31`→`identity_lin` (0.00461), `-32`→`identity_lin` (0.0337), `-33`→`identity_lin` (-0.0904), `-36`→`identity_lin` (0.0685), `-40`→`identity_lin` (0.0534), `-43`→`identity_lin` (0.18), `-52`→`identity_lin` (0.106), `-54`→`identity_lin` (0.00889), `-56`→`identity_`

lin (0.805), -58→identity_lin (0.679), -64→identity_lin (0.398), -60_a→identity_lin (0.126), -57_a→identity_lin (-1.05), -51_a→identity_lin (-0.0419), -42_a→identity_lin (-0.311), -34_a→identity_lin (-0.404), -28_a→identity_lin (0.144), -23_a→identity_lin (-0.07).

- **Covers** (coverage, own and as part of the hierarchy): 28 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: state:39 -0.40 per SD (-2.85 logit, but state_offsets gives back +2.30), state:11 -0.33 (-2.50), state:46 -0.31 (-2.22, of which state_offsets gives back +0.77), state:47 +0.18 (+1.35), state:33 +0.14 (+1.05, of which state_offsets takes back -0.68), state:0 -0.97, state:14 -0.94, state:38 -0.77, state:21 +0.71, state:10 -0.64, state:40 +0.54, state:29 -0.53, then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.05, international plan 0.80, voicemail messages 0.68, state:39 0.40, evening charge 0.40).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
& + 0.0685x_{state:40} + 0.00461x_{state:36} + 0.18x_{state:47} + 0.398x_{totalEveCharge} \\
& + 0.103x_{state:21} + 0.126x_{totalDayCalls_a} + 0.00838x_{state:28} \\
& + 0.00889x_{areaCode:415} + 0.0459x_{state:18} + 0.805x_{internationalPlan} \\
& + 0.0513x_{state:19} + 0.106x_{accountLength} + 0.0534x_{state:44} \\
& - 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 0.404x_{state:39_a} \\
& - 0.311x_{state:46_a} - 0.0781x_{state:10} - 0.0904x_{state:38} - 0.00327x_{state:3} \\
& - 0.114x_{state:0} - 0.07x_{state:29_a} - 0.0181x_{state:27} - 0.0419x_{state:9_a} \\
& - 0.122x_{state:14}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00552, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: stat

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.3 A3: day_usage_hinges

Kind: Leaf annotation (depth 1); parent A1 (section 4.6.1).

- **Entry nodes:** -57_b (voiceMailPlan_b), -59 (totalDayMinutes), -61 (totalDayCharge).
- **Exit nodes:** 3118, 4000, 10131.
- **Internal nodes:** none.

- **Edges (weight):** -59→4000 (0.7), -61→3118 (0.364), -59→10131 (0.789), -57_b→10131 (-0.347).

- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at about 209 day minutes without a voicemail plan (about 263 with one), 4000 at about 224 minutes, and 3118 at a day charge of about 51.7 (about 304 minutes; on for only 1% of customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.05 at 200-220, +0.51 at 220-240, +1.31 at 240-260, +2.19 at 260-280, +3.10 at 280-300, +4.69 above 300 (voicemail-plan holders about 1.1-1.7 lower from 240 minutes up); churn by band 0.06/0.14/0.26/0.46/0.58/0.79 against model 0.04/0.08/0.15/0.24/0.37/0.64 (right shape, under-predicted, as is the whole model). Sensitivity: 10131 day minutes 0.18, voicemail plan 0.08; 4000 day minutes 0.14; 3118 day charge 0.004. Mean ablation AUC 0.8083 (the largest single drop); linear replacement 0.8657 (R2 0.47/0.44/0.05); restored alone onto all-linear 0.8713 (all-linear 0.8352).

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.836 + 0.365 x_{totalDayCharge}\big) \\
y_1 &= \mathrm{relu}\big(-0.565 + 0.7 x_{totalDayMinutes}\big) \\
y_2 &= \mathrm{relu}\big(-0.625 + 0.789 x_{totalDayMinutes} \\
&\qquad - 0.347 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

where y_0 is node 3118; y_1 is node 4000; y_2 is node 10131; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00207, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).

- EMPIRICAL, ablation_study; data split: train. 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at ab

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.4 A4: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 4.6.1).

- **Entry nodes:** -60_b (totalDayCalls_b), -28_b (state:33_b), -29 (state:34), -20 (state:26), -41 (state:45), -38 (state:42), -9 (state:16), -51_b (state:9_b), -50 (state:8), -42_b (state:46_b), -8 (state:15), -19 (state:25), -34_b (state:39_b).

- **Exit nodes:** 1295, 7872, 1390, 9954, 11361, 12690, 13068, 15362, 15490, 15904, 17102, 5786, 5533.

- **Internal nodes:** none.

- **Edges (weight):** -8→9954 (1.01), -19→1390 (-0.574), -20→5786 (-1.5), -29→5786 (-0.732), -9→15362 (0.999), -38→15490 (0.984), -41→17102 (0.995), -50→12690 (1), 1295→7872 (1.02), 5786→5533 (0.655), 15490→15904 (0.993), -60_b→5533 (-0.00543), -51_b→13068 (1.02), -42_b→11361 (1.65), -34_b→1295 (-1.28), -28_b→5786 (0.163).

- **Covers** (coverage, own and as part of the hierarchy): 13 candidate nodes.

Hypothesis. Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 5786 (state:26/33/34: +0.38 for state:26 and state:34, -0.68 for state:33 against its direct +1.05), 1390 (state:25 +0.39), 17102 (state:45 -0.66), 15490 -> 15904 (state:42 +0.54), 15362 (state:16 +0.38), 9954 (state:15 +0.37), 12690 (state:8 -0.20), 13068 (state:9 -0.14, plus direct -0.30), and 5533, which also reads day calls but is never active on train (pre-activation at most -0.30), so it contributes exactly 0. Evidence: sensitivities are on the matching state only (state:39 1.25/1.28, state:46 1.65, state:42 0.98); linear replacement AUC 0.8884 (unchanged); mean ablation 0.8856.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(0.78 - 1.28 x_{state:39_b}\big) \\
y_1 &= \mathrm{relu}\big(0.0571 + 1.02\,\mathrm{relu}\big(0.78 \\
&\qquad - 1.28 x_{state:39_b}\big)\big) \\
y_2 &= \mathrm{relu}\big(0.252 - 0.574 x_{state:25}\big) \\
y_3 &= \mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big) \\
y_4 &= \mathrm{relu}\big(1.36 + 1.65 x_{state:46_b}\big) \\
y_5 &= \mathrm{relu}\big(-0.0337 + 1.0 x_{state:8}\big) \\
y_6 &= \mathrm{relu}\big(-0.372 + 1.02 x_{state:9_b}\big) \\
y_7 &= \mathrm{relu}\big(-0.581 + 0.999 x_{state:16}\big) \\
y_8 &= \mathrm{relu}\big(0.361 + 0.984 x_{state:42}\big) \\
y_9 &= \mathrm{relu}\big(-0.0356 + 0.993\,\mathrm{relu}\big(0.361 \\
&\qquad + 0.984 x_{state:42}\big)\big) \\
y_{10} &= \mathrm{relu}\big(-1.42 + 0.995 x_{state:45}\big) \\
y_{11} &= \mathrm{relu}\big(0.383 + 0.163 x_{state:33_b} - 1.5 x_{state:26} \\
&\qquad - 0.732 x_{state:34}\big) \\
y_{12} &= \mathrm{relu}\big(-1.54 + 0.655\,\mathrm{relu}\big(0.383 \\
&\qquad + 0.163 x_{state:33_b} - 1.5 x_{state:26} - 0.732 x_{state:34}\big) \\
&\qquad - 0.00543 x_{totalDayCalls_b}\big)
\end{aligned}
$$

where y_0 is node `1295`; y_1 is node `7872`; y_2 is node `1390`; y_3 is node `9954`; y_4 is node `11361`; y_5 is node `12690`; y_6 is node `13068`; y_7 is node `15362`; y_8 is node `15490`; y_9 is node `15904`; y_{10} is node `17102`; y_{11} is node `5786`; y_{12} is node `5533`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0277, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 578

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.5 A5: service_calls

Kind: Composition annotation (depth 1); parent A1 (section 4.6.1); children A6 (section 4.6.6), A7 (section 4.6.7).

- **Entry nodes:** `-71_ac` (numberCustomerServiceCalls_ac), `-46` (state:5), `-35` (state:4), `-23_b` (state:29_b), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `5268`→`identity_svc` (1.71), `9436`→`identity_svc` (1.88), `6713`→`identity_svc` (1.73).
- **With its descendants:** 11 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 10 candidate nodes.

Hypothesis. Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 calls, then about +2.3 logit per extra call (+2.06 at 4, +4.40 at 5, +6.67 at 6, +9.28 at 7); the early hinge carries most of it, the late chain adds an extra bend from 5 calls. For state:29 customers only the late chain remains. Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.058/0.055/0.050/0.062/0.209/0.573/0.890/0.979: the threshold is right, but the model under-predicts up to 4 calls and is over-confident from 6 calls (42 customers). Sensitivity of identity_svc: state:29 1.04, calls (early copy) 0.48, state:4 0.13. Mean-ablating both children gives AUC 0.8016, linear replacement 0.8713; restoring service_calls alone onto the all-linear model gives 0.8657 (all-linear 0.8352).

Closed form (children as named functions):

$$\begin{aligned} y_0 = {} & 1.88\,\mathrm{csc_early_hinge_0}\big(x_{state:29_b}, \\ & x_{numberCustomerServiceCalls_b}\big) \\ & + 1.71\,\mathrm{csc_late_chain}\big(x_{numberCustomerServiceCalls_ac}, x_{state:5}, \\ & x_{state:4}\big) + 1.73\,\mathrm{csc_early_hinge_1}\big(x_{state:29_b}, \\ & x_{numberCustomerServiceCalls_b}\big) \end{aligned}$$

Expanded over the children:

$$\begin{aligned} y_0 = {} & 1.88\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\ & - 1.12x_{state:29_b}\big) + 1.71\,\mathrm{relu}\big(-0.557 \\ & + 0.256\,\mathrm{relu}\big(0.476 + 0.74x_{numberCustomerServiceCalls_ac}\big) \\ & + 0.327x_{numberCustomerServiceCalls_ac} - 1.39\,\mathrm{relu}\big(0.219 \\ & + 0.414x_{state:4} - 1.86x_{state:5}\big)\big) + 1.73\,\mathrm{relu}\big(-0.0961 \\ & + 1.49\,\mathrm{relu}\big(-0.615 + 0.52x_{numberCustomerServiceCalls_b} \\ & - 1.12x_{state:29_b}\big)\big) \end{aligned}$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0157, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).

- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 ca

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.6 A6: csc_late_chain

Kind: Leaf annotation (depth 2); parent A5 (section 4.6.5).

- **Entry nodes:** `-71_ac` (numberCustomerServiceCalls_ac), `-46` (state:5), `-35` (state:4).
- **Exit nodes:** `5268`.
- **Internal nodes:** `349`, `12374`.
- **Edges (weight):** `-35`→`349` (0.414), `-46`→`349` (-1.86), `349`→`5268` (-1.39), `12374`→`5268` (0.256), `-71_ac`→`5268` (0.327), `-71_ac`→`12374` (0.741).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service calls that is off for 0-4 calls for most customers and adds about +0.65 logit per call from 5 calls (+0.55 at 5, +1.24 at 6, +1.88 at 7); in state:5 it starts one call earlier (on from 3 calls) and in state:4 it is always off. Evidence (train, n=4000): 5268 is active on 3.4% of customers (on for 3% at 4 calls, 99% at 5); sensitivity of 5268 is 0.074 to state:4, 0.017 to calls, 0.017 to state:5; line plot correlation 0.51. Mean ablation AUC 0.8889 (full 0.8884): it adds nothing to ranking on its own; linear replacement 0.8887 (R2 0.27); restored alone onto the all-linearised model 0.8401 (all-linear 0.8352).

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-0.557 + 0.256\operatorname{relu}\big(0.476 \\
&+ 0.74x_{numberCustomerServiceCalls_ac}\big) + 0.327x_{numberCustomerServiceCalls_ac} \\
&- 1.39\operatorname{relu}\big(0.219 + 0.414x_{state:4} - 1.86x_{state:5}\big)\big)
\end{aligned}
$$

where y_0 is node `5268`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00148, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service ca

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

4.6.7 A7: csc_early_hinge

Kind: Leaf annotation (depth 2); parent A5 (section 4.6.5).

- **Entry nodes:** `-23_b` (state:29_b), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `9436`, `6713`.
- **Internal nodes:** none.
- **Edges (weight):** `9436`→`6713` (1.49), `-71_b`→`9436` (0.52), `-23_b`→`9436` (-1.12).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all with 3+ calls, outside state:29); empirical churn jumps from 0.11 at 3 calls to 0.44 at 4. Evidence: sensitivity 0.35 (6713) / 0.23 (9436) to state:29 and 0.16 / 0.11 to calls; the line_plot sweeps state:29 (first entry), so it is flat at 0 (tool quirk: line plots vary only the first entry). Mean ablation AUC 0.8281; linear replacement 0.8736 (R2 0.57/0.50); restored alone onto all-linear 0.8654.

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.615 + 0.52 x_{numberCustomerServiceCalls_b} \\
&\qquad - 1.12 x_{state:29_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.0961 + 1.49 \operatorname{relu}\big(-0.615 \\
&\qquad + 0.52 x_{numberCustomerServiceCalls_b} - 1.12 x_{state:29_b}\big)\big)
\end{aligned}
$$

where y_0 is node `9436`; y_1 is node `6713`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00608, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all wit

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4.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

5. churn100-s7-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	1.22e-15	1.22e-15	1.22e-15	agree
Closed-form max abs error (all leaves)	0.03806	0.03806	0.03806	agree
Numeric-function max abs error	n/a	8.88e-16	8.88e-16	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	churn_logit	churn_logit	agree
Leaf / composition counts	5 / 2	n/a	5 / 2	agree
Operation counts	5 splits, 2 identity	add identity node 2, annotate 7, rename node 71, split node 5	add identity node 2, annotate 7, rename node 71, split node 5	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `state_offsets` (0.0381). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `state_offsets`: *"always active, constant +0.008), 5667 state:2 is never active (dead unit, zero effect)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `intl_plan_switch`: *"linear replacement 0.9189 (identical, exact switch)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 5 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): None new. Tool calls driven from scratch Python scripts that run experiments/species_leads/xn.py per call (all logged), because the worktree-isolated Bash refuses compound shell commands. Ablations at output-term level on the original phenotype (local numpy).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `churn100-s7-c1-plain-v5.pdf` in this genome's folder):

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up five parts: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment and state adjustments. [C047] This part combines a usage score, built from daytime, evening and night-time minutes and the voice-mail plan, with calls to customer service, not as a simple sum. [C060] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan they add nothing until a point, then steadily: going from 0 to 4 calls takes a typical customer from about 1% to about 15%. [C055] Through it, daytime minutes add to the chance only above about 187 minutes for customers without a voice-mail plan, in an uneven way: for a typical customer, going from 0 to 271 minutes takes it from about 1% to about 11%. [C056] Daytime minutes count more with more than 201 evening minutes, in an uneven way: from about 1% to about 11% between 112 and 249 minutes, against about 1% to about 2% with fewer. [C067] This part reads the international plan alone and acts on the 380 of 4000 training customers who have one, 161 of whom left; for the typical

customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance of leaving from about 1% to about 15%. [C077] This part adds up many inputs with fixed weights, each moving a typical customer's chance by a few points at most: compared with most other states, Washington raises it to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] A higher international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and more voice-mail messages and a longer account also raise it steadily, by less than a point. [C083, C086, C098] More international calls lower it, each extra call taking away about the same, from about 1% at 2 to under 1% at 8 calls, and more evening and night-time minutes lower it steadily through this part, by less than a point. [C087, C092, C096] This part reads the voice-mail plan and the state: through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also lowers a typical customer's chance to under 1%. [C108, C109] This part reads only the state: it moves a typical customer's chance from about 1% in West Virginia to about 5% in South Dakota and about 2% in Indiana and in most other states, and keeps it under 1% in Virginia (66 to 122 training customers per named state; the full list is in the technical detail). [C111, C112, C114, C115]

Worth checking: Worth checking: the state moves the chance by several points (for example Washington about 5% and Vermont under 1% across the whole model), based on 72 to 122 training customers per state. [C002, C022] Worth checking: South Dakota gets a rise through the state adjustments (to about 5%) and a fall through the voice-mail and South Dakota adjustment (to under 1%), so the two pull against each other. [C111, C108] Worth checking: more evening and night-time minutes raise the chance through the usage score but lower it through the fixed-weight inputs. [C063, C064, C092, C096] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C027, C023]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C047	structure: made of linear_backbone, usage_response, intl_plan_switch, vmail_state41_unit, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C055	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 86% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 15% (typical patient with a fold)	fold 2.24 calls (IQR 2.24 to 2.96); past it log-odds 2.20, OR 9.02; range across patients 0.00 to 2.90	3833, 614	train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000)	yes	clearly
C056	threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 11% (typical patient with a fold)	fold 187 minutes (IQR 158 to 214); past it log-odds 2.80, OR 16.4; range across patients 0.00 to 6.18	3014, 1382	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	clearly
C060	structure: made of usage_hinge and inputs number customer service calls; combined non-additively		log-odds n/a, OR n/a	4000	train	yes	
C067	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening minutes: total evening minutes above 201 minutes minus total evening minutes at or below 201 minutes; stronger in total evening minutes above 201 minutes *shape: accelerating; 75% of patients same; fallback; risk saturates near 0%*	total evening minutes above 201 minutes: 1% to 11%; total evening minutes at or below 201 minutes: 1% to 2%	log-odds difference 1.54, OR 4.66	1998, 2002	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000); group difference on test 1.71 [1.48, 1.94]	yes	clearly
C077	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	3620, 380	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C078	subgroup: who this part acts on: 380 of 4000 patients	2% to 26% (typical patient this part acts on (risk without the part, then with it))	share 0.10; log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	380, 3620	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C080	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	log-odds 1.10, OR 2.99; range across patients 1.10 to 1.10	2529, 72	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C081	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.81, OR 2.24; range across patients 0.81 to 0.81	2529, 75	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C082	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -1.89, OR 0.15; range across patients -1.89 to -1.89	2529, 80	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -1.09, OR 0.34; range across patients -1.09 to -1.09	2529, 78	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C086	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47	4000	train; test unclear (AUC gain 0.0008 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C087	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	4000	train; test unclear (AUC gain 0.002 [-0.004, 0.008], n=1000)	yes	slightly; on the training data only
C092	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.35, OR 0.70; range across patients -0.35 to -0.35	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.0005], n=1000)	yes	slightly; on the training data only
C096	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.29, OR 0.75; range across patients -0.29 to -0.29	4000	train; test disagrees (AUC gain -0.002 [-0.006, 0.0006], n=1000)	yes	slightly; on the training data only
C098	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.001], n=1000)	yes	slightly; on the training data only
C108	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: state other than South Dakota, South Dakota*	1% to <1%	log-odds -1.04, OR 0.35; range across patients -1.04 to -1.04	3934, 66	train; test agrees (AUC gain 0.001 [0.0004, 0.002], n=1000)	yes	slightly
C109	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	2939, 1061	train; test unclear (AUC gain 0.002 [-0.003, 0.008], n=1000)	yes	slightly; on the training data only
C111	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 5%	log-odds 1.73, OR 5.62; range across patients 1.73 to 1.73	122, 66	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C112	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 2%	log-odds 0.83, OR 2.30; range across patients 0.83 to 0.83	122, 82	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C114	direction: state: West Virginia to state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia …*	1% to 2%	log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69	122, 3505	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C115	direction: state: West Virginia to Virginia (through this part)	1% to 1%	log-odds -0.43, OR 0.65; range across patients -0.43 to -0.43	122, 91	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004],	yes	slightly; on the training

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict churn100-s7-c1 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/churn100-s7-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict churn100-s7-c1 --verified --minutes N --changes "..." --notes "..."`

5 Explanation certificate: churn100-s7-c1

This section is the explanation certificate of the churn100-s7-c1 network. It is rendered from the canonical JSON record `churn100-s7-c1.json` (schema 1.4.0, SHA-256 `55f21bd4a4575c9c...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 5 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s7, seed 7
Genome	`32d24e98-266e-450c-88c8-5bdc1abbf5dd` (NEAT key 20708, generation 99)
Explanation	`b370983e-3d7d-40c8-8b8f-f37c41a96cc1` (85 operations)
Train AUC / accuracy	0.9189 / 0.921 (n = 4000)
Test AUC / accuracy	0.9007 / 0.921 (n = 1000)
Output activation	sigmoid
Phenotype	86 nodes (71 inputs, 14 hidden, 1 output), 59 enabled edges
Explained model state	95 nodes, 61 enabled edges
Refactoring operations	add_identity_node 2, split_node 5
Function preservation	max $\|\Delta$ output$\| = 1.22 \times 10^{-15}$ on the training split
Annotations	7 (5 leaf, 2 composition)
Structural coverage	1 (94 of 94 candidate nodes; 35 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`55f21bd4a4575c9c...`

Table 5: Explanation certificate summary: churn100-s7-c1.

Drafter's summary. logit P(churn) = -1.96 + a small linear backbone over 27 inputs (all |w| <= 0.32 per SD; state offsets up to -1.9 logit) + an international-plan switch (+2.91 logit) + an always-on voicemail-plan/state:41 offset unit (-0.54, -1.04) + exact single-state offsets + usage_response, all additive in the logit. All nonlinearity is in usage_response: a usage hinge (day/evening/night minutes, never active for voicemail-plan customers, on from about 190 day minutes), a steeper

second hinge for heavy users (from about 240 day minutes), and a service-calls gate (shut for 0-2 calls, opening from 3) that is closed again by usage, so service calls matter only for light users. Train AUC 0.919, accuracy 0.921. This retrained elite copy of the 10-seed churn100-s7 genome (same key 20708, generation 98) has the same three-part usage response and intl-plan switch; the weights differ slightly (e.g. intl plan +2.91 here vs +2.88).

What resisted explanation. Nothing structural (5 input splits, no hidden splits, no consolidation). The service-calls gate (4344) and the heavy-use hinge (14351 -> 10552) take the usage hinge 879 as their only hidden input, so they sit as own nodes of the usage_response composition rather than in leaves (a leaf cannot take another annotation's exit as an entry). Calibration gaps: the gate opens at 3 calls (+0.80 logit; model 0.150 vs empirical 0.106 churn, 530 customers), under-predicts at 4 calls (0.32 vs 0.44) and is over-confident from 6 calls (0.85-0.94 vs 0.55-0.68; 39 customers). State offsets are overstated for state:46 (-1.89 vs empirical -0.91), state:5 (-1.09 vs -0.51), state:10 (-0.63 vs -0.22) and state:49 (-0.69 vs -0.32). 5667 (state:2) is a dead unit (never active). closed_form_max_abs_err 0.038 (state_offsets, 13164 pre-activation up to 7.7) is 3-significant-figure rounding; numeric error 8.9e-16.

5.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up five parts: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment and state adjustments. [C047] This part combines a usage score, built from daytime, evening and night-time minutes and the voice-mail plan, with calls to customer service, not as a simple sum. [C060] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan they add nothing until a point, then steadily: going from 0 to 4 calls takes a typical customer from about 1% to about 15%. [C055] Through it, daytime minutes add to the chance only above about 187 minutes for customers without a voice-mail plan, in an uneven way: for a typical customer, going from 0 to 271 minutes takes it from about 1% to about 11%. [C056] Daytime minutes count more with more than 201 evening minutes, in an uneven way: from about 1% to about 11% between 112 and 249 minutes, against about 1% to about 2% with fewer. [C067] This part reads the international plan alone and acts on the 380 of 4000 training customers who have one, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance of leaving from about 1% to about 15%. [C077] This part adds up many inputs with fixed weights, each moving a typical customer's chance by a few points at most: compared with most other states, Washington raises it to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] A higher international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and more voice-mail messages and a longer account also raise it steadily, by less than a point. [C083, C086, C098] More international calls lower it, each extra call taking away about the same, from about 1% at 2 to under 1% at 8 calls, and more evening and night-time minutes lower it steadily through this part, by less than a point. [C087, C092, C096] This part reads the voice-mail plan and the state: through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also lowers a typical customer's chance to under 1%. [C108, C109] This part reads only the state: it moves a typical customer's chance from about 1% in West Virginia to about 5% in South Dakota and about 2% in Indiana and in most other states,

and keeps it under 1% in Virginia (66 to 122 training customers per named state; the full list is in the technical detail). [C111, C112, C114, C115]

The whole model. The model adds up five parts into one score: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment, and state adjustments. [C047] *Evidence.* It is made up of these five parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C047] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 15%, and more daytime minutes raise it in an uneven way, from about 1% at 112 to about 4% at 249 minutes. [C001, C004] In Washington the chance is about 5% and in Montana about 4%, against about 1% in West Virginia and under 1% in Vermont. [C002, C003, C022] The model ignores daytime calls, evening calls, the evening charge, international minutes and the night-time charge. [C042, C043, C044, C045, C046]

Usage and service calls. This part raises the chance of leaving for heavy daytime users without a voice-mail plan and for customers who call customer service 3 times or more: daytime minutes in an uneven way, and calls for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily. [C056, C055] *Evidence.* It is made up of the usage score (explained below) together with customer-service calls, combined in a way that is not a simple sum. [C060] Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes for a typical customer. [C056] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily, from about 1% to about 15% between 0 and 4 calls. [C055]

Usage score. This part builds a usage score: heavy daytime use raises the chance of leaving, in an uneven way and more so with high evening minutes, and a voice-mail plan lowers it. [C071, C067, C062] *Evidence.* Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes. [C071] Daytime minutes count more with more than 201 evening minutes (about 1% to about 11%, against about 1% to about 2%) and with more than 200 night-time minutes (about 1% to about 7%, against about 1% to about 3%), in an uneven way. [C067, C068] A voice-mail plan lowers the chance through it from about 3% to about 1% for the typical customer it acts on. [C062] Its formula reproduces the part exactly on all 4000 training customers. [C076]

International plan. This part raises the chance of leaving for customers with an international plan. [C077] *Evidence.* It acts on the 380 of 4000 training customers with an international plan, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance from about 1% to about 15%. [C077] Its formula reproduces the part exactly on all 4000 training customers. [C079]

Fixed-weight inputs. This part adds small fixed-weight effects: a higher international charge, more voice-mail messages and a longer account raise the chance steadily, more international calls lower it, each extra call taking away about the same, and some states shift it up or down. [C083, C086, C098, C087, C080, C082] *Evidence.* Through it, compared with most other states, Washington raises a typical customer's chance to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] The international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and international calls lower it from about 1% at 2 to under 1% at 8 calls, each extra call taking away about the same. [C083, C087] More evening and night-time minutes lower it steadily through this part, by less than a point. [C092, C096] Its formula reproduces the part exactly on all 4000 training customers. [C107]

Voice mail and South Dakota. This part lowers the chance of leaving for customers in South Dakota and for customers with a voice-mail plan. [C108, C109] *Evidence.* Through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also takes a typical customer to under 1%. [C108, C109] Its formula reproduces the part exactly on all 4000 training customers. [C110]

State adjustments. This part shifts the chance of leaving by state: it is lowest in West Virginia and Virginia and highest in South Dakota. [C111, C114, C115] *Evidence.* Through it, a typical customer's chance is about 5% in South Dakota and about 2% in Indiana and in most other states, against about 1% in West Virginia and under 1% in Virginia. [C111, C112, C114, C115] Its formula reproduces the part exactly on all 4000 training customers. [C117]

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example Washington about 5% and Vermont under 1% across the whole model), based on 72 to 122 training customers per state. [C002, C022] Worth checking: South Dakota gets a rise through the state adjustments (to about 5%) and a fall through the voice-mail and South Dakota adjustment (to under 1%), so the two pull against each other. [C111, C108] Worth checking: more evening and night-time minutes raise the chance through the usage score but lower it through the fixed-weight inputs. [C063, C064, C092, C096] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C027, C023]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	3620, 380	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 5%	log-odds 1.79, OR 5.98; range across patients 1.79 to 1.79	122, 72	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C003	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 4%	log-odds 1.50, OR 4.47; range across patients 1.50 to 1.50	122, 75	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32	4000	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	slightly
C005	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	log-odds 0.86, OR 2.37; range across patients 0.00 to 1.25	4000	train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000)	yes	slightly
C006	direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 3%	log-odds 1.12, OR 3.06; range across patients 1.12 to 1.12	122, 66	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C007	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 3%	log-odds 0.94, OR 2.55; range across patients 0.79 to 1.12	122, 66	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C008	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 3%	log-odds 1.02, OR 2.78; range across patients 1.02 to 1.02	122, 90	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C009	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 2%	log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98	122, 87	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C010	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 2%	log-odds 0.97, OR 2.64; range across patients 0.97 to 0.97	122, 74	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C011	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 2%	log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96	122, 84	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C012	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 2%	log-odds 0.91, OR 2.47; range across patients 0.91 to 0.91	122, 81	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.83, OR 2.30; range across patients 0.83 to 0.83	122, 82	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C014	direction: state: West Virginia to Mississippi (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.80, OR 2.22; range across patients 0.80 to 0.80	122, 77	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C015	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.70, OR 2.02; range across patients 0.70 to 0.70	122, 82	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C016	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: dip at 212 minutes; 36% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.67, OR 1.96; range across patients -0.35 to 3.65	4000	train; test agrees (AUC gain 0.030 [0.010, 0.050], n=1000)	no	slightly
C017	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69	122, 78	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C018	direction: state: West Virginia to state other than Alaska, Georgia, Indiana, Kentucky, Arkansas, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69	122, 2191	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C019	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.80, OR 2.23; range across patients 0.80 to 0.80	4000	train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.58, OR 1.78; range across patients 0.58 to 0.58	122, 86	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C021	direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.55, OR 1.74; range across patients 0.55 to 0.55	122, 72	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -1.20, OR 0.30; range across patients -1.20 to -1.20	122, 80	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C023	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47	4000	train; test unclear (AUC gain 0.0008 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47	122, 68	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	4000	train; test unclear (AUC gain 0.002 [-0.004, 0.008], n=1000)	yes	slightly; on the training data only
C026	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	122, 82	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C027	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -1.37, OR 0.25; range across patients -3.46 to -0.54	2939, 1061	train; test agrees (AUC gain 0.019 [0.005, 0.031], n=1000)	yes	slightly
C028	direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40	122, 78	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C029	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 40.06 US dollars; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 38.81 US dollars*	1% to 1%	log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11	4000	train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C030	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds -0.28, OR 0.76; range across patients -0.28 to -0.28	122, 91	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.001], n=1000)	yes	slightly; on the training data only
C032	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: dip at 218 minutes; 30% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.34, OR 1.40; range across patients -0.29 to 2.38	4000	train; test agrees (AUC gain 0.007 [0.0006, 0.013], n=1000)	no	slightly
C033	direction: area code: area code other than area code 510 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 510, area code other than area code 510*	1% to 1%	log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10	3040, 960	train; test unclear (AUC gain 8e-05 [-0.001, 0.002], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07	122, 60	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C035	direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.06, OR 1.07; range across patients 0.06 to 0.06	4000	train; test disagrees (AUC gain -9e-05 [-0.0008, 0.0006], n=1000)	yes	slightly; on the training data only
C036	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02	122, 56	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	log-odds 2.91, OR 18.3	3620, 380	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C038	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...*	1% to 5%	log-odds 1.79, OR 5.98	122, 72	train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000)	yes	slightly; on the training data only
C039	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	log-odds 1.47, OR 4.37	4000	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	slightly
C040	magnitude: customer-service calls: 0 calls to 3 calls, typical patient *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	log-odds 1.25, OR 3.50	4000	train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000)	yes	slightly
C041	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.80, OR 2.23	4000	train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000)	yes	slightly; on the training data only
C042	null: model ignores total daytime calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C043	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C044	null: model ignores total evening charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C045	null: model ignores total international minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C046	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C047	structure: made of linear_backbone, usage_response, intl_plan_switch, vmail_state41_unit, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C048	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32	4000	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 3%	log-odds 0.86, OR 2.37; range across patients 0.00 to 1.25	4000	train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000)	yes	slightly
C050	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 1.03, OR 2.79; range across patients 0.00 to 4.01	4000	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000)	yes	slightly
C051	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.63, OR 1.87; range across patients 0.00 to 2.67	4000	train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000)	yes	slightly
C052	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 1%	log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11	4000	train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C053	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	1% to 1%	log-odds -0.18, OR 0.83; range across patients -0.32 to 0.00	3934, 66	train; test unclear (AUC gain 0.0002 [-0.0002, 0.0007], n=1000)	yes	slightly; on the training data only
C054	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: a voice-mail plan, no voice-mail plan*	1% to 1%	log-odds -0.83, OR 0.44; range across patients -2.92 to 0.00	2939, 1061	train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000)	yes	slightly
C055	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 86% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%*	1% to 15% (typical patient with a fold)	fold 2.24 calls (IQR 2.24 to 2.96); past it log-odds 2.20, OR 9.02; range across patients 0.00 to 2.90	3833, 614	train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000)	yes	clearly
C056	threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 11% (typical patient with a fold)	fold 187 minutes (IQR 158 to 214); past it log-odds 2.80, OR 16.4; range across patients 0.00 to 6.18	3014, 1382	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	clearly
C057	threshold: total daytime charge: fold at about 38.81 US dollars (starts) for customers without a voice-mail plan *shape: hinge at 38.81 US dollars; 45% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 36.87 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 2% (typical patient with a fold)	fold 38.8 US dollars (IQR 37.2 to 47.6); past it log-odds 0.73, OR 2.07; range across patients 0.00 to 2.28	2578, 579	train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C058	threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%*	1% to 2% (typical patient with a fold)	fold 220 minutes (IQR 150 to 296); past it log-odds 0.75, OR 2.11; range across patients -0.06 to 2.45	2340, 1023	train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000)	no	slightly
C059	threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%*	2% to 2% (typical patient with a fold (customers with a voice-mail plan))	fold 318 minutes (IQR 294 to 327); past it log-odds -0.01, OR 0.99; range across patients 0.00 to 0.00	43, 0 (small n)	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000)	no	slightly; the model has little data on this
C060	structure: made of usage_hinge and inputs number customer service calls; combined non-additively		log-odds n/a, OR n/a	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%*	1% to 4%	log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32	4000	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	slightly
C062	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to 1% (typical patient this part acts on)	log-odds -0.83, OR 0.44; range across patients -2.92 to 0.00	2939, 1061	train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000)	yes	slightly
C063	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 1.03, OR 2.79; range across patients 0.00 to 4.01	4000	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000)	yes	slightly
C064	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.63, OR 1.87; range across patients 0.00 to 2.67	4000	train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000)	yes	slightly
C065	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C072 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)*	1% to 1%	log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11	4000	train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C066	direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota*	3% to 3% (typical patient this part acts on)	log-odds -0.18, OR 0.83; range across patients -0.32 to 0.00	3934, 66	train; test unclear (AUC gain 0.0002 [-0.0002, 0.0007], n=1000)	yes	slightly; on the training data only
C067	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening minutes: total evening minutes above 201 minutes minus total evening minutes at or below 201 minutes; stronger in total evening minutes above 201 minutes *shape: accelerating; 75% of patients same; fallback; risk saturates near 0%*	total evening minutes above 201 minutes: 1% to 11%; total evening minutes at or below 201 minutes: 1% to 2%	log-odds difference 1.54, OR 4.66	1998, 2002	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000); group difference on test 1.71 [1.48, 1.94]	yes	clearly
C068	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total night-time minutes: total night-time minutes above 200 minutes minus total night-time minutes at or below 200 minutes; stronger in total night-time minutes above 200 minutes *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total night-time minutes above 200 minutes: 1% to 7%; total night-time minutes at or below 200 minutes: 1% to 3%	log-odds difference 1.00, OR 2.71	2000, 2000	train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000); group difference on test 1.09 [0.85, 1.33]	yes	slightly
C069	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 70% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: 1% to 1%; no voice-mail plan: 1% to 4%	log-odds difference -1.86, OR 0.16	1061, 2939	train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000); group difference on test -1.89 [-2.04, -1.73]	no	slightly
C070	interaction: effect of total evening minutes (138 minutes to 266 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: dip at 212 minutes; 48% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: 1% to <1%; no voice-mail plan: 1% to 2%	log-odds difference -1.40, OR 0.25	1061, 2939	train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000); group difference on test -1.35 [-1.49, -1.21]	no	slightly
C071	threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of*	1% to 11% (typical patient with a fold)	fold 187 minutes (IQR 158 to 214); past it log-odds 2.80, OR 16.4; range across patients 0.00 to 6.18	3014, 1382	train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%*	1% to 2% (typical patient with a fold)	fold 220 minutes (IQR 150 to 296); past it log-odds 0.75, OR 2.11; range across patients -0.06 to 2.45	2340, 1023	train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000)	no	slightly
C074	threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%*	2% to 2% (typical patient with a fold (customers with a voice-mail plan))	fold 318 minutes (IQR 294 to 327); past it log-odds -0.01, OR 0.99; range across patients 0.00 to 0.00	43, 0 (small n)	train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000)	no	slightly; the model has little data on this
C075	subgroup: who this part acts on: 1382 of 4000 patients	1% to 3% (typical patient this part acts on (risk without the part, then with it))	share 0.35; log-odds 1.77, OR 5.86; range across patients -0.34 to 5.50	1382, 2618	train; test agrees (AUC gain 0.214 [0.160, 0.273], n=1000)	no	slightly
C076	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	4000	train	yes	
C077	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 15%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	3620, 380	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C078	subgroup: who this part acts on: 380 of 4000 patients	2% to 26% (typical patient this part acts on (risk without the part, then with it))	share 0.10; log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	380, 3620	train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000)	yes	clearly
C079	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C080	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 3%	log-odds 1.10, OR 2.99; range across patients 1.10 to 1.10	2529, 72	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C081	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 2%	log-odds 0.81, OR 2.24; range across patients 0.81 to 0.81	2529, 75	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C082	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -1.89, OR 0.15; range across patients -1.89 to -1.89	2529, 80	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C083	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.80, OR 2.23; range across patients 0.80 to 0.80	4000	train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000)	yes	slightly; on the training data only
C084	direction: state: state	1% to <1%	log-odds -1.23, OR	2529,	train; test disagrees	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -1.09, OR 0.34; range across patients -1.09 to -1.09	2529, 78	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C086	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47	4000	train; test unclear (AUC gain 0.0008 [-0.004, 0.006], n=1000)	yes	slightly; on the training data only
C087	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	4000	train; test unclear (AUC gain 0.002 [-0.004, 0.008], n=1000)	yes	slightly; on the training data only
C088	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.43, OR 1.53; range across patients 0.43 to 0.43	2529, 66	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C089	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.43, OR 1.53; range across patients 0.43 to 0.43	2529, 66	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C090	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Georgia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to <1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	2529, 60	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C091	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	2529, 90	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C092	direction: total evening	1% to 1%	log-odds -0.35, OR	4000	train; test disagrees	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.21, OR 1.24; range across patients 0.21 to 0.21	2529, 81	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C098	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.001], n=1000)	yes	slightly; on the training data only
C099	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds -0.22, OR 0.80; range across patients -0.22 to -0.22	2529, 68	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C100	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Virginia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.15, OR 1.16; range across patients 0.15 to 0.15	2529, 91	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C101	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Mexico (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds -0.14, OR 0.87; range across patients -0.14 to -0.14	2529, 72	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C102	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Mississippi (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …*	1% to 1%	log-odds 0.11, OR 1.11; range across patients 0.11 to 0.11	2529, 77	train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000)	yes	slightly; on the training data only
C103	direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 510, area code other than area code 510*	1% to 1%	log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10	3040, 960	train; test unclear (AUC gain 8e-05 [-0.001, 0.002], n=1000)	yes	slightly; on the training data only
C104	direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana,	1% to 1%	log-odds -0.11, OR 0.89; range across patients -0.11 to -0.11	2529, 86	train; test disagrees (AUC gain -0.004 [-0.016, 0.006],	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to 1%	log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	2939, 1061	train; test unclear (AUC gain 0.002 [-0.003, 0.008], n=1000)	yes	slightly; on the training data only
C110	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C111	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 5%	log-odds 1.73, OR 5.62; range across patients 1.73 to 1.73	122, 66	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C112	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 2%	log-odds 0.83, OR 2.30; range across patients 0.83 to 0.83	122, 82	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C113	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 2%	log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69	122, 78	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C114	direction: state: West Virginia to state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 2%	log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69	122, 3505	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C115	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 1%	log-odds -0.43, OR 0.65; range across patients -0.43 to -0.43	122, 91	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C116	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...*	1% to 1%	log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02	122, 56	train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000)	yes	slightly; on the training data only
C117	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C118	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C119	boundary: boundary		log-odds 1.00, OR n/a	4000	train	yes	

Not supported by the model.

- C016: direction claim about total evening minutes in the whole model: not supported by the model (58% of patients have the opposite sign (allowed: 10%)).
- C032: direction claim about total night-time minutes in the whole model: not supported by the model (63% of patients have the opposite sign (allowed: 10%)).

- C058: threshold claim about total night-time minutes in part ann_83: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).
- C059: threshold claim about total evening minutes in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C069: interaction claim about total daytime minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C070: interaction claim about total evening minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C073: threshold claim about total night-time minutes in part ann_79: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).
- C074: threshold claim about total evening minutes in part ann_79: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C075: subgroup claim about state and voice-mail plan in part ann_79: not supported by the model (its effect goes both ways across the patients it acts on).
- C118: boundary claim about number of calls to customer service and total daytime minutes in part ann_83: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

5.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (35):** `-11` (state:18), `-12` (state:19), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-2` (state:1), `-23` (state:29), `-24` (state:3), `-25` (state:30), `-28` (state:33), `-29` (state:34), `-31` (state:36), `-33` (state:38), `-35` (state:4), `-38` (state:42), `-39` (state:43), `-4` (state:11), `-40` (state:44), `-47` (state:50), `-48` (state:6), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-51` (state:9), `-53` (areaCode:408), `-54` (areaCode:415), `-6` (state:13), `-60` (totalDayCalls), `-63` (totalEveCalls), `-64` (totalEveCharge), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-7` (state:14), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

5.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, ...) are used in the figures.

▷ A1: churn_logit (composition, section 5.6.1)
  ▷ A2: linear_backbone (leaf, section 5.6.2)
  ▷ A3: vmail_state41_unit (leaf, section 5.6.3)
  ▷ A4: intl_plan_switch (leaf, section 5.6.4)
  ▷ A5: state_offsets (leaf, section 5.6.5)

▷ A6: usage_response (composition, section 5.6.6)
▷ A7: usage_hinge (leaf, section 5.6.7)

5.4 Network figures

5.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 1.22×10^{-15} (tolerance 1.00×10^{-9}).

- #71: split `-37` into `-37_a`, `-37_b`, `-37_c`, `-37_d`.
- #72: split `-57` into `-57_a`, `-57_b`.
- #73: split `-62` into `-62_a`, `-62_b`.
- #74: split `-65` into `-65_a`, `-65_b`.
- #75: split `-41` into `-41_a`, `-41_b`.
- #76: identity node `identity_lin` before `0` (27 connection(s)).
- #77: identity node `identity_usage` before `0` (3 connection(s)).

5.6 Annotations

5.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 5.6.2), A3 (section 5.6.3), A4 (section 5.6.4), A5 (section 5.6.5), A6 (section 5.6.6).

- **Entry nodes:** `-71` (numberCustomerServiceCalls), `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-64` (totalEveCharge), `-63` (totalEveCalls), `-61` (totalDayCharge), `-60` (totalDayCalls), `-59` (totalDayMinutes), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-43` (state:47), `-42` (state:46), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-36` (state:40), `-35` (state:4), `-34` (state:39), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-28` (state:33), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-23` (state:29), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-11` (state:18), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), … (18 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `582`→`0` (-1.4), `2067`→`0` (-0.262), `5667`→`0` (0.0263), `5821`→`0` (-0.0979), `6711`→`0` (1.68), `12726`→`0` (0.0229), `13164`→`0` (0.135), `13420`→`0` (-0.122), `identity_lin`→`0` (1), `identity_usage`→`0` (1).
- **With its descendants:** 95 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 94 candidate nodes.

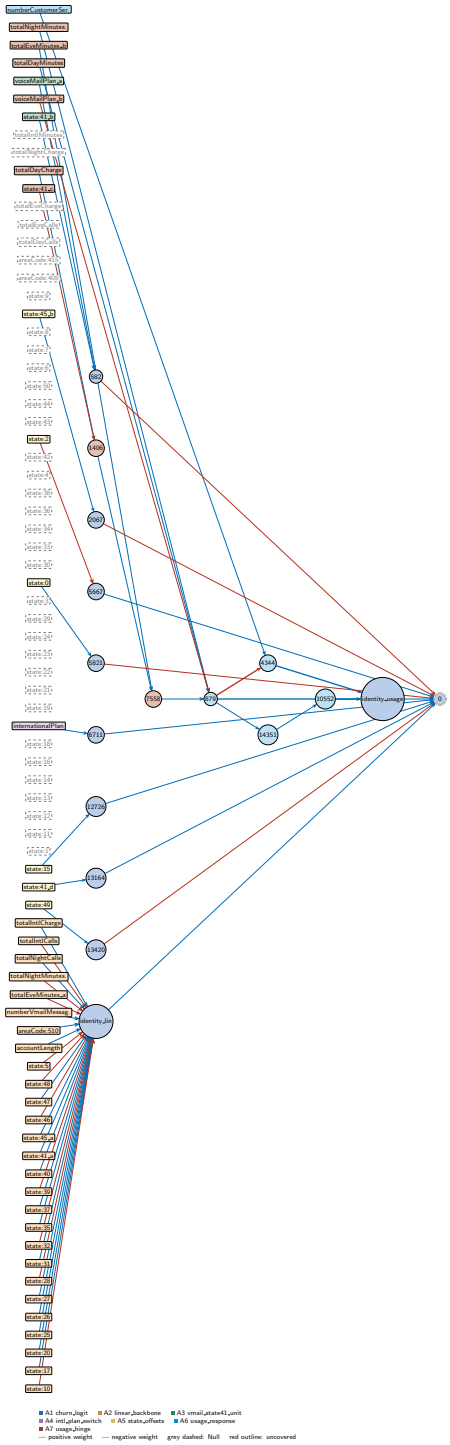


Figure 14: churn100-s7-c1: full view of the explained model state.

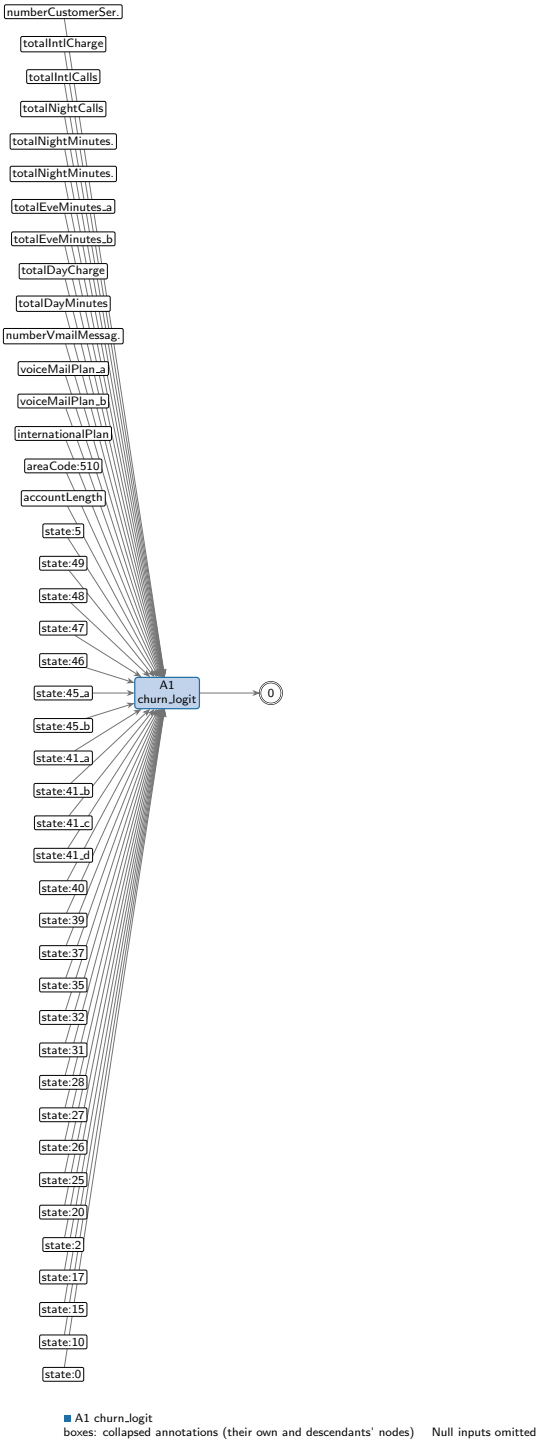


Figure 15: churn100-s7-c1: collapsed view at hierarchy level 0 (fully recomposed).

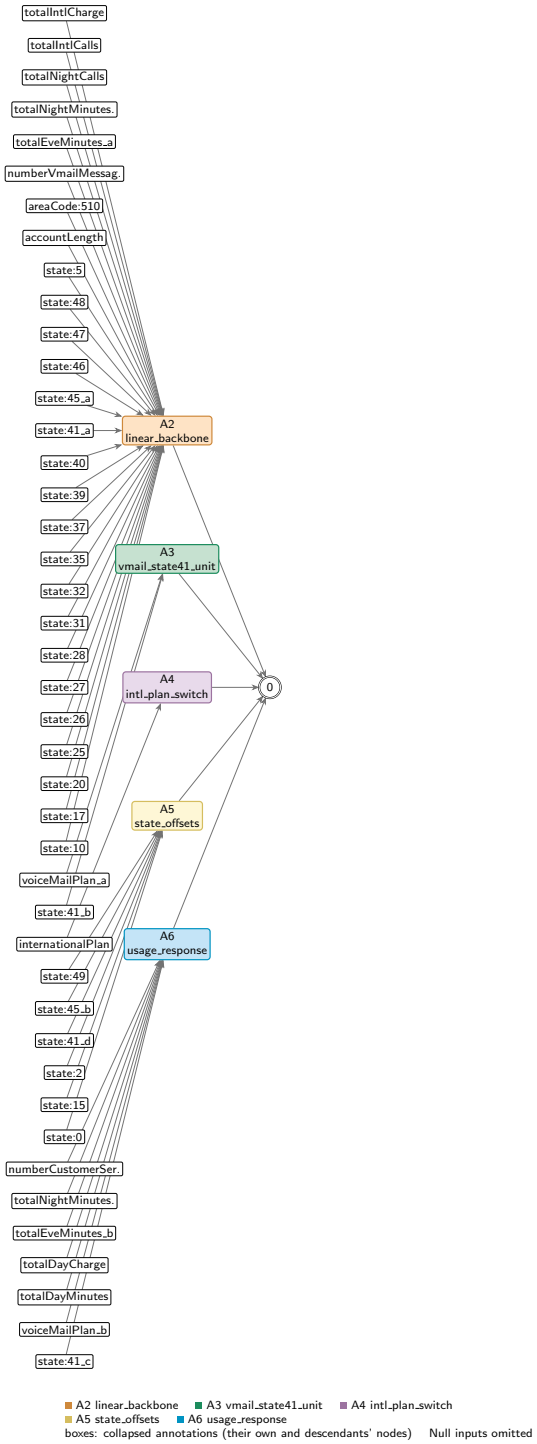


Figure 16: churn100-s7-c1: collapsed view at hierarchy level 1.

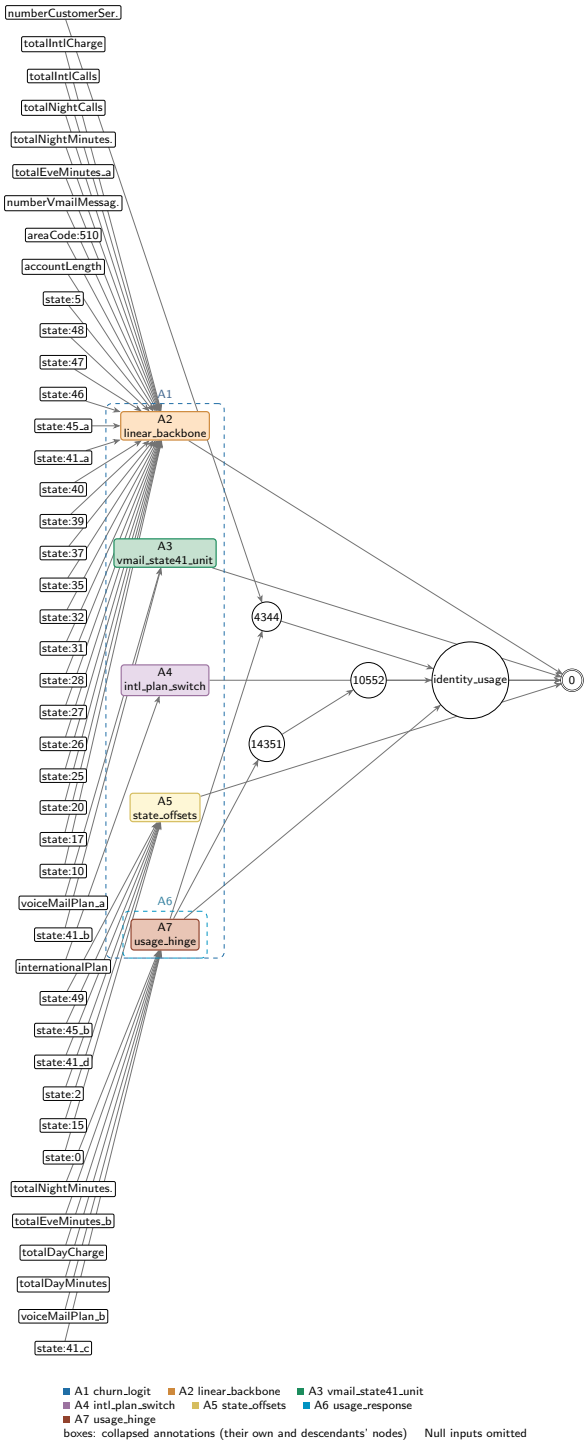


Figure 17: churn100-s7-c1: collapsed view at hierarchy level 2.

Hypothesis. Root: logit P(churn) = -1.96 + linear_backbone (identity_lin) + usage_response (identity_usage) + 1.68*6711 (intl_plan_switch) - 1.40*582 (vmail_state41_unit) + state_offsets (-0.26*2067 - 0.10*5821 + 0.02*12726 + 0.13*13164 - 0.12*13420 + 0.03*5667), purely additive in the logit; the sigmoid is the only coupling. 35 inputs are unconnected (international minutes, night charge, evening charge and calls, day calls, area codes 408/415 and 29 state indicators) and have zero effect. Train AUC 0.9189, accuracy 0.921. Performance sources (train): all units replaced by least-squares linear fits gives AUC 0.8336 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring usage_response alone gives 0.9189 (all the gain); its parts alone: usage hinge 0.8803, heavy-use hinge 0.8870, service-calls gate 0.8731. The international-plan, voicemail and state units are exact switches/offsets and restore nothing. Rare-category switches: state:46 -1.89 logit (80 rows, empirical log-odds offset -0.91); state:39 -1.23 (82, -0.93); state:47 +1.10 (72, +0.92); state:5 -1.09 (78, -0.51); state:45 -0.97 (91, -0.85); state:26 +0.81 (75, +0.50); state:49 -0.69 (122, -0.32); state:0 -0.68 (56, -0.76); state:10 -0.63 (60, -0.22); state:41 +0.25 mean, range -3.85..+0.43 (66, -0.04: it switches off the 1406 day-charge hinge, so heavy users lose up to 3.85).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.95 + 1.0\,\text{linear_backbone}\big(x_{state:17}, x_{state:20}, \\
&\quad x_{state:25}, x_{state:26}, x_{state:27}, x_{state:28}, x_{state:31}, x_{state:32}, \\
&\quad x_{state:10}, x_{state:35}, x_{state:37}, x_{state:39}, x_{state:40}, x_{state:41_a}, \\
&\quad x_{state:45_a}, x_{state:46}, x_{state:47}, x_{state:48}, x_{state:5}, x_{accountLength}, \\
&\quad x_{areaCode:510}, x_{numberVmailMessages}, x_{totalEveMinutes_a}, x_{totalNightMinutes_a}, \\
&\quad x_{totalNightCalls}, x_{totalIntlCalls}, x_{totalIntlCharge}\big) \\
&\quad + 1.0\,\text{usage_response}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&\quad x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, x_{totalDayCharge}, \\
&\quad x_{numberCustomerServiceCalls}\big) \\
&\quad + 0.0229\,\text{state_offsets_2}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big) \\
&\quad + 0.0263\,\text{state_offsets_5}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big) \\
&\quad + 0.135\,\text{state_offsets_3}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big) \\
&\quad + 1.68\,\text{intl_plan_switch}\big(x_{internationalPlan}\big) \\
&\quad - 0.262\,\text{state_offsets_0}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big) \\
&\quad - 1.4\,\text{vmail_state41_unit}\big(x_{voiceMailPlan_a}, x_{state:41_b}\big) \\
&\quad - 0.0979\,\text{state_offsets_1}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big) \\
&\quad - 0.122\,\text{state_offsets_4}\big(x_{state:2}, x_{state:49}, x_{state:41_d}, \\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.95 + 3.06\,\text{relu}\big(-0.638 + 1.17\,\text{relu}\big(-0.158 \\
&\quad + 1.08\,\text{relu}\big(-0.993 + 0.219 x_{totalNightMinutes_b} \\
&\quad + 0.527 x_{totalDayMinutes} + 0.554\,\text{relu}\big(0.515 \\
&\quad + 1.21\,\text{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&\quad - 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&\quad - 1.01 x_{voiceMailPlan_b}\big)\big)\big) + 0.321 x_{totalIntlCharge}
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.0229\,\mathrm{relu}\big(0.47 + 0.878x_{state:15}\big) \\
&+ 0.199x_{numberVmailMessages} + 2.57\,\mathrm{relu}\big(-0.993 \\
&+ 0.219x_{totalNightMinutes_b} + 0.527x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545x_{totalDayCharge} - 0.885x_{state:41_c}\big) + 0.665x_{totalEveMinutes_b}\big) \\
&- 1.01x_{voiceMailPlan_b}\big) + 0.042x_{state:35} + 0.0442x_{areaCode:510} \\
&+ 0.0249x_{totalNightCalls} + 0.0263\,\mathrm{relu}\big(-0.325 \\
&- 0.0317x_{state:2}\big) + 0.0544x_{state:41_a} + 0.0545x_{state:40} \\
&+ 0.0145x_{state:25} + 0.00187x_{state:20} + 0.0301x_{state:17} \\
&+ 0.135\,\mathrm{relu}\big(-0.236 + 1.02x_{state:41_d}\big) + 0.146x_{state:47} \\
&+ 0.0375x_{state:27} + 0.0379x_{state:37} + 0.0871x_{accountLength} \\
&+ 0.0223x_{state:45_a} + 2.88\,\mathrm{relu}\big(-0.384 \\
&+ 0.748x_{numberCustomerServiceCalls} - 3.24\,\mathrm{relu}\big(-0.993 \\
&+ 0.219x_{totalNightMinutes_b} + 0.527x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545x_{totalDayCharge} - 0.885x_{state:41_c}\big) + 0.665x_{totalEveMinutes_b}\big) \\
&- 1.01x_{voiceMailPlan_b}\big)\big) + 0.049x_{state:31} \\
&+ 1.68\,\mathrm{relu}\big(-0.228 + 0.635x_{internationalPlan}\big) \\
&+ 0.109x_{state:26} - 0.0289x_{state:28} - 0.139x_{totalEveMinutes_a} - 0.175x_{state:39} \\
&- 0.262\,\mathrm{relu}\big(-0.691 + 0.757x_{state:45_b}\big) \\
&- 0.264x_{totalIntlCalls} - 0.0184x_{state:32} - 0.0761x_{state:10} \\
&- 0.114x_{totalNightMinutes_a} - 0.0165x_{state:48} - 0.264x_{state:46} \\
&- 0.151x_{state:5} - 1.4\,\mathrm{relu}\big(1.51 + 0.17x_{voiceMailPlan_a} \\
&+ 0.0942x_{state:41_b}\big) - 0.0979\,\mathrm{relu}\big(-1.7 + 1.02x_{state:0}\big) \\
&- 0.122\,\mathrm{relu}\big(0.00575 + 1.01x_{state:49}\big)
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00775, and the exact function by at most 7.77×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.96 + linear_backbone (identity_lin) + usage_response (identity_usage) + 1.68*6711 (intl_plan_switch) - 1.40*582 (vmail_state41_unit) + state_offsets (-0.26*2067 - 0.10*5821 + 0.02*12726 + 0.13*13164 - 0.12*13420 + 0.03*5667), purely additive in the logit; the sigmoid is the only coupling. 35 inputs are unconnected (international minutes, night charge, evening charge and c

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 5.6.1).

- **Entry nodes:** `-10` (state:17), `-14` (state:20), `-19` (state:25), `-20` (state:26), `-21` (state:27), `-22` (state:28), `-26` (state:31), `-27` (state:32), `-3` (state:10), `-30` (state:35), `-32` (state:37), `-34` (state:39), `-36` (state:40), `-37_a` (state:41_a), `-41_a` (state:45_a), `-42` (state:46), `-43` (state:47), `-44` (state:48), `-46` (state:5), `-52` (accountLength), `-55` (areaCode:510), `-58` (numberVmailMessages), `-62_a` (totalEveMinutes_a), `-65_a` (totalNightMinutes_a), `-66` (totalNightCalls), `-69` (totalIntlCalls), `-70` (totalIntlCharge).

- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-3`→`identity_lin` (-0.0761), `-10`→`identity_lin` (0.0301), `-14`→`identity_lin` (0.00187), `-19`→`identity_lin` (0.0145), `-20`→`identity_lin` (0.109), `-21`→`identity_lin` (0.0375), `-22`→`identity_lin` (-0.0289), `-26`→`identity_lin` (0.049), `-27`→`identity_lin` (-0.0184), `-30`→`identity_lin` (0.042), `-32`→`identity_lin` (0.0379), `-34`→`identity_lin` (-0.175), `-36`→`identity_lin` (0.0545), `-42`→`identity_lin` (-0.264), `-43`→`identity_lin` (0.146), `-44`→`identity_lin` (-0.0165), `-46`→`identity_lin` (-0.151), `-52`→`identity_lin` (0.0871), `-55`→`identity_lin` (0.0442), `-58`→`identity_lin` (0.199), `-66`→`identity_lin` (0.0249), `-69`→`identity_lin` (-0.264), `-70`→`identity_lin` (0.321), `-37_a`→`identity_lin` (0.0544), `-62_a`→`identity_lin` (-0.139), `-65_a`→`identity_lin` (-0.114), `-41_a`→`identity_lin` (0.0223).
- **Covers** (coverage, own and as part of the hierarchy): 27 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 27 inputs, all with small weights (at most 0.32 per SD): international charge +0.32, international calls -0.26, voicemail messages +0.20, evening minutes -0.14 and night minutes -0.11 (these two also feed the usage hinge with positive weights), account length +0.09, area code 510 +0.04 (+0.10 logit), and state offsets: state:46 -1.89 logit (80 customers), state:39 -1.23 (82), state:47 +1.10 (72), state:5 -1.09 (78), state:26 +0.81, state:10 -0.63, state:40 +0.43, state:41 +0.43, state:31 +0.33, then smaller. International plan, voicemail plan, day minutes and customer-service calls have NO direct edge. Evidence: closed form exact (identity, R2 1.0); train sensitivities equal the weights (international charge 0.32, international calls 0.26, state:46 0.26, voicemail messages 0.20). Mean ablation AUC 0.9076 (full 0.9189).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.321x_{totalIntlCharge} + 0.199x_{numberVmailMessages} + 0.042x_{state:35} \\
& + 0.0442x_{areaCode:510} + 0.0249x_{totalNightCalls} + 0.0544x_{state:41_a} \\
& + 0.0545x_{state:40} + 0.0145x_{state:25} + 0.00187x_{state:20} + 0.0301x_{state:17} \\
& + 0.146x_{state:47} + 0.0375x_{state:27} + 0.0379x_{state:37} + 0.0871x_{accountLength} \\
& + 0.0223x_{state:45_a} + 0.049x_{state:31} + 0.109x_{state:26} - 0.0289x_{state:28} \\
& - 0.139x_{totalEveMinutes_a} - 0.175x_{state:39} - 0.264x_{totalIntlCalls} \\
& - 0.0184x_{state:32} - 0.0761x_{state:10} - 0.114x_{totalNightMinutes_a} \\
& - 0.0165x_{state:48} - 0.264x_{state:46} - 0.151x_{state:5}
\end{aligned}
$$

where y_0 is node `identity_lin`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00352, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 27 inputs, all with small weights (at most 0.32 per SD): international charge +0.32, international calls -0.26, voicemail messages +0.20, evening minutes -0.14 and night minutes -0.11 (these two also feed the usage hinge with positive weights), account length +0.09, area code 510 +0.04 (+0.10 logit),

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5.6.3 A3: vmail_state41_unit

Kind: Leaf annotation (depth 1); parent A1 (section 5.6.1).

- **Entry nodes:** `-57_a` (voiceMailPlan_a), `-37_b` (state:41_b).
- **Exit nodes:** `582`.
- **Internal nodes:** none.
- **Edges (weight):** `-37_b`→`582` (0.0942), `-57_a`→`582` (0.17).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 582 = relu(1.51 + 0.17 vmailPlan + 0.09 state:41), logit weight -1.40. The pre-activation stays between 1.40 and 2.52 on train, so the unit is always active and exactly affine (R2 1.0): a constant -1.96 logit for customers without a voicemail plan outside state:41, -0.54 extra for a voicemail plan and -1.04 extra for state:41. Hypothesis: a fixed offset plus a voicemail-plan and state:41 discount. Evidence: sensitivity 0.17 (voicemail plan), 0.09 (state:41); linear replacement AUC 0.9189 (unchanged), mean ablation 0.9184.

Closed form:

$$y_0 = \mathrm{relu}\big(1.51 + 0.17 x_{voiceMailPlan_a} + 0.0942 x_{state:41_b}\big)$$

where y_0 is node `582`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00187, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 582 = relu(1.51 + 0.17 vmailPlan + 0.09 state:41), logit weight -1.40. The pre-activation stays between 1.40 and 2.52 on train, so the unit is always active and exactly affine (R2 1.0): a constant -1.96 logit for customers without a voicemail plan outside state:41, -0.54 extra for a voicemail plan and -1.04 extra for state:41. Hypothesis: a fixed offset plus a voicemail-plan and state:41 discount.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.6.4 A4: intl_plan_switch

Kind: Leaf annotation (depth 1); parent A1 (section 5.6.1).

- **Entry nodes:** `-56` (internationalPlan).
- **Exit nodes:** `6711`.
- **Internal nodes:** none.
- **Edges (weight):** `-56`→`6711` (0.635).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 6711 = relu(-0.23 + 0.64 intlPlan), logit weight +1.68: exactly 0 for customers without an international plan and +2.91 logit for the 380 with one (R2 1.0). Hypothesis: an international-plan switch, the model's only route for the plan (no direct edge). Empirical churn 0.42 with the plan vs 0.11 without; model 0.45 vs 0.11. No interaction with international usage (international minutes are unconnected; international charge and calls act linearly in the backbone).

Evidence: line plot correlation 1.00; mean ablation AUC 0.8611; linear replacement 0.9189 (identical, exact switch).

Closed form:

$$y_0 = \text{relu}\big(-0.228 + 0.635 x_{internationalPlan}\big)$$

where y_0 is node `6711`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00131, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 6711 = relu(-0.23 + 0.64 intlPlan), logit weight +1.68: exactly 0 for customers without an international plan and +2.91 logit for the 380 with one (R2 1.0). Hypothesis: an international-plan switch, the model's only route for the plan (no direct edge). Empirical churn 0.42 with the plan vs 0.11 without; model 0.45 vs 0.11. No interaction with international usage (international minutes are unconnec

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.6.5 A5: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 5.6.1).

- **Entry nodes:** `-13` (state:2), `-45` (state:49), `-37_d` (state:41_d), `-8` (state:15), `-1` (state:0), `-41_b` (state:45_b).
- **Exit nodes:** `2067`, `5821`, `12726`, `13164`, `13420`, `5667`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`5821` (1.02), `-13`→`5667` (-0.0317), `-8`→`12726` (0.878), `-45`→`13420` (1.01), `-37_d`→`13164` (1.02), `-41_b`→`2067` (0.757).
- **Covers** (coverage, own and as part of the hierarchy): 6 candidate nodes.

Hypothesis. Six single-state units, each a function of one binary indicator and so exactly an offset (R2 1.0): 2067 state:45 -1.12 logit (91 customers; with its direct +0.15, net -0.97), 13164 state:41 +1.04 (66; it offsets most of state:41's -1.04 in vmail_state41_unit), 13420 state:49 -0.69 (122), 5821 state:0 -0.68 (56), 12726 state:15 +0.14 (82; always active, constant +0.008), 5667 state:2 is never active (dead unit, zero effect). Hypothesis: per-state offsets. Evidence: sensitivities as listed; linear replacement AUC 0.9189 (unchanged); mean ablation 0.9167.

Closed form:

$$\begin{aligned}
y_0 &= \text{relu}\big(-0.691 + 0.757 x_{state:45_b}\big)\\
y_1 &= \text{relu}\big(-1.7 + 1.02 x_{state:0}\big)\\
y_2 &= \text{relu}\big(0.47 + 0.878 x_{state:15}\big)\\
y_3 &= \text{relu}\big(-0.236 + 1.02 x_{state:41_d}\big)\\
y_4 &= \text{relu}\big(0.00575 + 1.01 x_{state:49}\big)\\
y_5 &= \text{relu}\big(-0.325 - 0.0317 x_{state:2}\big)
\end{aligned}$$

where y_0 is node `2067`; y_1 is node `5821`; y_2 is node `12726`; y_3 is node `13164`; y_4 is node `13420`; y_5 is node `5667`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0381, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Six single-state units, each a function of one binary indicator and so exactly an offset (R2 1.0): 2067 state:45 -1.12 logit (91 customers; with its direct +0.15, net -0.97), 13164 state:41 +1.04 (66; it offsets most of state:41's -1.04 in vmail_state41_unit), 13420 state:49 -0.69 (122), 5821 state:0 -0.68 (56), 12726 state:15 +0.14 (82; always active, constant +0.008), 5667 state:2 is never activ

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.6.6 A6: usage_response

Kind: Composition annotation (depth 1); parent A1 (section 5.6.1); children A7 (section 5.6.7).

- **Entry nodes:** `-65_b` (totalNightMinutes_b), `-57_b` (voiceMailPlan_b), `-59` (totalDayMinutes), `-62_b` (totalEveMinutes_b), `-37_c` (state:41_c), `-61` (totalDayCharge), `-71` (numberCustomerServiceCalls).
- **Exit nodes:** `identity_usage`.
- **Internal nodes:** `4344`, `14351`, `10552`.
- **Edges (weight):** `-71`→`4344` (0.748), `879`→`4344` (-3.24), `879`→`14351` (1.08), `14351`→`10552` (1.17), `879`→`identity_usage` (2.57), `4344`→`identity_usage` (2.88), `10552`→`identity_usage` (3.06).
- **With its descendants:** 14 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 13 candidate nodes.

Hypothesis. Composition on usage_hinge: identity_usage = 2.57*879 + 3.06*10552 + 2.88*4344 enters the logit with weight 1. 10552 = relu(-0.64 + 1.17*relu(-0.16 + 1.08*879)) is a second, steeper hinge that opens once 879 exceeds 0.65 (about 240 day minutes at median other use; active for 12% of customers), so the usage effect accelerates. 4344 = relu(-0.38 + 0.75 calls - 3.24*879) is a service-calls gate: shut for 0-2 calls, open from 3 calls, and closed again by usage (at 3 calls once 879 > 0.14, at 5 calls once 879 > 0.49). Hypothesis: usage and service calls interact non-additively: service calls raise churn only for low-usage customers (or voicemail-plan customers, for whom usage is off), heavy users get the usage hinges instead. Evidence (train): usage logit (879+10552) by day minutes 0.14 (<200), 0.58 (200-220), 1.08 (220-240), 2.02 (240-260), 3.43 (260-280), 6.37 (>=280); churn 0.10/0.06/0.14/0.26/0.46/0.67 vs model 0.09/0.09/0.17/0.30/0.49/0.69. Without a voicemail plan churn reaches 0.84 above 280 minutes (model 0.86), with a plan 0.04 (model 0.09). Gate logit at 0-7 calls: 0/0/0/0.80/2.11/3.59/5.30/7.08 (1.25/2.90/4.55/6.20 at 3-6 calls when usage is off, 0.10/0.50/1.05/2.01 when it is on). Churn by calls 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 vs model 0.116/0.111/0.109/0.150/0.319/0.598/0.849/0.940: the gate opens one call too early (3 calls: +0.80 logit, empirical churn unchanged) and is over-confident from 6 calls (39 customers). Mean ablation of all three terms AUC 0.6557 (the largest drop: full 0.9189); gate alone 0.8202, heavy-use hinge alone 0.9093; linear replacement 0.8336 (R2 0.53/0.29/0.53); restoring usage_response alone onto the all-linear model recovers 0.9189 (all-linear 0.8336), i.e. all of the gain.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 = 3.06\,\mathrm{relu}\big(&-0.638 + 1.17\,\mathrm{relu}\big(-0.158 \\
&+ 1.08\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, \\
&x_{totalDayCharge}\big)\big)\big) \\
&+ 2.57\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, x_{totalDayCharge}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 2.88\,\mathrm{relu}\big(-0.384 + 0.748 x_{numberCustomerServiceCalls} \\
&- 3.24\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, x_{totalDayCharge}\big)\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = 3.06\,\mathrm{relu}\big(&-0.638 + 1.17\,\mathrm{relu}\big(-0.158 \\
&+ 1.08\,\mathrm{relu}\big(-0.993 + 0.219 x_{totalNightMinutes_b} \\
&+ 0.527 x_{totalDayMinutes} + 0.554\,\mathrm{relu}\big(0.515 \\
&+ 1.21\,\mathrm{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&- 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)\big)\big) + 2.57\,\mathrm{relu}\big(-0.993 \\
&+ 0.219 x_{totalNightMinutes_b} + 0.527 x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545 x_{totalDayCharge} - 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big) + 2.88\,\mathrm{relu}\big(-0.384 \\
&+ 0.748 x_{numberCustomerServiceCalls} - 3.24\,\mathrm{relu}\big(-0.993 \\
&+ 0.219 x_{totalNightMinutes_b} + 0.527 x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545 x_{totalDayCharge} - 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)\big)
\end{aligned}
$$

where y_0 is node `identity_usage`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0254, and the exact function by at most 4.44×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition on usage_hinge: identity_usage = 2.57*879 + 3.06*10552 + 2.88*4344 enters the logit with weight 1. 10552 = relu(-0.64 + 1.17*relu(-0.16 + 1.08*879)) is a second, steeper hinge that opens once 879 exceeds 0.65 (about 240 day minutes at median other use; active for 12% of customers), so the usage effect accelerates. 4344 = relu(-0.38 + 0.75 calls - 3.24*879) is a service-calls gate: shut

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.6.7 A7: usage_hinge

Kind: Leaf annotation (depth 2); parent A6 (section 5.6.6).

- **Entry nodes:** `-65_b` (totalNightMinutes_b), `-57_b` (voiceMailPlan_b), `-59` (totalDayMinutes), `-62_b` (totalEveMinutes_b), `-37_c` (state:41_c), `-61` (totalDayCharge).
- **Exit nodes:** `879`.
- **Internal nodes:** `1406`, `7558`.
- **Edges (weight):** `-59`→`879` (0.527), `-61`→`1406` (0.545), `7558`→`879` (0.554), `1406`→`7558` (1.21), `-37_c`→`1406` (-0.885), `-57_b`→`879` (-1.01), `-62_b`→`7558` (0.665), `-65_b`→`879` (0.219).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

Hypothesis. 879 = relu(-0.99 + 0.53 dayMin + 0.22 nightMin - 1.01 vmailPlan + 0.55*7558), 7558 = relu(0.51 + 0.67 eveMin + 1.21*1406), 1406 = relu(-0.50 + 0.54 dayCharge - 0.88 state:41). Day minutes and day charge are collinear (r = 1.0000), so 879 is a hinge in total usage, dominated by day use. Hypothesis: a usage hinge that is never active for voicemail-plan customers (0 of 1061) and active for 47% of the rest; at median evening/night use it switches on at about 190 day minutes and rises steadily, with a second slope from the 1406 day-charge hinge (on for 24% of customers, never in state:41). Its exit feeds the logit directly (weight 2.57) and drives the heavy-use hinge and the service-calls gate in usage_response. Evidence (train, n=4000): sensitivity of 879 to voicemail plan 0.35, day minutes 0.18, evening minutes 0.12, state:41 0.09, night minutes 0.08, day charge 0.05. Mean ablation of its direct logit term AUC 0.9017; linear replacement 0.9107 (R2 0.53); restored alone onto the all-linearised model 0.8803 (all-linear 0.8336).

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-0.993 + 0.219 x_{totalNightMinutes_b} \\
&+ 0.527 x_{totalDayMinutes} + 0.554 \operatorname{relu}\big(0.515 \\
&+ 1.21 \operatorname{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&- 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

where y_0 is node `879`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00297, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 879 = relu(-0.99 + 0.53 dayMin + 0.22 nightMin - 1.01 vmailPlan + 0.55*7558), 7558 = relu(0.51 + 0.67 eveMin + 1.21*1406), 1406 = relu(-0.50 + 0.54 dayCharge - 0.88 state:41). Day minutes and day charge are collinear (r = 1.0000), so 879 is a hinge in total usage, dominated by day use. Hypothesis: a usage hinge that is never active for voicemail-plan customers (0 of 1061) and active for 47% of the

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

5.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

6. heart-s1-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	3.33e-16	3.33e-16	3.33e-16	agree
Closed-form max abs error (all leaves)	0.00514	0.00514	0.00514	agree
Numeric-function max abs error	n/a	0	0	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	5 / 1	n/a	5 / 1	agree
Operation counts	3 splits, 0 identity	annotate 6, rename node 26, split node 3	annotate 6, rename node 26, split node 3	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `st_depression_normal_ecg_gate` (0.00514). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `ca_slope2_relay`: *"Evidence (train, n=242): pre-activation 0.28..2.19, never clips, so the part is exactly affine on the data;"* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `st_depression_normal_ecg_gate`: *"Hypothesis: node 1801 = ReLU(1.542 restecg:0 + 0.817 oldpeak - 1.521) is a gate in which ST depression (oldpeak) raises predicted risk only for patients with a normal resting ECG (restecg:0), once oldpeak is above about its mean (..."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 3 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): Root annotation via apply_operation must exclude child subgraph nodes (passing the whole graph is refused as 'already covered'); not a bug, but undocumented in the protocol. Otherwise none observed with v3 tooling.

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s1-c1-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and five parts add to that sum: two add more for each extra vessel coloured, one lowers the risk for typical angina, one lowers it for non-anginal chest pain, one acts only on the 42 of 242 training patients with a normal resting ECG and marked ST depression, and one lowers it for a higher maximum heart rate, with nothing until about 145 beats per minute, then steadily. [C29, C30, C40, C31, C43, C38, C46] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to below 1% and fasting blood sugar above 120 mg/dL to about 8%; through this sum alone, compared with no chest pain symptoms, typical angina or non-anginal chest pain lowers it to about 2% and atypical angina to about 5% (chest pain type also acts through the parts below). [C10, C19, C12, C17] A reversible defect (or the undefined code 4) on the thallium scan raises it to about 55%; higher cholesterol (from about 8% at 193 mg/dL to

about 45% at 308 mg/dL), higher resting blood pressure (from about 9% at 110 mm Hg to about 35% at 155 mm Hg) and older age (from about 10% at 42 years to about 25% at 66 years) raise it steadily. [C05, C07, C08, C13] Through this sum alone, each extra vessel coloured by fluoroscopy adds about the same (from about 18% at 0 to about 80% at 2 vessels) and a flat ST segment raises it to about 87%, while the maximum heart rate barely changes it; all three also act through the parts below. [C04, C03, C21] Rarer results also count (the model has little data on this, n = 15 and 17): a fixed defect on the thallium scan lowers the risk to about 3%, and through this sum alone a downsloping ST segment raises it to about 32%; the model ignores exercise-induced angina. [C14, C15, C28] This part combines chest pain type with the number of vessels coloured: through it, typical angina lowers a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, more vessels raise the predicted risk, and even more so with typical angina, each extra vessel adding about the same: going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, and from about 18% to about 99% with any other chest pain type. [C32] Across the whole model, typical angina takes a typical patient's predicted risk to below 1%. [C09] This part combines the resting ECG with ST depression on exercise and acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, for patients with a normal resting ECG, in an uneven way: for a typical such patient from about 13% at 0 mm to about 63% at 3.5 mm. [C37] More ST depression therefore raises the predicted risk, but only if the resting ECG is normal: going from 0 to 2.8 mm moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] This part combines the number of vessels coloured with a flat ST segment: through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Across the whole model, each extra vessel takes a typical patient from about 18% at 0 to about 99% at 2 vessels, about the same for each, flattening out near the top because the predicted risk cannot go much higher, and a flat ST segment takes it to about 91%. [C01, C02] This part reads only non-anginal chest pain: through it, non-anginal chest pain lowers a typical patient's predicted risk from about 18% to about 8%. [C43] Across the whole model, non-anginal chest pain takes it to about 1%. [C11] This part reads only the maximum heart rate: a higher maximum heart rate lowers the predicted risk with nothing until about 145 beats per minute, then steadily, for a typical patient from about 19% at 118 to about 14% at 175 beats per minute. [C45, C46] Across the whole model the same holds, from about 19% at 118 to about 14% at 175 beats per minute. [C20]

Worth checking: Worth checking: patients with typical angina or non-anginal chest pain get the lowest predicted risk of all chest pain types (below 1% and about 1% for a typical patient), below patients with no chest pain symptoms. [C09, C11] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C14] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 8%). [C19] Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 18% to about 99%. [C01] Worth checking: ST depression on exercise raises the predicted risk only when the resting ECG is normal. [C36] Worth checking: the model ignores exercise-induced angina. [C28]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	log-odds 5.89, OR 362; range across patients 5.83 to 5.83	242	train; test unclear (AUC gain 0.041 [-0.032, 0.123], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	log-odds 3.81, OR 44.9; range across patients 3.81 to 3.81	115, 110	train; test disagrees (AUC gain -0.011 [-0.065, 0.036], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 87%	log-odds 3.46, OR 31.9; range across patients 3.46 to 3.46	115, 110	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	on the training data only
C04	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (direct contribution only, not through any part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 80%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	242	train; test disagrees (AUC gain -0.001 [-0.039, 0.032], n=61)	yes	on the training data only
C05	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	log-odds 1.74, OR 5.71; range across patients 1.74 to 1.74	135, 92	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	on the training data only
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	8% to 45%	log-odds 2.22, OR 9.23; range across patients 2.22 to 2.22	242	train; test disagrees (AUC gain -0.022 [-0.066, 0.011], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	9% to 35%	log-odds 1.65, OR 5.20; range across patients 1.65 to 1.65	242	train; test unclear (AUC gain 0.000 [-0.022, 0.020], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to <1%	log-odds -4.55, OR 0.01; range across patients -4.78 to -4.08	108, 22	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to <1%	log-odds -3.94, OR 0.02; range across patients -3.94 to -3.94	159, 83	train; test unclear (AUC gain 0.023 [-0.014, 0.068], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 1%	log-odds -3.43, OR 0.03; range across patients -3.43 to -3.43	108, 70	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C12	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than atypical angina, no chest pain (asymptomatic) (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 2%	log-odds -2.51, OR 0.08; range across patients -2.51 to -2.51	108, 92	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly
C13	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of*	10% to 25%	log-odds 1.05, OR 2.87; range across patients 1.05 to 1.05	242	train; test disagrees (AUC gain -0.008 [-0.018, 0.001], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 3%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	135, 15 (small n)	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	the model has little data on this; on the training data only
C15	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	115, 17 (small n)	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	the model has little data on this; on the training data only
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 5%	log-odds -1.34, OR 0.26; range across patients -1.34 to -1.34	108, 42	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly
C19	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	209, 33	train; test disagrees (AUC gain -0.001 [-0.010, 0.009], n=61)	yes	on the training data only
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: hinge at 145 beats per minute; 100% of patients same; verified; matches the fold at 145 beats per minute*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C21	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: flat; 100% of patients same; verified*	18% to 18%	log-odds 0.005, OR 1.00; range across patients 0.005 to 0.005	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of ca_ slope2_relay, cp3_relay, st_ depression_normal_ecg_ gate, max_heart_rate_ hinge, typical_angina_ca_ switch and inputs age, sex, cp, trestbps, chol, fbs, thalach, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 61%	log-odds 2.04, OR 7.73; range across patients 1.98 to 1.98	242	train; test disagrees (AUC gain -0.003 [-0.032, 0.019], n=61)	yes	on the training data only
C31	direction: chest pain type: chest pain type other than typical angina to typical angina (through this part) *shape: categorical; highest to lowest: chest pain type other than typical angina, typical angina*	18% to 2%	log-odds -2.04, OR 0.13; range across patients -2.26 to -1.57	220, 22	train; test unclear (AUC gain 0.001 [-0.005, 0.008], n=61)	yes	on the training data only
C32	interaction: effect of number of vessels coloured by fluoroscopy (0 vessels to 2 vessels) by chest pain type: typical angina minus chest pain type other than typical angina; stronger in typical angina *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	typical angina: <1% to 55%; chest pain type other than typical angina: 18% to 99%	log-odds difference 0.70, OR 2.00	22, 220	train; test agrees (AUC gain 0.001 [-0.005, 0.008], n=61); group difference on test 0.70 [0.70, 0.70]	yes	clearly
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by	resting ECG other than a	log-odds difference -1.74, OR 0.18	121, 121	train; test agrees (AUC gain -0.008	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	13% to 63% (typical patient with a fold (patients with a normal resting ECG))	fold 1.01 mm (IQR 1.01 to 1.01); past it log-odds 2.41, OR 11.1; range across patients 2.41 to 2.41	121, 41	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C38	subgroup: who this part acts on: 42 of 242 patients	98% to 99% (typical patient this part acts on (risk without the part, then with it))	share 0.17; log-odds 1.08, OR 2.95; range across patients 0.19 to 2.50	42, 200	train; test unclear (AUC gain 0.005 [-0.006, 0.025], n=61)	yes	slightly; on the training data only
C40	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 36%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	242	train; test disagrees (AUC gain -0.002 [-0.019, 0.011], n=61)	yes	on the training data only
C41	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	18% to 23%	log-odds 0.34, OR 1.41; range across patients 0.34 to 0.34	132, 110	train; test disagrees (AUC gain -0.003 [-0.016, 0.005], n=61)	yes	on the training data only
C43	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	172, 70	train; test unclear (AUC gain 0.001 [-0.008, 0.013], n=61)	yes	on the training data only
C45	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C46	threshold: maximum heart rate: fold at about 145 beats per minute (starts) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 13% (typical patient with a fold)	fold 145 beats per minute (IQR 145 to 145); past it log-odds -0.44, OR 0.64; range across patients -0.44 to -0.44	242, 155	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s1-c1 --verified --minutes N --notes "..."`.

Verdict.

Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s1-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`

```
uv run python -m explaneat export record-verdict heart-s1-c1 --verified --minutes N --changes "..."
--notes "..."
```


6 Explanation certificate: heart-s1-c1

This section is the explanation certificate of the heart-s1-c1 network. It is rendered from the canonical JSON record `heart-s1-c1.json` (schema 1.4.0, SHA-256 `555774df08fc19cc...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 6 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s1, seed 1
Genome	`a2df22ce-983d-47b6-bbd6-3176d38ed5ff` (NEAT key 9674, generation 39)
Explanation	`492866e6-a73b-4465-92a8-ecad82a2adcc` (35 operations)
Train AUC / accuracy	0.9392 / 0.8802 (n = 242)
Test AUC / accuracy	0.861 / 0.7705 (n = 61)
Output activation	sigmoid
Phenotype	34 nodes (26 inputs, 7 hidden, 1 output), 30 enabled edges
Explained model state	38 nodes, 30 enabled edges
Refactoring operations	split_node 3
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 3.33 \times 10^{-16}$ on the training split
Annotations	6 (5 leaf, 1 composition)
Structural coverage	1 (37 of 37 candidate nodes; 9 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`555774df08fc19cc...`

Table 6: Explanation certificate summary: heart-s1-c1.

Drafter's summary. Essentially logistic regression (all hidden units linearised: train AUC 0.9355 vs 0.9392; L2 logistic regression 0.9355). 13 direct edges carry logit SD 4.25 of 5.63; ca enters by three parallel routes (effective +2.60 logit per SD), cp:3 only via always-on relay 1173. The one consequential nonlinearity is gate 1801 = ReLU(1.54 restecg:0 + 0.82 oldpeak - 1.52): ST depression adds risk only with a normal resting ECG (+0.0034 AUC over linear). Typical angina (cp:1) lowers risk via switch 3210, less with more vessels; a small hinge lets higher max heart rate lower risk. Same design as the 10-seed heart-s1 record (gen-31 genome), with an added typical-angina switch and heart-rate hinge.

What resisted explanation. Nothing resisted coverage; only input fan-out needed splitting (ca x3, slope:2, thalach). Four of seven hidden ReLUs (664, 1173, 2245, 2650) never clip on train, so their annotations describe affine relays (reweighting of ca, slope:2, cp:3 and the heart-rate hinge) rather than distinct concepts. 3210 is a pure cp:1 switch (on for all 22 typical-angina patients, off for the rest), so its best linear fit is almost exact. Nine unconnected inputs (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7) listed as zero-effect root entries.

6.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and five parts add to that sum: two add more for each extra vessel coloured, one lowers the risk for typical angina, one lowers it for non-anginal chest pain, one acts only on the 42 of 242 training patients with a normal resting ECG and marked ST depression, and one lowers it for a higher maximum heart rate, with nothing until about 145 beats per minute, then steadily. [C29, C30, C40, C31, C43, C38, C46] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to below 1% and fasting blood sugar above 120 mg/dL to about 8%; through this sum alone, compared with no chest pain symptoms, typical angina or non-anginal chest pain lowers it to about 2% and atypical angina to about 5% (chest pain type also acts through the parts below). [C10, C19, C12, C17] A reversible defect (or the undefined code 4) on the thallium scan raises it to about 55%; higher cholesterol (from about 8% at 193 mg/dL to about 45% at 308 mg/dL), higher resting blood pressure (from about 9% at 110 mm Hg to about 35% at 155 mm Hg) and older age (from about 10% at 42 years to about 25% at 66 years) raise it steadily. [C05, C07, C08, C13] Through this sum alone, each extra vessel coloured by fluoroscopy adds about the same (from about 18% at 0 to about 80% at 2 vessels) and a flat ST segment raises it to about 87%, while the maximum heart rate barely changes it; all three also act through the parts below. [C04, C03, C21] Rarer results also count (the model has little data on this, n = 15 and 17): a fixed defect on the thallium scan lowers the risk to about 3%, and through this sum alone a downsloping ST segment raises it to about 32%; the model ignores exercise-induced angina. [C14, C15, C28] This part combines chest pain type with the number of vessels coloured: through it, typical angina lowers a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, more vessels raise the predicted risk, and even more so with typical angina, each extra vessel adding about the same: going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, and from about 18% to about 99% with any other chest pain type. [C32] Across the whole model, typical angina takes a typical patient's predicted risk to below 1%. [C09] This part combines the resting ECG with ST depression on exercise and acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, for patients with a normal resting ECG, in an uneven way: for a typical such patient from about 13% at 0 mm to about 63% at 3.5 mm. [C37] More ST depression therefore raises the predicted risk, but only if the resting ECG is normal: going from 0 to 2.8 mm moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] This part combines the number of vessels coloured with a flat ST segment: through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Across the whole model, each extra vessel takes a typical patient from about 18% at 0 to about 99% at 2 vessels, about the same

for each, flattening out near the top because the predicted risk cannot go much higher, and a flat ST segment takes it to about 91%. [C01, C02] This part reads only non-anginal chest pain: through it, non-anginal chest pain lowers a typical patient's predicted risk from about 18% to about 8%. [C43] Across the whole model, non-anginal chest pain takes it to about 1%. [C11] This part reads only the maximum heart rate: a higher maximum heart rate lowers the predicted risk with nothing until about 145 beats per minute, then steadily, for a typical patient from about 19% at 118 to about 14% at 175 beats per minute. [C45, C46] Across the whole model the same holds, from about 19% at 118 to about 14% at 175 beats per minute. [C20]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, five parts add to that sum, and exercise-induced angina is ignored. [C29, C01, C02, C10, C28] *Evidence.* It is made up of the inputs that feed the score directly and five parts: vessels with a flat ST segment, the non-anginal chest pain relay, ST depression with a normal ECG, the maximum heart rate part and typical angina with vessels (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, each extra vessel adds about the same, taking a typical patient's predicted risk from about 18% at 0 to about 99% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher, against about 80% through the sum alone. [C01, C04] More ST depression raises it in an uneven way across the whole model, from about 18% at 0 mm to about 55% at 2.8 mm; it enters the score only through the part with a normal ECG. [C06] The model ignores exercise-induced angina. [C28]

Typical angina with vessels. This part lowers the predicted risk for typical angina and adds risk for each extra vessel coloured, about the same for each, with the vessels counting more for patients with typical angina. [C31, C30, C32] *Evidence.* Through it, typical angina takes a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, each extra vessel adding about the same, and from about 18% to about 99% with any other chest pain type. [C32] Its formula reproduces the part exactly on all 242 training patients. [C33]

ST depression with a normal ECG. This part raises the predicted risk with ST depression on exercise, in an uneven way, only for patients with a normal resting ECG, once it rises above about 1.0 mm. [C37, C36] *Evidence.* It acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, in an uneven way: for a typical patient with a normal resting ECG from about 13% at 0 mm to about 63% at 3.5 mm. [C37] Going from 0 to 2.8 mm of ST depression moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] Its formula reproduces the part exactly on all 242 training patients. [C39]

Vessels with flat ST segment. This part adds risk for each extra vessel coloured, about the same for each, and for a flat ST segment. [C40, C41] *Evidence.* Through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Non-anginal chest pain relay. This part lowers the predicted risk for patients with non-anginal chest pain. [C43] *Evidence.* Through it, non-anginal chest pain takes a typical patient's predicted

risk from about 18% to about 8%. [C43] Its formula reproduces the part exactly on all 242 training patients. [C44]

Maximum heart rate. This part lowers the predicted risk for a higher maximum heart rate: nothing until about 145 beats per minute, then steadily. [C45, C46] *Evidence.* Through it, a higher maximum heart rate takes a typical patient from about 19% at 71 to about 13% at 182 beats per minute, with nothing until about 145 beats per minute, then steadily. [C46] Its formula reproduces the part exactly on all 242 training patients. [C47]

Worth checking against what is known. Worth checking: patients with typical angina or non-anginal chest pain get the lowest predicted risk of all chest pain types (below 1% and about 1% for a typical patient), below patients with no chest pain symptoms. [C09, C11] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C14] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 8%). [C19] Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 18% to about 99%. [C01] Worth checking: ST depression on exercise raises the predicted risk only when the resting ECG is normal. [C36] Worth checking: the model ignores exercise-induced angina. [C28]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	log-odds 5.89, OR 362; range across patients 5.83 to 5.83	242	train; test unclear (AUC gain 0.041 [-0.032, 0.123], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	log-odds 3.81, OR 44.9; range across patients 3.81 to 3.81	115, 110	train; test disagrees (AUC gain -0.011 [-0.065, 0.036], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 87%	log-odds 3.46, OR 31.9; range across patients 3.46 to 3.46	115, 110	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	on the training data only
C04	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (direct contribution only, not through any part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 80%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	242	train; test disagrees (AUC gain -0.001 [-0.039, 0.032], n=61)	yes	on the training data only
C05	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	log-odds 1.74, OR 5.71; range across patients 1.74 to 1.74	135, 92	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	on the training data only
C06	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	log-odds 0.87, OR 2.38; range across patients 0.00 to 1.74	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	8% to 45%	log-odds 2.22, OR 9.23; range across patients 2.22 to 2.22	242	train; test disagrees (AUC gain -0.022 [-0.066, 0.011], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	9% to 35%	log-odds 1.65, OR 5.20; range across patients 1.65 to 1.65	242	train; test unclear (AUC gain 0.000 [-0.022, 0.020], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to <1%	log-odds -4.55, OR 0.01; range across patients -4.78 to -4.08	108, 22	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to <1%	log-odds -3.94, OR 0.02; range across patients -3.94 to -3.94	159, 83	train; test unclear (AUC gain 0.023 [-0.014, 0.068], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 1%	log-odds -3.43, OR 0.03; range across patients -3.43 to -3.43	108, 70	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C12	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than atypical angina, no chest pain (asymptomatic) (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type*	18% to 2%	log-odds -2.51, OR 0.08; range across patients -2.51 to -2.51	108, 92	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	10% to 25%	log-odds 1.05, OR 2.87; range across patients 1.05 to 1.05	242	train; test disagrees (AUC gain -0.008 [-0.018, 0.001], n=61)	yes	on the training data only
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 3%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	135, 15 (small n)	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	the model has little data on this; on the training data only
C15	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	115, 17 (small n)	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	the model has little data on this; on the training data only
C16	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	115, 17 (small n)	train; test disagrees (AUC gain -0.011 [-0.065, 0.036], n=61)	yes	the model has little data on this; on the training data only
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 5%	log-odds -1.34, OR 0.26; range across patients -1.34 to -1.34	108, 42	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly
C18	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 5%	log-odds -1.34, OR 0.26; range across patients -1.34 to -1.34	108, 42	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C19	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	209, 33	train; test disagrees (AUC gain -0.001 [-0.010, 0.009], n=61)	yes	on the training data only
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: hinge at 145 beats per minute; 100% of patients same; verified; matches the fold at 145 beats per minute*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C21	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: flat; 100% of patients same; verified*	18% to 18%	log-odds 0.005, OR 1.00; range across patients 0.005 to 0.005	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C22	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 18%	log-odds -0.46, OR 0.63; range across patients -1.74 to 0.00	121, 121	train; test disagrees (AUC gain -0.008 [-0.021, 0.000], n=61)	yes	slightly; on the training data only
C23	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	log-odds 5.83, OR 340	242	train; test unclear (AUC gain 0.041 [-0.032, 0.123], n=61)	yes	on the training data only
C24	magnitude: slope of the ST segment at peak	18% to 91%	log-odds 3.81, OR 44.9	115, 110	train; test disagrees (AUC gain -0.011	yes	on the training

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	log-odds 1.74, OR 5.71	135, 92	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	on the training data only
C26	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	log-odds 1.74, OR 5.68	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	8% to 45%	log-odds 2.22, OR 9.23	242	train; test disagrees (AUC gain -0.022 [-0.066, 0.011], n=61)	yes	on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of ca_slope2_relay, cp3_relay, st_depression_normal_ecg_gate, max_heart_rate_hinge, typical_angina_ca_switch and inputs age, sex, cp, trestbps, chol, fbs, thalach, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 61%	log-odds 2.04, OR 7.73; range across patients 1.98 to 1.98	242	train; test disagrees (AUC gain -0.003 [-0.032, 0.019], n=61)	yes	on the training data only
C31	direction: chest pain type: chest pain type other than typical angina to typical angina (through this part) *shape: categorical; highest to lowest: chest pain type other than typical angina, typical angina*	18% to 2%	log-odds -2.04, OR 0.13; range across patients -2.26 to -1.57	220, 22	train; test unclear (AUC gain 0.001 [-0.005, 0.008], n=61)	yes	on the training data only
C32	interaction: effect of number of vessels coloured by fluoroscopy (0 vessels to 2 vessels) by chest pain type: typical angina minus chest pain type other than typical angina; stronger in typical angina *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	typical angina: <1% to 55%; chest pain type other than typical angina: 18% to 99%	log-odds difference 0.70, OR 2.00	22, 220	train; test agrees (AUC gain 0.001 [-0.005, 0.008], n=61); group difference on test 0.70 [0.70, 0.70]	yes	clearly
C33	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C34	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	18% to 55%	log-odds 0.87, OR 2.38; range across patients 0.00 to 1.74	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	99% to 98% (typical patient this part acts on)	log-odds -0.46, OR 0.63; range across patients -1.74 to 0.00	121, 121	train; test disagrees (AUC gain -0.008 [-0.021, 0.000], n=61)	yes	slightly; on the training data only
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback*	resting ECG other than a normal resting ECG: 18% to 18%; a normal resting ECG: 18% to 55%	log-odds difference -1.74, OR 0.18	121, 121	train; test agrees (AUC gain -0.008 [-0.021, 0.000], n=61); group difference on test -1.74 [-1.74, -1.74]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	13% to 63% (typical patient with a fold (patients with a normal resting ECG))	fold 1.01 mm (IQR 1.01 to 1.01); past it log-odds 2.41, OR 11.1; range across patients 2.41 to 2.41	121, 41	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C38	subgroup: who this part acts on: 42 of 242 patients	98% to 99% (typical patient this part acts on (risk without the part, then with it))	share 0.17; log-odds 1.08, OR 2.95; range across patients 0.19 to 2.50	42, 200	train; test unclear (AUC gain 0.005 [-0.006, 0.025], n=61)	yes	slightly; on the training data only
C39	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C40	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 36%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	242	train; test disagrees (AUC gain -0.002 [-0.019, 0.011], n=61)	yes	on the training data only
C41	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	18% to 23%	log-odds 0.34, OR 1.41; range across patients 0.34 to 0.34	132, 110	train; test disagrees (AUC gain -0.003 [-0.016, 0.005], n=61)	yes	on the training data only
C42	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C43	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	172, 70	train; test unclear (AUC gain 0.001 [-0.008, 0.013], n=61)	yes	on the training data only
C44	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C45	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C46	threshold: maximum heart rate: fold at about 145 beats per minute (starts) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 13% (typical patient with a fold)	fold 145 beats per minute (IQR 145 to 145); past it log-odds -0.44, OR 0.64; range across patients -0.44 to -0.44	242, 155	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	on the training data only
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model. Every claim passed its checks.

6.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (9):** `-10` (fbs:0), `-13` (restecg:1), `-14` (restecg:2), `-16` (exang:0), `-17` (exang:1), `-19` (slope:1), `-24` (thal:4), `-26` (thal:7), `-3` (sex:1).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

6.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 6.6.1)

▷ A2: ca_slope2_relay (leaf, section 6.6.2)
▷ A3: cp3_relay (leaf, section 6.6.3)
▷ A4: st_depression_normal_ecg_gate (leaf, section 6.6.4)
▷ A5: max_heart_rate_hinge (leaf, section 6.6.5)
▷ A6: typical_angina_ca_switch (leaf, section 6.6.6)

6.4 Network figures

6.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 3.33×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-22` into `-22_a`, `-22_b`, `-22_c`.
- #27: split `-20` into `-20_a`, `-20_b`.
- #28: split `-15` into `-15_a`, `-15_b`.

6.6 Annotations

6.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 6.6.2), A3 (section 6.6.3), A4 (section 6.6.4), A5 (section 6.6.5), A6 (section 6.6.6).

- **Entry nodes:** `-1` (age), `-2` (sex:0), `-3` (sex:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9` (chol), `-10` (fbs:0), `-11` (fbs:1), `-13` (restecg:1), `-14` (restecg:2), `-15_a` (thalach_a), `-16` (exang:0), `-17` (exang:1), `-19` (slope:1), `-20_a` (slope:2_a), `-21` (slope:3), `-22_a` (ca_a), `-23` (thal:3), `-24` (thal:4), `-25` (thal:6), `-26` (thal:7).
- **Exit nodes:** `0`.

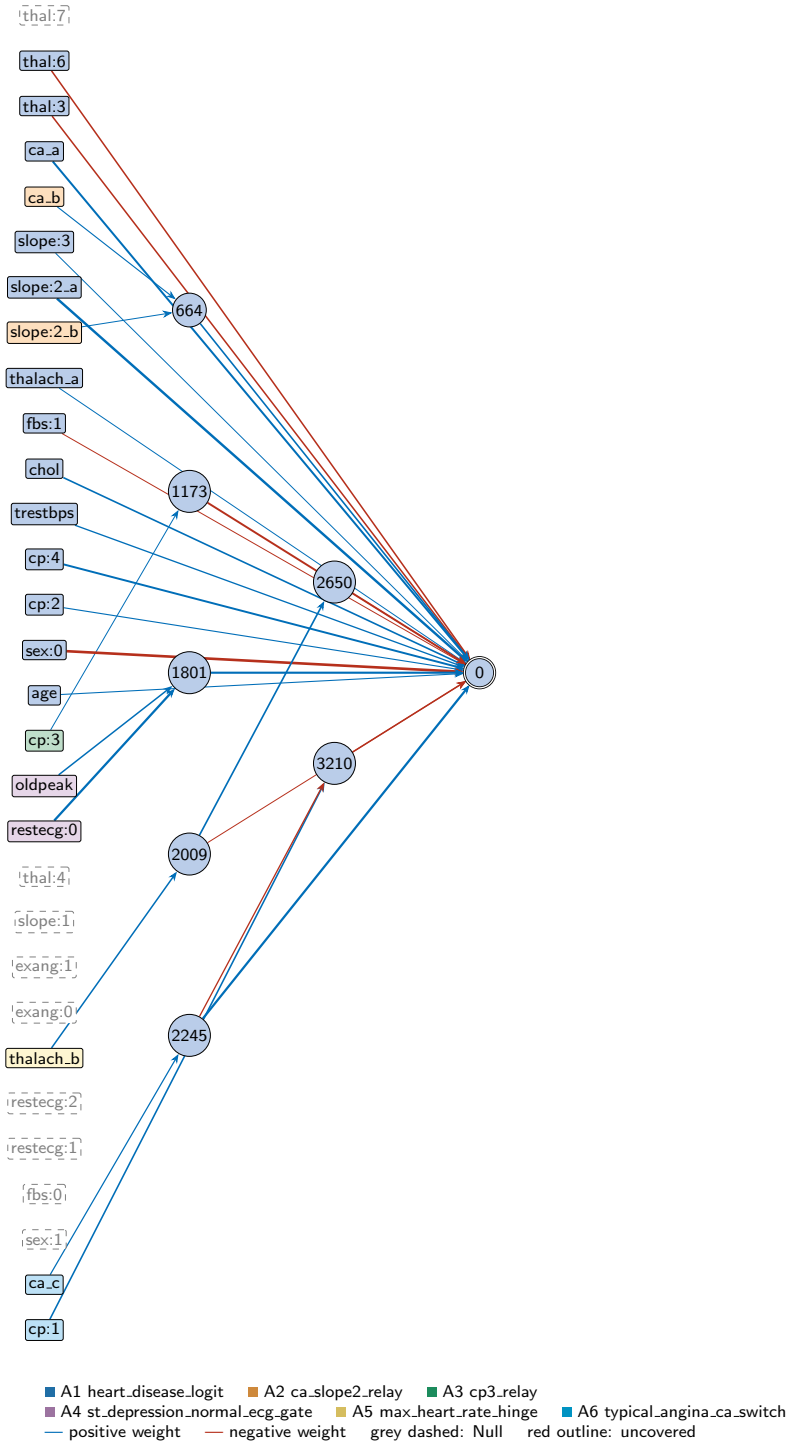


Figure 18: heart-s1-c1: full view of the explained model state.

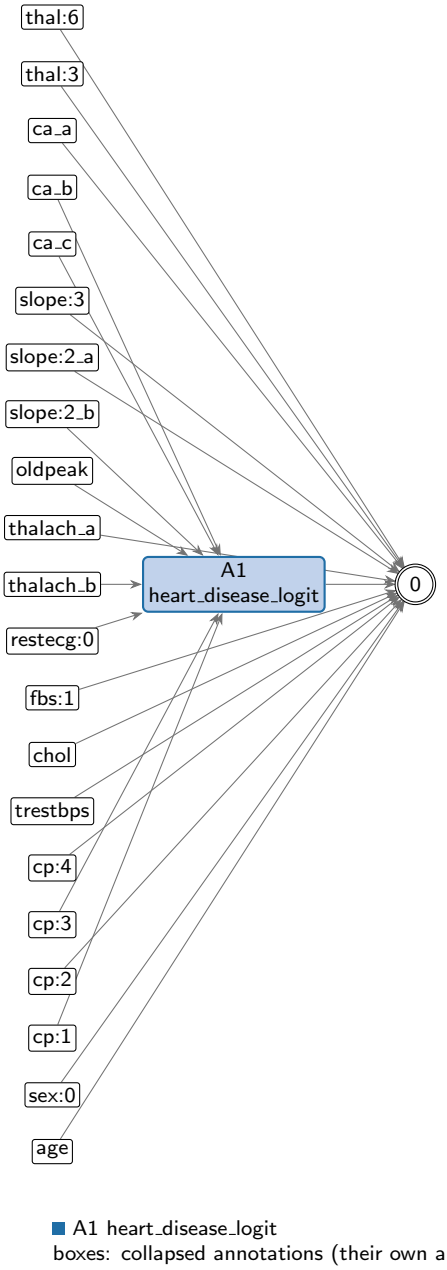


Figure 19: heart-s1-c1: collapsed view at hierarchy level 0 (fully recomposed).

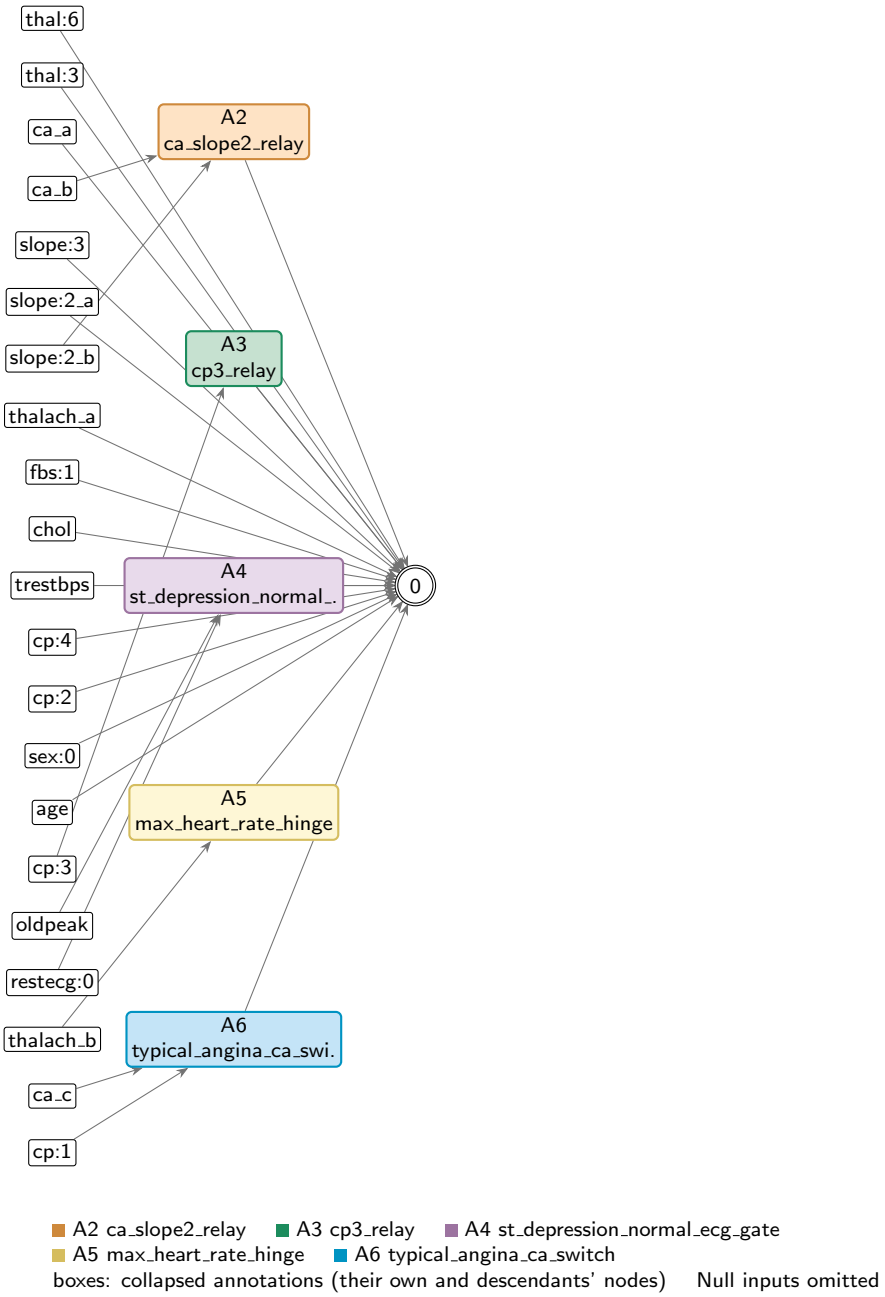


Figure 20: heart-s1-c1: collapsed view at hierarchy level 1.

- **Internal nodes:** none.
- **Edges (weight):** `-1`→`0` (0.399), `-2`→`0` (-1.87), `-5`→`0` (0.443), `-7`→`0` (1.25), `-8`→`0` (0.665), `-9`→`0` (0.931), `-11`→`0` (-0.315), `-21`→`0` (0.197), `-23`→`0` (-0.866), `-25`→`0` (-0.861), `664`→`0` (0.92), `1173`→`0` (-1.24), `1801`→`0` (1.4), `2009`→`0` (-0.151), `2245`→`0` (1.53), `2650`→`0` (-0.172), `3210`→`0` (-0.833), `-22_a`→`0` (1.3), `-20_a`→`0` (1.72), `-15_a`→`0` (0.00188).
- **With its descendants:** 38 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 37 candidate nodes.

Hypothesis. Hypothesis: the heart-disease logit is essentially a logistic regression. 13 direct input->output edges (largest: sex:0 -1.87, slope:2 +1.72, ca +1.30, cp:4 +1.25, chol +0.93, thal:3 -0.87, thal:6 -0.86, trestbps +0.67, cp:2 +0.44, age +0.40, fbs:1 -0.32, slope:3 +0.20, thalach +0.002) are added to five children: ca_slope2_relay and the 2245 relay (raise the effective ca weight to 2.60 per SD), cp3_relay (non-anginal pain lowers risk), st_depression_normal_ecg_gate (ST depression raises risk only with a normal resting ECG), typical_angina_ca_switch's 3210 (typical angina lowers risk, less with more vessels) and max_heart_rate_hinge (small). Nine inputs are unconnected (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7): exercise-induced angina is ignored. Evidence (train, n=242): AUC 0.9392; all hidden units replaced by their best linear fits 0.9355 (= L2 logistic regression 0.9355); all ReLUs as identity 0.9078; all hidden mean-ablated 0.9241. The gain over linear (+0.0037) comes from the ECG/ST gate (+0.0034 restored alone), the heart-rate hinge (+0.0008) and the typical-angina switch (+0.0003); the relays contribute none. Direct edges carry logit SD 4.25 of 5.63.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.276 + 1.3x_{ca_a} + 0.666x_{trestbps}\\
&\quad + 0.92\,\mathrm{ca_slope2_relay}\big(x_{ca_b}, x_{slope:2_b}\big)\\
&\quad + 0.00188x_{thalach_a} + 1.25x_{cp:4}\\
&\quad + 1.4\,\mathrm{st_depression_normal_ecg_gate}\big(x_{restecg:0}, x_{oldpeak}\big)\\
&\quad + 1.53\,\mathrm{typical_angina_ca_switch_0}\big(x_{ca_c}, x_{cp:1}\big)\\
&\quad + 0.197x_{slope:3} + 0.399x_{age} + 1.72x_{slope:2_a} + 0.443x_{cp:2} + 0.931x_{chol}\\
&\quad - 1.87x_{sex:0} - 1.24\,\mathrm{cp3_relay}\big(x_{cp:3}\big) - 0.315x_{fbs:1}\\
&\quad - 0.861x_{thal:6} - 0.151\,\mathrm{max_heart_rate_hinge_0}\big(x_{thalach_b}\big)\\
&\quad - 0.172\,\mathrm{max_heart_rate_hinge_1}\big(x_{thalach_b}\big)\\
&\quad - 0.833\,\mathrm{typical_angina_ca_switch_1}\big(x_{ca_c}, x_{cp:1}\big)\\
&\quad - 0.866x_{thal:3}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.276 + 1.3x_{ca_a} + 0.666x_{trestbps} + 0.92\,\mathrm{relu}\big(0.783\\
&\quad + 0.185x_{slope:2_b} + 0.456x_{ca_b}\big) + 0.00188x_{thalach_a} + 1.25x_{cp:4}\\
&\quad + 1.4\,\mathrm{relu}\big(-1.52 + 1.54x_{restecg:0} + 0.818x_{oldpeak}\big)\\
&\quad + 1.53\,\mathrm{relu}\big(1.28 + 0.578x_{ca_c}\big) + 0.197x_{slope:3}\\
&\quad + 0.399x_{age} + 1.72x_{slope:2_a} + 0.443x_{cp:2} + 0.931x_{chol} - 1.87x_{sex:0}\\
&\quad - 1.24\,\mathrm{relu}\big(1.82 + 0.336x_{cp:3}\big) - 0.315x_{fbs:1}\\
&\quad - 0.861x_{thal:6} - 0.151\,\mathrm{relu}\big(0.201 + 0.833x_{thalach_b}\big)\\
&\quad - 0.172\,\mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201\\
&\quad + 0.833x_{thalach_b}\big)\big) - 0.833\,\mathrm{relu}\big(0.42 + 0.901x_{cp:1}\\
&\quad - 0.644\,\mathrm{relu}\big(1.28 + 0.578x_{ca_c}\big)\big) - 0.866x_{thal:3}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00427, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=True) and effective linear weights through the always-on relays
- EMPIRICAL, `ablation_study`; data split: train. Train split, full sample; ablations and linear replacements on in-memory copies (no graph edits)

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.6.2 A2: ca_slope2_relay

Kind: Leaf annotation (depth 1); parent A1 (section 6.6.1).

- **Entry nodes:** `-22_b` (ca_b), `-20_b` (slope:2_b).
- **Exit nodes:** `664`.
- **Internal nodes:** none.
- **Edges (weight):** `-22_b`→`664` (0.456), `-20_b`→`664` (0.185).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: node 664 = ReLU(0.456 ca + 0.185 slope:2 + 0.783) is an affine relay that strengthens the direct ca and flat-ST-slope (slope:2) effects. Evidence (train, n=242): pre-activation 0.28..2.19, never clips, so the part is exactly affine on the data; it adds +0.42 logit per SD of ca and +0.17 per SD of slope:2 (output weight 0.92). Mean ablation AUC 0.9392 -> 0.9375; linear replacement leaves AUC at 0.9392. No concept beyond reweighting.

Closed form:

$$y_0 = \mathrm{relu}\big(0.783 + 0.185 x_{slope:2_b} + 0.456 x_{ca_b}\big)$$

where y_0 is node `664`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 4.47×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.6.3 A3: cp3_relay

Kind: Leaf annotation (depth 1); parent A1 (section 6.6.1).

- **Entry nodes:** `-6` (cp:3).
- **Exit nodes:** `1173`.
- **Internal nodes:** none.
- **Edges (weight):** `-6`→`1173` (0.336).

- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Hypothesis: node 1173 = ReLU(0.336 cp:3 + 1.818) is an always-on relay through which non-anginal chest pain (cp:3) lowers predicted risk. Evidence (train, n=242): pre-activation 1.60..2.34, never clips (entry-exit correlation 1.0); with output weight -1.24 it contributes -0.417 logit per SD of cp:3, about -0.92 logit for non-anginal pain vs other types. cp:3 has no direct edge, so this is its only route. Mean ablation AUC 0.9373; linear replacement 0.9392.

Closed form:

$$y_0 = \mathrm{relu}\big(1.82 + 0.336 x_{cp:3}\big)$$

where y_0 is node `1173`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00246, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.6.4 A4: st_depression_normal_ecg_gate

Kind: Leaf annotation (depth 1); parent A1 (section 6.6.1).

- **Entry nodes:** `-12` (restecg:0), `-18` (oldpeak).
- **Exit nodes:** `1801`.
- **Internal nodes:** none.
- **Edges (weight):** `-12`→`1801` (1.54), `-18`→`1801` (0.817).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: node 1801 = ReLU(1.542 restecg:0 + 0.817 oldpeak - 1.521) is a gate in which ST depression (oldpeak) raises predicted risk only for patients with a normal resting ECG (restecg:0), once oldpeak is above about its mean (threshold -0.03 SD); with an abnormal resting ECG the threshold is 3.75 SD and the gate is effectively closed. Evidence (train, n=242): active for 41 of 121 normal-ECG patients and 1 of 121 others; adds 1.40 x activation = +1.14 logit per SD oldpeak when open. It is the model's main nonlinearity: mean ablation AUC 0.9339, linear replacement 0.9362, and restoring it onto the all-linear model gains +0.0034 AUC. oldpeak has no direct edge, so the model ignores ST depression for patients with an abnormal resting ECG.

Closed form:

$$y_0 = \mathrm{relu}\big(-1.52 + 1.54 x_{restecg:0} + 0.818 x_{oldpeak}\big)$$

where y_0 is node `1801`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00514, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.6.5 A5: max_heart_rate_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 6.6.1).

- **Entry nodes:** `-15_b` (thalach_b).
- **Exit nodes:** `2009`, `2650`.
- **Internal nodes:** none.
- **Edges (weight):** `-15_b`→`2009` (0.833), `2009`→`2650` (0.982).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Hypothesis: a small hinge on maximum heart rate (thalach). Node 2009 = ReLU(0.833 thalach + 0.201) feeds the output (-0.151) and the always-on relay 2650 = ReLU(0.982*2009 + 0.859) (-0.172), a net -0.32 x 2009: higher maximum heart rate lowers predicted risk, but only above about -0.24 SD; below that it has no effect (the direct thalach edge, +0.002, is negligible). Evidence (train, n=242): 2009 active for 64% of patients; logit contribution SD 0.16 (small); mean ablation AUC 0.9391, linear replacement 0.9392, restore-onto-linear gain +0.0008.

Closed form:

$$y_0 = \mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big)$$
$$y_1 = \mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big)\big)$$

where y_0 is node `2009`; y_1 is node `2650`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 7.41×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.6.6 A6: typical_angina_ca_switch

Kind: Leaf annotation (depth 1); parent A1 (section 6.6.1).

- **Entry nodes:** `-22_c` (ca_c), `-4` (cp:1).
- **Exit nodes:** `2245`, `3210`.
- **Internal nodes:** none.
- **Edges (weight):** `-22_c`→`2245` (0.578), `-4`→`3210` (0.901), `2245`→`3210` (-0.644).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: (i) node 2245 = ReLU(0.578 ca + 1.276) is an always-on ca relay adding +0.88 logit per SD of ca (output weight 1.528); (ii) node 3210 = ReLU(0.901 cp:1 - 0.644*2245 + 0.420) is a typical-angina switch: it is on only for cp:1 patients and then lowers the logit by 0.833 x its activation, from about -2.26 at the lowest ca (-0.73 SD) to about -1.22 at the highest ca (2.64 SD). So typical angina lowers predicted risk, and the reduction shrinks with more

narrowed vessels. Evidence (train, n=242): 2245 never clips (0.86..2.80); 3210 active for all 22 cp:1 patients and none of the 220 others. Mean ablation AUC 2245 0.9339, 3210 0.9354; linear replacement 0.9392 / 0.9391 (3210 is a cp:1 switch whose within-group response is linear in ca).

Closed form:

$$\begin{aligned} y_0 &= \mathrm{relu}\big(1.28 + 0.578 x_{ca_c}\big) \\ y_1 &= \mathrm{relu}\big(0.42 + 0.901 x_{cp:1} - 0.644\,\mathrm{relu}\big(1.28 \\ &\qquad + 0.578 x_{ca_c}\big)\big) \end{aligned}$$

where y_0 is node `2245`; y_1 is node `3210`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00423, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

6.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

7. heart-s2-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.01311	0.01311	0.01311	agree
Numeric-function max abs error	n/a	8.88e-16	8.88e-16	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	3 / 1	n/a	3 / 1	agree
Operation counts	5 splits, 0 identity	annotate 4, rename node 26, split node 5	annotate 4, rename node 26, split node 5	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `oldpeak_hinges` (0.0131), then `age_hr_reserve_gate` (0.0114). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `age_hr_reserve_gate`: *"Hypothesis: R3195 is an always-on affine relay of thal:7, so the gate is a single ReLU, 1718 = ReLU(-1.524 - 1.502*age + 1.023*thalach - 1.561*thal:7), that opens when the achieved max heart rate is high for the patient's age and..."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `oldpeak_hinges`: *"it acts only here and in the flat-slope gate."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 5 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): No v3 tool bugs met. compute_shap on all 242 train rows took about a minute. Input fan-out splits (-26, -15, -9, -20, -18) taken from the topology. The agent sandbox refused shell loops calling uv with variables, so tools were called through a small Python subprocess wrapper in scratch (same xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s2-c1-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 29%; it adds up most of its inputs independently, and three parts add to that sum: one acts on 80 of 242 training patients through age and maximum heart rate, one on 121 through ST depression on exercise, and one on 117 through the ST segment, ST depression and cholesterol. [C29, C38, C42, C52] These add up independently, and these lower the risk: compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 29% to about 3%, non-anginal chest pain to about 5% and atypical angina to about 14%; being a woman lowers it to about 3% and fasting blood sugar above 120 mg/dL to about 14%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 32% at 118 to about 28% at 175 beats per minute (maximum heart rate also acts through the age and heart rate gate, below). [C06, C09, C17, C07, C18, C23] Each extra vessel coloured by fluoroscopy adds about the same, from about 29% at 0 to about 94% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; higher resting blood pressure raises it steadily, from about

20% at 110 mm Hg to about 44% at 155 mm Hg, and chest pain brought on by exercise raises it to about 43%. [C01, C08, C19] Through this sum alone, a flat rather than upsloping ST segment raises it to about 73%, higher cholesterol raises it steadily from about 18% at 193 mg/dL to about 49% at 308 mg/dL, and a reversible defect on the thallium scan raises it to about 42%; all three also act through the parts below. [C03, C04, C20] Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a downsloping ST segment raises the risk to about 50% and a fixed defect on the thallium scan lowers it to about 11%, and an ST-T wave abnormality on the resting ECG raises it to about 46%. [C10, C13, C16] This part reads age, maximum heart rate and the thallium scan, and acts on 80 of 242 training patients, 9 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 4% without it and about 1% with it. [C38] For patients without a reversible defect on the thallium scan, it switches on when age in years minus about 0.28 × the maximum heart rate in beats per minute is below about 11.4 (for maximum heart rates from about 71 to 202 beats per minute); 80 of 242 training patients are on that side, and 11% of them had heart disease, against 60% of the others. [C54] Through it, older age raises the predicted risk steadily up to about 56 years (between 45 and 61 years for different patients), then no further for most patients, but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 29% at 66 years. [C30, C37] Across the whole model, a higher maximum heart rate lowers the risk in an uneven way, more without a reversible defect on the thallium scan (from about 32% at 118 to about 21% at 175 beats per minute, against about 45% to about 40% with one). [C35] This part reads only ST depression on exercise and acts on 121 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 91% with it. [C42] Through it, more ST depression raises the predicted risk slowly at first, then faster, adding risk only once it rises above about 0.8 mm: for a typical patient from about 29% at 0 mm to about 48% at 2.8 mm. [C40, C41] This part reads the slope of the ST segment, ST depression on exercise and cholesterol, and acts on 117 of 242 training patients, 78 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 86% without it and about 90% with it. [C52] For patients whose ST segment is not flat, it switches on when ST depression on exercise is above about 2.23 mm (for cholesterol from about 126 to 186 mg/dL), or when ST depression minus about 0.012 × the cholesterol in mg/dL is above about 0.011 (for cholesterol from about 186 to 417 mg/dL); 117 of 242 training patients are on, and 67% of them had heart disease, against 23% of the others. [C55] Across the whole model, more ST depression raises the risk slowly at first, then faster, and even more so with a flat ST segment: from about 75% to about 91% then, against about 29% to about 48% otherwise. [C48] Higher cholesterol raises it steadily across the whole model, from about 67% to about 88% with a flat ST segment and from about 18% to about 49% otherwise, while through this part it lowers the risk in an uneven way, for the typical patient the part acts on from about 91% to about 88%. [C47, C46]

Worth checking: Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 29% to about 94%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 29% to about 14%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 29% to about 11%), from only 15 training patients. [C14] Worth checking: across the whole model, older age raises the predicted risk only for patients without a reversible defect on the thallium scan, and through one part only up to about 56 years. [C33, C37]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	29% to 94%	log-odds 3.71, OR 40.7; range across patients 3.71 to 3.71	242	train; test unclear (AUC gain 0.032 [-0.044, 0.111], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 73%	log-odds 1.90, OR 6.70; range across patients 1.90 to 1.90	115, 110	train; test disagrees (AUC gain -0.014 [-0.055, 0.021], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	18% to 49%	log-odds 1.46, OR 4.32; range across patients 1.46 to 1.46	242	train; test disagrees (AUC gain -0.030 [-0.068, -0.004], n=61)	yes	on the training data only
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 3%	log-odds -2.61, OR 0.07; range across patients -2.61 to -2.61	108, 22	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C07	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	29% to 3%	log-odds -2.51, OR 0.08; range across patients -2.51 to -2.51	159, 83	train; test unclear (AUC gain 0.030 [-0.011, 0.079], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 44%	log-odds 1.15, OR 3.16; range across patients 1.15 to 1.15	242	train; test disagrees (AUC gain -0.003 [-0.022, 0.014], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 5%	log-odds -1.98, OR 0.14; range across patients -1.98 to -1.98	108, 70	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C10	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 50%	log-odds 0.91, OR 2.48; range across patients 0.91 to 0.91	115, 17 (small n)	train; test disagrees (AUC gain -0.014 [-0.055, 0.021], n=61)	yes	the model has little data on this; on the training data only
C13	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 11%	log-odds -1.15, OR 0.32; range across patients -1.15 to -1.15	137, 15 (small n)	train; test unclear (AUC gain 0.002 [-0.016, 0.019], n=61)	yes	the model has little data on this; on the training data only
C16	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: an ST-T wave abnormality on the resting ECG, resting ECG other than an ST-T wave abnormality on the resting ECG*	29% to 46%	log-odds 0.73, OR 2.08; range across patients 0.73 to 0.73	240, 2 (small n)	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	the model has little data on this; on the training data only
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain*	29% to 14%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	108, 42	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C19	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	29% to 43%	log-odds 0.59, OR 1.80; range across patients 0.59 to 0.59	160, 82	train; test unclear (AUC gain 0.005 [-0.006, 0.022], n=61)	yes	on the training data only
C20	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 42%	log-odds 0.58, OR 1.78; range across patients 0.58 to 0.58	137, 90	train; test unclear (AUC gain 0.002 [-0.016, 0.019], n=61)	yes	on the training data only
C23	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	32% to 28%	log-odds -0.20, OR 0.82; range across patients -0.20 to -0.20	242	train; test unclear (AUC gain 0.003 [-0.004, 0.016], n=61)	yes	slightly; on the training data only
C29	structure: made of age_hr_reserve_gate, st_slope_chol_gate, oldpeak_hinges and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: age: 42 years to 66 years (through this part) *shape: hinge at 56 years; 86% of patients same; mixed (but step for patients with a reversible defect on the thallium scan); one point with C37 (median fold 56 years, typical patient 54 years, 10th-90th percentile 45 years to 61 years, wide)*	12% to 29%	log-odds 0.74, OR 2.11; range across patients 0.00 to 1.71	242	train; test disagrees (AUC gain -0.006 [-0.022, 0.008], n=61)	yes	on the training data only
C35	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in thallium stress test result other than a reversible defect on the thallium scan *shape: accelerating; 79% of patients same; fallback*	a reversible defect on the thallium scan: 45% to 40%; thallium stress test result other than a reversible defect on the thallium scan: 32% to 21%	log-odds difference 0.67, OR 1.95	90, 152	train; test agrees (AUC gain 0.01 [-0.003, 0.029], n=61); group difference on test 0.71 [0.53, 0.91]	yes	clearly
C37	threshold: age: fold at about 56 years (stops) for patients with a normal thallium scan *shape: hinge at 56 years; 99% of patients same; verified; risk saturates near 0%; one point with C37 (median fold 56 years, typical patient 56 years, 10th-90th percentile 45 years to 61 years, wide)*	<1% to <1% (typical patient with a fold (patients with a normal thallium scan))	fold 55.6 years (IQR 51.7 to 58.5); past it log-odds 1.35, OR 3.84; range across patients 0.44 to 1.98	135, 73	train; test disagrees (AUC gain -0.006 [-0.022, 0.008], n=61)	yes	slightly; on the training data only
C38	subgroup: who this part acts on: 80 of 242 patients	4% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.33; log-odds -1.06, OR 0.35; range across patients -1.99 to -0.20	80, 162	train; test unclear (AUC gain 0.017 [-0.001, 0.052], n=61)	yes	slightly; on the training data only
C40	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 100% of patients same; verified*	29% to 48%	log-odds 0.80, OR 2.22; range across patients 0.80 to 0.80	242	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	on the training data only
C41	threshold: ST depression on exercise: fold at about 0.8 mm (starts) *shape: accelerating; 100% of patients same; verified*	29% to 57% (typical patient with a fold)	fold 0.76 mm (IQR 0.76 to 0.76); past it log-odds 1.15, OR 3.16; range across patients 1.15 to 1.15	242, 121	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	on the training data only
C42	subgroup: who this part acts on: 121 of 242 patients	89% to 91% (typical patient this part acts on (risk without the part, then with it))	share 0.50; log-odds 0.43, OR 1.54; range across patients 0.03 to 1.15	121, 121	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	slightly; on the training data only
C46	direction: serum cholesterol: 193 mg/dl	91% to 88%	log-odds -0.11, OR	242	train; test unclear	yes	slightly; on

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C47	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in slope of the ST segment at peak exercise other than a flat ST segment *shape: steady; 100% of patients same; verified*	a flat ST segment: 67% to 88%; slope of the ST segment at peak exercise other than a flat ST segment: 18% to 49%	log-odds difference -0.21, OR 0.81	110, 132	train; test agrees (AUC gain -0.005 [-0.019, 0.003], n=61); group difference on test -0.21 [-0.22, -0.19]	yes	clearly
C48	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in a flat ST segment *shape: accelerating; 100% of patients same; verified*	a flat ST segment: 75% to 91%; slope of the ST segment at peak exercise other than a flat ST segment: 29% to 48%	log-odds difference 0.42, OR 1.53	110, 132	train; test agrees (AUC gain -0.005 [-0.019, 0.003], n=61); group difference on test 0.41 [0.37, 0.43]	yes	slightly
C52	subgroup: who this part acts on: 117 of 242 patients	86% to 90% (typical patient this part acts on (risk without the part, then with it))	share 0.48; log-odds 0.31, OR 1.36; range across patients 0.08 to 0.55	117, 125	train; test disagrees (AUC gain -0.001 [-0.016, 0.011], n=61)	yes	slightly; on the training data only
C54	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	
C55	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s2-c1 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s2-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s2-c1 --verified --minutes N --changes "..." --notes "..."`

7 Explanation certificate: heart-s2-c1

This section is the explanation certificate of the heart-s2-c1 network. It is rendered from the canonical JSON record `heart-s2-c1.json` (schema 1.4.0, SHA-256 `ec79581c37441832...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 7 summarises the network and its coverage.

Drafter's summary. Mostly logistic regression: 16 direct edges (logit SD 3.24; train AUC 0.930 alone) plus three additive nonlinear corrections (logit SD 0.95) to 0.943. Largest: a protective heart-rate-reserve gate ReLU(-1.52 - 1.50 age + 1.02 thalach - 1.56 thal:7) read out -0.61, open for half the patients without a reversible defect; then a flat-ST-slope gate scaled by oldpeak and down by cholesterol, and a convex two-kink oldpeak dose-response. Age and oldpeak act only through

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s2, seed 2
Genome	`22757c22-3eac-4248-961b-ed0cd500a680` (NEAT key 9083, generation 39)
Explanation	`eac2ad3d-7d0e-4112-9d34-7f0f217a2c31` (35 operations)
Train AUC / accuracy	0.9432 / 0.8802 (n = 242)
Test AUC / accuracy	0.8966 / 0.8361 (n = 61)
Output activation	sigmoid
Phenotype	33 nodes (26 inputs, 6 hidden, 1 output), 31 enabled edges
Explained model state	39 nodes, 31 enabled edges
Refactoring operations	split_node 5
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 2.22 \times 10^{-16}$ on the training split
Annotations	4 (3 leaf, 1 composition)
Structural coverage	1 (38 of 38 candidate nodes; 8 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`ec79581c37441832...`

Table 7: Explanation certificate summary: heart-s2-c1.

these gates. Same mechanisms as the generation-38 heart-s2 record, weights shifted by at most about 0.03 after retraining.

What resisted explanation. Nothing resisted coverage. Relays 3195 (thal:7, active 100%) and 3464 (chol, active 93%) are affine on the data, so each gate is effectively one ReLU. Counter-directional signs reported, not explained: cp:2 +0.40 and fbs:1 -0.31 in the linear core, negative chol inside the flat-slope gate.

7.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 29%; it adds up most of its inputs independently, and three parts add to that sum: one acts on 80 of 242 training patients through age and maximum heart rate, one on 121 through ST depression on exercise, and one on 117 through the ST segment, ST depression and cholesterol. [C29, C38, C42, C52] These add up independently, and these lower the risk: compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 29% to about 3%, non-anginal chest pain to about 5% and atypical angina to about 14%; being a woman lowers it to about 3% and fasting blood sugar above 120 mg/dL to about 14%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 32% at 118 to about 28% at 175 beats per minute (maximum heart rate also acts through the age and heart rate gate, below). [C06, C09, C17, C07, C18, C23] Each extra vessel coloured by fluoroscopy adds about the same, from about 29% at 0 to about 94% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; higher resting blood pressure raises it steadily, from about 20% at 110 mm Hg to about 44% at 155 mm Hg, and chest pain brought on by exercise raises it to about 43%. [C01, C08, C19] Through this sum alone, a flat rather than upsloping ST segment raises it to about 73%, higher cholesterol raises it

steadily from about 18% at 193 mg/dL to about 49% at 308 mg/dL, and a reversible defect on the thallium scan raises it to about 42%; all three also act through the parts below. [C03, C04, C20] Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a downsloping ST segment raises the risk to about 50% and a fixed defect on the thallium scan lowers it to about 11%, and an ST-T wave abnormality on the resting ECG raises it to about 46%. [C10, C13, C16] This part reads age, maximum heart rate and the thallium scan, and acts on 80 of 242 training patients, 9 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 4% without it and about 1% with it. [C38] For patients without a reversible defect on the thallium scan, it switches on when age in years minus about 0.28 × the maximum heart rate in beats per minute is below about 11.4 (for maximum heart rates from about 71 to 202 beats per minute); 80 of 242 training patients are on that side, and 11% of them had heart disease, against 60% of the others. [C54] Through it, older age raises the predicted risk steadily up to about 56 years (between 45 and 61 years for different patients), then no further for most patients, but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 29% at 66 years. [C30, C37] Across the whole model, a higher maximum heart rate lowers the risk in an uneven way, more without a reversible defect on the thallium scan (from about 32% at 118 to about 21% at 175 beats per minute, against about 45% to about 40% with one). [C35] This part reads only ST depression on exercise and acts on 121 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 91% with it. [C42] Through it, more ST depression raises the predicted risk slowly at first, then faster, adding risk only once it rises above about 0.8 mm: for a typical patient from about 29% at 0 mm to about 48% at 2.8 mm. [C40, C41] This part reads the slope of the ST segment, ST depression on exercise and cholesterol, and acts on 117 of 242 training patients, 78 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 86% without it and about 90% with it. [C52] For patients whose ST segment is not flat, it switches on when ST depression on exercise is above about 2.23 mm (for cholesterol from about 126 to 186 mg/dL), or when ST depression minus about 0.012 × the cholesterol in mg/dL is above about 0.011 (for cholesterol from about 186 to 417 mg/dL); 117 of 242 training patients are on, and 67% of them had heart disease, against 23% of the others. [C55] Across the whole model, more ST depression raises the risk slowly at first, then faster, and even more so with a flat ST segment: from about 75% to about 91% then, against about 29% to about 48% otherwise. [C48] Higher cholesterol raises it steadily across the whole model, from about 67% to about 88% with a flat ST segment and from about 18% to about 49% otherwise, while through this part it lowers the risk in an uneven way, for the typical patient the part acts on from about 91% to about 88%. [C47, C46]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, and three parts add to that sum: one for age and maximum heart rate, one for ST depression and one for the ST segment with cholesterol. [C29, C01, C02, C06, C07] *Evidence.* It is made up of the inputs that feed the score directly and three parts, the age and heart rate gate, the ST depression hinges and the flat ST segment with cholesterol part (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Age and ST depression on exercise do not feed the score directly, so they act only through these parts. [C29, C30, C40] Changing one input at a time across the whole model, a flat rather than upsloping ST segment takes a typical patient's predicted risk from about 29% to about 77%, against about 73% through the sum alone. [C02, C03] Across the whole model, a higher maximum heart rate lowers it in an uneven way, from about 32% at 118 to about 21% at 175 beats per minute, against about 28% through the sum alone, where it falls steadily. [C22, C23] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15), and a reversible defect raises it to about 42%. [C14, C21]

Age and heart rate gate. This part lowers the predicted risk for patients whose age is low relative to their maximum heart rate; for the typical patient it acts on, a reversible defect on the

thallium scan raises the risk through it from about 1% to about 4%. [C54, C32] *Evidence.* For patients without a reversible defect on the thallium scan, it switches on when age in years minus about 0.28 × the maximum heart rate in beats per minute is below about 11.4 (for maximum heart rates from about 71 to 202 beats per minute); 80 of 242 training patients are on that side, and 11% of them had heart disease, against 60% of the others. [C54] Through it, older age raises the predicted risk steadily up to about 56 years (between 45 and 61 years for different patients), then no further for most patients, but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 29% at 66 years. [C30, C37] Across the whole model, older age raises the risk, then no further, only if the thallium scan shows no reversible defect: from about 12% to about 29%, while with one it stays at about 42%. [C33] A higher maximum heart rate lowers the risk through it in an uneven way, from about 29% at 118 to about 22% at 175 beats per minute, and across the whole model it lowers it more without a reversible defect (from about 32% to about 21%, against about 45% to about 40%). [C31, C35] Its formula reproduces the part exactly on all 242 training patients. [C39]

ST depression hinges. This part raises the predicted risk with ST depression on exercise, slowly at first, then faster, adding risk only once it rises above about 0.8 mm. [C40, C41] *Evidence.* It acts on 121 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 91% with it. [C42] Through it, more ST depression raises the risk slowly at first, then faster, adding risk only once it rises above about 0.8 mm: for a typical patient from about 29% at 0 mm to about 48% at 2.8 mm. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C43]

Flat ST segment with cholesterol. This part raises the predicted risk mainly for patients with a flat ST segment, and makes ST depression on exercise count more for them, rising slowly at first, then faster. [C44, C48] *Evidence.* It acts on 117 of 242 training patients, 78 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 86% without it and about 90% with it. [C52] For patients whose ST segment is not flat, it switches on when ST depression on exercise is above about 2.23 mm (for cholesterol from about 126 to 186 mg/dL), or when ST depression minus about 0.012 × the cholesterol in mg/dL is above about 0.011 (for cholesterol from about 186 to 417 mg/dL); 117 of 242 training patients are on, and 67% of them had heart disease, against 23% of the others. [C55] Through it, a flat ST segment raises a typical patient's predicted risk from about 29% to about 34%; for the typical patient it acts on, more ST depression raises it in an uneven way from about 87% to about 92%, and higher cholesterol lowers it in an uneven way from about 91% to about 88%. [C44, C45, C46] Across the whole model, more ST depression raises the risk slowly at first, then faster, from about 75% to about 91% with a flat ST segment, against about 29% to about 48% otherwise. [C48] Its formula reproduces the part exactly on all 242 training patients. [C53]

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 29% to about 94%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 29% to about 14%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 29% to about 11%), from only 15 training patients. [C14] Worth checking: across the whole model, older age raises the predicted risk only for patients without a reversible defect on the thallium scan, and through one part only up to about 56 years. [C33, C37]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	29% to 94%	log-odds 3.71, OR 40.7; range across patients 3.71 to 3.71	242	train; test unclear (AUC gain 0.032 [-0.044, 0.111], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 77%	log-odds 2.14, OR 8.54; range across patients 1.93 to 2.43	115, 110	train; test disagrees (AUC gain -0.013 [-0.060, 0.026], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 73%	log-odds 1.90, OR 6.70; range across patients 1.90 to 1.90	115, 110	train; test disagrees (AUC gain -0.014 [-0.055, 0.021], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	18% to 49%	log-odds 1.46, OR 4.32; range across patients 1.46 to 1.46	242	train; test disagrees (AUC gain -0.030 [-0.068, -0.004], n=61)	yes	on the training data only
C05	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	18% to 49%	log-odds 1.35, OR 3.87; range across patients 1.24 to 1.46	242	train; test disagrees (AUC gain -0.030 [-0.068, -0.004], n=61)	yes	on the training data only
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 3%	log-odds -2.61, OR 0.07; range across patients -2.61 to -2.61	108, 22	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C07	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	29% to 3%	log-odds -2.51, OR 0.08; range across patients -2.51 to -2.51	159, 83	train; test unclear (AUC gain 0.030 [-0.011, 0.079], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 44%	log-odds 1.15, OR 3.16; range across patients 1.15 to 1.15	242	train; test disagrees (AUC gain -0.003 [-0.022, 0.014], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 5%	log-odds -1.98, OR 0.14; range across patients -1.98 to -1.98	108, 70	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C10	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 50%	log-odds 0.91, OR 2.48; range across patients 0.91 to 0.91	115, 17 (small n)	train; test disagrees (AUC gain -0.014 [-0.055, 0.021], n=61)	yes	the model has little data on this; on the training data only
C11	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 50%	log-odds 0.91, OR 2.48; range across patients 0.91 to 0.91	115, 17 (small n)	train; test disagrees (AUC gain -0.013 [-0.060, 0.026], n=61)	yes	the model has little data on this; on the training data only
C12	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: accelerating; 100% of patients same; verified*	29% to 48%	log-odds 1.01, OR 2.74; range across patients 0.80 to 1.25	242	train; test unclear (AUC gain 0.005 [-0.009, 0.022], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 11%	log-odds -1.15, OR 0.32; range across patients -1.15 to -1.15	137, 15 (small n)	train; test unclear (AUC gain 0.002 [-0.016, 0.019], n=61)	yes	the model has little data on this; on the training data only
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	29% to 11%	log-odds -1.15, OR 0.32; range across patients -1.15 to -1.15	137, 15 (small n)	train; test unclear (AUC gain 0.015 [-0.01, 0.044], n=61)	yes	the model has little data on this; on the training data only
C15	direction: age: 42 years to 66 years (whole model) *shape: hinge at 54 years; 86% of patients same; mixed (but step for men and patients with a reversible defect on the thallium scan); matches the fold at 56 years*	12% to 29%	log-odds 0.74, OR 2.11; range across patients 0.00 to 1.71	242	train; test disagrees (AUC gain -0.006 [-0.022, 0.008], n=61)	yes	on the training data only
C16	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: an ST-T wave abnormality on the resting ECG, resting ECG other than an ST-T wave abnormality on the resting ECG*	29% to 46%	log-odds 0.73, OR 2.08; range across patients 0.73 to 0.73	240, 2 (small n)	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	the model has little data on this; on the training data only
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 14%	log-odds -0.94, OR 0.39; range across patients -0.94 to -0.94	108, 42	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	29% to 14%	log-odds -0.90, OR 0.41; range across patients -0.90 to -0.90	209, 33	train; test unclear (AUC gain 0.002 [-0.012, 0.020], n=61)	yes	on the training data only
C19	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	29% to 43%	log-odds 0.59, OR 1.80; range across patients 0.59 to 0.59	160, 82	train; test unclear (AUC gain 0.005 [-0.006, 0.022], n=61)	yes	on the training data only
C20	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed*	29% to 42%	log-odds 0.58, OR 1.78; range across patients 0.58 to 0.58	137, 90	train; test unclear (AUC gain 0.002 [-0.016, 0.019], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	29% to 77%	log-odds 2.12, OR 8.33	115, 110	train; test disagrees (AUC gain -0.013 [-0.060, 0.026], n=61)	yes	on the training data only
C26	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	18% to 49%	log-odds 1.46, OR 4.32	242	train; test disagrees (AUC gain -0.030 [-0.068, -0.004], n=61)	yes	on the training data only
C27	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	29% to 3%	log-odds -2.61, OR 0.07	108, 22	train; test agrees (AUC gain 0.033 [0.005, 0.072], n=61)	yes	clearly
C28	magnitude: sex: male to female, typical patient *shape: categorical; highest to lowest: male, female*	29% to 3%	log-odds -2.51, OR 0.08	159, 83	train; test unclear (AUC gain 0.030 [-0.011, 0.079], n=61)	yes	on the training data only
C29	structure: made of age_hr_reserve_gate, st_slope_chol_gate, oldpeak_hinges and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: age: 42 years to 66 years (through this part) *shape: hinge at 56 years; 86% of patients same; mixed (but step for patients with a reversible defect on the thallium scan); one point with C37 (median fold 56 years, typical patient 54 years, 10th-90th percentile 45 years to 61 years, wide)*	12% to 29%	log-odds 0.74, OR 2.11; range across patients 0.00 to 1.71	242	train; test disagrees (AUC gain -0.006 [-0.022, 0.008], n=61)	yes	on the training data only
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 165 beats per minute; 68% of patients same; fallback; one point with C36 (median fold 165 beats per minute, typical patient 162 beats per minute, 10th-90th percentile 125 beats per minute to 195 beats per minute, wide)*	29% to 22%	log-odds -0.43, OR 0.65; range across patients -1.57 to 0.00	242	train; test unclear (AUC gain 0.009 [-0.002, 0.028], n=61)	yes	on the training data only
C32	direction: thallium stress test result: thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	1% to 4% (typical patient this part acts on)	log-odds 0.41, OR 1.50; range across patients 0.00 to 1.56	152, 90	train; test unclear (AUC gain 0.01 [-0.003, 0.029], n=61)	yes	slightly; on the training data only
C33	interaction: effect of age (42 years to 66 years) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in thallium stress test result other than a reversible defect on the thallium scan *shape: hinge at 54 years; 97% of patients same; verified*	a reversible defect on the thallium scan: 42% to 42%; thallium stress test result other than a reversible defect on the thallium scan: 12% to 29%	log-odds difference -1.17, OR 0.31	90, 152	train; test agrees (AUC gain 0.01 [-0.003, 0.029], n=61); group difference on test -1.25 [-1.42, -1.05]	yes	clearly
C34	interaction: effect of age (42 years to 66 years) by maximum heart rate: maximum heart rate above 152 beats per minute minus maximum heart rate at or below 152 beats per minute; stronger in maximum heart rate above 152 beats per minute *shape: hinge at 58 years; 92% of patients same;*	maximum heart rate above 152 beats per minute: 8% to 28%; maximum heart rate at or below 152 beats per minute: 18% to 30%	log-odds difference 0.83, OR 2.30	121, 121	train; test agrees (AUC gain 0.009 [-0.002, 0.028], n=61); group difference on test 0.85 [0.54, 1.17]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	threshold: age: fold at about 56 years (stops) for patients with a normal thallium scan *shape: hinge at 56 years; 99% of patients same; verified; risk saturates near 0%; one point with C37 (median fold 56 years, typical patient 56 years, 10th-90th percentile 45 years to 61 years, wide)*	<1% to <1% (typical patient with a fold (patients with a normal thallium scan))	fold 55.6 years (IQR 51.7 to 58.5); past it log-odds 1.35, OR 3.84; range across patients 0.44 to 1.98	135, 73	train; test disagrees (AUC gain -0.006 [-0.022, 0.008], n=61)	yes	slightly; on the training data only
C38	subgroup: who this part acts on: 80 of 242 patients	4% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.33; log-odds -1.06, OR 0.35; range across patients -1.99 to -0.20	80, 162	train; test unclear (AUC gain 0.017 [-0.001, 0.052], n=61)	yes	slightly; on the training data only
C39	structure: its formula reproduces it on 242 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	242	train	yes	
C40	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 100% of patients same; verified*	29% to 48%	log-odds 0.80, OR 2.22; range across patients 0.80 to 0.80	242	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	on the training data only
C41	threshold: ST depression on exercise: fold at about 0.8 mm (starts) *shape: accelerating; 100% of patients same; verified*	29% to 57% (typical patient with a fold)	fold 0.76 mm (IQR 0.76 to 0.76); past it log-odds 1.15, OR 3.16; range across patients 1.15 to 1.15	242, 121	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	on the training data only
C42	subgroup: who this part acts on: 121 of 242 patients	89% to 91% (typical patient this part acts on (risk without the part, then with it))	share 0.50; log-odds 0.43, OR 1.54; range across patients 0.03 to 1.15	121, 121	train; test unclear (AUC gain 0.005 [-0.008, 0.020], n=61)	yes	slightly; on the training data only
C43	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C44	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	29% to 34%	log-odds 0.24, OR 1.27; range across patients 0.03 to 0.53	132, 110	train; test disagrees (AUC gain -0.005 [-0.019, 0.003], n=61)	yes	slightly; on the training data only
C45	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: steady; 43% of patients same; fallback*	87% to 92% (typical patient this part acts on)	log-odds 0.21, OR 1.24; range across patients 0.00 to 0.45	242	train; test disagrees (AUC gain -0.001 [-0.009, 0.006], n=61)	yes	slightly; on the training data only
C46	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (through this part) *shape: hinge at 184 mg/dL (4.8 mmol/L); 46% of patients same; fallback; one point with C50 (median fold 184 mg/dL (4.8 mmol/L), typical patient 183 mg/dL (4.7 mmol/L), 10th-90th percentile 184 mg/dL (4.8 mmol/L) to 184 mg/dL (4.8 mmol/L))*	91% to 88% (typical patient this part acts on)	log-odds -0.11, OR 0.90; range across patients -0.23 to 0.00	242	train; test unclear (AUC gain 0.003 [-0.007, 0.019], n=61)	yes	slightly; on the training data only
C47	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in slope of the ST segment at peak exercise other than a flat ST segment *shape: steady; 100% of patients same; verified*	a flat ST segment: 67% to 88%; slope of the ST segment at peak exercise other than a flat ST segment: 18% to 49%	log-odds difference -0.21, OR 0.81	110, 132	train; test agrees (AUC gain -0.005 [-0.019, 0.003], n=61); group difference on test -0.21 [-0.22, -0.19]	yes	clearly
C48	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by slope of the ST segment at peak exercise: a flat ST segment minus slope of the ST segment at peak exercise other than a flat ST segment; stronger in a flat ST segment *shape: accelerating; 100% of patients same; verified*	a flat ST segment: 75% to 91%; slope of the ST segment at peak exercise other than a flat ST segment: 29% to 48%	log-odds difference 0.42, OR 1.53	110, 132	train; test agrees (AUC gain -0.005 [-0.019, 0.003], n=61); group difference on test 0.41 [0.37, 0.43]	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C49	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by ST depression on exercise: ST depression on exercise above 0.8 mm minus ST depression on exercise at or below 0.8 mm; stronger in ST depression on exercise at or below 0.8 mm *shape: steady; 100% of patients same; verified*	ST depression on exercise above 0.8 mm: 22% to 54%; ST depression on exercise at or below 0.8 mm: 18% to 49%	log-odds difference -0.10, OR 0.91	121, 121	train; test agrees (AUC gain -0.001 [-0.009, 0.006], n=61); group difference on test -0.11 [-0.16, -0.06]	no	slightly
C50	threshold: serum cholesterol: fold at about 184 mg/dL (4.8 mmol/L) (starts) *shape: hinge at 184 mg/dL (4.8 mmol/L); 46% of patients same; fallback; one point with C50 (median fold 184 mg/dL (4.8 mmol/L), typical patient 183 mg/dL (4.7 mmol/L), 10th-90th percentile 184 mg/dL (4.8 mmol/L) to 184 mg/dL (4.8 mmol/L))*	91% to 89% (typical patient with a fold)	fold 184 mg/dL (IQR 184 to 184); past it log-odds -0.26, OR 0.77; range across patients -0.28 to -0.21	123, 114	train; test unclear (AUC gain 0.003 [-0.007, 0.019], n=61)	yes	slightly; on the training data only
C51	threshold: ST depression on exercise: fold at about 2.8 mm (starts) *shape: hinge at 2.8 mm; 88% of patients same; fallback; one point with C51 (median fold 2.8 mm, typical patient 3.0 mm, 10th-90th percentile 1.5 mm to 3.7 mm, wide)*	27% to 29% (typical patient with a fold)	fold 2.84 mm (IQR 2.45 to 3.28); past it log-odds 0.13, OR 1.14; range across patients 0.00 to 0.33	148, 23	train; test disagrees (AUC gain -0.001 [-0.009, 0.006], n=61)	yes	slightly; on the training data only
C52	subgroup: who this part acts on: 117 of 242 patients	86% to 90% (typical patient this part acts on (risk without the part, then with it))	share 0.48; log-odds 0.31, OR 1.36; range across patients 0.08 to 0.55	117, 125	train; test disagrees (AUC gain -0.001 [-0.016, 0.011], n=61)	yes	slightly; on the training data only
C53	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	242	train	yes	
C54	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	
C55	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	

Not supported by the model.

- C49: interaction claim about serum cholesterol and ST depression during exercise (compared with rest) in part ann_32: not supported by the model (the group difference is below 0.1 log-odds).

7.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (8):** `-10` (fbs:0), `-12` (restecg:0), `-14` (restecg:2), `-16` (exang:0), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4), `-6` (cp:3).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

7.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 7.6.1)

- ▷ A2: age_hr_reserve_gate (leaf, section 7.6.2)
- ▷ A3: st_slope_chol_gate (leaf, section 7.6.3)
- ▷ A4: oldpeak_hinges (leaf, section 7.6.4)

7.4 Network figures

7.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-26` into `-26_a`, `-26_b`, `-26_c`.
- #27: split `-15` into `-15_a`, `-15_b`.
- #28: split `-9` into `-9_a`, `-9_b`.
- #29: split `-20` into `-20_a`, `-20_b`.
- #30: split `-18` into `-18_a`, `-18_b`.

7.6 Annotations

7.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 7.6.2), A3 (section 7.6.3), A4 (section 7.6.4).

- **Entry nodes:** `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9_a` (chol_a), `-11` (fbs:1), `-13` (restecg:1), `-15_a` (thalach_a), `-17` (exang:1), `-19` (slope:1), `-20_a` (slope:2_a), `-22` (ca), `-25` (thal:6), `-26_a` (thal:7_a), `-6` (cp:3), `-10` (fbs:0), `-12` (restecg:0), `-14` (restecg:2), `-16` (exang:0), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4), `-1` (age), `-15_b` (thalach_b), `-26_b` (thal:7_b), `-26_c` (thal:7_c), `-9_b` (chol_b), `-18_a` (oldpeak_a), `-20_b` (slope:2_b), `-18_b` (oldpeak_b).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-2`→`0` (-0.742), `-3`→`0` (0.447), `-4`→`0` (-0.18), `-5`→`0` (0.397), `-7`→`0` (0.986), `-8`→`0` (0.465), `-9_a`→`0` (0.613), `-11`→`0` (-0.308), `-13`→`0` (0.0662), `-15_a`→`0` (-0.0791), `-17`→`0` (0.278), `-19`→`0` (-0.453), `-20_a`→`0` (0.495), `-22`→`0` (1.65), `-25`→`0` (-0.278), `-26_a`→`0` (0.279), `1718`→`0` (-0.605), `1476`→`0` (0.248), `1946`→`0` (0.378), `3284`→`0` (0.419).
- **With its descendants:** 39 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 38 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = -0.227 + LINEAR CORE - 0.605*age_hr_reserve_gate + 0.248*st_slope_chol_gate + 0.378*oldpeak_hinges[1946] + 0.419*oldpeak_hinges[3284]. LINEAR CORE (16 direct input->output edges, z-scored inputs): ca +1.653, cp:4 +0.986, sex:0 -0.742, chol +0.613, slope:2 +0.495, trestbps +0.465, slope:1 -0.453, sex:1 +0.447, cp:2 +0.397, fbs:1 -0.308, thal:7 +0.279, exang:1 +0.278, thal:6 -0.278, cp:1 -0.180, thalach -0.079, restecg:1 +0.066. Eight inputs are unused (cp:3, fbs:0, restecg:0, restecg:2, exang:0, slope:3, thal:3, thal:4; zero-effect

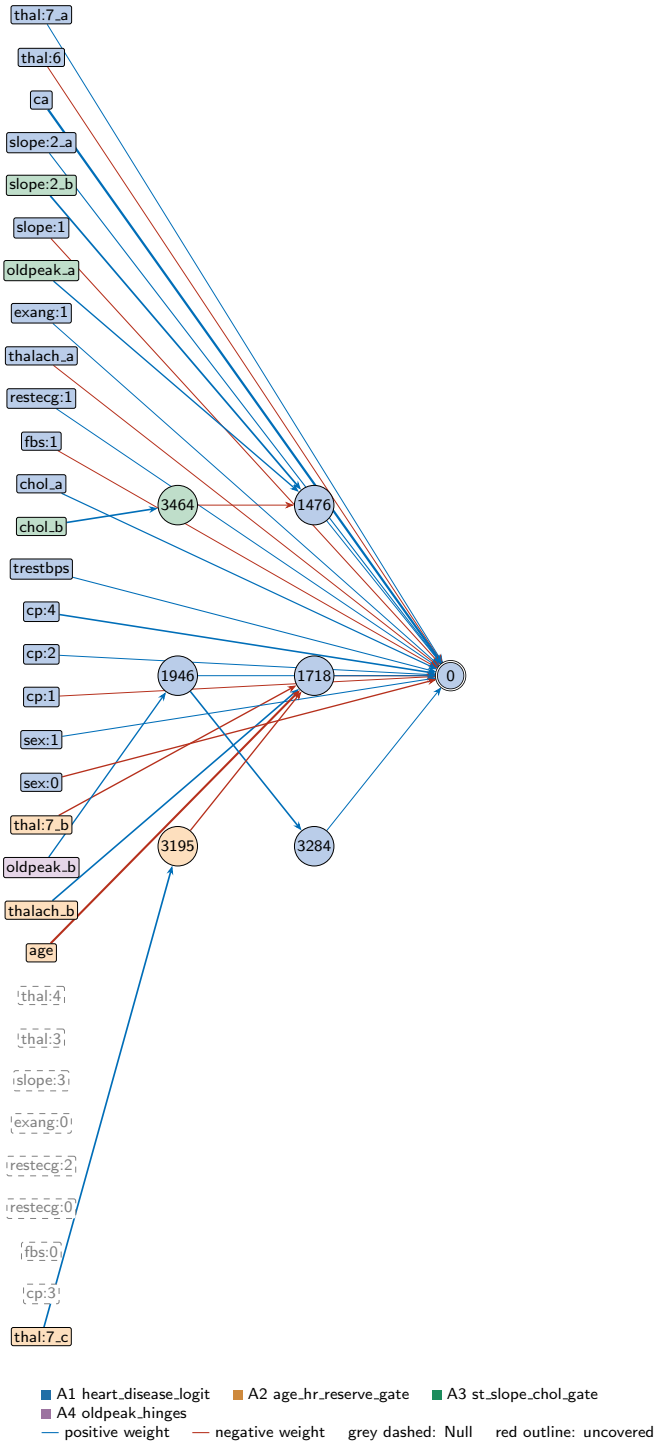


Figure 21: heart-s2-c1: full view of the explained model state.

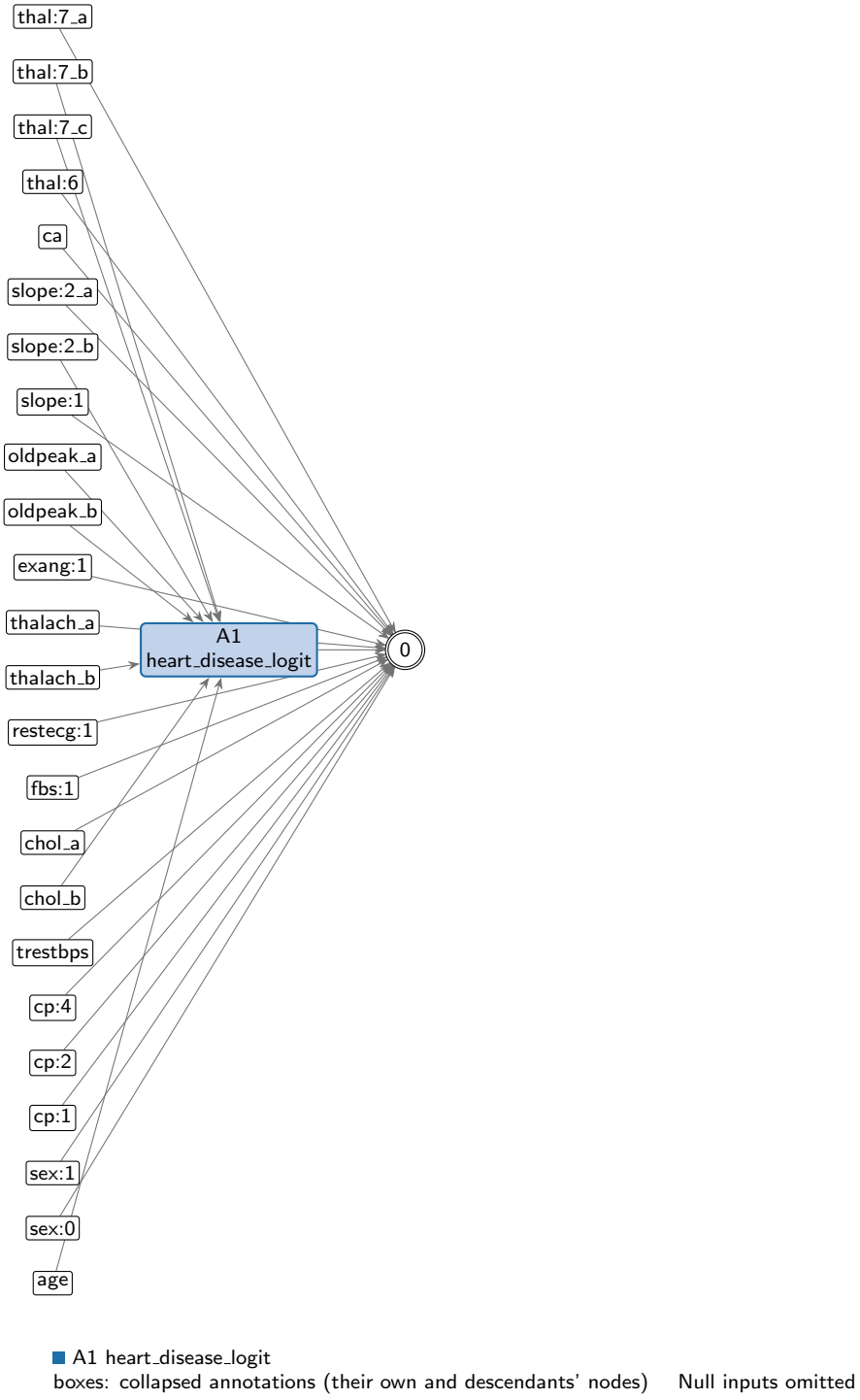


Figure 22: heart-s2-c1: collapsed view at hierarchy level 0 (fully recomposed).

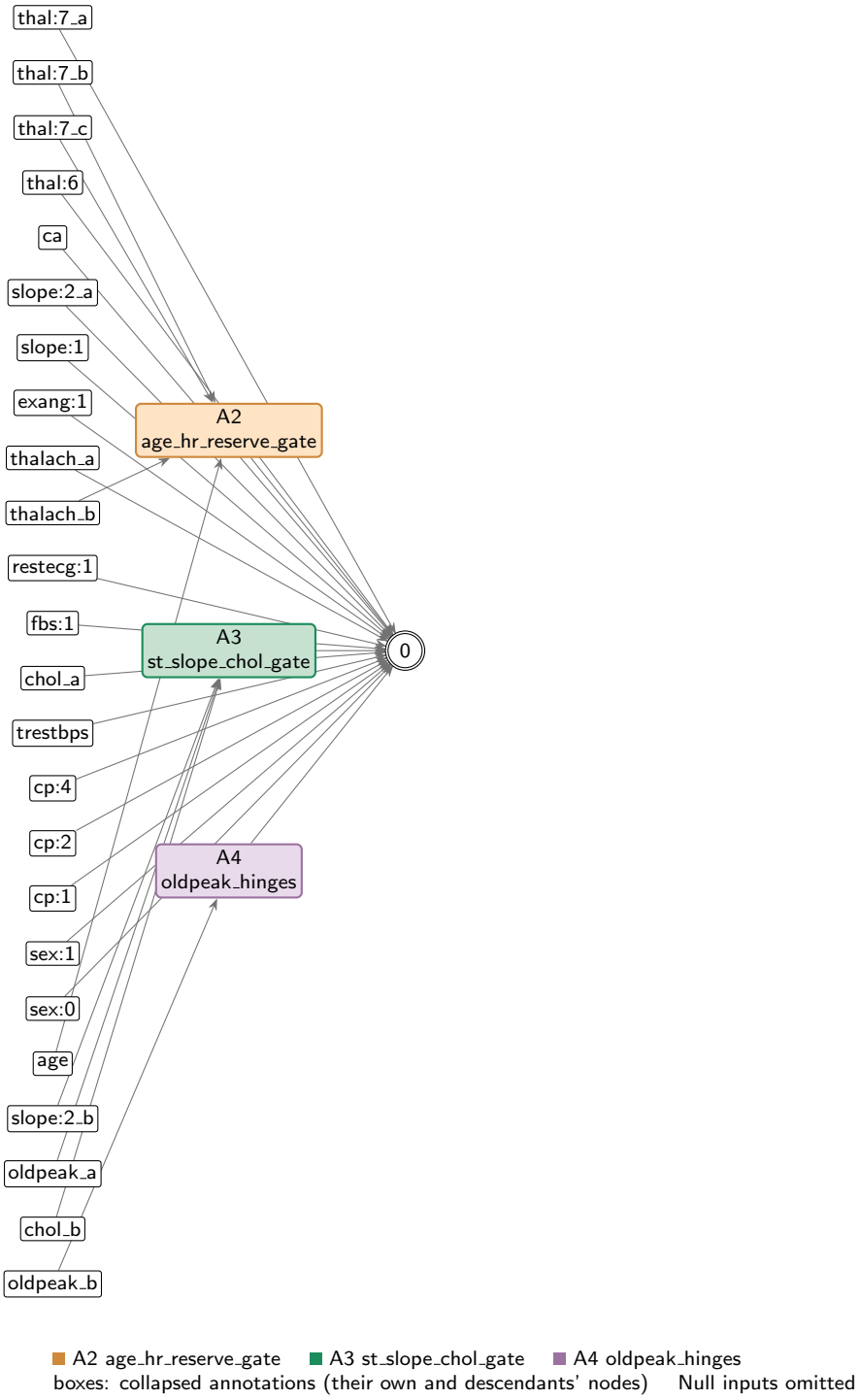


Figure 23: heart-s2-c1: collapsed view at hierarchy level 1.

entries of this root). Age and oldpeak have no direct edge; they act only through the children. COMPOSITION: purely additive in the logit. The three children add nonlinear corrections to the linear core: (1) age_hr_reserve_gate, a protective ReLU on when max heart rate is high for age and there is no reversible thallium defect; (2) st_slope_chol_gate, a flat-ST-slope indicator scaled up by oldpeak and down by cholesterol; (3) oldpeak_hinges, a convex two-kink ST-depression dose-response. The children share no hidden nodes; they are coupled only through shared inputs (thal:7, thalach, chol and slope:2 also have direct edges, oldpeak feeds (2) and (3)). EVIDENCE (train, n=242): full model AUC 0.9432, acc 0.880 (compute_performance). Linear core alone AUC 0.930; logit SD 3.24 (core) vs 0.95 (all nonlinear readouts). Zero-ablating one child at a time: hr gate 0.937, flat-slope gate 0.942, oldpeak hinges 0.942. Replacing all hidden readouts with one least-squares linear fit in their six inputs gives 0.937; an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.152, cp:4 0.111, sex:0 0.071, slope:2 0.066, slope:1 0.049, thal:7 0.048, chol 0.044. Interpretation: close to logistic regression plus a protective heart-rate-reserve-for-age gate (blocked by a reversible defect) and two small oldpeak/flat-slope corrections. Counter-directional linear signs (cp:2 +0.40, fbs:1 -0.31, negative chol inside the flat-slope gate) are reported, not explained. Retrained copy of the 10-seed heart-s2 genome (same NEAT key 9083): same mechanisms, weights within about 0.03.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= -0.227 + 0.397x_{cp:2} + 0.278x_{exang:1} + 0.613x_{chol_a} \\
&\quad + 0.419\,\mathrm{oldpeak_hinges_1}\big(x_{oldpeak_b}\big) + 0.448x_{sex:1} \\
&\quad + 0.0662x_{restecg:1} + 0.986x_{cp:4} + 0.279x_{thal:7_a} \\
&\quad + 0.378\,\mathrm{oldpeak_hinges_0}\big(x_{oldpeak_b}\big) + 1.65x_{ca} \\
&\quad + 0.465x_{trestbps} + 0.495x_{slope:2_a} \\
&\quad + 0.248\,\mathrm{st_slope_chol_gate}\big(x_{chol_b}, x_{oldpeak_a}, \\
&\quad x_{slope:2_b}\big) - 0.278x_{thal:6} - 0.308x_{fbs:1} - 0.18x_{cp:1} - 0.453x_{slope:1} \\
&\quad - 0.605\,\mathrm{age_hr_reserve_gate}\big(x_{age}, x_{thalach_b}, x_{thal:7_b}, \\
&\quad x_{thal:7_c}\big) - 0.0791x_{thalach_a} - 0.742x_{sex:0}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= -0.227 + 0.397x_{cp:2} + 0.278x_{exang:1} + 0.613x_{chol_a} \\
&\quad + 0.419\,\mathrm{relu}\big(-0.582 + 1.0\,\mathrm{relu}\big(0.177 \\
&\quad + 0.757x_{oldpeak_b}\big)\big) + 0.448x_{sex:1} + 0.0662x_{restecg:1} + 0.986x_{cp:4} \\
&\quad + 0.279x_{thal:7_a} + 0.378\,\mathrm{relu}\big(0.177 + 0.757x_{oldpeak_b}\big) \\
&\quad + 1.65x_{ca} + 0.465x_{trestbps} + 0.495x_{slope:2_a} \\
&\quad + 0.248\,\mathrm{relu}\big(0.255 + 0.776x_{oldpeak_a} + 1.15x_{slope:2_b} \\
&\quad - 0.38\,\mathrm{relu}\big(1.3 + 1.01x_{chol_b}\big)\big) - 0.278x_{thal:6} \\
&\quad - 0.308x_{fbs:1} - 0.18x_{cp:1} - 0.453x_{slope:1} - 0.605\,\mathrm{relu}\big(-0.809 \\
&\quad + 1.02x_{thalach_b} - 1.5x_{age} - 0.772x_{thal:7_b} \\
&\quad - 0.73\,\mathrm{relu}\big(0.979 + 1.08x_{thal:7_c}\big)\big) - 0.0791x_{thalach_a} \\
&\quad - 0.742x_{sex:0}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00201, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=true); output is sigmoid.

- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, ablations, linear replacements and SHAP.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

7.6.2 A2: age_hr_reserve_gate

Kind: Leaf annotation (depth 1); parent A1 (section 7.6.1).

- **Entry nodes:** `-1` (age), `-15_b` (thalach_b), `-26_b` (thal:7_b), `-26_c` (thal:7_c).
- **Exit nodes:** `1718`.
- **Internal nodes:** `3195`.
- **Edges (weight):** `-1`→`1718` (-1.5), `-15_b`→`1718` (1.02), `-26_b`→`1718` (-0.772), `-26_c`→`3195` (1.08), `3195`→`1718` (-0.73).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. Heart-rate-reserve gate (protective). 1718 = ReLU(-0.809 - 1.502*age + 1.023*thalach - 0.772*thal:7 - 0.730*R3195), relay R3195 = ReLU(0.979 + 1.081*thal:7); read out -0.605 into the logit. Hypothesis: R3195 is an always-on affine relay of thal:7, so the gate is a single ReLU, 1718 = ReLU(-1.524 - 1.502*age + 1.023*thalach - 1.561*thal:7), that opens when the achieved max heart rate is high for the patient's age and there is no reversible thallium defect, and then lowers the predicted risk. EVIDENCE (train, n=242): R3195 is active for 100% (pre-activation >= 0.148), so the reduction is exact on the data. The gate is active for 33% (51% without a reversible defect, 3% with one). Disease rate 11% when active vs 60% when inactive. Mean |sensitivity| age 0.50 > thalach 0.34 > thal:7 copies 0.26/0.26. Zero-ablating its readout lowers train AUC 0.943 -> 0.937, the largest single nonlinear effect; a linear refit in its inputs recovers only to 0.939. Supported. Same mechanism and nearly the same weights as the 10-seed heart-s2 record (its generation-38 parent), re-derived here from this genome's own weights.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.809 + 1.02x_{thalach_b} - 1.5x_{age} \\ - 0.772x_{thal:7_b} - 0.73\,\mathrm{relu}\big(0.979 + 1.08x_{thal:7_c}\big)\big)$$

where y_0 is node `1718`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.0114, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

7.6.3 A3: st_slope_chol_gate

Kind: Leaf annotation (depth 1); parent A1 (section 7.6.1).

- **Entry nodes:** `-9_b` (chol_b), `-18_a` (oldpeak_a), `-20_b` (slope:2_b).
- **Exit nodes:** `1476`.

- **Internal nodes:** 3464.
- **Edges (weight):** -9_b→3464 (1.01), 3464→1476 (-0.38), -18_a→1476 (0.776), -20_b→1476 (1.15).
- **Covers** (coverage, own and as part of the hierarchy): 4 candidate nodes.

Hypothesis. Flat-ST-slope gate (raises risk). 1476 = ReLU(0.255 + 1.150*slope:2 + 0.776*oldpeak - 0.380*C3464), C3464 = ReLU(1.302 + 1.011*chol); read out +0.248. Hypothesis: a one-sided detector that is on for patients with a flat peak-exercise ST segment, its height raised by ST depression and lowered by cholesterol through an almost-always-active chol relay. EVIDENCE (train, n=242): C3464 is active for 93% (off only for z-chol < -1.29), so 1476 is nearly ReLU(-0.239 + 0.776*oldpeak + 1.150*slope:2 - 0.384*chol). Active for 98% of the 110 flat-slope patients and 7% of the others; corr with oldpeak 0.69. Disease rate 67% when active vs 23% when inactive. Mean |sensitivity| slope:2 0.56 > oldpeak 0.38 > chol 0.17. Zero-ablating its readout lowers train AUC 0.943 -> 0.942 (linear refit 0.943). Supported but small. The negative chol term partly offsets chol's direct +0.613 edge for flat-slope patients; no clinical reading is offered.

Closed form:

$$y_0 = \mathrm{relu}\big(0.255 + 0.776 x_{oldpeak_a} + 1.15 x_{slope:2_b} - 0.38\,\mathrm{relu}\big(1.3 + 1.01 x_{chol_b}\big)\big)$$

where y_0 is node 1476; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00231, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, empirical_viz (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

7.6.4 A4: oldpeak_hinges

Kind: Leaf annotation (depth 1); parent A1 (section 7.6.1).

- **Entry nodes:** -18_b (oldpeak_b).
- **Exit nodes:** 1946, 3284.
- **Internal nodes:** none.
- **Edges (weight):** -18_b→1946 (0.757), 1946→3284 (1).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Oldpeak hinge pair (raises risk). 1946 = ReLU(0.177 + 0.757*oldpeak), 3284 = ReLU(-0.582 + 1.004*1946); read out +0.378 and +0.419. Hypothesis: a convex piecewise-linear dose-response in ST depression alone: no effect for the lowest oldpeak values, a moderate rise from z-oldpeak -0.23, and a steeper rise (slope about doubles) above z-oldpeak 0.53. EVIDENCE (train, n=242): 1946 is active for 50% (disease rate 64%), 3284 for 24% (disease rate 79%). The readout 0.378*1946 + 0.419*3284 has slope 0, 0.286 and 0.605 per z-unit in the three pieces. Removing both readouts lowers train AUC 0.943 -> 0.942; a straight-line refit in oldpeak gives 0.9425. Supported: a small convex ST-depression term. Oldpeak has no direct edge; it acts only here and in the flat-slope gate.

Closed form:

$$y_0 = \mathrm{relu}\big(0.177 + 0.757 x_{oldpeak_b}\big)$$
$$y_1 = \mathrm{relu}\big(-0.582 + 1.0\,\mathrm{relu}\big(0.177 + 0.757 x_{oldpeak_b}\big)\big)$$

where y_0 is node `1946`; y_1 is node `3284`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.0131, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

7.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

8. heart-s2-c2: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.005339	0.005339	0.005339	agree
Numeric-function max abs error	n/a	4.44e-16	4.44e-16	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	2 / 1	n/a	2 / 1	agree
Operation counts	2 splits, 0 identity	annotate 3, rename node 26, split node 2	annotate 3, rename node 26, split node 2	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `heart_rate_nonanginal_gate` (0.00534). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `heart_rate_nonanginal_gate`: *"The gate opens once max heart rate passes a threshold set by the other two inputs: about -2.3 SD with non-anginal pain (open for 66 of 68), about +0.78 SD for other pain types without a fixed defect (38 of 159), and never with a f..."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `restecg2_relay`: *"exactly affine in the binary input on the data."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 2 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): No v3 tool bugs met. compute_shap on all 242 train rows took about a minute; unconnected inputs get mean |SHAP| about 0.0009 (KernelSHAP noise, exact effect zero). The agent sandbox refuses compound shell commands with variables, so tools were called through a small Python subprocess wrapper in scratch (same species_leads/xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s2-c2-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 32%; it adds up most of its inputs independently, and two parts add to that sum: one acts on 104 of 242 training patients through chest pain type, maximum heart rate and the thallium scan, and one through the resting ECG barely changes the predictions. [C29, C37, C41] These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 14%; being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 20%; and a higher maximum heart rate lowers it steadily, from about 37% at 118 to about 28% at 175 beats per minute (chest pain type and maximum heart rate also act through the heart rate gate, below). [C04, C12, C16, C08, C18, C20] Each extra vessel coloured by fluoroscopy adds about the same, from about 32% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted

risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 81%; and ST depression on exercise, cholesterol and resting blood pressure each raise it steadily, from about 25% at 0 mm to about 55% at 2.8 mm, from about 22% at 193 mg/dL to about 50% at 308 mg/dL, and from about 21% at 110 mm Hg to about 48% at 155 mm Hg. [C01, C02, C03, C07, C09] Through this sum alone a reversible defect on the thallium scan raises it to about 57% (the thallium scan also acts through the heart rate gate, below); older age raises it steadily but only from about 31% at 42 to about 32% at 66 years; and the model ignores chest pain brought on by exercise. [C10, C21, C28] Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54%, and through this sum alone a fixed defect on the thallium scan lowers it to about 12%. [C13, C14] This part reads chest pain type, maximum heart rate and the thallium scan, and acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Across the whole model, non-anginal chest pain lowers the predicted risk, and even more so without a fixed defect on the thallium scan: from about 32% to about 3% then, against about 12% to about 2% with one (the model has little data on this, n = 15). [C33] Across the whole model, a higher maximum heart rate lowers the risk, and even more so with non-anginal chest pain, where it falls steadily from about 7% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower; for other chest pain types it falls from about 37% to about 26%. [C35] Through this part alone, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] This part reads only the resting ECG: signs of left ventricular hypertrophy take a typical patient's predicted risk from about 32% to about 31%, and switching the part off changes any prediction by at most about half a percentage point. [C39, C41]

Worth checking: Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 32% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C05, C06, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 32% to about 20%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 32% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores whether exercise brings on chest pain. [C28]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	32% to 96%	log-odds 3.99, OR 54.0; range across patients 3.99 to 3.99	242	train; test unclear (AUC gain 0.037 [-0.038, 0.119], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 81%	log-odds 2.21, OR 9.14; range across patients 2.21 to 2.21	115, 110	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	25% to 55%	log-odds 1.31, OR 3.72; range across patients 1.31 to 1.31	242	train; test unclear (AUC gain 0.011 [-0.005, 0.034], n=61)	yes	on the training data only
C04	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05; range across patients -3.03 to -3.03	108, 22	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	22% to 50%	log-odds 1.28, OR 3.58; range across patients 1.28 to 1.28	242	train; test disagrees (AUC gain -0.011 [-0.039, 0.012], n=61)	yes	on the training data only
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	32% to 4%	log-odds -2.34, OR 0.10; range across patients -2.34 to -2.34	159, 83	train; test unclear (AUC gain 0.027 [-0.01, 0.074], n=61)	yes	on the training data only
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.22, OR 3.38; range across patients 1.22 to 1.22	242	train; test unclear (AUC gain 0.009 [-0.013, 0.034], n=61)	yes	on the training data only
C10	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	log-odds 1.04, OR 2.82; range across patients 1.04 to 1.04	137, 90	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	on the training data only
C12	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 7%	log-odds -1.75, OR 0.17; range across patients -1.75 to -1.75	108, 70	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 54%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	115, 17 (small n)	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	the model has little data on this; on the training data only
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed*	32% to 12%	log-odds -1.24, OR 0.29; range across patients -1.24 to -1.24	137, 15 (small n)	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	the model has little data on this; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	32% to 20%	log-odds -0.65, OR 0.52; range across patients -0.65 to -0.65	209, 33	train; test unclear (AUC gain 0.000 [-0.011, 0.013], n=61)	yes	on the training data only
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	37% to 28%	log-odds -0.39, OR 0.68; range across patients -0.39 to -0.39	242	train; test unclear (AUC gain 0.006 [0.000, 0.020], n=61)	yes	on the training data only
C21	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	31% to 32%	log-odds 0.07, OR 1.08; range across patients 0.07 to 0.07	242	train; test disagrees (AUC gain -0.001 [-0.005, 0.000], n=61)	yes	slightly; on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of heart_rate_nonanginal_gate, restecg2_relay and inputs age, sex, cp, trestbps, chol, fbs, thalach, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 168 beats per minute; 70% of patients same; mixed (but steady for patients with non-anginal chest pain); one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	32% to 29%	log-odds -0.34, OR 0.71; range across patients -0.93 to -0.12	242	train; test unclear (AUC gain 0.008 [0.000, 0.022], n=61)	yes	slightly; on the training data only
C33	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	a fixed defect on the thallium scan: 12% to 2%; thallium stress test result other than a fixed defect on the thallium scan: 32% to 3%	log-odds difference 0.79, OR 2.20	15, 227 (small n)	train; test agrees (AUC gain 0.001 [0.000, 0.006], n=61); group difference on test 0.73 [0.60, 0.85]	yes	the model has little data on this
C35	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 97% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 7% to 2%; chest pain type other than non-anginal chest pain: 37% to 26%	log-odds difference -0.80, OR 0.45	70, 172	train; test agrees (AUC gain 0.008 [-0.012, 0.030], n=61); group difference on test -0.82 [-0.83, -0.81]	yes	clearly
C36	threshold: maximum heart rate: fold at about 168 beats per minute (starts) for patients with no chest pain symptoms (asymptomatic) *shape: hinge at 168 beats per minute; 100% of patients same; verified; risk saturates near 100%; one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	99% to 99% (typical patient with a fold (patients with no chest pain symptoms (asymp-tomatic)))	fold 168 beats per minute (IQR 168 to 168); past it log-odds -0.23, OR 0.79; range across patients -0.23 to -0.23	99, 14 (small n)	train; test unclear (AUC gain 0.008 [0.000, 0.022], n=61)	yes	slightly; the model has little data on this; on the training data only
C37	subgroup: who this part acts on: 104 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.43; log-odds -0.67, OR 0.51; range across patients -1.22 to -0.05	104, 138	train; test unclear (AUC gain 0.018 [0.000, 0.044], n=61)	yes	slightly; on the training data only
C39	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the	32% to 31%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	123, 119	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s2-c2 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s2-c2.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s2-c2 --verified --minutes N --changes "..." --notes "..."`

8 Explanation certificate: heart-s2-c2

This section is the explanation certificate of the heart-s2-c2 network. It is rendered from the canonical JSON record `heart-s2-c2.json` (schema 1.4.0, SHA-256 `1276e9a13f62dbc0...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 8 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s2, seed 2
Genome	`d4bbc960-816a-4781-8979-fdd4165debf2` (NEAT key 9685, generation 39)
Explanation	`3531d96c-aa00-4141-863a-2c85155afe8f` (31 operations)
Train AUC / accuracy	0.9403 / 0.8967 (n = 242)
Test AUC / accuracy	0.9009 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	31 nodes (26 inputs, 4 hidden, 1 output), 24 enabled edges
Explained model state	33 nodes, 24 enabled edges
Refactoring operations	split_node 2
Function preservation	max $\lvert\Delta$ output$\rvert = 2.22 \times 10^{-16}$ on the training split
Annotations	3 (2 leaf, 1 composition)
Structural coverage	1 (32 of 32 candidate nodes; 8 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`1276e9a13f62dbc0...`

Table 8: Explanation certificate summary: heart-s2-c2.

Drafter's summary. Mostly logistic regression: 16 direct edges (logit SD 3.66; train AUC 0.934 alone, L2 logistic regression 0.9355) plus one consequential nonlinear part to 0.940. The protective gate 3842 = ReLU(1.02 ReLU(0.688 thalach - 0.374 thal:6 + 2.15 [non-anginal pain] - 0.347) - 0.295), read out -0.519, opens for nearly all non-anginal-pain patients, for others only above about +0.8 SD max heart rate, and never with a fixed thallium defect; once open a higher max heart rate

lowers risk 3.4x as steeply. A restecg:2 relay is negligible. Unlike the generation-38 heart-s2 record (age/thalach/thal:7 gate, flat-slope gate, oldpeak hinges), age and oldpeak here act only linearly and the gate is keyed on chest-pain type and fixed defect instead.

What resisted explanation. Nothing resisted coverage; only input fan-out needed splitting (thalach, thal:6). Relay 3607 is a step on the binary restecg:2 input, so exactly affine on the data and negligible (read out -0.021). Eight inputs are unconnected (thal:3, thal:4, slope:3, exang:0, exang:1, restecg:0, restecg:1, fbs:0) and listed as zero-effect root entries; exercise angina is ignored entirely. Counter-directional signs reported, not explained: cp:2 +0.26 and fbs:1 -0.22 in the linear core, and thal:6 (fixed defect) -0.30 directly plus closing the protective gate.

8.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 32%; it adds up most of its inputs independently, and two parts add to that sum: one acts on 104 of 242 training patients through chest pain type, maximum heart rate and the thallium scan, and one through the resting ECG barely changes the predictions. [C29, C37, C41] These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 14%; being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 20%; and a higher maximum heart rate lowers it steadily, from about 37% at 118 to about 28% at 175 beats per minute (chest pain type and maximum heart rate also act through the heart rate gate, below). [C04, C12, C16, C08, C18, C20] Each extra vessel coloured by fluoroscopy adds about the same, from about 32% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 81%; and ST depression on exercise, cholesterol and resting blood pressure each raise it steadily, from about 25% at 0 mm to about 55% at 2.8 mm, from about 22% at 193 mg/dL to about 50% at 308 mg/dL, and from about 21% at 110 mm Hg to about 48% at 155 mm Hg. [C01, C02, C03, C07, C09] Through this sum alone a reversible defect on the thallium scan raises it to about 57% (the thallium scan also acts through the heart rate gate, below); older age raises it steadily but only from about 31% at 42 to about 32% at 66 years; and the model ignores chest pain brought on by exercise. [C10, C21, C28] Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54%, and through this sum alone a fixed defect on the thallium scan lowers it to about 12%. [C13, C14] This part reads chest pain type, maximum heart rate and the thallium scan, and acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Across the whole model, non-anginal chest pain lowers the predicted risk, and even more so without a fixed defect on the thallium scan: from about 32% to about 3% then, against about 12% to about 2% with one (the model has little data on this, n = 15). [C33] Across the whole model, a higher maximum heart rate lowers the risk, and even more so with non-anginal chest pain, where it falls steadily from about 7% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower; for other chest pain types it falls from about 37% to about 26%. [C35] Through this part alone, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] This part reads only the resting ECG: signs of left ventricular hypertrophy

take a typical patient's predicted risk from about 32% to about 31%, and switching the part off changes any prediction by at most about half a percentage point. [C39, C41]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, and two parts add to that sum: a heart rate gate that lowers the risk, and a small step for the resting ECG. [C29, C01, C02, C05, C08] *Evidence.* It is made up of the inputs that feed the score directly and two parts, the heart rate gate and the resting ECG step (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Chest pain brought on by exercise feeds nothing, so the model ignores it. [C28] Across the whole model, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 32% to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%. [C05, C06, C17] Across the whole model, a higher maximum heart rate lowers the risk, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% at 118 to about 26% at 175 beats per minute, against about 28% through the sum alone. [C19, C20] Across the whole model, a reversible defect on the thallium scan raises it to about 57%, and a fixed defect lowers it to about 12% (the model has little data on this, n = 15). [C11, C15]

Heart rate gate. This part lowers the predicted risk for patients with non-anginal chest pain or a high maximum heart rate, unless the thallium scan shows a fixed defect (the model has little data on this, n = 15). [C37, C33] *Evidence.* It acts on 104 of 242 training patients, 24 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C37] Through it, non-anginal chest pain takes a typical patient's predicted risk from about 32% to about 16%, and for the typical patient it acts on a fixed defect on the thallium scan raises it from about 2% to about 4% (the model has little data on this, n = 15). [C30, C32] Through it, for most patients a higher maximum heart rate does nothing until about 168 beats per minute, then lowers the risk steadily, but for patients with non-anginal chest pain it lowers it steadily all along: for a typical patient from about 32% at 118 to about 29% at 175 beats per minute (the model has little data on this, n = 14). [C31, C36] Across the whole model, a higher maximum heart rate lowers the risk more without a fixed defect on the thallium scan, for most patients slowly at first, then faster, but for patients with non-anginal chest pain steadily: from about 37% to about 26%, against about 14% to about 10% with one (the model has little data on this, n = 15). [C34] Its formula reproduces the part exactly on all 242 training patients. [C38]

Resting ECG step. This part barely changes the predictions: signs of left ventricular hypertrophy on the resting ECG take a typical patient's predicted risk from about 32% to about 31%. [C39, C41] *Evidence.* Switching it off changes any training patient's predicted risk by at most about half a percentage point. [C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 32% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C05, C06, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 32% to about 20%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 32% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores whether exercise brings on chest pain. [C28]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	32% to 96%	log-odds 3.99, OR 54.0; range across patients 3.99 to 3.99	242	train; test unclear (AUC gain 0.037 [-0.038, 0.119], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 81%	log-odds 2.21, OR 9.14; range across patients 2.21 to 2.21	115, 110	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	25% to 55%	log-odds 1.31, OR 3.72; range across patients 1.31 to 1.31	242	train; test unclear (AUC gain 0.011 [-0.005, 0.034], n=61)	yes	on the training data only
C04	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05; range across patients -3.03 to -3.03	108, 22	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05; range across patients -3.03 to -3.03	108, 22	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C06	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 3%	log-odds -2.53, OR 0.08; range across patients -2.88 to -1.96	108, 70	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	22% to 50%	log-odds 1.28, OR 3.58; range across patients 1.28 to 1.28	242	train; test disagrees (AUC gain -0.011 [-0.039, 0.012], n=61)	yes	on the training data only
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	32% to 4%	log-odds -2.34, OR 0.10; range across patients -2.34 to -2.34	159, 83	train; test unclear (AUC gain 0.027 [-0.01, 0.074], n=61)	yes	on the training data only
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.22, OR 3.38; range across patients 1.22 to 1.22	242	train; test unclear (AUC gain 0.009 [-0.013, 0.034], n=61)	yes	on the training data only
C10	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	log-odds 1.04, OR 2.82; range across patients 1.04 to 1.04	137, 90	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	on the training data only
C11	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 57%	log-odds 1.04, OR 2.82; range across patients 1.04 to 1.04	137, 90	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	on the training data only
C12	direction: chest pain type:	32% to 7%	log-odds -1.75, OR	108, 70	train; test agrees	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	32% to 54%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	115, 17 (small n)	train; test disagrees (AUC gain -0.004 [-0.044, 0.034], n=61)	yes	the model has little data on this; on the training data only
C14	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	log-odds -1.24, OR 0.29; range across patients -1.24 to -1.24	137, 15 (small n)	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	the model has little data on this; on the training data only
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	32% to 12%	log-odds -1.00, OR 0.37; range across patients -1.24 to -0.42	137, 15 (small n)	train; test unclear (AUC gain 0.018 [-0.012, 0.053], n=61)	yes	the model has little data on this; on the training data only
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	log-odds -1.05, OR 0.35; range across patients -1.05 to -1.05	108, 42	train; test agrees (AUC gain 0.039 [0.009, 0.076], n=61)	yes	clearly
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 14%	log-odds -1.05, OR 0.35; range across patients -1.05 to -1.05	108, 42	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	32% to 20%	log-odds -0.65, OR 0.52; range across patients -0.65 to -0.65	209, 33	train; test unclear (AUC gain 0.000 [-0.011, 0.013], n=61)	yes	on the training data only
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 67% of patients same; mixed (but steady for patients with non-anginal chest pain)*	37% to 26%	log-odds -0.73, OR 0.48; range across patients -1.32 to -0.51	242	train; test agrees (AUC gain 0.014 [0.002, 0.037], n=61)	yes	clearly
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	37% to 28%	log-odds -0.39, OR 0.68; range across patients -0.39 to -0.39	242	train; test unclear (AUC gain 0.006 [0.000, 0.020], n=61)	yes	on the training data only
C21	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	31% to 32%	log-odds 0.07, OR 1.08; range across patients 0.07 to 0.07	242	train; test disagrees (AUC gain -0.001 [-0.005, 0.000], n=61)	yes	slightly; on the training data only
C22	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest*	32% to 31%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	123, 119	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: steady; 100% of patients same; verified*	25% to 55%	log-odds 1.31, OR 3.72	242	train; test unclear (AUC gain 0.011 [-0.005, 0.034], n=61)	yes	on the training data only
C26	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	32% to 2%	log-odds -3.03, OR 0.05	108, 22	train; test agrees (AUC gain 0.057 [0.017, 0.104], n=61)	yes	clearly
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	22% to 50%	log-odds 1.28, OR 3.58	242	train; test disagrees (AUC gain -0.011 [-0.039, 0.012], n=61)	yes	on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of heart_rate_nonanginal_gate, restecg2_relay and inputs age, sex, cp, trestbps, chol, fbs, thalach, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	32% to 16%	log-odds -0.79, OR 0.46; range across patients -1.14 to -0.21	172, 70	train; test unclear (AUC gain 0.008 [-0.012, 0.030], n=61)	yes	on the training data only
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 168 beats per minute; 70% of patients same; mixed (but steady for patients with non-anginal chest pain); one point with C36 (median fold 168 beats per minute, typical patient 166 beats per minute, 10th-90th percentile 168 beats per minute to 168 beats per minute)*	32% to 29%	log-odds -0.34, OR 0.71; range across patients -0.93 to -0.12	242	train; test unclear (AUC gain 0.008 [0.000, 0.022], n=61)	yes	slightly; on the training data only
C32	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	2% to 4% (typical patient this part acts on)	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.82	227, 15 (small n)	train; test unclear (AUC gain 0.001 [0.000, 0.006], n=61)	yes	slightly; the model has little data on this; on the training data only
C33	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	a fixed defect on the thallium scan: 12% to 2%; thallium stress test result other than a fixed defect on the thallium scan: 32% to 3%	log-odds difference 0.79, OR 2.20	15, 227 (small n)	train; test agrees (AUC gain 0.001 [0.000, 0.006], n=61); group difference on test 0.73 [0.60, 0.85]	yes	the model has little data on this
C34	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: accelerating; 70% of patients same; mixed (but steady for patients with non-anginal chest pain)*	a fixed defect on the thallium scan: 14% to 10%; thallium stress test result other than a fixed defect on the thallium scan: 37% to 26%	log-odds difference 0.30, OR 1.35	15, 227 (small n)	train; test agrees (AUC gain 0.001 [0.000, 0.006], n=61); group difference on test 0.34 [0.26, 0.44]	yes	the model has little data on this

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	subgroup: who this part acts on: 104 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.43; log-odds -0.67, OR 0.51; range across patients -1.22 to -0.05	104, 138	train; test unclear (AUC gain 0.018 [0.000, 0.044], n=61)	yes	slightly; on the training data only
C38	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	242	train	yes	
C39	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part) *shape: categorical; highest to lowest: resting ECG other than signs of left ventricular hypertrophy on the resting ECG, signs of left ventricular hypertrophy on the resting ECG*	32% to 31%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	123, 119	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C40	subgroup: who this part acts on: 119 of 242 patients	85% to 84% (typical patient this part acts on (risk without the part, then with it))	share 0.49; log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	119, 123	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	no	slightly; on the training data only
C41	null: this part barely changes predictions (switched off)		max change 0.005 (max |change in predicted risk|)	242	train	yes	
C42	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model.

- C40: subgroup claim about resting ECG in part ann_29: not supported by the model (its effect is negligible).

8.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (8):** `-10` (fbs:0), `-12` (restecg:0), `-13` (restecg:1), `-16` (exang:0), `-17` (exang:1), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

8.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 8.6.1)

▷ A2: heart_rate_nonanginal_gate (leaf, section 8.6.2)

▷ A3: restecg2_relay (leaf, section 8.6.3)

8.4 Network figures

8.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-15` into `-15_a`, `-15_b`.
- #27: split `-25` into `-25_a`, `-25_b`.

8.6 Annotations

8.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 8.6.2), A3 (section 8.6.3).

- **Entry nodes:** `-1` (age), `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9` (chol), `-10` (fbs:0), `-11` (fbs:1), `-12` (restecg:0), `-13` (restecg:1), `-15_a` (thalach_a), `-16` (exang:0), `-17` (exang:1), `-18` (oldpeak), `-19` (slope:1), `-20` (slope:2), `-21` (slope:3), `-22` (ca), `-23` (thal:3), `-24` (thal:4), `-25_a` (thal:6_a), `-26` (thal:7).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`0` (0.0279), `-2`→`0` (-0.771), `-3`→`0` (0.34), `-4`→`0` (-0.369), `-5`→`0` (0.264), `-7`→`0` (0.869), `-8`→`0` (0.492), `-9`→`0` (0.535), `-11`→`0` (-0.222), `-18`→`0` (0.556), `-19`→`0` (-0.469), `-20`→`0` (0.634), `-22`→`0` (1.78), `-26`→`0` (0.501), `3607`→`0` (-0.0209), `3842`→`0` (-0.519), `-15_a`→`0` (-0.153), `-25_a`→`0` (-0.3).
- **With its descendants:** 33 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 32 candidate nodes.

Hypothesis. Root: heart-disease logit at output 0 (sigmoid). Essentially logistic regression: 16 direct edges (ca +1.78, cp:4 +0.87, sex:0 -0.77, slope:2 +0.63, oldpeak +0.56, chol +0.54, thal:7 +0.50, trestbps +0.49, slope:1 -0.47, cp:1 -0.37, sex:1 +0.34, thal:6 -0.30, cp:2 +0.26, fbs:1 -0.22, thalach -0.15, age +0.03) carry logit SD 3.66 and train AUC 0.9343 alone (L2 logistic regression on all 26 inputs: 0.9355). The two children are added to the logit: heart_rate_nonanginal_gate (x -0.519; +0.006 AUC, the only consequential nonlinearity) and restecg2_relay (x -0.021, negligible). Eight inputs are unconnected, zero effect: thal:3, thal:4, slope:3, exang:0, exang:1, restecg:0, restecg:1, fbs:0 (exercise angina is ignored). Evidence (train, n=242): composed closed form; sensitivity and mean |SHAP| both rank ca first (0.16), then cp:4, sex:0, slope:2; full-model train AUC 0.9403, accuracy 0.897.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= 0.324 + 0.501x_{thal:7} + 0.556x_{oldpeak} + 0.34x_{sex:1} + 0.0279x_{age} \\
&\quad + 0.634x_{slope:2} + 0.264x_{cp:2} + 0.535x_{chol} + 0.869x_{cp:4} + 1.78x_{ca} \\
&\quad + 0.492x_{trestbps} - 0.771x_{sex:0} - 0.153x_{thalach_a} \\
&\quad - 0.0209\,\mathrm{restecg2_relay}(x_{restecg:2}) - 0.369x_{cp:1} \\
&\quad - 0.519\,\mathrm{heart_rate_nonanginal_gate}(x_{cp:3}, x_{thalach_b}, \\
&\quad x_{thal:6_b}) - 0.3x_{thal:6_a} - 0.222x_{fbs:1} - 0.469x_{slope:1}
\end{aligned}
$$

Expanded over the children:

$$y_0 = \sigma(z_0)$$

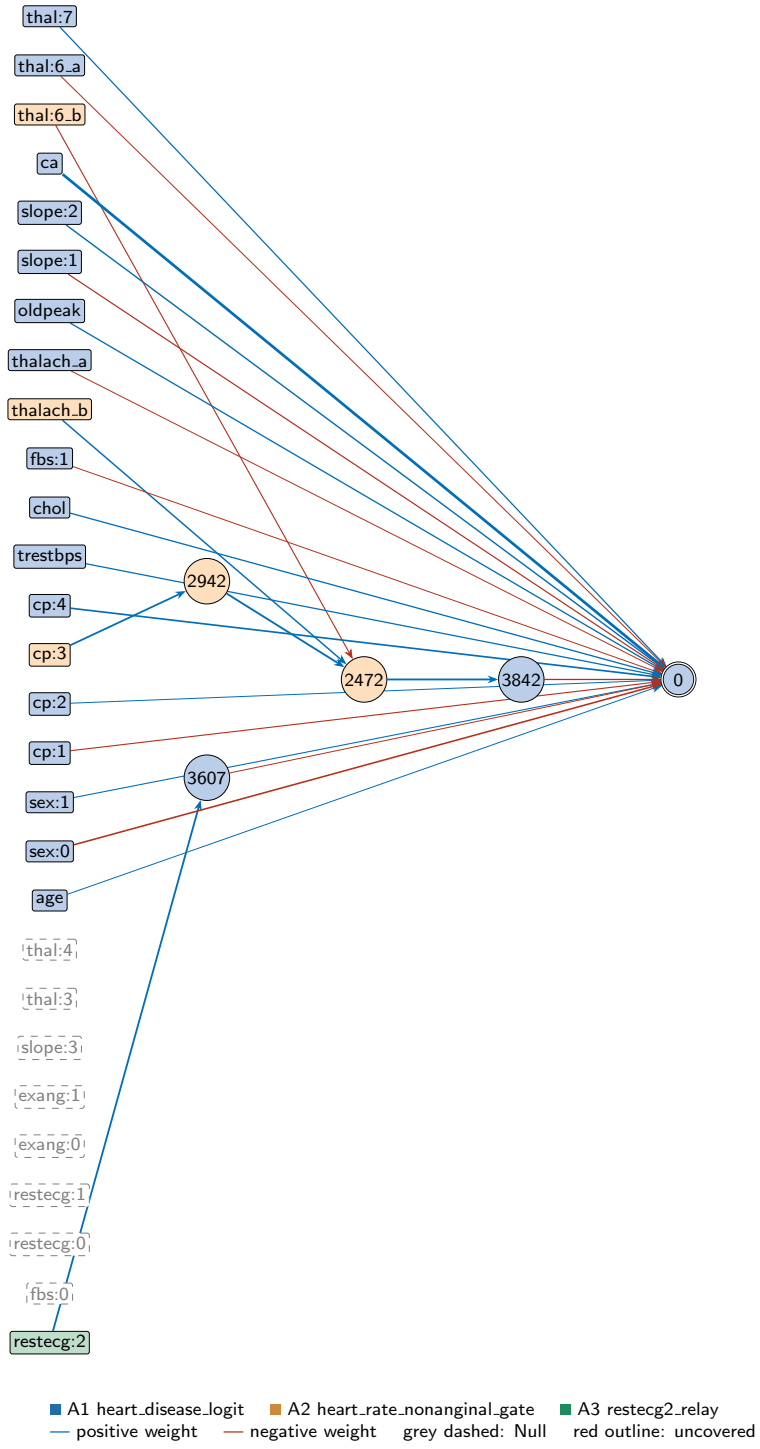


Figure 24: heart-s2-c2: full view of the explained model state.

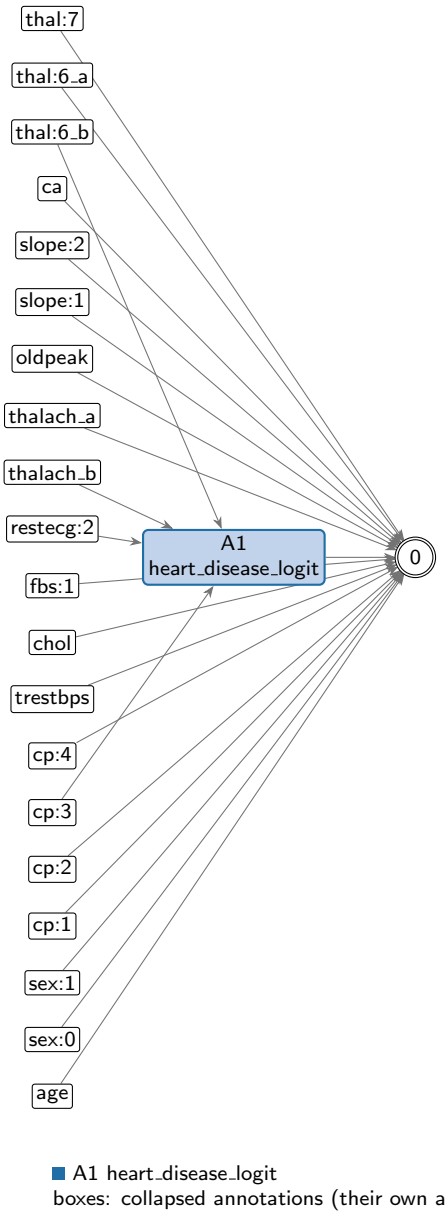


Figure 25: heart-s2-c2: collapsed view at hierarchy level 0 (fully recomposed).

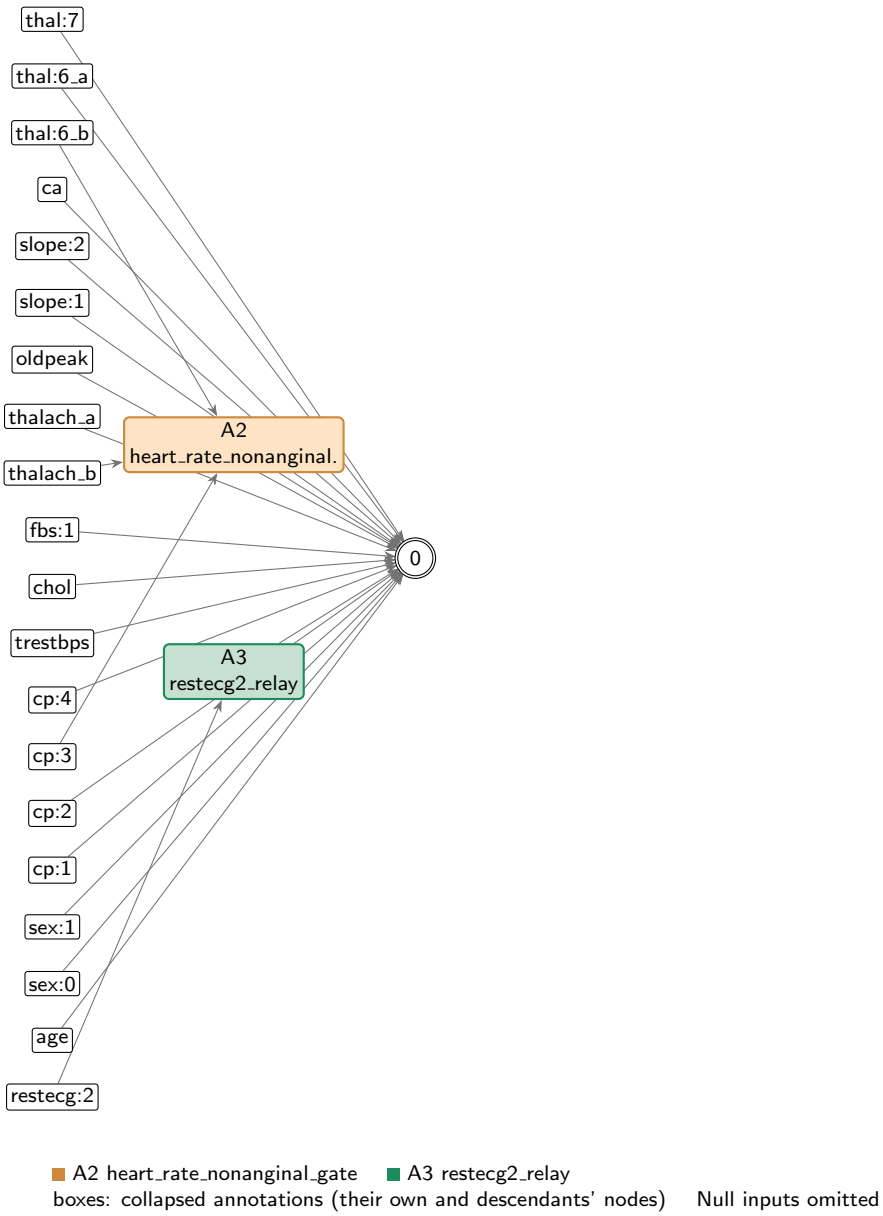


Figure 26: heart-s2-c2: collapsed view at hierarchy level 1.

$$
\begin{aligned}
z_0 = 0.324 &+ 0.501x_{thal:7} + 0.556x_{oldpeak} + 0.34x_{sex:1} + 0.0279x_{age} \\
&+ 0.634x_{slope:2} + 0.264x_{cp:2} + 0.535x_{chol} + 0.869x_{cp:4} + 1.78x_{ca} \\
&+ 0.492x_{trestbps} - 0.771x_{sex:0} - 0.153x_{thalach_a} \\
&- 0.0209\,\mathrm{relu}\big(-0.016 + 0.997x_{restecg:2}\big) - 0.369x_{cp:1} \\
&- 0.519\,\mathrm{relu}\big(-0.295 + 1.02\,\mathrm{relu}\big(-0.347 \\
&+ 0.688x_{thalach_b} + 1.03\,\mathrm{relu}\big(0.571 + 0.968x_{cp:3}\big) \\
&- 0.374x_{thal:6_b}\big)\big) - 0.3x_{thal:6_a} - 0.222x_{fbs:1} - 0.469x_{slope:1}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 8.94×10^{-4}, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=True): direct logistic terms plus the two child functions
- EMPIRICAL, `ablation_study`; data split: train. Train split, full sample; ablations and linear replacements on in-memory copies (no graph edits); SHAP on all 242 train rows

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

8.6.2 A2: heart_rate_nonanginal_gate

Kind: Leaf annotation (depth 1); parent A1 (section 8.6.1).

- **Entry nodes:** `-6` (cp:3), `-15_b` (thalach_b), `-25_b` (thal:6_b).
- **Exit nodes:** `3842`.
- **Internal nodes:** `2942`, `2472`.
- **Edges (weight):** `-6`→`2942` (0.968), `-15_b`→`2472` (0.688), `-25_b`→`2472` (-0.374), `2942`→`2472` (1.03), `2472`→`3842` (1.02).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. Protective heart-rate gate, gated by chest-pain type and thallium. 2942 = ReLU(0.968 cp:3 + 0.571) is a non-anginal-pain indicator (2.09 for the 70 cp:3 patients, 0 otherwise). 3842 = ReLU(1.021*ReLU(0.688 thalach - 0.374 thal:6 + 1.028*2942 - 0.347) - 0.295) is read out -0.519 into the logit. The gate opens once max heart rate passes a threshold set by the other two inputs: about -2.3 SD with non-anginal pain (open for 66 of 68), about +0.78 SD for other pain types without a fixed defect (38 of 159), and never with a fixed thallium defect (0 of 15). Once open, each SD of thalach lowers the logit by 0.36 more (total -0.52 vs -0.15 closed), and non-anginal pain lowers it by about 1.1. Evidence (train, n=242): closed form from get_formula; gate open for 104 patients (disease rate 23% vs 60% when closed); logit contribution SD 0.45; removing it drops train AUC 0.9403 -> 0.9344, a linear refit on cp:3, thalach, thal:6 recovers 0.9378 (R2 0.87), so most of its value is the gating rather than the linear trend. It is the model's only consequential nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.295 + 1.02\,\mathrm{relu}\big(-0.347 \\
&+ 0.688x_{thalach_b} + 1.03\,\mathrm{relu}\big(0.571 + 0.968x_{cp:3}\big) \\
&- 0.374x_{thal:6_b}\big)\big)
\end{aligned}
$$

where y_0 is node `3842`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00534, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

8.6.3 A3: restecg2_relay

Kind: Leaf annotation (depth 1); parent A1 (section 8.6.1).

- **Entry nodes:** `-14` (restecg:2).
- **Exit nodes:** `3607`.
- **Internal nodes:** none.
- **Edges (weight):** `-14`→`3607` (0.997).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. restecg:2 relay: 3607 = ReLU(0.997 restecg:2 - 0.016) is 0.998 for the 119 patients with left ventricular hypertrophy on the resting ECG and 0 for the other 123, i.e. exactly affine in the binary input on the data. Read out -0.021, it lowers the logit by 0.02 for LVH: negligible. Evidence (train, n=242): closed form; entry-exit correlation 1.0; logit SD 0.01; removing it changes train AUC 0.9403 -> 0.9401.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.016 + 0.997 x_{restecg:2}\big)$$

where y_0 is node `3607`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 1.39×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

8.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

9. heart-s2-c3: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.006749	0.006749	0.006749	agree
Numeric-function max abs error	n/a	0	0	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	3 / 1	n/a	3 / 1	agree
Operation counts	4 splits, 0 identity	annotate 4, rename node 26, split node 4	annotate 4, rename node 26, split node 4	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `nonanginal_thalach_gate` (0.00675). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `nonanginal_thalach_gate`: *"...(train, n=242): active for 97% of the 70 non-anginal-pain patients (closed only for z-thalach < -1.97) and 5% of the 172 others (only z-thalach > 1.53)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 4 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): No v3 tool bugs met. compute_shap on all 242 train rows took about a minute; it gives the nine unconnected inputs mean |SHAP| about 0.001 (estimator noise, not an effect). The agent sandbox refused shell loops calling uv with variables, so tools were called through a small Python subprocess wrapper in scratch (same xn.py). Input fan-out splits (-6, -15, -18, -9) taken from the topology and detect_splits.

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s2-c3-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 30%; it adds up most of its inputs independently, and three parts that all read chest pain type add to that sum: one acts on 76 of 242 training patients through maximum heart rate, one on 61 through cholesterol, and one barely changes the predictions. [C29, C34, C40, C50] These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 30% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 13% (chest pain type also acts through the parts below); being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 19%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 36% at 118 to about 27% at 175 beats per minute (it also acts through the parts below). [C05, C14, C16, C10, C18, C19] Each extra vessel coloured by fluoroscopy adds about the same, from about 30% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 80%, higher resting blood pressure raises it steadily from about 20% at 110 mm Hg to about 46% at 155 mm Hg, and a reversible defect on the thallium scan raises it to about 55%. [C01,

C02, C11, C12] Through this sum alone, more ST depression on exercise raises it steadily from about 23% at 0 mm to about 53% at 2.8 mm, and higher cholesterol raises it steadily from about 21% at 193 mg/dL to about 48% at 308 mg/dL; both also act through the parts below. [C03, C07] Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54% and a fixed defect on the thallium scan lowers it to about 12%, while signs of left ventricular hypertrophy on the resting ECG raise it from about 30% to about 33%. [C13, C15, C21] This part reads chest pain type and maximum heart rate, and acts on 76 of 242 training patients, mostly patients with non-anginal chest pain, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C34] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 17%. [C30] Across the whole model, a higher maximum heart rate lowers the risk slowly at first, then faster, and more with non-anginal chest pain: from about 7% at 118 to about 2% at 175 beats per minute then, flattening out near the bottom because the predicted risk cannot go much lower, against about 36% to about 27% for other chest pain types. [C32] This part reads chest pain type and cholesterol, and acts on 61 of 242 training patients, all of them with non-anginal chest pain, 13 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 3% without it and about 2% with it. [C40] Through it, for patients with non-anginal chest pain, higher cholesterol raises the predicted risk steadily up to about 282 mg/dL, where the part switches off, then no further: for the typical patient it acts on, from about 2% at 193 mg/dL to about 3% at 308 mg/dL. [C37, C39] Across the whole model, higher cholesterol raises the risk in an uneven way from about 2% to about 7% with non-anginal chest pain, against about 21% to about 48% for other chest pain types. [C38] This part reads chest pain type, maximum heart rate and ST depression on exercise, and barely changes the predictions (by at most 0.5 percentage points). [C50] It switches on when ST depression on exercise minus about 0.019 × the maximum heart rate exceeds about -1.71, which never happens with non-anginal chest pain; 73 of 242 training patients are on, and 79% of them had heart disease, against 29% of the others. [C52]

Worth checking: Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 30% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types (about 2%), below non-anginal chest pain (about 3%), atypical angina (about 13%) and no chest pain symptoms (about 30%). [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 30% to about 19%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 30% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores age. [C27] Worth checking: with non-anginal chest pain, one part lowers the predicted risk for cholesterol below about 282 mg/dL, so cholesterol moves the risk from about 2% to about 7% for these patients, against about 21% to about 48% for others. [C38, C39]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	30% to 96%	log-odds 3.99, OR 54.1; range across patients 3.99 to 3.99	242	train; test unclear (AUC gain 0.039 [-0.034, 0.120], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 80%	log-odds 2.20, OR 9.04; range across patients 2.20 to 2.20	115, 110	train; test disagrees (AUC gain -0.005 [-0.050, 0.037], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	23% to 53%	log-odds 1.32, OR 3.74; range across patients 1.32 to 1.32	242	train; test unclear (AUC gain 0.011 [-0.01, 0.038], n=61)	yes	on the training data only
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	log-odds -3.11, OR 0.04; range across patients -3.11 to -3.11	108, 22	train; test agrees (AUC gain 0.038 [0.011, 0.072], n=61)	yes	clearly
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	242	train; test disagrees (AUC gain -0.012 [-0.044, 0.013], n=61)	yes	on the training data only
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	30% to 4%	log-odds -2.33, OR 0.10; range across patients -2.33 to -2.33	159, 83	train; test unclear (AUC gain 0.025 [-0.011, 0.071], n=61)	yes	on the training data only
C11	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 46%	log-odds 1.22, OR 3.39; range across patients 1.22 to 1.22	242	train; test unclear (AUC gain 0.004 [-0.019, 0.032], n=61)	yes	on the training data only
C12	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	30% to 55%	log-odds 1.03, OR 2.81; range across patients 1.03 to 1.03	137, 90	train; test unclear (AUC gain 0.020 [-0.008, 0.054], n=61)	yes	on the training data only
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 54%	log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98	115, 17 (small n)	train; test disagrees (AUC gain -0.005 [-0.050, 0.037], n=61)	yes	the model has little data on this; on the training data only
C14	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 7%	log-odds -1.75, OR 0.17; range across patients -1.75 to -1.75	108, 70	train; test agrees (AUC gain 0.038 [0.011, 0.072], n=61)	yes	clearly
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on*	30% to 12%	log-odds -1.17, OR 0.31; range across patients -1.17 to -1.17	137, 15 (small n)	train; test unclear (AUC gain 0.020 [-0.008, 0.054], n=61)	yes	the model has little data on this; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	30% to 19%	log-odds -0.61, OR 0.54; range across patients -0.61 to -0.61	209, 33	train; test unclear (AUC gain 0.001 [-0.012, 0.016], n=61)	yes	on the training data only
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	36% to 27%	log-odds -0.45, OR 0.64; range across patients -0.45 to -0.45	242	train; test unclear (AUC gain 0.005 [-0.004, 0.020], n=61)	yes	on the training data only
C21	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, resting ECG other than signs of left ventricular hypertrophy on the resting ECG*	30% to 33%	log-odds 0.11, OR 1.11; range across patients 0.11 to 0.11	123, 119	train; test disagrees (AUC gain -0.002 [-0.010, 0.000], n=61)	yes	slightly; on the training data only
C29	structure: made of nonanginal_thalach_gate, oldpeak_gate_without_nonanginal, nonanginal_low_chol_gate and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	30% to 17%	log-odds -0.69, OR 0.50; range across patients -1.07 to -0.19	172, 70	train; test unclear (AUC gain 0.008 [-0.009, 0.028], n=61)	yes	on the training data only
C32	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 7% to 2%; chest pain type other than non-anginal chest pain: 36% to 27%	log-odds difference -0.89, OR 0.41	70, 172	train; test agrees (AUC gain 0.008 [-0.009, 0.028], n=61); group difference on test -0.89 [-0.89, -0.88]	yes	slightly
C34	subgroup: who this part acts on: 76 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.31; log-odds -0.71, OR 0.49; range across patients -1.06 to -0.14	76, 166	train; test unclear (AUC gain 0.013 [-0.001, 0.038], n=61)	yes	slightly; on the training data only
C37	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (through this part) *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of patients same; verified; risk saturates near 0%; one point with C39 (median fold 282 mg/dL (7.3 mmol/L), typical patient 283 mg/dL (7.3 mmol/L), 10th-90th percentile 282 mg/dL (7.3 mmol/L) to 282 mg/dL (7.3 mmol/L))*	2% to 3% (typical patient this part acts on)	log-odds 0.05, OR 1.05; range across patients 0.00 to 0.16	242	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only
C38	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 100% of patients same; fallback; risk saturates near 0%*	non-anginal chest pain: 2% to 7%; chest pain type other than non-anginal chest pain: 21% to 48%	log-odds difference 0.16, OR 1.18	70, 172	train; test agrees (AUC gain 0.002 [0.000, 0.009], n=61); group difference on test 0.16 [0.16, 0.16]	yes	clearly
C39	threshold: serum cholesterol: fold at about 282 mg/dL (7.3 mmol/L) (stops) for patients with non-anginal chest pain *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of*	2% to 3% (typical patient with a fold)	fold 282 mg/dL (IQR 282 to 282), past it log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	70, 61	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C52	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s2-c3 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s2-c3.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s2-c3 --verified --minutes N --changes "..." --notes "..."`

9 Explanation certificate: heart-s2-c3

This section is the explanation certificate of the heart-s2-c3 network. It is rendered from the canonical JSON record `heart-s2-c3.json` (schema 1.4.0, SHA-256 `6b36c4ae4f911613...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 9 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s2, seed 2
Genome	`1a5255ec-4f4d-4ff4-916a-1b5c4622ccc9` (NEAT key 9600, generation 39)
Explanation	`6565cffd-9c19-4ae2-b635-a2e35f3d485a` (34 operations)
Train AUC / accuracy	0.9381 / 0.8884 (n = 242)
Test AUC / accuracy	0.8987 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	30 nodes (26 inputs, 3 hidden, 1 output), 26 enabled edges
Explained model state	36 nodes, 26 enabled edges
Refactoring operations	split_node 4
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 2.22 \times 10^{-16}$ on the training split
Annotations	4 (3 leaf, 1 composition)
Structural coverage	1 (35 of 35 candidate nodes; 9 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`6b36c4ae4f911613...`

Table 9: Explanation certificate summary: heart-s2-c3.

Drafter's summary. Mostly logistic regression: 16 direct edges (logit SD 3.67; train AUC 0.934 alone) plus three additive ReLU gates (logit SD 0.42) to 0.938. All three gates are switched by non-anginal chest pain (cp:3), which has no direct edge: a protective gate ReLU(-0.35 + 1.08 cp:3 + 0.68

thalach) read out -0.51, open for 97% of non-anginal-pain patients, and a smaller low-cholesterol gate for the same patients (-0.24), plus a near-null oldpeak hinge for everyone else. Age is not used at all. Unlike the 10-seed heart-s2 record (key 9083, an age/heart-rate-reserve gate and oldpeak hinges), this genome's only nonlinear effect is a chest-pain-type-specific level and thalach/cholesterol slope.

What resisted explanation. Nothing resisted coverage. The ST-depression gate for patients without non-anginal pain is a near-null part (readout -0.018, SD 0.009 logit units; ablating it raises train AUC by 0.0001) and is annotated as such. Counter-directional signs reported, not explained: cp:2 +0.27 and fbs:1 -0.21 in the linear core, extra protection for non-anginal-pain patients with lower cholesterol, and the gate that slightly lowers risk with more ST depression.

9.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 30%; it adds up most of its inputs independently, and three parts that all read chest pain type add to that sum: one acts on 76 of 242 training patients through maximum heart rate, one on 61 through cholesterol, and one barely changes the predictions. [C29, C34, C40, C50] These add up independently, and these lower the risk: through this sum alone, compared with no chest pain symptoms, typical angina takes a typical patient's predicted risk from about 30% to about 2%, non-anginal chest pain to about 7% and atypical angina to about 13% (chest pain type also acts through the parts below); being a woman lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 19%; and through this sum alone a higher maximum heart rate lowers it steadily, from about 36% at 118 to about 27% at 175 beats per minute (it also acts through the parts below). [C05, C14, C16, C10, C18, C19] Each extra vessel coloured by fluoroscopy adds about the same, from about 30% at 0 to about 96% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 80%, higher resting blood pressure raises it steadily from about 20% at 110 mm Hg to about 46% at 155 mm Hg, and a reversible defect on the thallium scan raises it to about 55%. [C01, C02, C11, C12] Through this sum alone, more ST depression on exercise raises it steadily from about 23% at 0 mm to about 53% at 2.8 mm, and higher cholesterol raises it steadily from about 21% at 193 mg/dL to about 48% at 308 mg/dL; both also act through the parts below. [C03, C07] Some rarer results also count (the model has little data on this, n = 15 to 17): a downsloping ST segment raises the risk to about 54% and a fixed defect on the thallium scan lowers it to about 12%, while signs of left ventricular hypertrophy on the resting ECG raise it from about 30% to about 33%. [C13, C15, C21] This part reads chest pain type and maximum heart rate, and acts on 76 of 242 training patients, mostly patients with non-anginal chest pain, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C34] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 17%. [C30] Across the whole model, a higher maximum heart rate lowers the risk slowly at first, then faster, and more with non-anginal chest pain: from about 7% at 118 to about 2% at 175 beats per minute then, flattening out near the bottom because the predicted risk cannot go much lower, against about 36% to about 27% for other chest pain types. [C32] This part reads chest pain type and cholesterol, and acts on 61 of 242 training patients, all of them with non-anginal chest pain, 13 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 3% without it and about 2% with it. [C40] Through it, for patients with non-anginal chest pain, higher cholesterol raises the predicted risk steadily up to about 282 mg/dL, where the part switches off, then no further: for the typical patient it acts

on, from about 2% at 193 mg/dL to about 3% at 308 mg/dL. [C37, C39] Across the whole model, higher cholesterol raises the risk in an uneven way from about 2% to about 7% with non-anginal chest pain, against about 21% to about 48% for other chest pain types. [C38] This part reads chest pain type, maximum heart rate and ST depression on exercise, and barely changes the predictions (by at most 0.5 percentage points). [C50] It switches on when ST depression on exercise minus about 0.019 × the maximum heart rate exceeds about -1.71, which never happens with non-anginal chest pain; 73 of 242 training patients are on, and 79% of them had heart disease, against 29% of the others. [C52]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and three parts add to that sum, two of which lower the risk further for patients with non-anginal chest pain. [C29, C01, C02, C10, C09] *Evidence.* It is made up of the inputs that feed the score directly and three parts, the non-anginal chest pain parts with heart rate and with cholesterol and an ST depression gate (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, non-anginal chest pain rather than no chest pain symptoms takes a typical patient's predicted risk from about 30% to about 3%, against about 7% through the sum alone. [C09, C14] Across the whole model, a higher maximum heart rate lowers it slowly at first, then faster, from about 36% at 118 to about 27% at 175 beats per minute, where through the sum alone it falls steadily over the same range. [C20, C19] Across the whole model, more ST depression raises it steadily from about 23% to about 53%, and higher cholesterol steadily from about 21% to about 48%. [C04, C08] It ignores age and chest pain brought on by exercise. [C27, C28]

Non-anginal chest pain with heart rate. This part lowers the predicted risk for patients with non-anginal chest pain, and for them a higher maximum heart rate lowers it further, slowly at first, then faster. [C30, C32] *Evidence.* It acts on 76 of 242 training patients, mostly patients with non-anginal chest pain, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 5% without it and about 2% with it. [C34] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 17%. [C30] Through it, a higher maximum heart rate lowers the risk in an uneven way, for the typical patient it acts on from about 4% at 118 to about 2% at 175 beats per minute, flattening out near the bottom because the predicted risk cannot go much lower. [C31] Across the whole model, a higher maximum heart rate lowers the risk slowly at first, then faster, from about 7% to about 2% with non-anginal chest pain, against about 36% to about 27% for other chest pain types. [C32] Its formula reproduces the part exactly on all 242 training patients. [C35]

Non-anginal chest pain with cholesterol. This part lowers the predicted risk a little further for patients with non-anginal chest pain whose cholesterol stays below about 282 mg/dL. [C40, C39] *Evidence.* It acts on 61 of 242 training patients, all of them with non-anginal chest pain, 13 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 3% without it and about 2% with it. [C40] Through it, higher cholesterol raises the predicted risk steadily up to about 282 mg/dL, where the part switches off, then no further: for the typical patient it acts on, from about 2% at 193 mg/dL to about 3% at 308 mg/dL. [C37, C39] Through it, non-anginal chest pain rather than any other chest pain type takes a typical patient's predicted risk from about 30% to about 29%. [C36] Across the whole model, higher cholesterol raises the risk in an uneven way from about 2% to about 7% with non-anginal chest pain, against about 21% to about 48% for other chest pain types. [C38] Its formula reproduces the part exactly on all 242 training patients. [C41]

ST depression gate with little effect. This part barely changes the predictions. [C50] *Evidence.* It switches on when ST depression on exercise minus about 0.019 × the maximum heart rate exceeds about -1.71, which never happens with non-anginal chest pain; 73 of 242 training patients are on,

and 79% of them had heart disease, against 29% of the others. [C52] It changes the predicted risk by at most 0.5 percentage points: through it, ST depression on exercise and maximum heart rate barely move a typical patient's predicted risk, which stays at about 30%. [C50, C42, C43] Its formula reproduces the part exactly on all 242 training patients. [C51]

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 30% to about 96%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types (about 2%), below non-anginal chest pain (about 3%), atypical angina (about 13%) and no chest pain symptoms (about 30%). [C06, C09, C17] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 30% to about 19%). [C18] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk (for a typical patient from about 30% to about 12%), from only 15 training patients. [C15] Worth checking: the model ignores age. [C27] Worth checking: with non-anginal chest pain, one part lowers the predicted risk for cholesterol below about 282 mg/dL, so cholesterol moves the risk from about 2% to about 7% for these patients, against about 21% to about 48% for others. [C38, C39]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	30% to 96%	log-odds 3.99, OR 54.1; range across patients 3.99 to 3.99	242	train; test unclear (AUC gain 0.039 [-0.034, 0.120], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 80%	log-odds 2.20, OR 9.04; range across patients 2.20 to 2.20	115, 110	train; test disagrees (AUC gain -0.005 [-0.050, 0.037], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	23% to 53%	log-odds 1.32, OR 3.74; range across patients 1.32 to 1.32	242	train; test unclear (AUC gain 0.011 [-0.01, 0.038], n=61)	yes	on the training data only
C04	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	23% to 53%	log-odds 1.31, OR 3.69; range across patients 1.30 to 1.32	242	train; test unclear (AUC gain 0.011 [-0.01, 0.038], n=61)	yes	on the training data only
C05	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	log-odds -3.11, OR 0.04; range across patients -3.11 to -3.11	108, 22	train; test agrees (AUC gain 0.038 [0.011, 0.072], n=61)	yes	clearly
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	log-odds -3.11, OR 0.04; range across patients -3.11 to -3.11	108, 22	train; test agrees (AUC gain 0.055 [0.014, 0.101], n=61)	yes	clearly
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	242	train; test disagrees (AUC gain -0.012 [-0.044, 0.013], n=61)	yes	on the training data only
C08	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.31, OR 3.71; range across patients 1.26 to 1.43	242	train; test disagrees (AUC gain -0.012 [-0.044, 0.013], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 3%	log-odds -2.51, OR 0.08; range across patients -2.96 to -2.01	108, 70	train; test agrees (AUC gain 0.055 [0.014, 0.101], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	30% to 4%	log-odds -2.33, OR 0.10; range across patients -2.33 to -2.33	159, 83	train; test unclear (AUC gain 0.025 [-0.011, 0.071], n=61)	yes	on the training data only
C11	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	20% to 46%	log-odds 1.22, OR 3.39; range across patients 1.22 to 1.22	242	train; test unclear (AUC gain 0.004 [-0.019, 0.032], n=61)	yes	on the training data only
C12	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	30% to 55%	log-odds 1.03, OR 2.81; range across patients 1.03 to 1.03	137, 90	train; test unclear (AUC gain 0.020 [-0.008, 0.054], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	30% to 54%	log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98	115, 17 (small n)	train; test disagrees (AUC gain -0.005 [-0.050, 0.037], n=61)	yes	the model has little data on this; on the training data only
C14	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 7%	log-odds -1.75, OR 0.17; range across patients -1.75 to -1.75	108, 70	train; test agrees (AUC gain 0.038 [0.011, 0.072], n=61)	yes	clearly
C15	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan, a reversible defect on the thallium scan, a fixed defect on the thallium scan*	30% to 12%	log-odds -1.17, OR 0.31; range across patients -1.17 to -1.17	137, 15 (small n)	train; test unclear (AUC gain 0.020 [-0.008, 0.054], n=61)	yes	the model has little data on this; on the training data only
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 13%	log-odds -1.03, OR 0.36; range across patients -1.03 to -1.03	108, 42	train; test agrees (AUC gain 0.038 [0.011, 0.072], n=61)	yes	clearly
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 13%	log-odds -1.03, OR 0.36; range across patients -1.03 to -1.03	108, 42	train; test agrees (AUC gain 0.055 [0.014, 0.101], n=61)	yes	clearly
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	30% to 19%	log-odds -0.61, OR 0.54; range across patients -0.61 to -0.61	209, 33	train; test unclear (AUC gain 0.001 [-0.012, 0.016], n=61)	yes	on the training data only
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	36% to 27%	log-odds -0.45, OR 0.64; range across patients -0.45 to -0.45	242	train; test unclear (AUC gain 0.005 [-0.004, 0.020], n=61)	yes	on the training data only
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: accelerating; 100% of patients same; verified*	36% to 27%	log-odds -0.70, OR 0.50; range across patients -1.33 to -0.44	242	train; test unclear (AUC gain 0.012 [-0.002, 0.034], n=61)	yes	on the training data only
C21	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, resting ECG other than signs of left ventricular hypertrophy on the resting ECG*	30% to 33%	log-odds 0.11, OR 1.11; range across patients 0.11 to 0.11	123, 119	train; test disagrees (AUC gain -0.002 [-0.010, 0.000], n=61)	yes	slightly; on the training data only
C22	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	30% to 96%	log-odds 3.99, OR 54.1	242	train; test unclear (AUC gain 0.039 [-0.034, 0.120], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	30% to 2%	log-odds -3.11, OR 0.04	108, 22	train; test agrees (AUC gain 0.055 [0.014, 0.101], n=61)	yes	clearly
C26	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	21% to 48%	log-odds 1.26, OR 3.54	242	train; test disagrees (AUC gain -0.012 [-0.044, 0.013], n=61)	yes	on the training data only
C27	null: model ignores age (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of nonanginal_thalach_gate, oldpeak_gate_without_nonanginal, nonanginal_low_chol_gate and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	30% to 17%	log-odds -0.69, OR 0.50; range across patients -1.07 to -0.19	172, 70	train; test unclear (AUC gain 0.008 [-0.009, 0.028], n=61)	yes	on the training data only
C31	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 106 beats per minute; 29% of patients same; fallback; risk saturates near 0%*	4% to 2% (typical patient this part acts on)	log-odds -0.25, OR 0.78; range across patients -0.88 to 0.00	242	train; test unclear (AUC gain 0.006 [0.000, 0.020], n=61)	yes	slightly; on the training data only
C32	interaction: effect of maximum heart rate (118 beats per minute to 175 beats per minute) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 7% to 2%; chest pain type other than non-anginal chest pain: 36% to 27%	log-odds difference -0.89, OR 0.41	70, 172	train; test agrees (AUC gain 0.008 [-0.009, 0.028], n=61); group difference on test -0.89 [-0.89, -0.88]	yes	slightly
C33	threshold: maximum heart rate: fold at about 184 beats per minute (starts) for patients with no chest pain symptoms (asymptomatic) *shape: step at 190 beats per minute; 100% of patients same; verified; risk saturates near 100%*	99% to 99% (typical patient with a fold (patients with no chest pain symptoms (asymptomatic)))	fold 184 beats per minute (IQR 184 to 184); past it log-odds 0.00, OR 1.00; range across patients 0.00 to 0.00	108, 3 (small n)	train; test unclear (AUC gain 0.006 [0.000, 0.020], n=61)	no	slightly; the model has little data on this; on the training data only
C34	subgroup: who this part acts on: 76 of 242 patients	5% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.31; log-odds -0.71, OR 0.49; range across patients -1.06 to -0.14	76, 166	train; test unclear (AUC gain 0.013 [-0.001, 0.038], n=61)	yes	slightly; on the training data only
C35	structure: its formula reproduces it on 242 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	242	train	yes	
C36	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	30% to 29%	log-odds -0.08, OR 0.92; range across patients -0.16 to 0.00	172, 70	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (through this part) *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of patients same; verified; risk saturates near 0%; one point with C39 (median fold 282 mg/dL (7.3 mmol/L), typical patient 283 mg/dL (7.3 mmol/L), 10th-90th percentile 282 mg/dL (7.3 mmol/L) to 282 mg/dL (7.3 mmol/L))*	2% to 3% (typical patient this part acts on)	log-odds 0.05, OR 1.05; range across patients 0.00 to 0.16	242	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only
C38	interaction: effect of serum cholesterol (193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L)) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 100% of patients same; fallback; risk saturates near 0%*	non-anginal chest pain: 2% to 7%; chest pain type other than non-anginal chest pain: 21% to 48%	log-odds difference 0.16, OR 1.18	70, 172	train; test agrees (AUC gain 0.002 [0.000, 0.009], n=61); group difference on test 0.16 [0.16, 0.16]	yes	clearly
C39	threshold: serum cholesterol: fold at about 282 mg/dL (7.3 mmol/L) (stops) for patients with non-anginal chest pain *shape: hinge at 282 mg/dL (7.3 mmol/L); 100% of patients same; verified; risk saturates near 0%; one point with C39 (median fold 282 mg/dL (7.3 mmol/L), typical patient 283 mg/dL (7.3 mmol/L), 10th-90th percentile 282 mg/dL (7.3 mmol/L) to 282 mg/dL (7.3 mmol/L))*	2% to 3% (typical patient with a fold)	fold 282 mg/dL (IQR 282 to 282); past it log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	70, 61	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only
C40	subgroup: who this part acts on: 61 of 242 patients	3% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.25; log-odds -0.10, OR 0.91; range across patients -0.18 to -0.02	61, 181	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C41	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C42	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: flat; 100% of patients same; verified*	30% to 30%	log-odds -0.01, OR 0.99; range across patients -0.02 to 0.00	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C43	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: flat; 100% of patients same; verified*	30% to 30%	log-odds 0.004, OR 1.00; range across patients 0.00 to 0.01	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C44	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: non-anginal chest pain, chest pain type other than non-anginal chest pain*	>99% to >99% (typical patient this part acts on)	log-odds 0.005, OR 1.01; range across patients 0.00 to 0.02	172, 70	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C45	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by chest pain type: non-anginal chest pain minus chest pain type other than non-anginal chest pain; stronger in non-anginal chest pain *shape: steady; 100% of patients same; verified; risk saturates near 0%*	non-anginal chest pain: 2% to 8%; chest pain type other than non-anginal chest pain: 23% to 53%	log-odds difference 0.02, OR 1.02	70, 172	train; test agrees (AUC gain 0.000 [0.000, 0.000], n=61); group difference on test 0.02 [0.02, 0.02]	no	clearly
C46	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by maximum heart rate: maximum heart rate above 152 beats per minute minus maximum heart rate at or below 152 beats per minute; stronger in maximum heart rate above 152 beats per minute *shape: steady; 100% of patients same; verified*	maximum heart rate above 152 beats per minute: 21% to 50%; maximum heart rate at or below 152 beats per minute: 26% to 56%	log-odds difference 0.007, OR 1.01	121, 121	train; test agrees (AUC gain 0.000 [0.000, 0.000], n=61); group difference on test 0.008 [0.004, 0.01]	no	slightly
C47	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with no chest	99% to 99% (typical patient with a fold (patients with	fold 1.01 mm (IQR 0.67 to 1.31); past it log-odds -0.03, OR 0.97; range across	106, 58	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	no	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C49	subgroup: who this part acts on: 73 of 242 patients	>99% to >99% (typical patient this part acts on (risk without the part, then with it))	share 0.30; log-odds -0.02, OR 0.98; range across patients -0.03 to -0.002	73, 169	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	no	slightly; on the training data only
C50	null: this part barely changes predictions (switched off)		max change 0.005 (max |change in predicted risk|)	242	train	yes	
C51	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C52	boundary: boundary		log-odds 1.00, OR n/a	242	train	yes	

Not supported by the model.

- C33: threshold claim about maximum heart rate reached during the exercise test in part ann_30: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C45: interaction claim about ST depression during exercise (compared with rest) and chest pain type in part ann_31: not supported by the model (the group difference is below 0.1 log-odds).
- C46: interaction claim about ST depression during exercise (compared with rest) and maximum heart rate reached during the exercise test in part ann_31: not supported by the model (the group difference is below 0.1 log-odds).
- C47: threshold claim about ST depression during exercise (compared with rest) in part ann_31: not supported by the model (the fold changes the curve by only 0.011 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C48: threshold claim about maximum heart rate reached during the exercise test in part ann_31: not supported by the model (the fold changes the curve by only 0.022 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C49: subgroup claim about chest pain type and maximum heart rate reached during the exercise test in part ann_31: not supported by the model (its effect is negligible).

9.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (9):** `-1` (age), `-10` (fbs:0), `-12` (restecg:0), `-13` (restecg:1), `-16` (exang:0), `-17` (exang:1), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

9.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 9.6.1)

▷ A2: nonanginal_thalach_gate (leaf, section 9.6.2)

▷ A3: oldpeak_gate_without_nonanginal (leaf, section 9.6.3)

▷ A4: nonanginal_low_chol_gate (leaf, section 9.6.4)

9.4 Network figures

9.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-6` into `-6_a`, `-6_b`, `-6_c`.
- #27: split `-15` into `-15_a`, `-15_b`, `-15_c`.
- #28: split `-18` into `-18_a`, `-18_b`.
- #29: split `-9` into `-9_a`, `-9_b`.

9.6 Annotations

9.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 9.6.2), A3 (section 9.6.3), A4 (section 9.6.4).

- **Entry nodes:** `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9_a` (chol_a), `-11` (fbs:1), `-14` (restecg:2), `-15_a` (thalach_a), `-18_a` (oldpeak_a), `-19` (slope:1), `-20` (slope:2), `-22` (ca), `-25` (thal:6), `-26` (thal:7), `-1` (age), `-10` (fbs:0), `-12` (restecg:0), `-13` (restecg:1), `-16` (exang:0), `-17` (exang:1), `-21` (slope:3), `-23` (thal:3), `-24` (thal:4), `-6_a` (cp:3_a), `-15_b` (thalach_b), `-6_b` (cp:3_b), `-15_c` (thalach_c), `-18_b` (oldpeak_b), `-6_c` (cp:3_c), `-9_b` (chol_b).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-2`→`0` (-0.76), `-3`→`0` (0.347), `-4`→`0` (-0.393), `-5`→`0` (0.273), `-7`→`0` (0.869), `-8`→`0` (0.493), `-9_a`→`0` (0.529), `-11`→`0` (-0.209), `-14`→`0` (0.053), `-15_a`→`0` (-0.176), `-18_a`→`0` (0.559), `-19`→`0` (-0.488), `-20`→`0` (0.61), `-22`→`0` (1.78), `-25`→`0` (-0.282), `-26`→`0` (0.5), `2472`→`0` (-0.511), `2596`→`0` (-0.0184), `2863`→`0` (-0.235).
- **With its descendants:** 36 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 35 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = 0.323 + LINEAR CORE - 0.511*nonanginal_thalach_gate - 0.018*oldpeak_gate_without_nonanginal - 0.235*nonanginal_low_chol_gate. LINEAR CORE (16 direct input->output edges, z-scored inputs): ca +1.779, cp:4 +0.869, sex:0 -0.760, slope:2 +0.610, oldpeak +0.559, chol +0.529, thal:7 +0.500, trestbps +0.493, slope:1 -0.488, cp:1 -0.393, sex:1 +0.347, thal:6 -0.282, cp:2 +0.273, fbs:1 -0.209, thalach -0.176, restecg:2 +0.053. Nine inputs are unused (age, fbs:0, restecg:0, restecg:1, exang:0, exang:1, slope:3, thal:3, thal:4; zero-effect entries of this root). Non-anginal pain (cp:3) has no direct edge; it acts only through the three children, all of which are switched by it. COMPOSITION: purely additive in the logit. The children share no hidden nodes and are coupled only through shared inputs (cp:3 feeds all three; thalach, oldpeak and chol also have direct edges). (1) nonanginal_thalach_gate: open for 97% of non-anginal-pain patients, lowers their risk, more with higher max heart rate; (2) nonanginal_low_chol_gate: open for 87% of non-anginal-pain patients (those with z-chol < 0.76), lowers their risk more the lower the cholesterol; (3) oldpeak_gate_without_nonanginal: a near-null oldpeak hinge for the other patients (readout -0.018). Together (1)+(2) give non-anginal pain an average logit shift of -0.85 and make thalach and chol slopes depend on chest-pain type. EVIDENCE (train, n=242): full

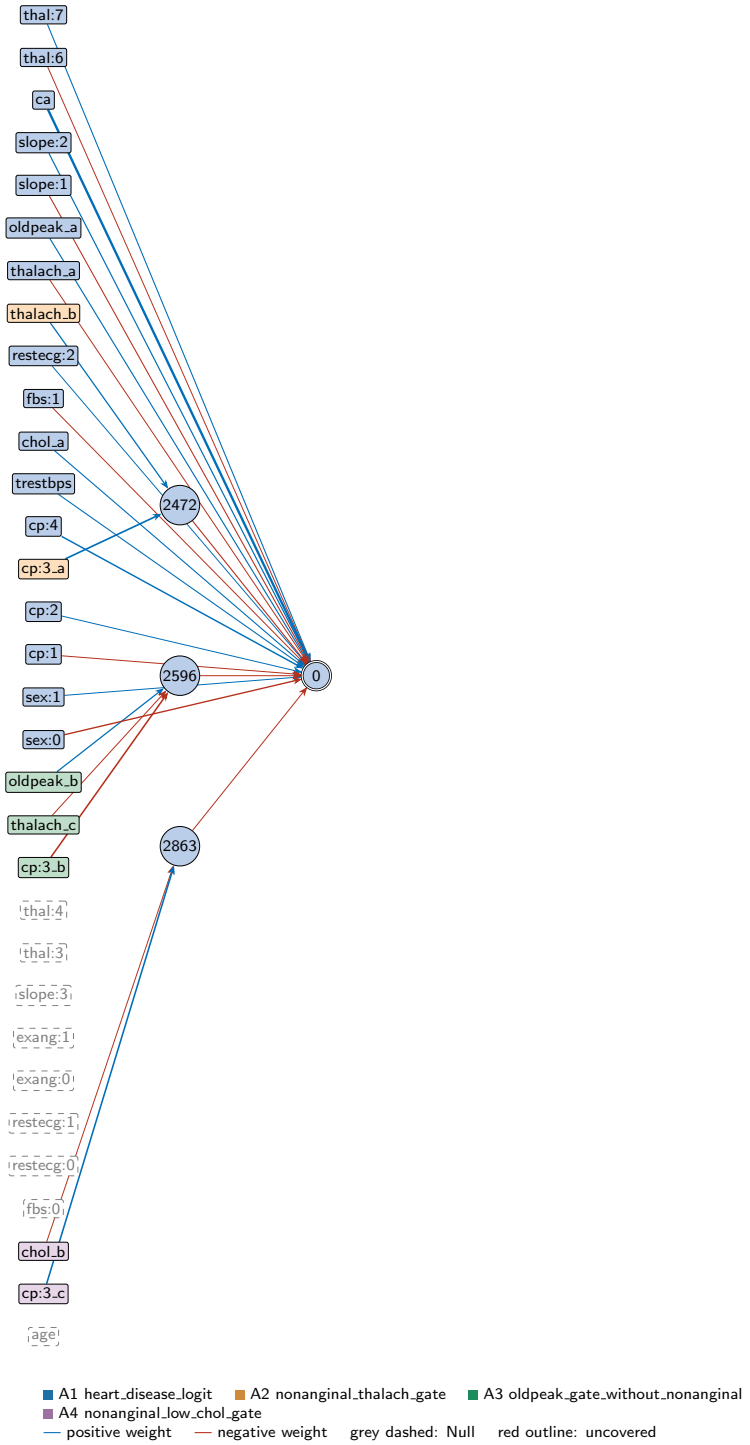


Figure 27: heart-s2-c3: full view of the explained model state.

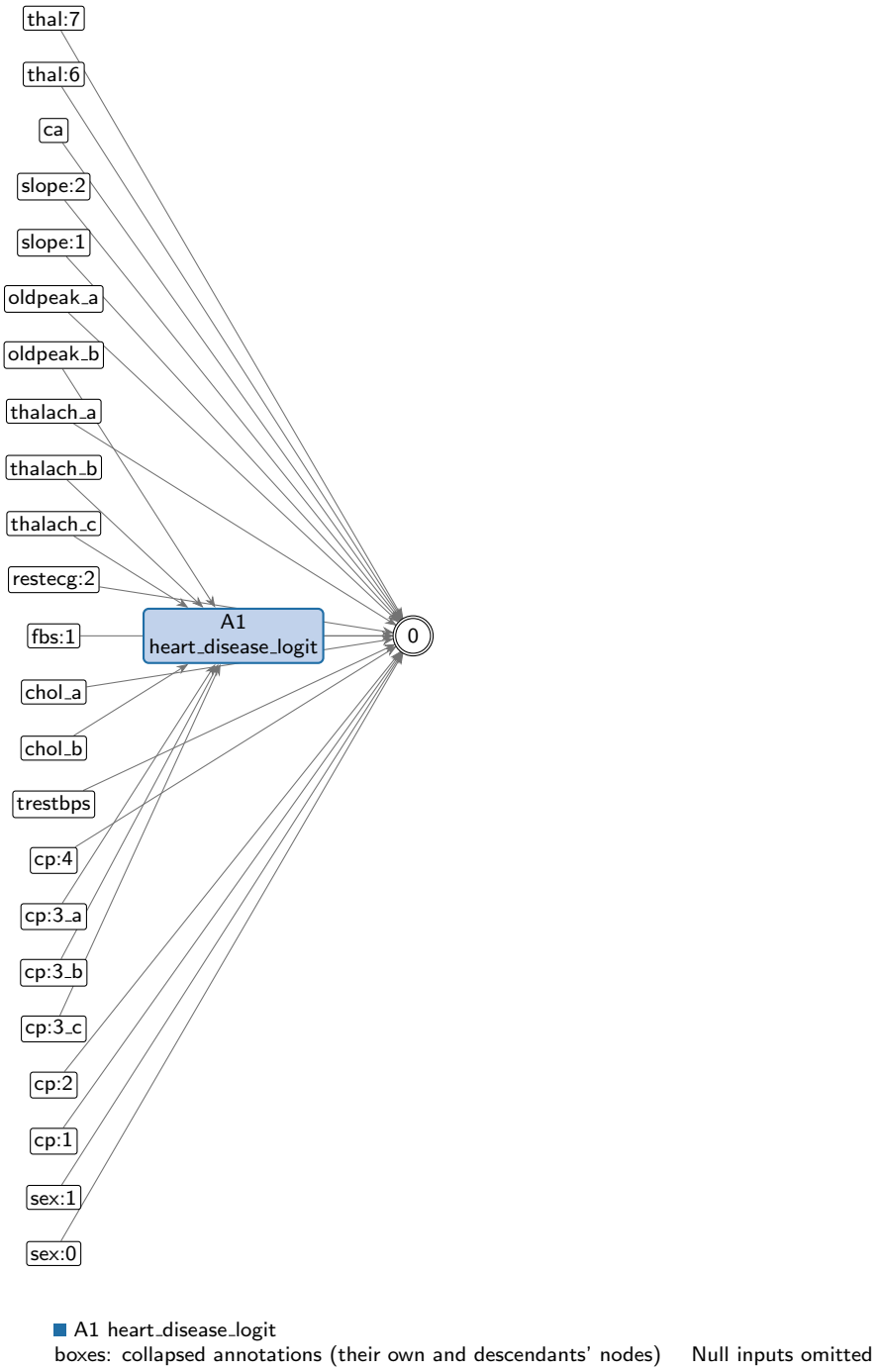


Figure 28: heart-s2-c3: collapsed view at hierarchy level 0 (fully recomposed).

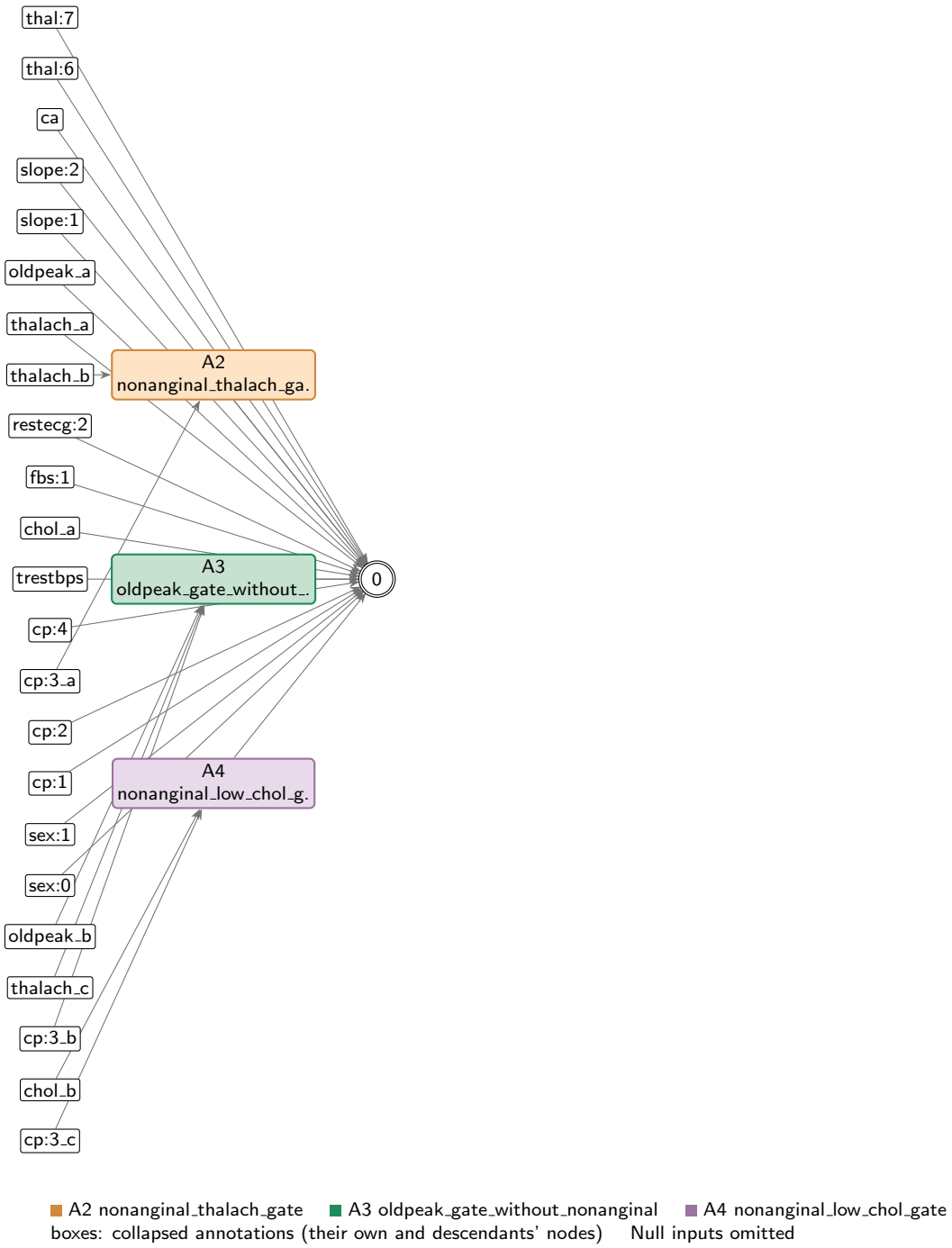


Figure 29: heart-s2-c3: collapsed view at hierarchy level 1.

model AUC 0.9381, acc 0.888 (compute_performance). Linear core alone AUC 0.934; logit SD 3.67 (core) vs 0.42 (all nonlinear readouts). Zero-ablating one child at a time: thalach gate 0.934, oldpeak gate 0.938, low-chol gate 0.938. Replacing all hidden readouts with one least-squares linear fit in their four inputs gives 0.938; an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242 rows): ca 0.161, cp:4 0.099, sex:0 0.073, slope:2 0.066, thal:7 0.054, slope:1 0.053, oldpeak 0.047, chol 0.041, trestbps 0.040, cp:3 (thalach gate copy) 0.029; unused inputs about 0.001 (estimator noise). Interpretation: logistic regression on 16 inputs plus a non-anginal-pain block whose nonlinearity is small (a linear refit loses nothing on train AUC). Counter-directional signs reported, not explained: cp:2 +0.27 and fbs:1 -0.21 in the core, the low-cholesterol protection for non-anginal pain. Compared with the 10-seed heart-s2 record (key 9083, gen 38, a different cluster of the same run): that genome has an age/heart-rate-reserve gate and oldpeak hinges and uses age; this one ignores age entirely and puts its only nonlinear effect on non-anginal chest pain.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= 0.323 + 0.559x_{oldpeak_a} + 0.529x_{chol_a} + 0.053x_{restecg:2} + 1.78x_{ca} \\
&\quad + 0.347x_{sex:1} + 0.273x_{cp:2} + 0.61x_{slope:2} + 0.869x_{cp:4} + 0.493x_{trestbps} \\
&\quad + 0.5x_{thal:7} - 0.176x_{thalach_a} \\
&\quad - 0.235\,\mathrm{nonanginal_low_chol_gate}(x_{cp:3_c}, x_{chol_b}) \\
&\quad - 0.488x_{slope:1} - 0.511\,\mathrm{nonanginal_thalach_gate}(x_{cp:3_a}, \\
&\quad x_{thalach_b}) - 0.393x_{cp:1} - 0.209x_{fbs:1} - 0.282x_{thal:6} \\
&\quad - 0.0184\,\mathrm{oldpeak_gate_without_nonanginal}(x_{cp:3_b}, x_{thalach_c}, \\
&\quad x_{oldpeak_b}) - 0.76x_{sex:0}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= 0.323 + 0.559x_{oldpeak_a} + 0.529x_{chol_a} + 0.053x_{restecg:2} + 1.78x_{ca} \\
&\quad + 0.347x_{sex:1} + 0.273x_{cp:2} + 0.61x_{slope:2} + 0.869x_{cp:4} + 0.493x_{trestbps} \\
&\quad + 0.5x_{thal:7} - 0.176x_{thalach_a} - 0.235\,\mathrm{relu}(-1.25 \\
&\quad + 0.983x_{cp:3_c} - 0.376x_{chol_b}) - 0.488x_{slope:1} \\
&\quad - 0.511\,\mathrm{relu}(-0.348 + 1.08x_{cp:3_a} + 0.678x_{thalach_b}) \\
&\quad - 0.393x_{cp:1} - 0.209x_{fbs:1} - 0.282x_{thal:6} - 0.0184\,\mathrm{relu}(-0.688 \\
&\quad + 0.673x_{oldpeak_b} - 0.241x_{thalach_c} - 0.985x_{cp:3_b}) - 0.76x_{sex:0}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00141, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=true); output is sigmoid.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, ablations, linear replacements and SHAP.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

9.6.2 A2: nonanginal_thalach_gate

Kind: Leaf annotation (depth 1); parent A1 (section 9.6.1).

- **Entry nodes:** `-6_a` (cp:3_a), `-15_b` (thalach_b).
- **Exit nodes:** `2472`.

- **Internal nodes:** none.
- **Edges (weight):** `-6_a`→`2472` (1.08), `-15_b`→`2472` (0.678).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Non-anginal-pain gate (protective). 2472 = ReLU(-0.348 + 1.076*cp:3 + 0.678*thalach), read out -0.511 into the logit. Hypothesis: this gate is how the model gives non-anginal chest pain (cp:3, which has no direct edge) a lower risk: it is open for nearly every non-anginal-pain patient and almost never otherwise, and once open a higher achieved max heart rate lowers the risk further (on top of thalach's small direct -0.176 edge). EVIDENCE (train, n=242): active for 97% of the 70 non-anginal-pain patients (closed only for z-thalach < -1.97) and 5% of the 172 others (only z-thalach > 1.53). Disease rate 20% when active vs 55% when inactive. For a non-anginal-pain patient it adds -0.684 - 0.346*z-thalach to the logit (mean -0.76). Mean |sensitivity| cp:3 0.34 > thalach 0.21. Zero-ablating its readout lowers train AUC 0.938 -> 0.934, the only nonlinear part that matters; a linear refit in (cp:3, thalach) recovers 0.938 (R2 0.87), so the gate acts almost as a cp:3 main effect plus a cp:3-specific thalach slope. Supported.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.348 + 1.08 x_{cp{:}3_a} + 0.678 x_{thalach_b}\big)$$

where y_0 is node `2472`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00675, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data reading of the gate.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000, sample_fraction 1.0)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

9.6.3 A3: oldpeak_gate_without_nonanginal

Kind: Leaf annotation (depth 1); parent A1 (section 9.6.1).

- **Entry nodes:** `-6_b` (cp:3_b), `-15_c` (thalach_c), `-18_b` (oldpeak_b).
- **Exit nodes:** `2596`.
- **Internal nodes:** none.
- **Edges (weight):** `-6_b`→`2596` (-0.985), `-15_c`→`2596` (-0.241), `-18_b`→`2596` (0.673).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. ST-depression gate for patients without non-anginal pain (negligible). 2596 = ReLU(-0.688 - 0.985*cp:3 - 0.241*thalach + 0.673*oldpeak), read out -0.018. Hypothesis: a hinge on ST depression (and lower max heart rate) for patients without non-anginal pain, whose readout is too small to change predictions. EVIDENCE (train, n=242): active for 42% of patients without non-anginal pain and never with it; disease rate 80% when active vs 29% when inactive, so it tracks risk, but its readout has SD 0.009 logit units against 3.67 for the linear core, and removing it changes train AUC by +0.0001. Its sign (slightly lowering the risk with more ST depression) runs against oldpeak's direct +0.559 edge, which dominates. Supported as a near-null part; a vestige rather than a mechanism.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.688 + 0.673x_{oldpeak_b} - 0.241x_{thalach_c} - 0.985x_{cp{:}3_b}\big)$$

where y_0 is node `2596`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00186, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data reading of the gate.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000, sample_fraction 1.0)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

9.6.4 A4: nonanginal_low_chol_gate

Kind: Leaf annotation (depth 1); parent A1 (section 9.6.1).

- **Entry nodes:** `-6_c` (cp:3_c), `-9_b` (chol_b).
- **Exit nodes:** `2863`.
- **Internal nodes:** none.
- **Edges (weight):** `-6_c`→`2863` (0.983), `-9_b`→`2863` (-0.376).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Non-anginal-pain, low-cholesterol gate (protective, small). 2863 = ReLU(-1.255 + 0.983*cp:3 - 0.376*chol), read out -0.235. Hypothesis: open only for non-anginal-pain patients whose cholesterol is not high (z-chol below 0.76); it lowers their risk, by more the lower the cholesterol, so for these patients cholesterol's direct +0.529 edge is partly undone. EVIDENCE (train, n=242): active for 87% of the 70 non-anginal-pain patients and none of the others. Disease rate 21% when active vs 52% when inactive. Readout up to -0.29 logit units (SD 0.053). Mean |sensitivity| cp:3 0.25 > chol 0.09. Zero-ablating it leaves train AUC unchanged (0.9381 -> 0.9382): it shifts the level of the non-anginal-pain group rather than ranking patients. Supported but small.

Closed form:

$$y_0 = \mathrm{relu}\big(-1.25 + 0.983x_{cp{:}3_c} - 0.376x_{chol_b}\big)$$

where y_0 is node `2863`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00471, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data reading of the gate.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000, sample_fraction 1.0)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

9.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

---

10. heart-s4-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	3.33e-16	3.33e-16	3.33e-16	agree
Closed-form max abs error (all leaves)	0.001728	0.001728	0.001728	agree
Numeric-function max abs error	n/a	0	0	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	4 / 1	n/a	4 / 1	agree
Operation counts	5 splits, 0 identity	annotate 5, rename node 26, split node 5	annotate 5, rename node 26, split node 5	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `no_asymptomatic_pain_indicator` (0.00173). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `fixed_defect_normal_sugar_gate`: *"EVIDENCE (train, n=242): exact on-data reduction 2065 = 3.639*[thal:6=1 AND fbs=0] (fbs=1 drives the pre-activation to <= -6.39)."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `no_asymptomatic_pain_indicator`: *"because cp:4 is binary it is exactly affine on the data, so it only adds to the cp:4 direct edge (+0.308 per z-unit): together, asymptomatic chest pain raises the logit by 1.47 relative to the other pain types."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 5 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): No v3 tool bugs met. compute_shap on all 242 train rows (about a minute) gives mean |SHAP| about 0.001 to the six unconnected inputs, which have exactly zero effect (KernelSHAP sampling noise). Input fan-out splits (-10, -11, -15, -21, -7) taken from the topology. The agent sandbox refused shell loops calling uv with variables, so tools were called through a small Python subprocess wrapper in scratch (same species_leads/xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s4-c1-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 24%; it adds up most of its inputs independently, and four small parts add to that sum: one for a fixed defect on the thallium scan with normal fasting blood sugar, one for chest pain type, one for a downsloping ST segment and one for maximum heart rate. [C31] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 24% to about 1%; through this sum alone, compared with no chest pain symptoms, typical angina lowers it to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%, fasting blood sugar above 120 mg/dL lowers it to about 7%, and a higher maximum heart rate lowers it steadily, from about 27% at 118 to about 21% at 175 beats per minute; chest pain type, fasting blood sugar and maximum heart rate also act through

the parts below. [C12, C13, C15, C22, C17, C24] Each extra vessel coloured by fluoroscopy adds about the same, from about 24% at 0 to about 98% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; through this sum alone a flat rather than upsloping ST segment raises it to about 82% and any thallium result other than a normal scan to about 48%, and chest pain brought on by exercise raises it to about 36% (the ST segment and the thallium scan also act through the parts below). [C01, C02, C06, C21] Higher cholesterol (from about 14% at 193 mg/dL to about 44% at 308 mg/dL), more ST depression on exercise (from about 18% at 0 mm to about 42% at 2.8 mm), higher resting blood pressure (from about 14% at 110 mm Hg to about 41% at 155 mm Hg) and older age (from about 21% at 42 to about 25% at 66 years) raise it steadily. [C04, C08, C05, C25] Through this sum alone a downsloping ST segment raises it to about 39% (the model has little data on this, n = 17), and an ST-T wave abnormality on the resting ECG raises it to about 38% (the model has little data on this, n = 2). [C19, C20] This part reads fasting blood sugar and the thallium scan and acts on 10 of 242 training patients, nearly all of them with a fixed defect on the thallium scan, 5 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 94% without it and about 24% with it (the model has little data on this, n = 10). [C35] Through it, a fixed defect on the thallium scan lowers a typical patient's predicted risk from about 24% to about 2% if fasting blood sugar is 120 mg/dL or less, but the other way if it is above 120 mg/dL, from about 7% to about 18% (the model has little data on this, n = 15). [C34] For the typical patient it acts on, fasting blood sugar above 120 mg/dL raises the risk through it from about 24% to about 94%. [C32] This part reads only chest pain type and acts on the 134 of 242 training patients who have some chest pain symptom, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 2% without it and about 1% with it. [C38] Through it, any chest pain type other than no chest pain symptoms lowers a typical patient's predicted risk from about 24% to about 12%. [C37] This part reads only the slope of the ST segment and acts on the 17 of 242 training patients with a downsloping ST segment, 8 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 66% without it and about 73% with it (the model has little data on this, n = 17). [C41] Through it, a downsloping ST segment raises a typical patient's predicted risk from about 24% to about 29% (the model has little data on this, n = 17). [C40] This part reads only maximum heart rate and acts on 190 of 242 training patients, 71 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 21% without it and about 20% with it. [C45] Through it, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the predicted risk steadily, for a typical patient from about 24% at 118 to about 23% at 175 beats per minute. [C43, C44]

Worth checking: Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 24% to about 98%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C10, C11, C16] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 24% to about 7%). [C18] Worth checking: through one part, the model treats a fixed defect on the thallium scan as lowering the risk for patients with normal fasting blood sugar (for a typical patient from about 24% to about 2%), from only 15 training patients. [C34]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	24% to 98%	log-odds 5.04, OR 154; range across patients 5.04 to 5.04	242	train; test unclear (AUC gain 0.044 [-0.038, 0.127], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	log-odds 2.72, OR 15.1; range across patients 2.72 to 2.72	115, 110	train; test disagrees (AUC gain -0.014 [-0.061, 0.025], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 44%	log-odds 1.60, OR 4.97; range across patients 1.60 to 1.60	242	train; test disagrees (AUC gain -0.019 [-0.049, 0.002], n=61)	yes	on the training data only
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	14% to 41%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	242	train; test unclear (AUC gain 0.002 [-0.018, 0.023], n=61)	yes	on the training data only
C06	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a normal thallium scan*	24% to 48%	log-odds 1.09, OR 2.98; range across patients 1.09 to 1.09	135, 107	train; test agrees (AUC gain 0.026 [0.005, 0.055], n=61)	yes	clearly
C08	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	18% to 42%	log-odds 1.16, OR 3.19; range across patients 1.16 to 1.16	242	train; test unclear (AUC gain 0.005 [-0.01, 0.024], n=61)	yes	on the training data only
C12	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	24% to 1%	log-odds -3.07, OR 0.05; range across patients -3.07 to -3.07	159, 83	train; test unclear (AUC gain 0.025 [-0.01, 0.069], n=61)	yes	on the training data only
C13	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 2%	log-odds -2.98, OR 0.05; range across patients -2.98 to -2.98	108, 22	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only
C15	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 3%	log-odds -2.27, OR 0.10; range across patients -2.27 to -2.27	108, 70	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	24% to 7%	log-odds -1.43, OR 0.24; range across patients -1.43 to -1.43	209, 33	train; test disagrees (AUC gain -0.002 [-0.020, 0.015], n=61)	yes	on the training data only
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 39%	log-odds 0.75, OR 2.11; range across patients 0.75 to 0.75	115, 17 (small n)	train; test disagrees (AUC gain -0.014 [-0.061, 0.025], n=61)	yes	the model has little data on this; on the training data only
C20	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest*	24% to 38%	log-odds 0.68, OR 1.98; range across patients 0.68 to 0.68	240, 2 (small n)	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	the model has little data on this; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C21	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	24% to 36%	log-odds 0.60, OR 1.81; range across patients 0.60 to 0.60	160, 82	train; test unclear (AUC gain 0.009 [-0.002, 0.026], n=61)	yes	on the training data only
C22	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 14%	log-odds -0.62, OR 0.54; range across patients -0.62 to -0.62	108, 42	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only
C24	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	27% to 21%	log-odds -0.32, OR 0.73; range across patients -0.32 to -0.32	242	train; test unclear (AUC gain 0.004 [0.000, 0.012], n=61)	yes	on the training data only
C25	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	21% to 25%	log-odds 0.23, OR 1.25; range across patients 0.23 to 0.23	242	train; test disagrees (AUC gain -0.002 [-0.009, 0.000], n=61)	yes	slightly; on the training data only
C31	structure: made of fixed_defect_normal_sugar_gate, no_asymptomatic_pain_indicator, downsloping_st_indicator, thalach_floor_relay and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C32	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (through this part) *shape: categorical; highest to lowest: fasting blood sugar above 120 mg/dL, fasting blood sugar of 120 mg/dL or less*	24% to 94% (typical patient this part acts on)	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.00	209, 33	train; test disagrees (AUC gain -0.004 [-0.017, 0.000], n=61)	yes	on the training data only
C34	interaction: effect of thallium stress test result (thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan) by fasting blood sugar: fasting blood sugar above 120 mg/dL minus fasting blood sugar of 120 mg/dL or less; stronger in fasting blood sugar of 120 mg/dL or less *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	fasting blood sugar above 120 mg/dL: 7% to 18%; fasting blood sugar of 120 mg/dL or less: 24% to 2%	log-odds difference 3.92, OR 50.3	33, 209, 15 (small n)	train; test agrees (AUC gain -0.004 [-0.017, 0.000], n=61); group difference on test 3.92 [3.92, 3.92]	yes	the model has little data on this
C35	subgroup: who this part acts on: 10 of 242 patients	94% to 24% (typical patient this part acts on (risk without the part, then with it))	share 0.04; log-odds -3.92, OR 0.02; range across patients -3.92 to -3.92	10, 232 (small n)	train; test disagrees (AUC gain -0.016 [-0.044, 0.000], n=61)	yes	the model has little data on this; on the training data only
C37	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than no chest pain (asymptomatic) (through this part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), chest pain type other than no chest pain (asymptomatic)*	24% to 12%	log-odds -0.85, OR 0.43; range across patients -0.85 to -0.85	108, 134	train; test unclear (AUC gain 0.017 [-0.004, 0.043], n=61)	yes	on the training data only
C38	subgroup: who this part acts on: 134 of 242 patients	2% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.55; log-odds -0.85, OR 0.43; range across patients -0.85 to -0.85	134, 108	train; test unclear (AUC gain 0.017 [-0.004, 0.043], n=61)	yes	slightly; on the training data only
C40	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a downsloping ST segment to a downsloping ST segment (through this part) *shape: categorical; highest to lowest: a downsloping ST segment, slope of the*	24% to 29%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	225, 17 (small n)	train; test unclear (AUC gain 0.001 [0.000, 0.006], n=61)	yes	the model has little data on this; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C43	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23%	log-odds -0.09, OR 0.92; range across patients -0.09 to -0.09	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C44	threshold: maximum heart rate: fold at about 131 beats per minute (starts) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23% (typical patient with a fold)	fold 131 beats per minute (IQR 131 to 131); past it log-odds -0.10, OR 0.90; range across patients -0.10 to -0.10	242, 190	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C45	subgroup: who this part acts on: 190 of 242 patients	21% to 20% (typical patient this part acts on (risk without the part, then with it))	share 0.79; log-odds -0.06, OR 0.95; range across patients -0.09 to -0.02	190, 52	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s4-c1 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s4-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s4-c1 --verified --minutes N --changes "..." --notes "..."`

10 Explanation certificate: heart-s4-c1

This section is the explanation certificate of the heart-s4-c1 network. It is rendered from the canonical JSON record `heart-s4-c1.json` (schema 1.4.0, SHA-256 `5bb264e6720c1a25`...), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 10 summarises the network and its coverage.

Drafter's summary. Essentially logistic regression: 19 direct edges (logit SD 4.55; train AUC 0.933 alone; ca +2.25, slope:2 +1.12, sex:1 +0.90) plus four additive hidden readouts (logit SD 0.86) to 0.942. The one meaningful nonlinearity is an AND gate 2065 = 3.64*[thal:6 AND fbs=0] read out -1.08, lowering the logit by 3.9 for the 10 fixed-defect patients with normal fasting sugar (core mean predicted 0.83, disease rate 50%). The cp:4 and slope:3 hidden indicators are exactly affine on their binary inputs and a floored thalach relay is near-inert. Same gate and linear core as the

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s4, seed 4
Genome	`c85804fa-08bd-4207-ac69-a31a8567473c` (NEAT key 9679, generation 39)
Explanation	`10565a81-f55b-47f4-b51e-21641ec08556` (36 operations)
Train AUC / accuracy	0.9423 / 0.8884 (n = 242)
Test AUC / accuracy	0.8858 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	31 nodes (26 inputs, 4 hidden, 1 output), 29 enabled edges
Explained model state	36 nodes, 29 enabled edges
Refactoring operations	split_node 5
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 3.33 \times 10^{-16}$ on the training split
Annotations	5 (4 leaf, 1 composition)
Structural coverage	1 (35 of 35 candidate nodes; 6 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`5bb264e6720c1a25...`

Table 10: Explanation certificate summary: heart-s4-c1.

10-seed heart-s4 record (key 9650, gen 38); here cp:4 and slope:3 indicators replace that genome's cp:2 indicator.

What resisted explanation. Nothing resisted coverage. The one real interaction (fixed thallium defect AND normal fasting sugar, gate 2065) rests on 10 training patients and may be partly noise-fitted. Hidden nodes 1754 (cp:4) and 3504 (slope:3) are exactly affine on their binary inputs, so they are re-parameterisations of linear terms, not mechanisms. Counter-directional signs reported, not explained: fbs:1 -0.58 in the linear core, and the gate making a fixed defect protective for normal-sugar patients.

10.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 24%; it adds up most of its inputs independently, and four small parts add to that sum: one for a fixed defect on the thallium scan with normal fasting blood sugar, one for chest pain type, one for a downsloping ST segment and one for maximum heart rate. [C31] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 24% to about 1%; through this sum alone, compared with no chest pain symptoms, typical angina lowers it to about 2%, non-anginal chest pain to about 3% and atypical angina to about 14%, fasting blood sugar above 120 mg/dL lowers it to about 7%, and a higher maximum heart rate lowers it steadily, from about 27% at 118 to about 21% at 175 beats per minute; chest pain type, fasting blood sugar and maximum heart rate also act through the parts below. [C12, C13, C15, C22, C17, C24] Each extra vessel coloured by fluoroscopy adds about the same, from about 24% at 0 to about 98% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; through this sum alone a flat rather than upsloping ST segment raises it to about 82% and any thallium result other than a normal scan

to about 48%, and chest pain brought on by exercise raises it to about 36% (the ST segment and the thallium scan also act through the parts below). [C01, C02, C06, C21] Higher cholesterol (from about 14% at 193 mg/dL to about 44% at 308 mg/dL), more ST depression on exercise (from about 18% at 0 mm to about 42% at 2.8 mm), higher resting blood pressure (from about 14% at 110 mm Hg to about 41% at 155 mm Hg) and older age (from about 21% at 42 to about 25% at 66 years) raise it steadily. [C04, C08, C05, C25] Through this sum alone a downsloping ST segment raises it to about 39% (the model has little data on this, n = 17), and an ST-T wave abnormality on the resting ECG raises it to about 38% (the model has little data on this, n = 2). [C19, C20] This part reads fasting blood sugar and the thallium scan and acts on 10 of 242 training patients, nearly all of them with a fixed defect on the thallium scan, 5 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 94% without it and about 24% with it (the model has little data on this, n = 10). [C35] Through it, a fixed defect on the thallium scan lowers a typical patient's predicted risk from about 24% to about 2% if fasting blood sugar is 120 mg/dL or less, but the other way if it is above 120 mg/dL, from about 7% to about 18% (the model has little data on this, n = 15). [C34] For the typical patient it acts on, fasting blood sugar above 120 mg/dL raises the risk through it from about 24% to about 94%. [C32] This part reads only chest pain type and acts on the 134 of 242 training patients who have some chest pain symptom, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 2% without it and about 1% with it. [C38] Through it, any chest pain type other than no chest pain symptoms lowers a typical patient's predicted risk from about 24% to about 12%. [C37] This part reads only the slope of the ST segment and acts on the 17 of 242 training patients with a downsloping ST segment, 8 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 66% without it and about 73% with it (the model has little data on this, n = 17). [C41] Through it, a downsloping ST segment raises a typical patient's predicted risk from about 24% to about 29% (the model has little data on this, n = 17). [C40] This part reads only maximum heart rate and acts on 190 of 242 training patients, 71 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 21% without it and about 20% with it. [C45] Through it, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the predicted risk steadily, for a typical patient from about 24% at 118 to about 23% at 175 beats per minute. [C43, C44]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and four small parts add to that sum: one for a fixed thallium defect with normal fasting blood sugar, one for chest pain type, one for a downsloping ST segment and one for maximum heart rate. [C31, C01, C02, C12, C10] *Evidence.* It is made up of the inputs that feed the score directly and four parts, the fixed thallium defect part, the chest pain offset, the downsloping ST offset and the maximum heart rate relay (each explained below); the score is their sum, which was checked on all 242 training patients. [C31] Changing one input at a time across the whole model, compared with no chest pain symptoms, typical angina lowers a typical patient's predicted risk from about 24% to about 1%, non-anginal chest pain to about 1% and atypical angina to about 7%, against about 2% for typical angina through the sum alone. [C10, C11, C16, C13] Across the whole model, fasting blood sugar above 120 mg/dL lowers it from about 24% to about 7%, the same as through the sum alone. [C18, C17] Across the whole model, a higher maximum heart rate lowers it slowly at first, then faster, from about 28% at 118 to about 21% at 175 beats per minute, against about 27% to about 21% through the sum alone, where it falls steadily. [C23, C24] Across the whole model, a downsloping rather than upsloping ST segment raises it to about 47% (the model has little data on this, n = 17), and a reversible defect on the thallium scan or a thallium result coded 4 raises it to about 48%. [C09, C07]

Fixed thallium defect with normal sugar. This part lowers the predicted risk for patients with a fixed defect on the thallium scan when their fasting blood sugar is 120 mg/dL or less (the model has little data on this, n = 15). [C34, C35] *Evidence.* It acts on 10 of 242 training patients, nearly all of them with a fixed defect on the thallium scan, 5 of whom had heart disease; for the

typical patient it acts on, the predicted risk is about 94% without it and about 24% with it (the model has little data on this, n = 10). [C35] Through it, a fixed defect on the thallium scan lowers a typical patient's predicted risk from about 24% to below 1% (the model has little data on this, n = 15). [C33] Through it, a fixed defect lowers the risk from about 24% to about 2% if fasting blood sugar is 120 mg/dL or less, but raises it from about 7% to about 18% if it is above 120 mg/dL (the model has little data on this, n = 15). [C34] For the typical patient it acts on, fasting blood sugar above 120 mg/dL raises the risk through it from about 24% to about 94%. [C32] Its formula reproduces the part exactly on all 242 training patients. [C36]

Chest pain offset. This part lowers the predicted risk for patients with any chest pain symptom, compared with patients who have none. [C37] *Evidence.* It acts on the 134 of 242 training patients who have some chest pain symptom, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 2% without it and about 1% with it. [C38] Through it, any chest pain type other than no chest pain symptoms takes a typical patient's predicted risk from about 24% to about 12%. [C37] Its formula reproduces the part exactly on all 242 training patients. [C39]

Downsloping ST offset. This part raises the predicted risk a little for patients with a downsloping ST segment (the model has little data on this, n = 17). [C40] *Evidence.* It acts on the 17 of 242 training patients with a downsloping ST segment, 8 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 66% without it and about 73% with it (the model has little data on this, n = 17). [C41] Through it, a downsloping ST segment takes a typical patient's predicted risk from about 24% to about 29% (the model has little data on this, n = 17). [C40] Its formula reproduces the part exactly on all 242 training patients. [C42]

Maximum heart rate relay. Through this part, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the predicted risk steadily. [C43, C44] *Evidence.* It acts on 190 of 242 training patients, 71 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 21% without it and about 20% with it. [C45] Through it, a higher maximum heart rate changes nothing until about 131 beats per minute, then lowers the risk steadily, for a typical patient from about 24% at 118 to about 23% at 175 beats per minute. [C43, C44] Its formula reproduces the part exactly on all 242 training patients. [C47]

Worth checking against what is known. Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 24% to about 98%. [C01] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C10, C11, C16] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 24% to about 7%). [C18] Worth checking: through one part, the model treats a fixed defect on the thallium scan as lowering the risk for patients with normal fasting blood sugar (for a typical patient from about 24% to about 2%), from only 15 training patients. [C34]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	24% to 98%	log-odds 5.04, OR 154; range across patients 5.04 to 5.04	242	train; test unclear (AUC gain 0.044 [-0.038, 0.127], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	log-odds 2.72, OR 15.1; range across patients 2.72 to 2.72	115, 110	train; test disagrees (AUC gain -0.014 [-0.061, 0.025], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	log-odds 2.72, OR 15.1; range across patients 2.72 to 2.72	115, 110	train; test disagrees (AUC gain -0.013 [-0.058, 0.026], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 44%	log-odds 1.60, OR 4.97; range across patients 1.60 to 1.60	242	train; test disagrees (AUC gain -0.019 [-0.049, 0.002], n=61)	yes	on the training data only
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	14% to 41%	log-odds 1.48, OR 4.37; range across patients 1.48 to 1.48	242	train; test unclear (AUC gain 0.002 [-0.018, 0.023], n=61)	yes	on the training data only
C06	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a normal thallium scan*	24% to 48%	log-odds 1.09, OR 2.98; range across patients 1.09 to 1.09	135, 107	train; test agrees (AUC gain 0.026 [0.005, 0.055], n=61)	yes	clearly
C07	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 48%	log-odds 1.09, OR 2.98; range across patients 1.09 to 1.09	135, 92	train; test unclear (AUC gain 0.005 [-0.028, 0.039], n=61)	yes	on the training data only
C08	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: steady; 100% of patients same; verified*	18% to 42%	log-odds 1.16, OR 3.19; range across patients 1.16 to 1.16	242	train; test unclear (AUC gain 0.005 [-0.01, 0.024], n=61)	yes	on the training data only
C09	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 47%	log-odds 1.05, OR 2.85; range across patients 1.05 to 1.05	115, 17 (small n)	train; test disagrees (AUC gain -0.013 [-0.058, 0.026], n=61)	yes	the model has little data on this; on the training data only
C10	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 1%	log-odds -3.84, OR 0.02; range across patients -3.84 to -3.84	108, 22	train; test agrees (AUC gain 0.057 [0.016, 0.105], n=61)	yes	clearly
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 1%	log-odds -3.12, OR 0.04; range across patients -3.12 to -3.12	108, 70	train; test agrees (AUC gain 0.057 [0.016, 0.105], n=61)	yes	clearly
C12	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	24% to 1%	log-odds -3.07, OR 0.05; range across patients -3.07 to -3.07	159, 83	train; test unclear (AUC gain 0.025 [-0.01, 0.069], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 2%	log-odds -2.98, OR 0.05; range across patients -2.98 to -2.98	108, 22	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 2%	log-odds -2.29, OR 0.10; range across patients -2.83 to 1.09	135, 15 (small n)	train; test unclear (AUC gain 0.005 [-0.028, 0.039], n=61)	no	the model has little data on this; on the training data only
C15	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 3%	log-odds -2.27, OR 0.10; range across patients -2.27 to -2.27	108, 70	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only
C16	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	24% to 7%	log-odds -1.47, OR 0.23; range across patients -1.47 to -1.47	108, 42	train; test agrees (AUC gain 0.057 [0.016, 0.105], n=61)	yes	clearly
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	24% to 7%	log-odds -1.43, OR 0.24; range across patients -1.43 to -1.43	209, 33	train; test disagrees (AUC gain -0.002 [-0.020, 0.015], n=61)	yes	on the training data only
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	24% to 7%	log-odds -1.18, OR 0.31; range across patients -1.43 to -1.43	209, 33	train; test disagrees (AUC gain -0.005 [-0.025, 0.013], n=61)	yes	on the training data only
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 39%	log-odds 0.75, OR 2.11; range across patients 0.75 to 0.75	115, 17 (small n)	train; test disagrees (AUC gain -0.014 [-0.061, 0.025], n=61)	yes	the model has little data on this; on the training data only
C20	direction: resting ECG: resting ECG other than an ST-T wave abnormality on the resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: an ST-T wave abnormality on the resting ECG, resting ECG other than an ST-T wave abnormality on the resting ECG*	24% to 38%	log-odds 0.68, OR 1.98; range across patients 0.68 to 0.68	240, 2 (small n)	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	the model has little data on this; on the training data only
C21	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	24% to 36%	log-odds 0.60, OR 1.81; range across patients 0.60 to 0.60	160, 82	train; test unclear (AUC gain 0.009 [-0.002, 0.026], n=61)	yes	on the training data only
C22	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not	24% to 14%	log-odds -0.62, OR 0.54; range across patients -0.62 to -0.62	108, 42	train; test unclear (AUC gain 0.029 [0.000, 0.065], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	21% to 25%	log-odds 0.23, OR 1.25; range across patients 0.23 to 0.23	242	train; test disagrees (AUC gain -0.002 [-0.009, 0.000], n=61)	yes	slightly; on the training data only
C26	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	24% to 98%	log-odds 5.04, OR 154	242	train; test unclear (AUC gain 0.044 [-0.038, 0.127], n=61)	yes	on the training data only
C27	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	24% to 82%	log-odds 2.72, OR 15.1	115, 110	train; test disagrees (AUC gain -0.013 [-0.058, 0.026], n=61)	yes	on the training data only
C28	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	14% to 44%	log-odds 1.60, OR 4.97	242	train; test disagrees (AUC gain -0.019 [-0.049, 0.002], n=61)	yes	on the training data only
C29	magnitude: resting blood pressure: 110 mm Hg to 155 mm Hg, typical patient *shape: steady; 100% of patients same; verified*	14% to 41%	log-odds 1.48, OR 4.37	242	train; test unclear (AUC gain 0.002 [-0.018, 0.023], n=61)	yes	on the training data only
C30	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	24% to 48%	log-odds 1.09, OR 2.98	135, 92	train; test unclear (AUC gain 0.005 [-0.028, 0.039], n=61)	yes	on the training data only
C31	structure: made of fixed_defect_normal_sugar_gate, no_asymptomatic_pain_indicator, downsloping_st_indicator, thalach_floor_relay and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C32	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (through this part) *shape: categorical; highest to lowest: fasting blood sugar above 120 mg/dL, fasting blood sugar of 120 mg/dL or less*	24% to 94% (typical patient this part acts on)	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.00	209, 33	train; test disagrees (AUC gain -0.004 [-0.017, 0.000], n=61)	yes	on the training data only
C33	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	24% to 1%	log-odds -3.38, OR 0.03; range across patients -3.92 to 0.00	227, 15 (small n)	train; test disagrees (AUC gain -0.016 [-0.044, 0.001], n=61)	yes	the model has little data on this; on the training data only
C34	interaction: effect of thallium stress test result (thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan) by fasting blood sugar: fasting blood sugar above 120 mg/dL minus fasting blood sugar of 120 mg/dL or less; stronger in fasting blood sugar of 120 mg/dL or less *shape: categorical; highest to lowest: thallium stress test result other than a fixed defect on the thallium scan, a fixed defect on the thallium scan*	fasting blood sugar above 120 mg/dL: 7% to 18%; fasting blood sugar of 120 mg/dL or less: 24% to 2%	log-odds difference 3.92, OR 50.3	33, 209, 15 (small n)	train; test agrees (AUC gain -0.004 [-0.017, 0.000], n=61); group difference on test 3.92 [3.92, 3.92]	yes	the model has little data on this
C35	subgroup: who this part acts on: 10 of 242 patients	94% to 24% (typical patient this part acts	share 0.04; log-odds -3.92, OR 0.02; range	10, 232 (small	train; test disagrees (AUC gain -0.016	yes	the model has little

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than no chest pain (asymptomatic) (through this part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), chest pain type other than no chest pain (asymptomatic)*	24% to 12%	log-odds -0.85, OR 0.43; range across patients -0.85 to -0.85	108, 134	train; test unclear (AUC gain 0.017 [-0.004, 0.043], n=61)	yes	on the training data only
C38	subgroup: who this part acts on: 134 of 242 patients	2% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.55; log-odds -0.85, OR 0.43; range across patients -0.85 to -0.85	134, 108	train; test unclear (AUC gain 0.017 [-0.004, 0.043], n=61)	yes	slightly; on the training data only
C39	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C40	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a downsloping ST segment to a downsloping ST segment (through this part) *shape: categorical; highest to lowest: a downsloping ST segment, slope of the ST segment at peak exercise other than a downsloping ST segment*	24% to 29%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	225, 17 (small n)	train; test unclear (AUC gain 0.001 [0.000, 0.006], n=61)	yes	the model has little data on this; on the training data only
C41	subgroup: who this part acts on: 17 of 242 patients	66% to 73% (typical patient this part acts on (risk without the part, then with it))	share 0.07; log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	17, 225 (small n)	train; test unclear (AUC gain 0.001 [0.000, 0.006], n=61)	yes	the model has little data on this; on the training data only
C42	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C43	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23%	log-odds -0.09, OR 0.92; range across patients -0.09 to -0.09	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C44	threshold: maximum heart rate: fold at about 131 beats per minute (starts) *shape: hinge at 131 beats per minute; 100% of patients same; verified; one point with C44 (median fold 131 beats per minute, typical patient 130 beats per minute, 10th-90th percentile 131 beats per minute to 131 beats per minute)*	24% to 23% (typical patient with a fold)	fold 131 beats per minute (IQR 131 to 131); past it log-odds -0.10, OR 0.90; range across patients -0.10 to -0.10	242, 190	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C45	subgroup: who this part acts on: 190 of 242 patients	21% to 20% (typical patient this part acts on (risk without the part, then with it))	share 0.79; log-odds -0.06, OR 0.95; range across patients -0.09 to -0.02	190, 52	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C46	null: this part barely changes predictions (switched off)		max change 0.026 (max |change in predicted risk|)	242	train	no	
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model.

- C14: direction claim about thallium stress test result in the whole model: not supported by the model (14% of patients have the opposite sign (allowed: 10%)).

- C46: null claim about maximum heart rate reached during the exercise test in part ann_34: not supported by the model (predictions move by up to 0.026).

10.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (6):** `-12` (restecg:0), `-14` (restecg:2), `-16` (exang:0), `-24` (thal:4), `-26` (thal:7), `-5` (cp:2).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

10.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 10.6.1)
- ▷ A2: fixed_defect_normal_sugar_gate (leaf, section 10.6.2)
- ▷ A3: no_asymptomatic_pain_indicator (leaf, section 10.6.3)
- ▷ A4: downsloping_st_indicator (leaf, section 10.6.4)
- ▷ A5: thalach_floor_relay (leaf, section 10.6.5)

10.4 Network figures

10.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 3.33×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-10` into `-10_a`, `-10_b`.
- #27: split `-11` into `-11_a`, `-11_b`.
- #28: split `-15` into `-15_a`, `-15_b`.
- #29: split `-21` into `-21_a`, `-21_b`.
- #30: split `-7` into `-7_a`, `-7_b`.

10.6 Annotations

10.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 10.6.2), A3 (section 10.6.3), A4 (section 10.6.4), A5 (section 10.6.5).

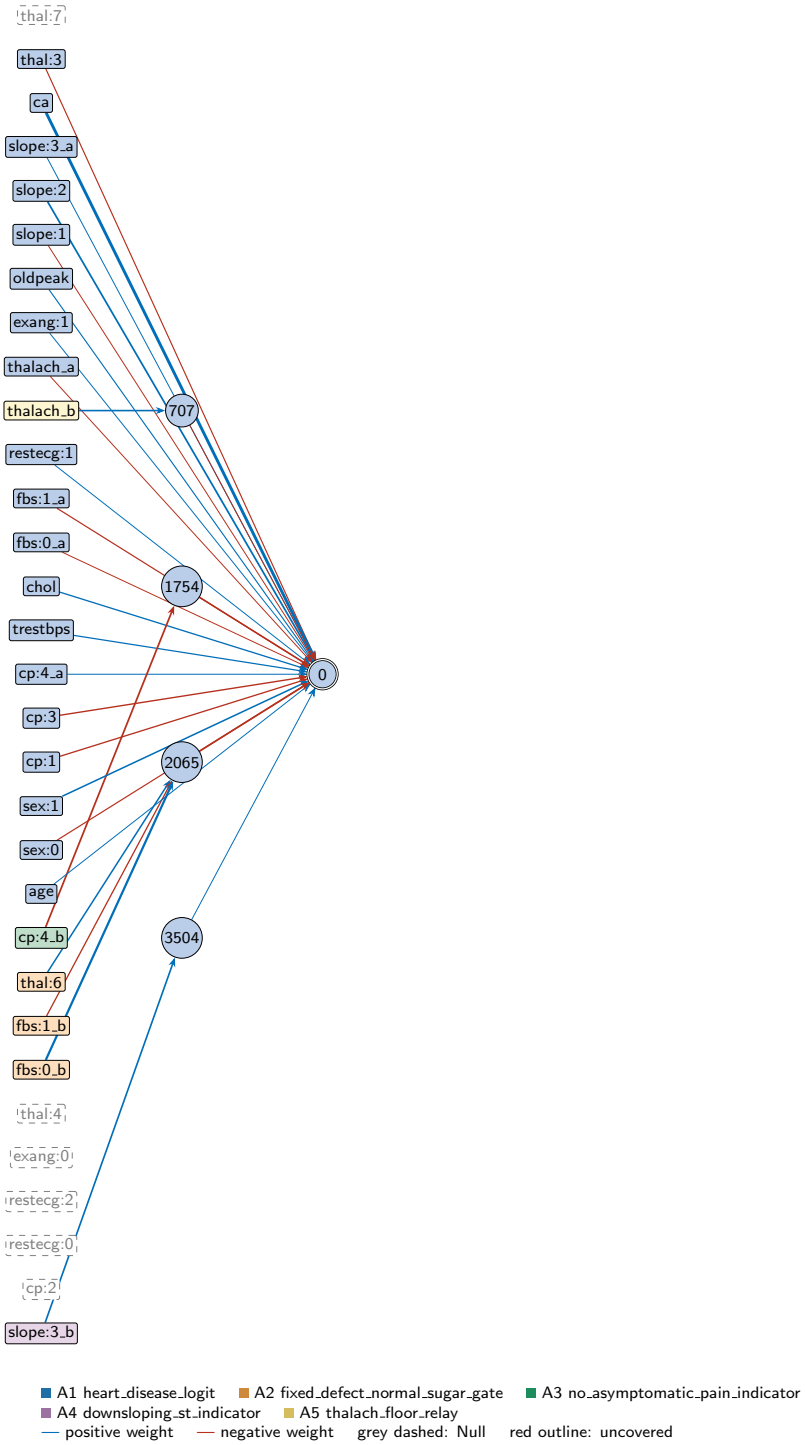


Figure 30: heart-s4-c1: full view of the explained model state.

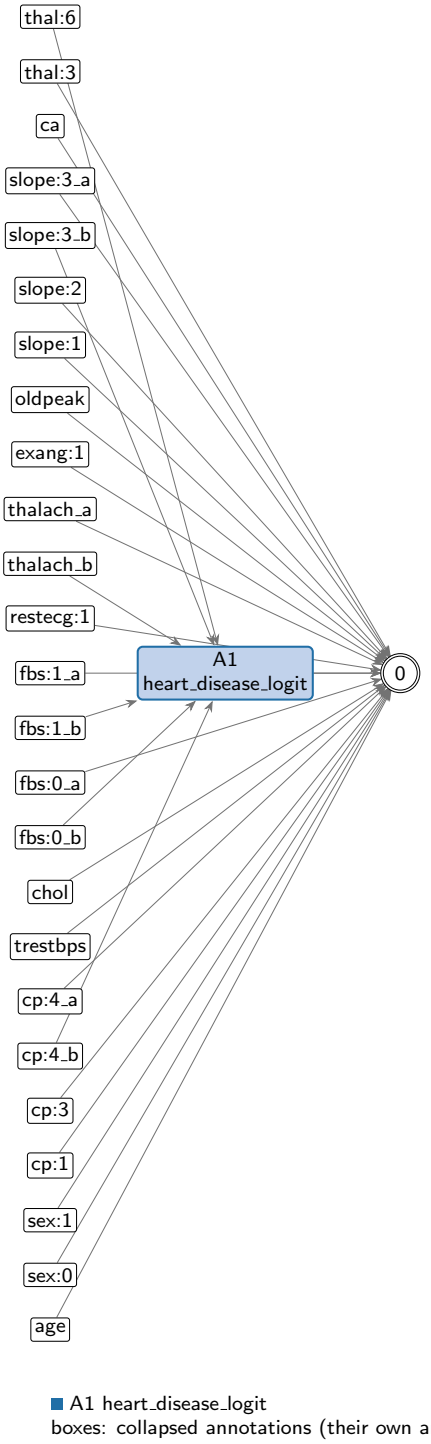


Figure 31: heart-s4-c1: collapsed view at hierarchy level 0 (fully recomposed).

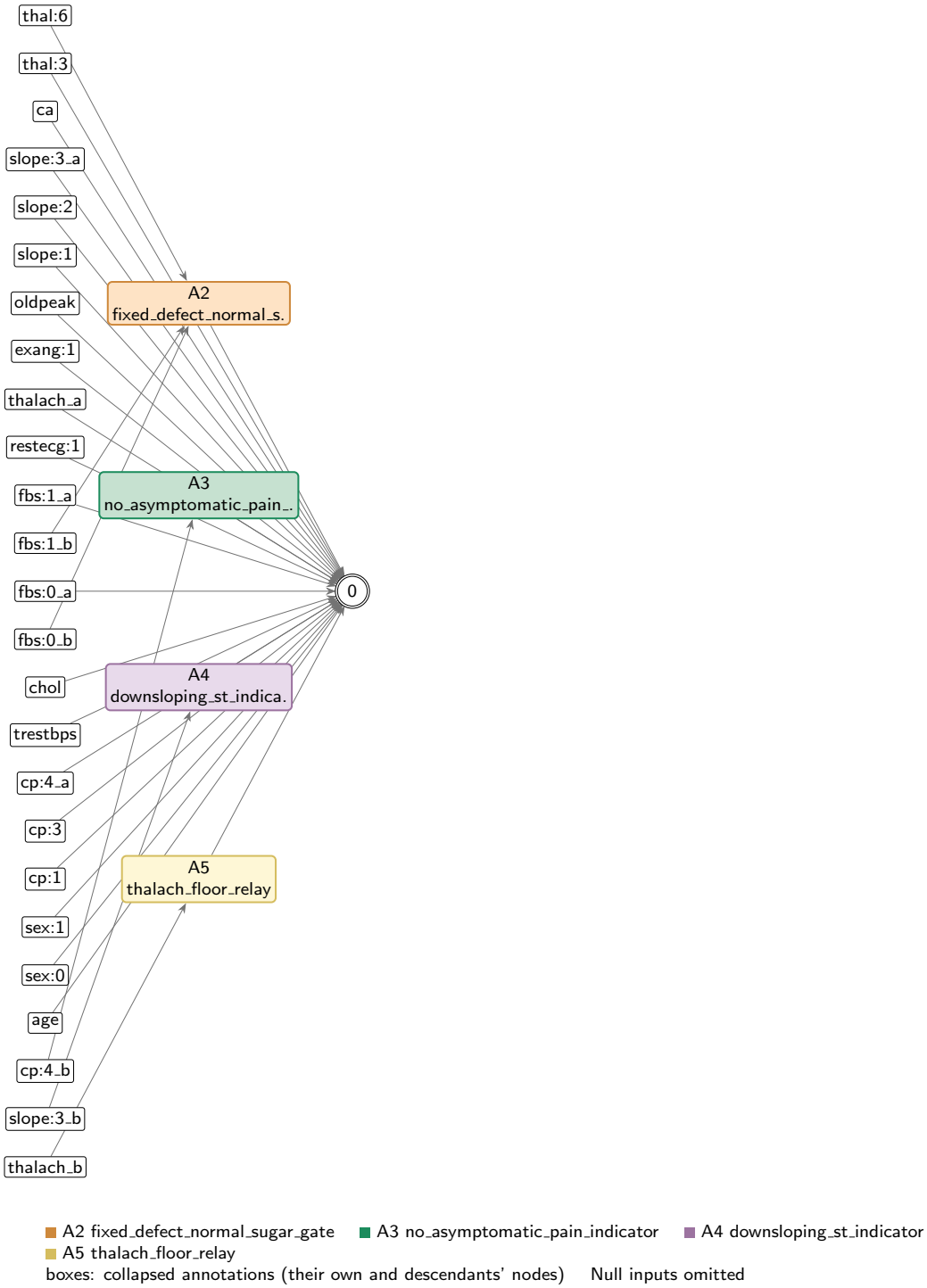


Figure 32: heart-s4-c1: collapsed view at hierarchy level 1.

- **Entry nodes:** -1 (age), -2 (sex:0), -3 (sex:1), -4 (cp:1), -6 (cp:3), -8 (trestbps), -9 (chol), -13 (restecg:1), -17 (exang:1), -18 (oldpeak), -19 (slope:1), -20 (slope:2), -22 (ca), -23 (thal:3), -10_a (fbs:0_a), -11_a (fbs:1_a), -15_a (thalach_a), -21_a (slope:3_a), -7_a (cp:4_a), -26 (thal:7), -24 (thal:4), -16 (exang:0), -14 (restecg:2), -12 (restecg:0), -5 (cp:2), -10_b (fbs:0_b), -11_b (fbs:1_b), -25 (thal:6), -7_b (cp:4_b), -21_b (slope:3_b), -15_b (thalach_b).
- **Exit nodes:** 0.
- **Internal nodes:** none.
- **Edges (weight):** -1→0 (0.0855), -2→0 (-0.559), -3→0 (0.899), -4→0 (-0.679), -6→0 (-0.749), -8→0 (0.596), -9→0 (0.671), -13→0 (0.0618), -17→0 (0.282), -18→0 (0.491), -19→0 (-0.238), -20→0 (1.11), -22→0 (2.25), -23→0 (-0.541), -10_a→0 (-0.0936), -11_a→0 (-0.583), -15_a→0 (-0.125), -21_a→0 (0.0688), -7_a→0 (0.308), 2065→0 (-1.08), 1754→0 (-0.899), 3504→0 (0.114), 707→0 (-0.0469).
- **With its descendants:** 36 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 35 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = 0.466 + LINEAR CORE - 1.077*fixed_defect_normal_sugar_gate - 0.899*no_asymptomatic_pain_indicator + 0.114*downsloping_st_indicator - 0.047*thalach_floor_relay. LINEAR CORE (19 direct input->output edges, z-scored inputs): ca +2.246, slope:2 +1.115, sex:1 +0.899, cp:3 -0.749, cp:1 -0.679, chol +0.671, trestbps +0.596, fbs:1 -0.583, sex:0 -0.559, thal:3 -0.541, oldpeak +0.491, cp:4 +0.308, exang:1 +0.282, slope:1 -0.238, thalach -0.125, fbs:0 -0.094, age +0.085, slope:3 +0.069, restecg:1 +0.062. Six inputs are unconnected (cp:2, restecg:0, restecg:2, exang:0, thal:4, thal:7; zero-effect entries of this root). thal:6 has no direct edge; it acts only through the fixed-defect gate. COMPOSITION: purely additive in the logit. Children: (1) fixed_defect_normal_sugar_gate, the one genuine interaction: an AND of fixed thallium defect and normal fasting sugar that lowers the logit by 3.92 for 10 patients; (2) no_asymptomatic_pain_indicator and (3) downsloping_st_indicator, binary ReLUs exactly affine on their one-hot inputs (reparameterisations of the cp:4 and slope:3 linear terms; cp:4 total +1.47 logit, slope:3 total +0.57); (4) thalach_floor_relay, a near-inert hinge on max heart rate. The children share no hidden nodes; they are coupled only through inputs that also have direct edges (fbs:0, fbs:1, cp:4, slope:3, thalach). EVIDENCE (train, n=242): full model AUC 0.9423, acc 0.888 (compute_performance). Linear core alone (direct edges only) AUC 0.9328; logit SD 4.55 (core) vs 0.86 (all hidden readouts). Zero-ablating one child at a time: fixed-defect gate 0.9343, cp:4 indicator 0.9411, thalach relay 0.9416, slope:3 indicator 0.9421. Replacing all hidden readouts with one least-squares linear fit in their six inputs gives 0.9371; an L2 logistic regression on all 26 inputs gives 0.9355. Mean |SHAP| (train, all 242, probability scale): ca 0.172, slope:2 0.099, sex:1 0.069, cp:4 0.068, cp:3 0.056, thal:3 0.049, sex:0 0.042, chol 0.042. Interpretation: essentially logistic regression with one meaningful nonlinearity, the fixed-defect/normal-sugar gate, which cancels the core's over-prediction for that small group (10 patients). Counter-directional signs reported, not explained: fbs:1 -0.58 (high fasting sugar lowers the logit) and the gate itself making a fixed defect protective for normal-sugar patients. Same mechanism family as the 10-seed heart-s4 record (key 9650, gen 38): the same gate and linear core (ca +2.16 there), but here a cp:4 indicator and a slope:3 indicator replace that genome's cp:2 indicator, and the thalach relay feeds the output directly.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= 0.466 + 0.0688 x_{\text{slope:3_a}} + 0.282 x_{\text{exang:1}} + 0.308 x_{\text{cp:4_a}} \\
&\quad + 0.899 x_{\text{sex:1}} + 0.491 x_{\text{oldpeak}} \\
&\quad + 0.114\,\text{downsloping_st_indicator}\big(x_{\text{slope:3_b}}\big) + 1.11 x_{\text{slope:2}} \\
&\quad + 2.25 x_{\text{ca}} + 0.596 x_{\text{trestbps}} + 0.671 x_{\text{chol}} + 0.0855 x_{\text{age}} + 0.0618 x_{\text{restecg:1}} \\
&\quad - 0.899\,\text{no_asymptomatic_pain_indicator}\big(x_{\text{cp:4_b}}\big)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.542x_{thal:3} - 0.583x_{fbs:1_a} - 0.0936x_{fbs:0_a} - 0.749x_{cp:3} \\
&- 0.0469\,\text{thalach_floor_relay}\big(x_{thalach_b}\big) - 0.238x_{slope:1} \\
&- 1.08\,\text{fixed_defect_normal_sugar_gate}\big(x_{fbs:0_b}, x_{fbs:1_b}, \\
&x_{thal:6}\big) - 0.559x_{sex:0} - 0.679x_{cp:1} - 0.125x_{thalach_a}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= 0.466 + 0.0688x_{slope:3_a} + 0.282x_{exang:1} + 0.308x_{cp:4_a} \\
&\quad + 0.899x_{sex:1} + 0.491x_{oldpeak} + 0.114\,\text{relu}\big(-1.02 \\
&\quad + 1.01x_{slope:3_b}\big) + 1.11x_{slope:2} + 2.25x_{ca} + 0.596x_{trestbps} \\
&\quad + 0.671x_{chol} + 0.0855x_{age} + 0.0618x_{restecg:1} \\
&\quad - 0.899\,\text{relu}\big(-0.0387 - 1.1x_{cp:4_b}\big) - 0.542x_{thal:3} \\
&\quad - 0.583x_{fbs:1_a} - 0.0936x_{fbs:0_a} - 0.749x_{cp:3} \\
&\quad - 0.0469\,\text{relu}\big(0.812 + 0.959x_{thalach_b}\big) - 0.238x_{slope:1} \\
&\quad - 1.08\,\text{relu}\big(-0.703 + 1.58x_{fbs:0_b} + 0.885x_{thal:6} \\
&\quad - 0.683x_{fbs:1_b}\big) - 0.559x_{sex:0} - 0.679x_{cp:1} - 0.125x_{thalach_a}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00419, and the exact function by at most 2.78×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=true); output is sigmoid.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, ablations, linear replacements and SHAP.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

10.6.2 A2: fixed_defect_normal_sugar_gate

Kind: Leaf annotation (depth 1); parent A1 (section 10.6.1).

- **Entry nodes:** `-10_b` (fbs:0_b), `-11_b` (fbs:1_b), `-25` (thal:6).
- **Exit nodes:** `2065`.
- **Internal nodes:** none.
- **Edges (weight):** `-10_b`→`2065` (1.58), `-11_b`→`2065` (-0.683), `-25`→`2065` (0.885).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. Fixed-defect AND normal-sugar gate (lowers risk). 2065 = ReLU(-0.703 + 1.576*fbs:0 - 0.683*fbs:1 + 0.885*thal:6), read out -1.077. Hypothesis: a discrete AND detector that is on only for patients with a fixed thallium defect (thal 6) and fasting blood sugar <= 120 mg/dL, and then lowers their logit by 3.92, cancelling the linear core's over-calling of this group (thal:6 has no direct edge; the core gives these patients a mean predicted probability of 0.83 while their disease rate is 50%). EVIDENCE (train, n=242): exact on-data reduction 2065 = 3.639*[thal:6=1 AND fbs=0] (fbs=1 drives the pre-activation to <= -6.39). Active for exactly the 10 patients in that group; disease rate 5/10. Zero-ablating it lowers train AUC 0.9423 -> 0.9343, the largest nonlinear effect; a linear refit in its three inputs recovers only to 0.937 (R2 0.69), because an AND cannot be written additively. Mean |sensitivity| fbs:0 0.065 > thal:6 0.037 > fbs:1 0.028 (small because the

gate is off for 96%). Supported, but the correction rests on 10 training patients and may be partly noise-fitted. Same gate as the 10-seed heart-s4 record (gen 38), now also fed by fbs:0.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.703 + 1.58x_{fbs:0_b} + 0.885x_{thal:6} - 0.683x_{fbs:1_b}\big)$$

where y_0 is node `2065`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 4.34×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

10.6.3 A3: no_asymptomatic_pain_indicator

Kind: Leaf annotation (depth 1); parent A1 (section 10.6.1).

- **Entry nodes:** `-7_b` (cp:4_b).
- **Exit nodes:** `1754`.
- **Internal nodes:** none.
- **Edges (weight):** `-7_b`→`1754` (-1.1).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Not-asymptomatic chest-pain indicator (lowers risk). 1754 = ReLU(-0.039 - 1.098*cp:4), read out -0.899. Hypothesis: a 0/1 indicator for chest pain other than asymptomatic (typical, atypical or non-anginal pain); because cp:4 is binary it is exactly affine on the data, so it only adds to the cp:4 direct edge (+0.308 per z-unit): together, asymptomatic chest pain raises the logit by 1.47 relative to the other pain types. EVIDENCE (train, n=242): on for exactly the 134 patients without asymptomatic pain (value 0.947, logit -0.852); disease rate 22% when on vs 71% when off. A linear replacement in cp:4 reproduces it exactly (R2 1.00, AUC unchanged 0.9423); zero-ablation (which also removes cp:4's main effect) gives 0.9411. Supported: a re-parameterisation of a linear cp:4 term, no interaction.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.0387 - 1.1x_{cp:4_b}\big)$$

where y_0 is node `1754`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00173, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

10.6.4 A4: downsloping_st_indicator

Kind: Leaf annotation (depth 1); parent A1 (section 10.6.1).

- **Entry nodes:** `-21_b` (slope:3_b).
- **Exit nodes:** `3504`.
- **Internal nodes:** none.
- **Edges (weight):** `-21_b`→`3504` (1.01).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Downsloping-ST indicator (raises risk slightly). 3504 = ReLU(-1.020 + 1.010*slope:3), read out +0.114. Hypothesis: a 0/1 indicator for a downsloping peak-exercise ST segment; exactly affine on the binary input, adding +0.30 logit to slope:3's direct edge (total +0.57 logit for downsloping vs not). EVIDENCE (train, n=242): on for exactly the 17 downsloping patients; disease rate 47% vs 44%. Linear replacement exact (R2 1.00); zero-ablation 0.9423 -> 0.9421. Supported but nearly inert; rests on 17 patients.

Closed form:

$$y_0 = \mathrm{relu}\big(-1.02 + 1.01 x_{slope:3_b}\big)$$

where y_0 is node `3504`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 1.95×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

10.6.5 A5: thalach_floor_relay

Kind: Leaf annotation (depth 1); parent A1 (section 10.6.1).

- **Entry nodes:** `-15_b` (thalach_b).
- **Exit nodes:** `707`.
- **Internal nodes:** none.
- **Edges (weight):** `-15_b`→`707` (0.959).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Max-heart-rate floor relay (lowers risk a little). 707 = ReLU(0.812 + 0.959*thalach), read out -0.047. Hypothesis: a hinge that adds a small extra protective slope in maximum heart rate achieved for all but the lowest ~21% of thalach values (kink at z-thalach -0.85); combined with the direct edge (-0.125 per z) the thalach effect is -0.125 per z below the kink and -0.170 per z above it. EVIDENCE (train, n=242): active for 78.5%; disease rate 37% when active vs 69% when off; readout logit SD only 0.035. Zero-ablating it lowers train AUC 0.9423 -> 0.9416; a straight-line refit gives 0.9420 (R2 0.90). Supported: a near-linear, nearly inert extension of the thalach term.

Closed form:

$$y_0 = \mathrm{relu}\big(0.812 + 0.959 x_{thalach_b}\big)$$

where y_0 is node `707`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 8.60×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula with the on-data simplification.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

10.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

11. heart-s5-c1: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.01126	0.01126	0.01125	agree
Numeric-function max abs error	n/a	1.78e-15	1.78e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	3 / 1	n/a	3 / 1	agree
Operation counts	4 splits, 2 identity	add identity node 2, annotate 4, rename node 26, split node 4	add identity node 2, annotate 4, rename node 26, split node 4	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `ischaemia_gate` (0.0113). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `ischaemia_gate`: *"above z-oldpeak -0.09 with a normal ECG only, -0.33 with a defect only, +0.58 with neither), and then raises the logit with oldpeak, directly (0.143 per unit) and through the blood-pressure gate (a further 0.183 per unit while tha..."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `age_gate_without_ischaemia`: *"...tra age term (0.125 per SD of age, on top of age's direct +0.117) that acts only while the BP gate is closed or nearly closed, i.e."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 4 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): get_input_distribution does not accept sample_fraction (called with max_samples only; all 242 rows used). get_viz_summary requires viz_type (sensitivity used). compute_shap (kernel, all 242 train rows) gives the four unconnected inputs mean |SHAP| about 0.001 instead of exactly 0 (sampling noise). The agent sandbox refused compound shell commands, so tools were called through a small Python subprocess wrapper in scratch (same xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s5-c1-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 22%; it adds up most of its inputs independently, and a chain of three parts adds to that sum: a signal from ST depression on exercise, the thallium scan and the resting ECG that acts on 116 of 242 training patients, a blood pressure step that reads that signal, and an extra age term that acts on 85 patients. [C32, C40, C51] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 22% to about 3% and fasting blood sugar above 120 mg/dL to about 11%; a higher maximum heart rate lowers it steadily, from about 28% at 118 to about 18% at 175 beats per minute; and through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 5% and atypical angina to about 11% (chest pain type also acts through the blood pressure step, below). [C12, C17,

C22, C10, C13, C20] Each extra vessel coloured by fluoroscopy adds about the same, from about 22% at 0 to about 88% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 67%, higher cholesterol raises it steadily from about 14% at 193 mg/dL to about 37% at 308 mg/dL, and chest pain brought on by exercise raises it to about 33%. [C01, C04, C07, C16] Through this sum alone, higher resting blood pressure raises it steadily, from about 14% at 110 mm Hg to about 35% at 155 mm Hg, older age raises it steadily, from about 19% at 42 to about 24% at 66 years, and a reversible defect on the thallium scan raises it to about 35%; all three also act through the parts below. [C08, C24, C15] Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a thallium result coded 4 raises the risk to about 83% and a fixed defect on the thallium scan lowers it to about 11%, and a downsloping ST segment raises it to about 27%. [C02, C18, C25] This part reads ST depression on exercise, the thallium scan and the resting ECG, and acts on 116 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 96% with it. [C40] Through it, more ST depression adds to the predicted risk only once it rises above about 0.9 mm, rising in an uneven way: for a typical patient from about 22% at 0 mm to about 48% at 2.8 mm. [C33, C39] ST depression counts more with a reversible defect on the thallium scan (from about 36% to about 80%, against about 22% to about 48% otherwise) and with a normal resting ECG (from about 22% to about 48%, against about 22% to about 34% otherwise), in an uneven way. [C36, C37] A reversible defect on the thallium scan also raises the risk more with a normal resting ECG, from about 22% to about 47% against about 22% to about 36% otherwise. [C38] This part reads that signal together with resting blood pressure and chest pain type, and barely changes the predictions: for the typical patient it acts on, atypical angina raises the predicted risk through it from about 69% to about 70%, and non-anginal chest pain leaves it at about 69%. [C42, C44] This part acts on 85 of 242 training patients, 37 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 28% without it and about 30% with it. [C51] Through it, age adds nothing until about 53 years (between 53 and 66 years for different patients), then raises the risk steadily, for a typical patient from about 21% at 42 to about 24% at 66 years. [C49, C50]

Worth checking: Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 22% to about 11%). [C17] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C11, C14, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C19] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C03] Worth checking: ST depression on exercise and a reversible thallium defect raise the predicted risk more when the resting ECG is normal than when it is not. [C37, C38]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	22% to 88%	log-odds 3.32, OR 27.6; range across patients 3.32 to 3.32	242	train; test unclear (AUC gain 0.039 [-0.026, 0.111], n=61)	yes	on the training data only
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 83%	log-odds 2.85, OR 17.3; range across patients 2.85 to 2.85	135, 2 (small n)	train; test unclear (AUC gain 0.012 [-0.009, 0.035], n=61)	yes	the model has little data on this; on the training data only
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 67%	log-odds 2.02, OR 7.53; range across patients 2.02 to 2.02	115, 110	train; test disagrees (AUC gain -0.019 [-0.070, 0.025], n=61)	yes	on the training data only
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 37%	log-odds 1.30, OR 3.68; range across patients 1.30 to 1.30	242	train; test disagrees (AUC gain -0.022 [-0.063, 0.008], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	14% to 35%	log-odds 1.19, OR 3.28; range across patients 1.19 to 1.19	242	train; test unclear (AUC gain 0.000 [-0.024, 0.022], n=61)	yes	on the training data only
C10	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 2%	log-odds -2.48, OR 0.08; range across patients -2.48 to -2.48	108, 22	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C12	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	22% to 3%	log-odds -2.26, OR 0.10; range across patients -2.26 to -2.26	159, 83	train; test unclear (AUC gain 0.024 [-0.008, 0.064], n=61)	yes	on the training data only
C13	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 5%	log-odds -1.73, OR 0.18; range across patients -1.73 to -1.73	108, 70	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C15	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 35%	log-odds 0.67, OR 1.95; range across patients 0.67 to 0.67	135, 90	train; test unclear (AUC gain 0.012 [-0.009, 0.035], n=61)	yes	on the training data only
C16	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	22% to 33%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	160, 82	train; test unclear (AUC gain 0.012 [-0.003, 0.032], n=61)	yes	on the training data only
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	22% to 11%	log-odds -0.81, OR 0.44; range across patients -0.81 to -0.81	209, 33	train; test disagrees (AUC gain -0.002 [-0.020, 0.015], n=61)	yes	on the training data only
C18	direction: thallium stress	22% to 11%	log-odds -0.77, OR	135, 15	train; test unclear	yes	the model

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 11%	log-odds -0.76, OR 0.47; range across patients -0.76 to -0.76	108, 42	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C22	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	27% to 18%	log-odds -0.53, OR 0.59; range across patients -0.53 to -0.53	242	train; test unclear (AUC gain 0.006 [0.000, 0.022], n=61)	yes	on the training data only
C24	direction: age: 42 years to 66 years (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	19% to 24%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	242	train; test disagrees (AUC gain -0.004 [-0.018, 0.012], n=61)	yes	on the training data only
C25	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 27%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	115, 17 (small n)	train; test disagrees (AUC gain -0.019 [-0.070, 0.025], n=61)	yes	the model has little data on this; on the training data only
C32	structure: made of ischaemia_gate, bp_offset_gate, age_gate_without_ischaemia and inputs age, sex, cp, trestbps, chol, fbs, thalach, exang, slope, ca, thal; added; ann_32 feeds ann_33; ann_33 feeds ann_34		log-odds n/a, OR n/a	242	train	yes	
C33	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 48% of patients same; fallback*	22% to 48%	log-odds 1.16, OR 3.20; range across patients 0.55 to 1.91	242	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result other than a reversible defect on the thallium scan; stronger in a reversible defect on the thallium scan *shape: steady; 37% of patients same; fallback*	a reversible defect on the thallium scan: 36% to 80%; thallium stress test result other than a reversible defect on the thallium scan: 22% to 48%	log-odds difference 0.68, OR 1.98	90, 152	train; test agrees (AUC gain 0.005 [-0.009, 0.025], n=61); group difference on test 0.73 [0.59, 0.86]	yes	clearly
C37	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: accelerating; 33% of patients same; fallback*	resting ECG other than a normal resting ECG: 22% to 34%; a normal resting ECG: 22% to 48%	log-odds difference -0.59, OR 0.55	121, 121	train; test agrees (AUC gain -0.002 [-0.015, 0.009], n=61); group difference on test -0.59 [-0.76, -0.40]	yes	clearly
C38	interaction: effect of thallium stress test result (thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	resting ECG other than a normal resting ECG: 22% to 36%; a normal resting ECG: 22% to 47%	log-odds difference -0.11, OR 0.90	121, 121	train; test agrees (AUC gain -0.002 [-0.015, 0.009], n=61); group difference on test -0.18 [-0.34, -0.03]	yes	clearly
C39	threshold: ST depression on exercise: fold at about 0.9 mm (starts) *shape: accelerating; 55% of patients same; fallback*	22% to 46% (typical patient with a fold)	fold 0.93 mm (IQR 0.93 to 1.72); past it log-odds 1.50, OR 4.47; range across patients 1.00 to 1.86	198, 72	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C40	subgroup: who this part acts on: 116 of 242 patients	89% to 96% (typical patient this part acts on (risk without the part, then with it))	share 0.48; log-odds 0.77, OR 2.17; range across patients 0.08 to 1.75	116, 126	train; test unclear (AUC gain 0.016 [-0.005, 0.048], n=61)	yes	on the training data only
C42	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to atypical angina (through	69% to 70% (typical patient this part acts on)	log-odds 0.04, OR 1.04; range across patients 0.00 to 0.08	130, 42	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C44	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: atypical angina, non-anginal chest pain, chest pain type other than atypical angina, non-anginal chest pain*	69% to 69% (typical patient this part acts on)	log-odds 0.003, OR 1.00; range across patients 0.00 to 0.006	130, 70	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C49	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	21% to 24%	log-odds 0.10, OR 1.11; range across patients 0.00 to 0.18	242	train; test unclear (AUC gain 0.000 [-0.01, 0.011], n=61)	yes	slightly; on the training data only
C50	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	20% to 24% (typical patient with a fold)	fold 52.7 years (IQR 52.7 to 58.6); past it log-odds 0.17, OR 1.18; range across patients 0.03 to 0.21	174, 85	train; test unclear (AUC gain 0.000 [-0.01, 0.011], n=61)	yes	slightly; on the training data only
C51	subgroup: who this part acts on: 85 of 242 patients	28% to 30% (typical patient this part acts on (risk without the part, then with it))	share 0.35; log-odds 0.10, OR 1.11; range across patients 0.01 to 0.20	85, 157	train; test disagrees (AUC gain -0.002 [-0.01, 0.004], n=61)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s5-c1 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s5-c1.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s5-c1 --verified --minutes N --changes "..." --notes "..."`

11 Explanation certificate: heart-s5-c1

This section is the explanation certificate of the heart-s5-c1 network. It is rendered from the canonical JSON record `heart-s5-c1.json` (schema 1.4.0, SHA-256 `e77e32e67561b5c4`...), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 11 summarises the network and its coverage.

Drafter's summary. Mostly logistic regression: 19 direct edges (logit SD 3.14; train AUC 0.934 alone) plus one serial ischaemia chain (logit SD 0.67) to 0.942. An ischaemia gate ReLU(0.25 + 2.59 oldpeak + 0.87 restecg:0 + 1.15 thal:7) opens above an ST-depression threshold set by a normal resting ECG and a reversible defect and adds risk directly; it feeds a gate that passes the score on minus a resting-blood-pressure offset (more risk when BP is low), which in turn switches off a small

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s5, seed 5
Genome	`4f9f15fb-446b-4560-9704-c7b1a83617b5` (NEAT key 9259, generation 39)
Explanation	`39c7fe21-518b-4517-8765-7e6008dc7d98` (36 operations)
Train AUC / accuracy	0.9418 / 0.8719 (n = 242)
Test AUC / accuracy	0.8869 / 0.8033 (n = 61)
Output activation	sigmoid
Phenotype	30 nodes (26 inputs, 3 hidden, 1 output), 31 enabled edges
Explained model state	36 nodes, 33 enabled edges
Refactoring operations	add_identity_node 2, split_node 4
Function preservation	max $\|\Delta \text{output}\| = 2.22 \times 10^{-16}$ on the training split
Annotations	4 (3 leaf, 1 composition)
Structural coverage	1 (35 of 35 candidate nodes; 4 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`e77e32e67561b5c4...`

Table 11: Explanation certificate summary: heart-s5-c1.

extra age term. Oldpeak and restecg:0 act only through this chain. Same design as the 10-seed heart-s5 record (generation 37, key 9330), with three hidden nodes instead of six.

What resisted explanation. Nothing resisted coverage. The serial chain 204 -> 290 -> 1987 (each node also read out to the output) was made into three leaves by intercepting the two chain links with identity nodes (identity_290, identity_1987) that become entries of the downstream leaves; the root composes them. Counter-directional signs reported, not explained: restecg:0 (normal ECG) raises the ischaemia score, trestbps enters the risk-raising BP gate negatively (low resting BP opens it), fbs:0 +0.28 directly. The age gate barely matters (readout ablation -0.001 AUC).

11.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 22%; it adds up most of its inputs independently, and a chain of three parts adds to that sum: a signal from ST depression on exercise, the thallium scan and the resting ECG that acts on 116 of 242 training patients, a blood pressure step that reads that signal, and an extra age term that acts on 85 patients. [C32, C40, C51] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 22% to about 3% and fasting blood sugar above 120 mg/dL to about 11%; a higher maximum heart rate lowers it steadily, from about 28% at 118 to about 18% at 175 beats per minute; and through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 5% and atypical angina to about 11% (chest pain type also acts through the blood pressure step, below). [C12, C17, C22, C10, C13, C20] Each extra vessel coloured by fluoroscopy adds about the same, from about 22% at 0 to about 88% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 67%, higher cholesterol raises it steadily from about 14% at 193 mg/dL to about 37% at

308 mg/dL, and chest pain brought on by exercise raises it to about 33%. [C01, C04, C07, C16] Through this sum alone, higher resting blood pressure raises it steadily, from about 14% at 110 mm Hg to about 35% at 155 mm Hg, older age raises it steadily, from about 19% at 42 to about 24% at 66 years, and a reversible defect on the thallium scan raises it to about 35%; all three also act through the parts below. [C08, C24, C15] Some rarer results also count (the model has little data on this, n = 2 to 17): through this sum alone a thallium result coded 4 raises the risk to about 83% and a fixed defect on the thallium scan lowers it to about 11%, and a downsloping ST segment raises it to about 27%. [C02, C18, C25] This part reads ST depression on exercise, the thallium scan and the resting ECG, and acts on 116 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 96% with it. [C40] Through it, more ST depression adds to the predicted risk only once it rises above about 0.9 mm, rising in an uneven way: for a typical patient from about 22% at 0 mm to about 48% at 2.8 mm. [C33, C39] ST depression counts more with a reversible defect on the thallium scan (from about 36% to about 80%, against about 22% to about 48% otherwise) and with a normal resting ECG (from about 22% to about 48%, against about 22% to about 34% otherwise), in an uneven way. [C36, C37] A reversible defect on the thallium scan also raises the risk more with a normal resting ECG, from about 22% to about 47% against about 22% to about 36% otherwise. [C38] This part reads that signal together with resting blood pressure and chest pain type, and barely changes the predictions: for the typical patient it acts on, atypical angina raises the predicted risk through it from about 69% to about 70%, and non-anginal chest pain leaves it at about 69%. [C42, C44] This part acts on 85 of 242 training patients, 37 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 28% without it and about 30% with it. [C51] Through it, age adds nothing until about 53 years (between 53 and 66 years for different patients), then raises the risk steadily, for a typical patient from about 21% at 42 to about 24% at 66 years. [C49, C50]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and a chain of three parts adds to that sum: a signal from ST depression and the thallium scan, a blood pressure step that reads it, and an extra age term after that. [C32, C01, C04, C12, C10] *Evidence.* It is made up of the inputs that feed the score directly and three parts, the ST depression and thallium signal, the blood pressure step and the extra age term (each explained below); the first part feeds the second and the second feeds the third, and the score is their sum, which was checked on all 242 training patients. [C32] ST depression on exercise and the resting ECG do not feed the score directly, so they act only through these parts. [C32, C33] Changing one input at a time across the whole model, more ST depression raises a typical patient's predicted risk in an uneven way, from about 22% at 0 mm to about 48% at 2.8 mm, and a reversible defect on the thallium scan takes it to about 47%, against about 35% through the sum alone. [C05, C06, C15] Across the whole model, higher resting blood pressure raises it in an uneven way, from about 14% at 110 mm Hg to about 35% at 155 mm Hg, and older age raises it in an uneven way, from about 18% at 42 to about 27% at 66 years. [C09, C23] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15), and a resting ECG other than normal barely changes it for a typical patient. [C19, C26]

ST depression and thallium signal. This part raises the predicted risk with ST depression on exercise once it rises above about 0.9 mm, in an uneven way, and more so with a reversible defect on the thallium scan or a normal resting ECG. [C33, C39, C36, C37] *Evidence.* It acts on 116 of 242 training patients, 77 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 89% without it and about 96% with it. [C40] Through it, more ST depression adds to the risk only once it rises above about 0.9 mm, rising in an uneven way: for a typical patient from about 22% at 0 mm to about 48% at 2.8 mm. [C33, C39] ST depression counts more with a reversible defect on the thallium scan (from about 36% to about 80%, against about 22% to about 48% otherwise) and with a normal resting ECG (from about 22% to about 48%, against about 22% to about 34% otherwise), in an uneven way. [C36, C37] Through it, a reversible defect on the thallium scan raises

a typical patient's risk from about 22% to about 32%, and more with a normal resting ECG (to about 47%, against about 36% otherwise). [C34, C38] Its formula reproduces the part exactly on all 242 training patients. [C41]

Blood pressure step. This part reads the ST depression and thallium signal together with resting blood pressure and chest pain type, and barely changes the predictions for the typical patient it acts on. [C42, C44] *Evidence.* For the typical patient it acts on, atypical angina raises the predicted risk through it from about 69% to about 70%, and non-anginal chest pain leaves it at about 69%. [C42, C44] Its formula reproduces the part exactly on all 242 training patients. [C48]

Extra age term. This part adds a little to the predicted risk for older patients: nothing until about 53 years, then steadily. [C49, C50] *Evidence.* It acts on 85 of 242 training patients, 37 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 28% without it and about 30% with it. [C51] Through it, age adds nothing until about 53 years (between 53 and 66 years for different patients), then raises the risk steadily, for a typical patient from about 21% at 42 to about 24% at 66 years. [C49, C50] Its formula reproduces the part exactly on all 242 training patients. [C52]

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 22% to about 11%). [C17] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C11, C14, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C19] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C03] Worth checking: ST depression on exercise and a reversible thallium defect raise the predicted risk more when the resting ECG is normal than when it is not. [C37, C38]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	22% to 88%	log-odds 3.32, OR 27.6; range across patients 3.32 to 3.32	242	train; test unclear (AUC gain 0.039 [-0.026, 0.111], n=61)	yes	on the training data only
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 83%	log-odds 2.85, OR 17.3; range across patients 2.85 to 2.85	135, 2 (small n)	train; test unclear (AUC gain 0.012 [-0.009, 0.035], n=61)	yes	the model has little data on this; on the training data only
C03	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 83%	log-odds 2.85, OR 17.3; range across patients 2.85 to 2.85	135, 2 (small n)	train; test unclear (AUC gain 0.017 [-0.010, 0.052], n=61)	yes	the model has little data on this; on the training data only
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 67%	log-odds 2.02, OR 7.53; range across patients 2.02 to 2.02	115, 110	train; test disagrees (AUC gain -0.019 [-0.070, 0.025], n=61)	yes	on the training data only
C05	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: accelerating; 48% of patients same; fallback*	22% to 48%	log-odds 1.16, OR 3.20; range across patients 0.55 to 1.91	242	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C06	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 47%	log-odds 1.01, OR 2.73; range across patients 0.67 to 1.44	135, 90	train; test unclear (AUC gain 0.017 [-0.010, 0.052], n=61)	yes	on the training data only
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	14% to 37%	log-odds 1.30, OR 3.68; range across patients 1.30 to 1.30	242	train; test disagrees (AUC gain -0.022 [-0.063, 0.008], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	14% to 35%	log-odds 1.19, OR 3.28; range across patients 1.19 to 1.19	242	train; test unclear (AUC gain 0.000 [-0.024, 0.022], n=61)	yes	on the training data only
C09	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 43% of patients same; fallback*	14% to 35%	log-odds 1.04, OR 2.82; range across patients 0.84 to 1.21	242	train; test unclear (AUC gain 0.001 [-0.022, 0.023], n=61)	yes	on the training data only
C10	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 2%	log-odds -2.48, OR 0.08; range across patients -2.48 to -2.48	108, 22	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C11	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 2%	log-odds -2.48, OR 0.08; range across patients -2.48 to -2.48	108, 22	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C12	direction: sex: male to female (whole model) *shape: categorical; highest*	22% to 3%	log-odds -2.26, OR 0.10; range across patients -2.26 to -2.26	159, 83	train; test unclear (AUC gain 0.024 [-0.008, 0.064], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 5%	log-odds -1.73, OR 0.18; range across patients -1.73 to -1.73	108, 70	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C14	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 5%	log-odds -1.73, OR 0.18; range across patients -1.73 to -1.72	108, 70	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C15	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 35%	log-odds 0.67, OR 1.95; range across patients 0.67 to 0.67	135, 90	train; test unclear (AUC gain 0.012 [-0.009, 0.035], n=61)	yes	on the training data only
C16	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	22% to 33%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	160, 82	train; test unclear (AUC gain 0.012 [-0.003, 0.032], n=61)	yes	on the training data only
C17	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	22% to 11%	log-odds -0.81, OR 0.44; range across patients -0.81 to -0.81	209, 33	train; test disagrees (AUC gain -0.002 [-0.020, 0.015], n=61)	yes	on the training data only
C18	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 11%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	135, 15 (small n)	train; test unclear (AUC gain 0.012 [-0.009, 0.035], n=61)	yes	the model has little data on this; on the training data only
C19	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 11%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	135, 15 (small n)	train; test unclear (AUC gain 0.017 [-0.010, 0.052], n=61)	yes	the model has little data on this; on the training data only
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 11%	log-odds -0.76, OR 0.47; range across patients -0.76 to -0.76	108, 42	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C21	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	22% to 11%	log-odds -0.72, OR 0.49; range across patients -0.76 to -0.68	108, 42	train; test agrees (AUC gain 0.040 [0.008, 0.079], n=61)	yes	clearly
C22	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model)	27% to 18%	log-odds -0.53, OR 0.59; range across patients -0.53 to -0.53	242	train; test unclear (AUC gain 0.006 [0.000, 0.022], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 27%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	115, 17 (small n)	train; test disagrees (AUC gain -0.019 [-0.070, 0.025], n=61)	yes	the model has little data on this; on the training data only
C26	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	22% to 22%	log-odds -0.23, OR 0.80; range across patients -0.57 to 0.00	121, 121	train; test disagrees (AUC gain -0.002 [-0.015, 0.009], n=61)	yes	slightly; on the training data only
C27	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	22% to 88%	log-odds 3.32, OR 27.6	242	train; test unclear (AUC gain 0.039 [-0.026, 0.111], n=61)	yes	on the training data only
C28	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	22% to 67%	log-odds 2.02, OR 7.53	115, 110	train; test disagrees (AUC gain -0.019 [-0.070, 0.025], n=61)	yes	on the training data only
C29	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: accelerating; 48% of patients same; fallback*	22% to 48%	log-odds 1.21, OR 3.36	242	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C30	magnitude: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	22% to 47%	log-odds 1.19, OR 3.28	135, 90	train; test unclear (AUC gain 0.017 [-0.010, 0.052], n=61)	yes	on the training data only
C31	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	14% to 37%	log-odds 1.30, OR 3.68	242	train; test disagrees (AUC gain -0.022 [-0.063, 0.008], n=61)	yes	on the training data only
C32	structure: made of ischaemia_gate, bp_offset_gate, age_gate_without_ischaemia and inputs age, sex, cp, trestbps, chol, fbs, thalach, exang, slope, ca, thal; added; ann_32 feeds ann_33; ann_33 feeds ann_34		log-odds n/a, OR n/a	242	train	yes	
C33	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: accelerating; 48% of patients same; fallback*	22% to 48%	log-odds 1.16, OR 3.20; range across patients 0.55 to 1.91	242	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C34	direction: thallium stress test result: thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	22% to 32%	log-odds 0.34, OR 1.40; range across patients 0.00 to 0.77	152, 90	train; test unclear (AUC gain 0.005 [-0.009, 0.025], n=61)	yes	on the training data only
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	96% to 94% (typical patient this part acts on)	log-odds -0.23, OR 0.80; range across patients -0.57 to 0.00	121, 121	train; test disagrees (AUC gain -0.002 [-0.015, 0.009], n=61)	yes	slightly; on the training data only
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by thallium stress test result: a reversible defect on the thallium scan minus thallium stress test result	a reversible defect on the thallium scan: 36% to 80%; thallium stress test result other than a	log-odds difference 0.68, OR 1.98	90, 152	train; test agrees (AUC gain 0.005 [-0.009, 0.025], n=61); group difference on test 0.73 [0.59, 0.86]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: accelerating; 33% of patients same; fallback*	resting ECG other than a normal resting ECG: 22% to 34%; a normal resting ECG: 22% to 48%	log-odds difference -0.59, OR 0.55	121, 121	train; test agrees (AUC gain -0.002 [-0.015, 0.009], n=61); group difference on test -0.59 [-0.76, -0.40]	yes	clearly
C38	interaction: effect of thallium stress test result (thallium stress test result other than a reversible defect on the thallium scan to a reversible defect on the thallium scan) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result other than a reversible defect on the thallium scan*	resting ECG other than a normal resting ECG: 22% to 36%; a normal resting ECG: 22% to 47%	log-odds difference -0.11, OR 0.90	121, 121	train; test agrees (AUC gain -0.002 [-0.015, 0.009], n=61); group difference on test -0.18 [-0.34, -0.03]	yes	clearly
C39	threshold: ST depression on exercise: fold at about 0.9 mm (starts) *shape: accelerating; 55% of patients same; fallback*	22% to 46% (typical patient with a fold)	fold 0.93 mm (IQR 0.93 to 1.72); past it log-odds 1.50, OR 4.47; range across patients 1.00 to 1.86	198, 72	train; test unclear (AUC gain 0.008 [-0.011, 0.031], n=61)	yes	on the training data only
C40	subgroup: who this part acts on: 116 of 242 patients	89% to 96% (typical patient this part acts on (risk without the part, then with it))	share 0.48; log-odds 0.77, OR 2.17; range across patients 0.08 to 1.75	116, 126	train; test unclear (AUC gain 0.016 [-0.005, 0.048], n=61)	yes	on the training data only
C41	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	242	train	yes	
C42	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to atypical angina (through this part) *shape: categorical; highest to lowest: atypical angina, non-anginal chest pain, chest pain type other than atypical angina, non-anginal chest pain*	69% to 70% (typical patient this part acts on)	log-odds 0.04, OR 1.04; range across patients 0.00 to 0.08	130, 42	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C43	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (through this part) *shape: hinge at 114 mm Hg; 45% of patients same; fallback*	22% to 22%	log-odds -0.15, OR 0.86; range across patients -0.34 to 0.02	242	train; test disagrees (AUC gain -0.003 [-0.015, 0.006], n=61)	no	slightly; on the training data only
C44	direction: chest pain type: chest pain type other than atypical angina, non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: atypical angina, non-anginal chest pain, chest pain type other than atypical angina, non-anginal chest pain*	69% to 69% (typical patient this part acts on)	log-odds 0.003, OR 1.00; range across patients 0.00 to 0.006	130, 70	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C45	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by chest pain type: atypical angina minus chest pain type other than atypical angina, non-anginal chest pain; stronger in atypical angina *shape: accelerating; 90% of patients same; verified*	atypical angina: 8% to 20%; chest pain type other than atypical angina, non-anginal chest pain: 14% to 35%	log-odds difference 0.04, OR 1.04	42, 130	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61); group difference on test 0.06 [-0.03, 0.14]	no	on the training data only
C46	threshold: resting blood pressure: fold at about 128 mm Hg (stops) *shape: several pieces; 14% of patients same; fallback*	22% to 21% (typical patient with a fold)	fold 128 mm Hg (IQR 120 to 138); past it log-odds -0.16, OR 0.85; range across patients -0.42 to 0.03	203, 101	train; test disagrees (AUC gain -0.003 [-0.015, 0.006], n=61)	no	slightly; on the training data only
C47	subgroup: who this part acts on: 140 of 242 patients	65% to 69% (typical patient this part acts on (risk without the part, then with it))	share 0.58; log-odds 0.33, OR 1.39; range across patients -0.01 to 0.84	140, 102	train; test unclear (AUC gain 0.006 [-0.004, 0.022], n=61)	no	slightly; on the training data only
C48	structure: its formula reproduces it on 242 training patients (largest difference 8.88178e-16)		log-odds 9e-16, OR n/a	242	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C49	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	21% to 24%	log-odds 0.10, OR 1.11; range across patients 0.00 to 0.18	242	train; test unclear (AUC gain 0.000 [-0.01, 0.011], n=61)	yes	slightly; on the training data only
C50	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 99% of patients same; verified; one point with C50 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 66 years, wide)*	20% to 24% (typical patient with a fold)	fold 52.7 years (IQR 52.7 to 58.6); past it log-odds 0.17, OR 1.18; range across patients 0.03 to 0.21	174, 85	train; test unclear (AUC gain 0.000 [-0.01, 0.011], n=61)	yes	slightly; on the training data only
C51	subgroup: who this part acts on: 85 of 242 patients	28% to 30% (typical patient this part acts on (risk without the part, then with it))	share 0.35; log-odds 0.10, OR 1.11; range across patients 0.01 to 0.20	85, 157	train; test disagrees (AUC gain -0.002 [-0.01, 0.004], n=61)	yes	slightly; on the training data only
C52	structure: its formula reproduces it on 242 training patients (largest difference 1.11022e-16)		log-odds 1e-16, OR n/a	242	train	yes	

Not supported by the model.

- C43: direction claim about resting blood pressure in part ann_33: not supported by the model (17% of patients have the opposite sign (allowed: 10%)).
- C45: interaction claim about resting blood pressure and chest pain type in part ann_33: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C46: threshold claim about resting blood pressure in part ann_33: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C47: subgroup claim about chest pain type and resting blood pressure in part ann_33: not supported by the model (its effect goes both ways across the patients it acts on).

11.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (4):** `-11` (fbs:1), `-13` (restecg:1), `-14` (restecg:2), `-3` (sex:1).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

11.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 11.6.1)

▷ A2: ischaemia_gate (leaf, section 11.6.2)

▷ A3: bp_offset_gate (leaf, section 11.6.3)
▷ A4: age_gate_without_ischaemia (leaf, section 11.6.4)

11.4 Network figures

11.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-26` into `-26_a`, `-26_b`.
- #27: split `-6` into `-6_a`, `-6_b`.
- #28: split `-8` into `-8_a`, `-8_b`.
- #29: split `-1` into `-1_a`, `-1_b`.
- #30: identity node `identity_290` before `290` (1 connection(s)).
- #31: identity node `identity_1987` before `1987` (1 connection(s)).

11.6 Annotations

11.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 11.6.2), A3 (section 11.6.3), A4 (section 11.6.4).

- **Entry nodes:** `-1_a` (age_a), `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-6_a` (cp:3_a), `-7` (cp:4), `-8_a` (trestbps_a), `-9` (chol), `-10` (fbs:0), `-11` (fbs:1), `-13` (restecg:1), `-14` (restecg:2), `-15` (thalach), `-16` (exang:0), `-17` (exang:1), `-19` (slope:1), `-20` (slope:2), `-21` (slope:3), `-22` (ca), `-23` (thal:3), `-24` (thal:4), `-25` (thal:6), `-26_a` (thal:7_a).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-2`→`0` (-1.07), `-4`→`0` (-0.493), `-7`→`0` (0.377), `-9`→`0` (0.546), `-10`→`0` (0.278), `-15`→`0` (-0.207), `-16`→`0` (0.149), `-17`→`0` (0.431), `-19`→`0` (-0.855), `-20`→`0` (0.153), `-21`→`0` (-0.366), `-22`→`0` (1.48), `-23`→`0` (-0.41), `-24`→`0` (0.183), `-25`→`0` (-0.384), `204`→`0` (0.143), `290`→`0` (0.17), `1987`→`0` (0.143), `-26_a`→`0` (-0.0766), `-6_a`→`0` (-0.44), `-8_a`→`0` (0.48), `-1_a`→`0` (0.117), `204`→`identity_290` (1.08), `290`→`identity_1987` (-1.59).
- **With its descendants:** 36 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 35 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = -0.504 + LINEAR CORE + 0.143*ischaemia_gate + 0.170*bp_offset_gate + 0.143*age_gate_without_ischaemia. LINEAR CORE (19 direct input->output edges, z-scored inputs): ca +1.480, sex:0 -1.072, slope:1 -0.855, chol +0.546, cp:1 -0.493, trestbps +0.480, cp:3 -0.440, exang:1 +0.431, thal:3 -0.410, thal:6 -0.384, cp:4 +0.377, slope:3 -0.366, fbs:0 +0.278, thalach -0.207, thal:4 +0.183, slope:2 +0.153, exang:0 +0.149, age +0.117, thal:7 -0.077. Four inputs are unconnected (sex:1, fbs:1, restecg:1, restecg:2; zero-effect entries of this root). Oldpeak, restecg:0 and cp:2 have no direct edge and act only through the chain. COMPOSITION: serial and additive. The three children form a chain, each read out additively into z: ischaemia_gate (oldpeak, normal resting ECG, reversible defect) feeds bp_offset_gate (the ischaemia score minus a resting-BP offset), which in turn switches off age_gate_without_ischaemia (an extra age term for

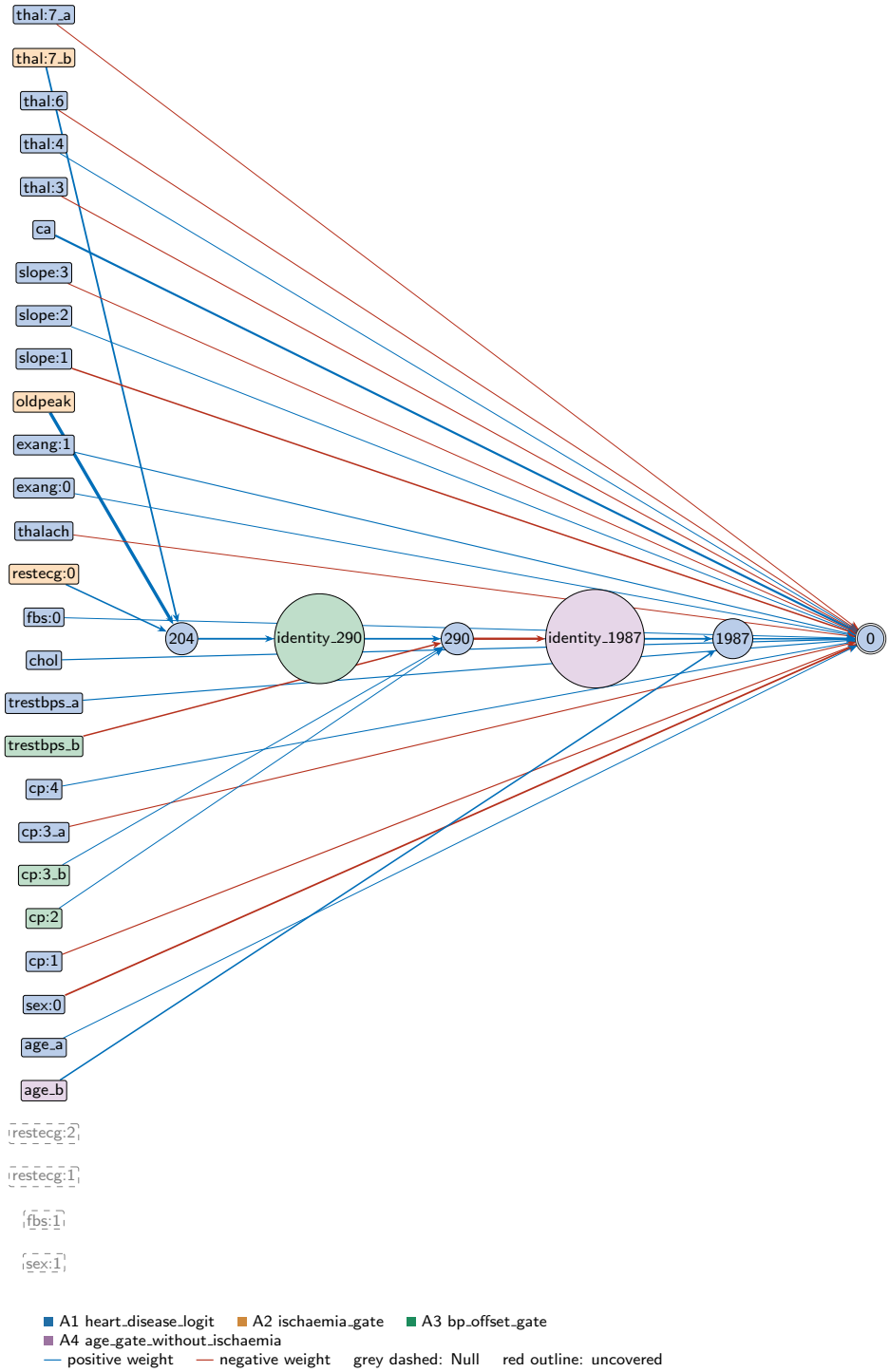


Figure 33: heart-s5-c1: full view of the explained model state.

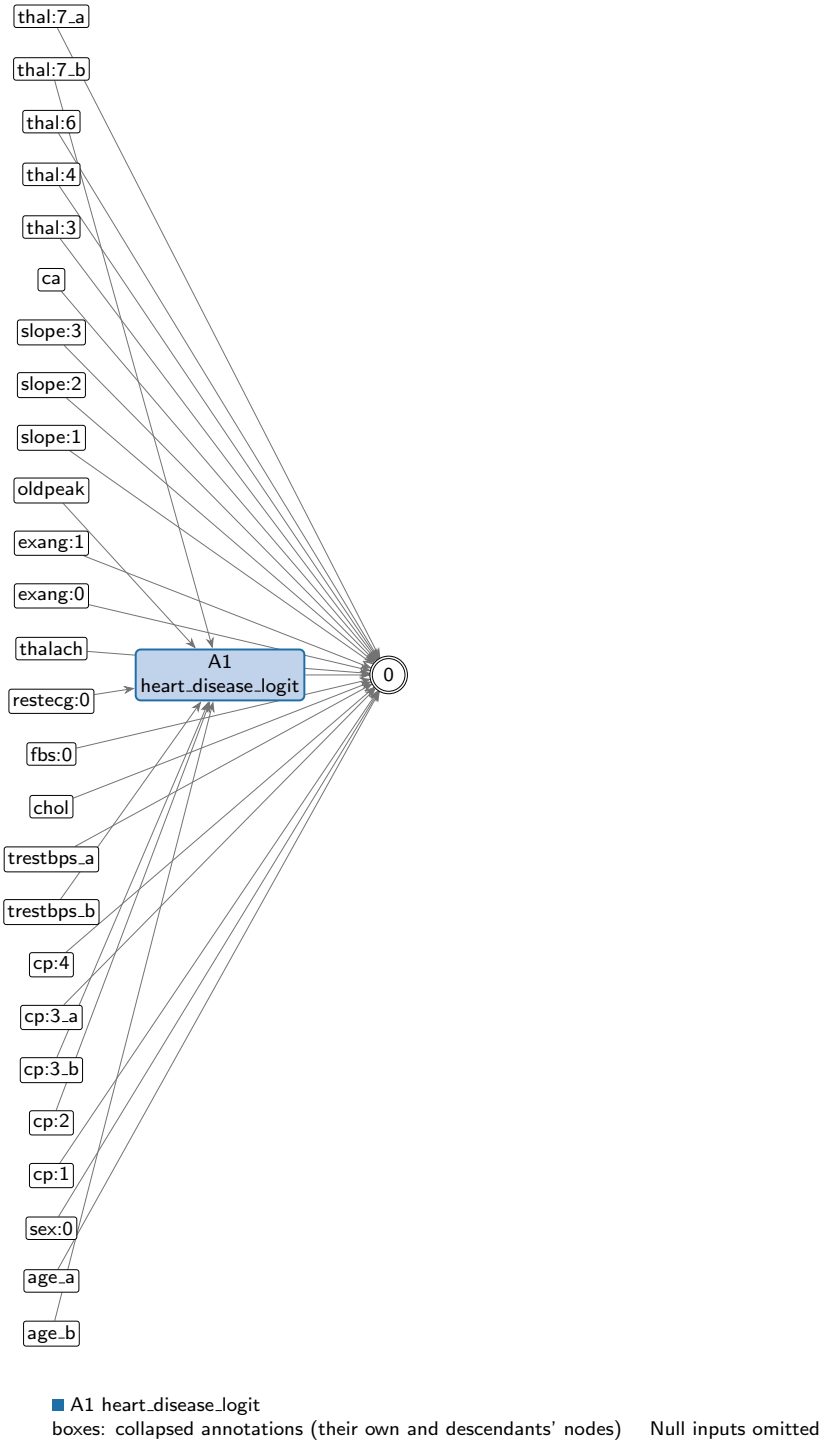


Figure 34: heart-s5-c1: collapsed view at hierarchy level 0 (fully recomposed).

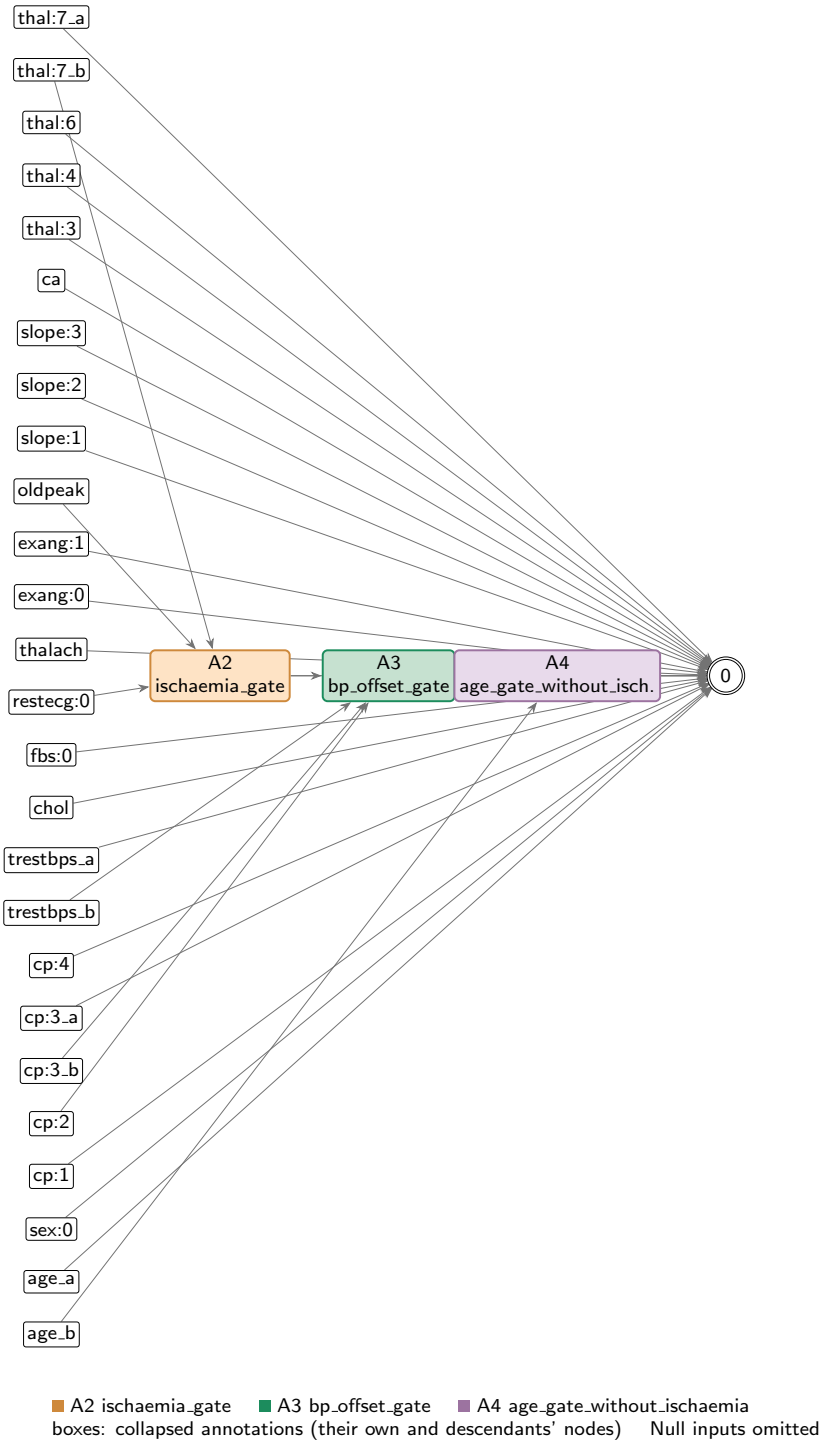


Figure 35: heart-s5-c1: collapsed view at hierarchy level 1.

patients without ischaemia evidence). The chain therefore raises risk with ST depression, more steeply where resting BP is low, and adds a small age term where the chain is quiet. Identity nodes identity_290 and identity_1987 carry the chain links (204 -> 290, 290 -> 1987) so that each gate is its own leaf. EVIDENCE (train, n=242): full model AUC 0.9418, acc 0.872 (compute_performance) . Linear core alone AUC 0.934; logit SD 3.14 (core) vs 0.67 (chain). Removing the whole chain (node 204 with downstream) gives 0.934; readout-only ablations: ischaemia 0.940, BP 0.939, age 0.941. Treating all three ReLUs as identities gives 0.936; one least-squares linear fit of the chain on its seven inputs gives 0.937 (R2 0.84); an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.144, sex:0 0.110, slope:1 0.100, thal:3 0.048, exang:1 0.047, chol 0.045, cp:4 0.044, cp:3 0.044. Interpretation: close to logistic regression plus one ischaemia correction: ST depression raises risk only above a threshold set by a normal resting ECG and a reversible defect, more so when resting BP is low; age gets a small extra weight when there is no such evidence. Counter-directional signs reported, not explained: restecg:0 positive in the ischaemia gate, trestbps negative in the BP gate, fbs:0 +0.28 directly. Same design as the 10-seed heart-s5 record (generation 37, key 9330: ischaemia gate -> resting-BP gate -> age term), here with three hidden nodes instead of six.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.504 + 0.143\,\mathrm{ischaemia_gate}\big(x_{restecg:0}, x_{oldpeak},\\
&\quad x_{thal:7_b}\big) + 1.48x_{ca}\\
&\quad + 0.143\,\mathrm{age_gate_without_ischaemia}\big(-1.59\,\mathrm{bp_offset_gate}\big(1.08\,\mathrm{ischaemia_gate}\big(x_{restecg:0},\\
&\quad x_{oldpeak}, x_{thal:7_b}\big), x_{cp:2}, x_{cp:3_b}, x_{trestbps_b}\big),\\
&\quad x_{age_b}\big) + 0.431x_{exang:1} + 0.48x_{trestbps_a} + 0.546x_{chol} + 0.278x_{fbs:0}\\
&\quad + 0.149x_{exang:0} + 0.153x_{slope:2}\\
&\quad + 0.17\,\mathrm{bp_offset_gate}\big(1.08\,\mathrm{ischaemia_gate}\big(x_{restecg:0},\\
&\quad x_{oldpeak}, x_{thal:7_b}\big), x_{cp:2}, x_{cp:3_b}, x_{trestbps_b}\big)\\
&\quad + 0.183x_{thal:4} + 0.377x_{cp:4} + 0.117x_{age_a} - 0.207x_{thalach} - 0.41x_{thal:3}\\
&\quad - 0.493x_{cp:1} - 0.0766x_{thal:7_a} - 0.384x_{thal:6} - 1.07x_{sex:0}\\
&\quad - 0.366x_{slope:3} - 0.855x_{slope:1} - 0.44x_{cp:3_a}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.504 + 0.143\,\mathrm{relu}\big(0.247 + 1.15x_{thal:7_b} + 2.59x_{oldpeak}\\
&\quad + 0.87x_{restecg:0}\big) + 1.48x_{ca} + 0.143\,\mathrm{relu}\big(0.203\\
&\quad + 0.875x_{age_b} - 1.59\,\mathrm{relu}\big(-0.495 + 0.0164x_{cp:3_b}\\
&\quad + 1.08\,\mathrm{relu}\big(0.247 + 1.15x_{thal:7_b} + 2.59x_{oldpeak}\\
&\quad + 0.87x_{restecg:0}\big) + 0.176x_{cp:2} - 0.815x_{trestbps_b}\big)\big)\\
&\quad + 0.431x_{exang:1} + 0.48x_{trestbps_a} + 0.546x_{chol} + 0.278x_{fbs:0}\\
&\quad + 0.149x_{exang:0} + 0.153x_{slope:2} + 0.17\,\mathrm{relu}\big(-0.495\\
&\quad + 0.0164x_{cp:3_b} + 1.08\,\mathrm{relu}\big(0.247 + 1.15x_{thal:7_b}\\
&\quad + 2.59x_{oldpeak} + 0.87x_{restecg:0}\big) + 0.176x_{cp:2} - 0.815x_{trestbps_b}\big)\\
&\quad + 0.183x_{thal:4} + 0.377x_{cp:4} + 0.117x_{age_a} - 0.207x_{thalach} - 0.41x_{thal:3}\\
&\quad - 0.493x_{cp:1} - 0.0766x_{thal:7_a} - 0.384x_{thal:6} - 1.07x_{sex:0}\\
&\quad - 0.366x_{slope:3} - 0.855x_{slope:1} - 0.44x_{cp:3_a}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00186, and the exact function by at most 2.78×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=true); output is sigmoid.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, ablations, linear replacements and SHAP.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

11.6.2 A2: ischaemia_gate

Kind: Leaf annotation (depth 1); parent A1 (section 11.6.1).

- **Entry nodes:** `-12` (restecg:0), `-18` (oldpeak), `-26_b` (thal:7_b).
- **Exit nodes:** `204`.
- **Internal nodes:** none.
- **Edges (weight):** `-12`→`204` (0.871), `-18`→`204` (2.59), `-26_b`→`204` (1.14).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. Ischaemia-evidence gate (raises risk). 204 = ReLU(0.247 + 2.593*oldpeak + 0.871*restecg:0 + 1.145*thal:7). Hypothesis: a one-sided ST-depression score that opens above an oldpeak threshold set by two flags (always open with a normal resting ECG and a reversible thallium defect; above z-oldpeak -0.09 with a normal ECG only, -0.33 with a defect only, +0.58 with neither), and then raises the logit with oldpeak, directly (0.143 per unit) and through the blood-pressure gate (a further 0.183 per unit while that gate is open). EVIDENCE (train, n=242): active for 48% (100% of the 44 patients with both flags, 20% of the 75 with neither); corr with oldpeak 0.88; disease rate 66% when active vs 24% when inactive. Mean |sensitivity| oldpeak 1.24 > thal:7 0.55 > restecg:0 0.42. This is the main nonlinear effect: removing it with its downstream effects lowers train AUC 0.942 -> 0.934 (to the linear-core level); an affine replacement recovers 0.938. Supported. Oldpeak and restecg:0 have no direct edge; they act only here. The positive restecg:0 weight (normal ECG raises the score) is counter-intuitive and is reported, not explained.

Closed form:

$$y_0 = \mathrm{relu}\big(0.247 + 1.15x_{thal:7_b} + 2.59x_{oldpeak} + 0.87x_{restecg:0}\big)$$

where y_0 is node `204`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.0113, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical); thresholds: thresholds = z-oldpeak at which 204 opens: always open with restecg:0 an.... Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

11.6.3 A3: bp_offset_gate

Kind: Leaf annotation (depth 1); parent A1 (section 11.6.1).

- **Entry nodes:** `identity_290`, `-5` (cp:2), `-6_b` (cp:3_b), `-8_b` (trestbps_b).
- **Exit nodes:** `290`.
- **Internal nodes:** none.
- **Edges (weight):** `identity_290`→`290` (1), `-5`→`290` (0.176), `-6_b`→`290` (0.0164), `-8_b`→`290` (-0.815).
- **Covers** (coverage, own and as part of the hierarchy): 4 candidate nodes.

Hypothesis. Blood-pressure offset gate on the ischaemia score (raises risk). 290 = ReLU(-0.495 + 1.079*ischaemia_gate - 0.815*trestbps + 0.176*cp:2 + 0.016*cp:3). Hypothesis: passes the ischaemia score on, minus an offset that grows with resting blood pressure; it adds 0.170 per unit to the logit and switches off the age gate. With the ischaemia gate closed it still opens for low resting blood pressure, so within this gate lower resting BP raises risk (0.170*0.815 = 0.139 per SD), partly offsetting trestbps's direct +0.48 edge. cp:2 (atypical angina) nudges it open; cp:3's weight 0.016 is negligible. EVIDENCE (train, n=242): active for 58% (83% with the ischaemia gate open, 35% without); with the ischaemia gate open, active for 100%, 87% and 59% of the low, middle and high resting-BP groups. Disease rate 51% when active vs 35% when inactive. Mean |sensitivity| ischaemia score 0.58 > trestbps 0.45 > cp:2 0.10 > cp:3 0.01. Readout ablation lowers train AUC 0.942 -> 0.939; an affine replacement fits it well (R2 0.97, AUC 0.941), so its nonlinearity matters little. Supported, small. The negative BP term inside a risk-raising gate is reported, not explained.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.495 + 1.0x_{identity_290} + 0.0164x_{cp:3_b} + 0.176x_{cp:2} - 0.815x_{trestbps_b}\big)$$

where y_0 is node `290`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 8.86×10^{-4}, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

11.6.4 A4: age_gate_without_ischaemia

Kind: Leaf annotation (depth 1); parent A1 (section 11.6.1).

- **Entry nodes:** `identity_1987`, `-1_b` (age_b).
- **Exit nodes:** `1987`.
- **Internal nodes:** none.
- **Edges (weight):** `identity_1987`→`1987` (1), `-1_b`→`1987` (0.875).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Age gate without ischaemia evidence (raises risk with age, small). 1987 = ReLU(0.203 + 0.875*age - 1.589*bp_offset_gate), read out +0.143. Hypothesis: an extra age term (0.125 per SD of age, on top of age's direct +0.117) that acts only while the BP gate is closed or nearly closed, i.e. for patients without much ischaemia evidence; any opening of the BP gate cancels it (each unit of the BP gate removes 0.227 from the logit through this node until it closes). EVIDENCE (train, n=242): active for 35% (65% when the BP gate is closed, rising with age: 0%, 80%, 100% of the youngest, middle and oldest groups; 14% when the BP gate is open). Disease rate barely differs (43% vs 45%). Mean |sensitivity| BP gate 0.35, age 0.31. Removing it lowers train AUC only 0.942 -> 0.941. Supported, near-negligible.

Closed form:

$$y_0 = \mathrm{relu}\big(0.203 + 1.0x_{identity_1987} + 0.875x_{age_b}\big)$$

where y_0 is node `1987`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 4.99×10^{-4}, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

11.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

12. heart-s5-c2: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.0173	0.0173	0.0173	agree
Numeric-function max abs error	n/a	1.78e-15	1.78e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	5 / 1	n/a	5 / 1	agree
Operation counts	5 splits, 2 identity	add identity node 2, annotate 6, rename node 26, split node 5	add identity node 2, annotate 6, rename node 26, split node 5	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `ischaemia_gate` (0.0173). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- Strong claim in `lvh_without_exercise_angina`: *"Hypothesis: a binary AND: open (constant 0.324, +0.236 logit) only for patients with left ventricular hypertrophy on the resting ECG and no exercise-induced angina, zero otherwise."* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- Strong claim in `high_fasting_sugar`: *"Hypothesis: a small additive term for high fasting sugar (exactly linear on the two values of fbs:1);"* Is this (exactness, threshold, rare category) supported by the evidence, or fitted to a few rows?
- 5 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): get_input_distribution does not accept sample_fraction (called with max_samples only; all 242 rows used). get_viz_summary requires viz_type (sensitivity used). compute_shap (kernel, all 242 train rows) gives the five unconnected inputs mean |SHAP| about 0.001 instead of exactly 0 (sampling noise). get_model_state's 'operations' key is absent, so op counts come from check_explanation.py. The agent sandbox refused compound shell commands, so tools were called through a small Python subprocess wrapper in scratch (same xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s5-c2-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 20%; it adds up most of its inputs independently, and five parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 109 of 242 training patients, a blood pressure step and a damping term that both read that signal, an extra term for left ventricular hypertrophy without chest pain on exercise, and a very small term for high fasting blood sugar. [C34, C42, C53, C61] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 20% to about 3%; through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 4%, fasting blood sugar above 120 mg/dL to about 10% and atypical angina to about 11%; and a

higher maximum heart rate lowers it steadily, from about 26% at 118 to about 16% at 175 beats per minute. [C08, C06, C12, C18, C20, C19] Each extra vessel coloured by fluoroscopy adds about the same, from about 20% at 0 to about 87% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 61%, higher cholesterol raises it steadily from about 13% at 193 mg/dL to about 34% at 308 mg/dL, and older age raises it steadily from about 17% at 42 to about 22% at 66 years. [C01, C02, C04, C27] Through this sum alone, higher resting blood pressure raises it steadily, from about 12% at 110 mm Hg to about 33% at 155 mm Hg, a thallium result other than a normal scan or a fixed defect raises it to about 36%, chest pain brought on by exercise raises it to about 33%, and a resting ECG other than normal raises it to about 25%; all four also act through the parts below. [C05, C09, C14, C26] Some rarer results also count (the model has little data on this, n = 15 to 17): through this sum alone a fixed defect on the thallium scan lowers the risk to about 9%, and a downsloping ST segment raises it to about 27%. [C17, C24] This part reads ST depression on exercise, the resting ECG and the thallium scan, and acts on 109 of 242 training patients, 65 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 89% with it. [C42] Through it, more ST depression adds to the predicted risk only once it rises above about 1.3 mm (between 0.4 and 1.3 mm for different patients), rising in an uneven way: for a typical patient from about 16% at 0 mm to about 50% at 2.8 mm. [C35, C41] ST depression counts more with a normal resting ECG (from about 16% to about 50%) than with any other resting ECG (from about 25% to about 48%), in an uneven way. [C39] Through it, a resting ECG other than normal lowers a typical patient's risk from about 20% to about 16%, and a thallium result coded 4 shuts the signal (the model has little data on this, n = 2). [C36, C37] This part reads that signal together with resting blood pressure, chest pain on exercise and chest pain type, and acts on 145 of 242 training patients, 73 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 51% without it and about 68% with it. [C53] Through it, higher resting blood pressure lowers the risk up to about 125 mm Hg (between 122 and 165 mm Hg for different patients), then no further, in an uneven way: for a typical patient from about 23% at 110 mm Hg to about 17% at 155 mm Hg. [C44, C52] Through it, chest pain on exercise lowers a typical patient's risk from about 20% to about 18%, and non-anginal chest pain to about 19%. [C45, C46] This part reads the same signal and the thallium scan, and acts on 234 of 242 training patients, 102 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 20% with it. [C56] Through it, a fixed defect on the thallium scan raises a typical patient's risk from about 20% to about 23% (the model has little data on this, n = 15). [C55] This part acts on 74 of 242 training patients, nearly all with left ventricular hypertrophy on the resting ECG and no chest pain on exercise, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 25% without it and about 30% with it. [C61] Through it, left ventricular hypertrophy raises a typical patient's risk from about 20% to about 24%, and for the typical patient it acts on, chest pain on exercise lowers it from about 30% to about 25%. [C59, C58] Across the model, chest pain on exercise raises a typical patient's risk from about 20% to about 31% without left ventricular hypertrophy, and from about 25% to about 35% with it. [C60] This part acts on 33 of 242 training patients, nearly all with fasting blood sugar above 120 mg/dL, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 25% with it. [C65] It barely changes the predictions, by at most 2 percentage points. [C66]

Worth checking: Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 20% to about 11%). [C22] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C07, C11, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C23] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C39] Worth checking: through the blood pressure step, higher resting blood pressure lowers the predicted risk, against its rise through the sum. [C44, C05] Worth checking: for the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define, the ST depression signal is shut. [C37]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	20% to 87%	log-odds 3.30, OR 27.2; range across patients 3.30 to 3.30	242	train; test unclear (AUC gain 0.025 [-0.049, 0.102], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 61%	log-odds 1.84, OR 6.32; range across patients 1.84 to 1.84	115, 110	train; test disagrees (AUC gain -0.017 [-0.061, 0.016], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	13% to 34%	log-odds 1.25, OR 3.48; range across patients 1.25 to 1.25	242	train; test disagrees (AUC gain -0.028 [-0.071, 0.002], n=61)	yes	on the training data only
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	12% to 33%	log-odds 1.23, OR 3.43; range across patients 1.23 to 1.23	242	train; test unclear (AUC gain 0.006 [-0.024, 0.037], n=61)	yes	on the training data only
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	log-odds -2.42, OR 0.09; range across patients -2.42 to -2.42	108, 22	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	20% to 3%	log-odds -2.24, OR 0.11; range across patients -2.24 to -2.24	159, 83	train; test unclear (AUC gain 0.022 [-0.011, 0.065], n=61)	yes	on the training data only
C09	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 36%	log-odds 0.85, OR 2.34; range across patients 0.85 to 0.85	135, 92	train; test unclear (AUC gain 0.013 [-0.011, 0.041], n=61)	yes	on the training data only
C12	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 4%	log-odds -1.70, OR 0.18; range across patients -1.70 to -1.70	108, 70	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C14	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (direct contribution only, not through any part) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	20% to 33%	log-odds 0.71, OR 2.03; range across patients 0.71 to 0.71	160, 82	train; test agrees (AUC gain 0.012 [0.001, 0.030], n=61)	yes	clearly
C17	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 9%	log-odds -0.86, OR 0.42; range across patients -0.86 to -0.86	135, 15 (small n)	train; test unclear (AUC gain 0.013 [-0.011, 0.041], n=61)	yes	the model has little data on this; on the training data only
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part)	20% to 10%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	209, 33	train; test disagrees (AUC gain -0.004 [-0.020, 0.009], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 11%	log-odds -0.72, OR 0.49; range across patients -0.72 to -0.72	108, 42	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C24	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 27%	log-odds 0.40, OR 1.48; range across patients 0.40 to 0.40	115, 17 (small n)	train; test disagrees (AUC gain -0.017 [-0.061, 0.016], n=61)	yes	the model has little data on this; on the training data only
C26	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	20% to 25%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	121, 121	train; test unclear (AUC gain 0.000 [-0.016, 0.015], n=61)	yes	on the training data only
C27	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	17% to 22%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	242	train; test disagrees (AUC gain -0.005 [-0.023, 0.01], n=61)	yes	on the training data only
C34	structure: made of ischaemia_gate, bp_offset_gate, ischaemia_damping_thal6, lvh_without_exercise_angina, high_fasting_sugar and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added; ann_33 feeds ann_34; ann_33 feeds ann_35		log-odds n/a, OR n/a	242	train	yes	
C35	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 50%	log-odds 1.22, OR 3.40; range across patients 0.77 to 1.69	242	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C36	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	20% to 16%	log-odds -0.26, OR 0.77; range across patients -0.64 to 0.00	121, 121	train; test disagrees (AUC gain -0.002 [-0.016, 0.012], n=61)	yes	slightly; on the training data only
C37	direction: thallium stress test result: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) to thallium stress test result coded 4 (meaning not recorded) (through this part) *shape: categorical; highest to lowest: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded), thallium stress test result coded 4 (meaning not recorded)*	20% to 16%	log-odds -0.33, OR 0.72; range across patients -1.07 to 0.00	240, 2 (small n)	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; the model has little data on this; on the training data only
C39	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.4 mm; 57% of patients same; fallback*	resting ECG other than a normal resting ECG: 25% to 48%; a normal resting ECG: 16% to 50%	log-odds difference -0.68, OR 0.50	121, 121	train; test agrees (AUC gain -0.002 [-0.016, 0.012], n=61); group difference on test -0.65 [-0.72, -0.58]	yes	clearly
C41	threshold: ST depression on exercise: fold at about 1.3 mm (starts) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 62% (typical patient with a fold)	fold 1.28 mm (IQR 0.35 to 1.28); past it log-odds 1.70, OR 5.47; range across patients 1.25 to 2.17	242, 109	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C42	subgroup: who this part acts on: 109 of 242 patients	76% to 89% (typical patient this part acts	share 0.45; log-odds 0.74, OR 2.09; range across patients 0.02 to	109, 133	train; test unclear (AUC gain 0.006 [-0.014, 0.032], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C45	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part) *shape: categorical; highest to lowest: no chest pain on exercise, chest pain brought on by exercise*	20% to 18%	log-odds -0.04, OR 0.96; range across patients -0.08 to 0.00	160, 82	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only
C46	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	20% to 19%	log-odds -0.02, OR 0.98; range across patients -0.04 to 0.00	172, 70	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C52	threshold: resting blood pressure: fold at about 125 mm Hg (stops) for patients without chest pain on exercise *shape: hinge at 125 mm Hg; 92% of patients same; verified; risk saturates near 0%; one point with C52 (median fold 125 mm Hg, typical patient 122 mm Hg, 10th-90th percentile 122 mm Hg to 165 mm Hg, wide)*	4% to 3% (typical patient with a fold (patients without chest pain on exercise))	fold 125 mm Hg (IQR 122 to 132); past it log-odds -0.30, OR 0.74; range across patients -0.63 to -0.17	143, 78	train; test disagrees (AUC gain -0.003 [-0.014, 0.004], n=61)	yes	slightly; on the training data only
C53	subgroup: who this part acts on: 145 of 242 patients	51% to 68% (typical patient this part acts on (risk without the part, then with it))	share 0.60; log-odds 0.52, OR 1.69; range across patients 0.03 to 1.22	145, 97	train; test unclear (AUC gain 0.004 [-0.013, 0.024], n=61)	yes	on the training data only
C55	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	20% to 23%	log-odds 0.18, OR 1.20; range across patients 0.12 to 0.29	227, 15 (small n)	train; test unclear (AUC gain 0.002 [0.000, 0.01], n=61)	yes	slightly; the model has little data on this; on the training data only
C56	subgroup: who this part acts on: 234 of 242 patients	24% to 20% (typical patient this part acts on (risk without the part, then with it))	share 0.97; log-odds -0.23, OR 0.79; range across patients -0.48 to -0.12	234, 8	train; test unclear (AUC gain 0.000 [-0.011, 0.01], n=61)	yes	slightly; on the training data only
C58	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part) *shape: categorical; highest to lowest: no chest pain on exercise, chest pain brought on by exercise*	30% to 25% (typical patient this part acts on)	log-odds -0.12, OR 0.89; range across patients -0.24 to 0.00	160, 82	train; test disagrees (AUC gain -0.003 [-0.013, 0.000], n=61)	yes	slightly; on the training data only
C59	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, resting ECG other than signs of left ventricular hypertrophy on the resting ECG*	20% to 24%	log-odds 0.16, OR 1.17; range across patients 0.00 to 0.24	123, 119	train; test unclear (AUC gain 0.001 [-0.01, 0.012], n=61)	yes	slightly; on the training data only
C60	interaction: effect of exercise-induced angina (no chest pain on exercise to chest pain brought on by exercise) by resting ECG: signs of left ventricular hypertrophy on the resting ECG minus resting ECG other than signs of left ventricular hypertrophy on the resting ECG; stronger in resting ECG other than signs of left ventricular hypertrophy on the resting ECG *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	signs of left ventricular hypertrophy on the resting ECG: 25% to 35%; resting ECG other than signs of left ventricular hypertrophy on the resting ECG: 20% to 31%	log-odds difference -0.22, OR 0.80	119, 123	train; test agrees (AUC gain 0.001 [-0.01, 0.012], n=61); group difference on test -0.22 [-0.24, -0.20]	yes	slightly
C61	subgroup: who this part acts on: 74 of 242 patients	25% to 30% (typical patient this part acts	share 0.31; log-odds 0.24, OR 1.27; range	74, 168	train; test disagrees (AUC gain -0.002	yes	slightly; on the

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s5-c2 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s5-c2.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s5-c2 --verified --minutes N --changes "..." --notes "..."`

12 Explanation certificate: heart-s5-c2

This section is the explanation certificate of the heart-s5-c2 network. It is rendered from the canonical JSON record `heart-s5-c2.json` (schema 1.4.0, SHA-256 `e5293773fce87fae...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 12 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s5, seed 5
Genome	`57c311c6-56f5-4c5d-a475-46433ef58bf2` (NEAT key 9858, generation 39)
Explanation	`dabdb4bb-e227-46eb-a995-d4123373f234` (39 operations)
Train AUC / accuracy	0.9389 / 0.8802 (n = 242)
Test AUC / accuracy	0.8922 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	32 nodes (26 inputs, 5 hidden, 1 output), 35 enabled edges
Explained model state	41 nodes, 37 enabled edges
Refactoring operations	add_identity_node 2, split_node 5
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 2.22 \times 10^{-16}$ on the training split
Annotations	6 (5 leaf, 1 composition)
Structural coverage	1 (40 of 40 candidate nodes; 5 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`e5293773fce87fae...`

Table 12: Explanation certificate summary: heart-s5-c2.

Drafter's summary. Mostly logistic regression: 17 direct edges (logit SD 3.19; train AUC 0.933 alone) plus a small ischaemia block (logit SD 0.57 for all hidden readouts) to 0.939. An ischaemia gate ReLU(0.39 + 2.54 oldpeak + 0.99 restecg:0 - 0.88 thal:4) opens above an ST-depression threshold that is lower with a normal resting ECG and raises risk directly and through a gate that passes the score on minus a resting-blood-pressure offset (more risk when BP is low); a near-linear node

takes back part of the ischaemia score and adds a fixed-defect term. Two small indicators add risk for LVH without exercise angina and (negligibly) for high fasting sugar. Same ischaemia-gate -> resting-BP-gate core as the 10-seed heart-s5 record (key 9330) and the sibling lead heart-s5-c1, but without their age gate.

What resisted explanation. Nothing resisted coverage. The ischaemia gate 204 feeds two downstream gates (290, 1416) as well as the output; the two links were intercepted with identity nodes (identity_290, identity_1416) so that each hidden node is its own leaf, and the root composes all five. Counter-directional signs reported, not explained: restecg:0 (normal ECG) raises the ischaemia score (its direct edge is -0.16), trestbps enters the risk-raising BP gate negatively (low resting BP opens it), thal:6 (fixed defect) has a direct -0.41 edge (partly offset by +0.07/SD through 1416), fbs:0 +0.26 directly so high fasting sugar lowers the logit overall despite the +0.07 indicator. thal:4 shuts the ischaemia gate but has only 2 training patients. The damping node 1416 and the fasting-sugar node 3884 are effectively linear (affine R2 0.99 and 1.0); the LVH indicator and fasting-sugar indicator barely affect ranking (readout ablation <= -0.0003 AUC).

12.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 20%; it adds up most of its inputs independently, and five parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 109 of 242 training patients, a blood pressure step and a damping term that both read that signal, an extra term for left ventricular hypertrophy without chest pain on exercise, and a very small term for high fasting blood sugar. [C34, C42, C53, C61] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 20% to about 3%; through this sum alone typical angina lowers it to about 2%, non-anginal chest pain to about 4%, fasting blood sugar above 120 mg/dL to about 10% and atypical angina to about 11%; and a higher maximum heart rate lowers it steadily, from about 26% at 118 to about 16% at 175 beats per minute. [C08, C06, C12, C18, C20, C19] Each extra vessel coloured by fluoroscopy adds about the same, from about 20% at 0 to about 87% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 61%, higher cholesterol raises it steadily from about 13% at 193 mg/dL to about 34% at 308 mg/dL, and older age raises it steadily from about 17% at 42 to about 22% at 66 years. [C01, C02, C04, C27] Through this sum alone, higher resting blood pressure raises it steadily, from about 12% at 110 mm Hg to about 33% at 155 mm Hg, a thallium result other than a normal scan or a fixed defect raises it to about 36%, chest pain brought on by exercise raises it to about 33%, and a resting ECG other than normal raises it to about 25%; all four also act through the parts below. [C05, C09, C14, C26] Some rarer results also count (the model has little data on this, n = 15 to 17): through this sum alone a fixed defect on the thallium scan lowers the risk to about 9%, and a downsloping ST segment raises it to about 27%. [C17, C24] This part reads ST depression on exercise, the resting ECG and the thallium scan, and acts on 109 of 242 training patients, 65 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 89% with it. [C42] Through it, more ST depression adds to the predicted risk only once it rises above about 1.3 mm (between 0.4 and 1.3 mm for different patients), rising in an uneven way: for a typical patient from about 16% at 0 mm to about 50% at 2.8 mm. [C35, C41] ST depression counts more with a normal resting ECG (from about 16% to about 50%) than with any other resting ECG (from about 25% to about 48%), in an uneven way. [C39] Through it, a resting ECG other than normal lowers a typical patient's risk from about 20% to about 16%, and a thallium result coded 4 shuts the signal

(the model has little data on this, n = 2). [C36, C37] This part reads that signal together with resting blood pressure, chest pain on exercise and chest pain type, and acts on 145 of 242 training patients, 73 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 51% without it and about 68% with it. [C53] Through it, higher resting blood pressure lowers the risk up to about 125 mm Hg (between 122 and 165 mm Hg for different patients), then no further, in an uneven way: for a typical patient from about 23% at 110 mm Hg to about 17% at 155 mm Hg. [C44, C52] Through it, chest pain on exercise lowers a typical patient's risk from about 20% to about 18%, and non-anginal chest pain to about 19%. [C45, C46] This part reads the same signal and the thallium scan, and acts on 234 of 242 training patients, 102 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 20% with it. [C56] Through it, a fixed defect on the thallium scan raises a typical patient's risk from about 20% to about 23% (the model has little data on this, n = 15). [C55] This part acts on 74 of 242 training patients, nearly all with left ventricular hypertrophy on the resting ECG and no chest pain on exercise, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 25% without it and about 30% with it. [C61] Through it, left ventricular hypertrophy raises a typical patient's risk from about 20% to about 24%, and for the typical patient it acts on, chest pain on exercise lowers it from about 30% to about 25%. [C59, C58] Across the model, chest pain on exercise raises a typical patient's risk from about 20% to about 31% without left ventricular hypertrophy, and from about 25% to about 35% with it. [C60] This part acts on 33 of 242 training patients, nearly all with fasting blood sugar above 120 mg/dL, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 25% with it. [C65] It barely changes the predictions, by at most 2 percentage points. [C66]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and five parts add to that sum: a signal from ST depression and the resting ECG, a blood pressure step and a damping term that read it, a hypertrophy term and a fasting sugar term. [C34, C01, C02, C08, C06] *Evidence.* It is made up of the inputs that feed the score directly and five parts (each explained below); the first part feeds the second and the third, and the score is their sum, which was checked on all 242 training patients. [C34] ST depression on exercise does not feed the score directly, so it acts only through these parts. [C34, C35] Changing one input at a time across the whole model, more ST depression raises a typical patient's predicted risk in an uneven way, from about 16% at 0 mm to about 50% at 2.8 mm, and higher resting blood pressure raises it in an uneven way, from about 15% at 110 mm Hg to about 28% at 155 mm Hg; through the sum alone it rises steadily, from about 12% to about 33%. [C03, C13, C05] Across the whole model, chest pain on exercise raises it to about 31%, non-anginal chest pain lowers it to about 4% and fasting blood sugar above 120 mg/dL to about 11%. [C15, C11, C22] Across the whole model, a fixed defect on the thallium scan lowers it to about 11% (the model has little data on this, n = 15). [C23]

ST depression signal. This part raises the predicted risk with ST depression on exercise once it rises above about 1.3 mm, in an uneven way, and more so with a normal resting ECG. [C35, C41, C39] *Evidence.* It acts on 109 of 242 training patients, 65 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 89% with it. [C42] Through it, more ST depression adds to the risk only once it rises above about 1.3 mm (between 0.4 and 1.3 mm for different patients), rising in an uneven way: for a typical patient from about 16% at 0 mm to about 50% at 2.8 mm. [C35, C41] ST depression counts more with a normal resting ECG (from about 16% to about 50%) than with any other resting ECG (from about 25% to about 48%), in an uneven way. [C39] Through it, a resting ECG other than normal lowers a typical patient's risk from about 20% to about 16%, and a thallium result coded 4 shuts the signal (the model has little data on this, n = 2). [C36, C37] Its formula reproduces the part exactly on all 242 training patients. [C43]

Blood pressure step. This part passes the ST depression signal on to the predicted risk, more for patients with lower resting blood pressure: through it, higher resting blood pressure lowers the risk in an uneven way up to about 125 mm Hg (between 122 mm Hg and 165 mm Hg for different patients), then no further. [C53, C44, C52] *Evidence.* It acts on 145 of 242 training patients, 73 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 51% without it and about 68% with it. [C53] Through it, higher resting blood pressure lowers the risk up to about 125 mm Hg (between 122 and 165 mm Hg for different patients), then no further, in an uneven way: for a typical patient from about 23% at 110 mm Hg to about 17% at 155 mm Hg. [C44, C52] Through it, chest pain on exercise lowers a typical patient's risk from about 20% to about 18%, and non-anginal chest pain to about 19%. [C45, C46] Its formula reproduces the part exactly on all 242 training patients. [C54]

Damping term. This part takes back a little of the ST depression signal for nearly every patient and adds a little for a fixed defect on the thallium scan (the model has little data on this, n = 15). [C56, C55] *Evidence.* It acts on 234 of 242 training patients, 102 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 20% with it. [C56] Through it, a fixed defect on the thallium scan raises a typical patient's risk from about 20% to about 23% (the model has little data on this, n = 15). [C55] Its formula reproduces the part exactly on all 242 training patients. [C57]

Hypertrophy without exercise pain. This part adds a little to the predicted risk for patients with left ventricular hypertrophy on the resting ECG and no chest pain on exercise. [C61, C59] *Evidence.* It acts on 74 of 242 training patients, nearly all with left ventricular hypertrophy and no chest pain on exercise, 30 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 25% without it and about 30% with it. [C61] Through it, left ventricular hypertrophy raises a typical patient's risk from about 20% to about 24%, and for the typical patient it acts on, chest pain on exercise lowers it from about 30% to about 25%. [C59, C58] Across the model, chest pain on exercise raises a typical patient's risk from about 20% to about 31% without left ventricular hypertrophy, and from about 25% to about 35% with it. [C60] Its formula reproduces the part exactly on all 242 training patients. [C63]

High fasting sugar term. This part adds a very small amount to the predicted risk for patients with fasting blood sugar above 120 mg/dL. [C65, C64] *Evidence.* It acts on 33 of 242 training patients, nearly all with fasting blood sugar above 120 mg/dL, 15 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 24% without it and about 25% with it. [C65] Through it, fasting blood sugar above 120 mg/dL raises a typical patient's risk from about 20% to about 21%, and it changes the predictions by at most 2 percentage points. [C64, C66] Its formula reproduces the part exactly on all 242 training patients. [C67]

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 20% to about 11%). [C22] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C07, C11, C21] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C23] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C39] Worth checking: through the blood pressure step, higher resting blood pressure lowers the predicted risk, against its rise through the sum. [C44, C05] Worth checking: for the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define, the ST depression signal is shut. [C37]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	20% to 87%	log-odds 3.30, OR 27.2; range across patients 3.30 to 3.30	242	train; test unclear (AUC gain 0.025 [-0.049, 0.102], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 61%	log-odds 1.84, OR 6.32; range across patients 1.84 to 1.84	115, 110	train; test disagrees (AUC gain -0.017 [-0.061, 0.016], n=61)	yes	on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 0.4 mm; 70% of patients same; fallback*	16% to 50%	log-odds 1.22, OR 3.40; range across patients 0.77 to 1.69	242	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C04	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	13% to 34%	log-odds 1.25, OR 3.48; range across patients 1.25 to 1.25	242	train; test disagrees (AUC gain -0.028 [-0.071, 0.002], n=61)	yes	on the training data only
C05	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (direct contribution only, not through any part) *shape: steady; 100% of patients same; verified*	12% to 33%	log-odds 1.23, OR 3.43; range across patients 1.23 to 1.23	242	train; test unclear (AUC gain 0.006 [-0.024, 0.037], n=61)	yes	on the training data only
C06	direction: chest pain type: no chest pain (asymptomatic) to typical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	log-odds -2.42, OR 0.09; range across patients -2.42 to -2.42	108, 22	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C07	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	log-odds -2.42, OR 0.09; range across patients -2.42 to -2.42	108, 22	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C08	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	20% to 3%	log-odds -2.24, OR 0.11; range across patients -2.24 to -2.24	159, 83	train; test unclear (AUC gain 0.022 [-0.011, 0.065], n=61)	yes	on the training data only
C09	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 36%	log-odds 0.85, OR 2.34; range across patients 0.85 to 0.85	135, 92	train; test unclear (AUC gain 0.013 [-0.011, 0.041], n=61)	yes	on the training data only
C10	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result coded 4 (meaning not recorded), a normal thallium scan, a fixed defect on the thallium scan*	20% to 36%	log-odds 0.85, OR 2.34; range across patients 0.85 to 0.85	135, 90	train; test unclear (AUC gain 0.014 [-0.009, 0.042], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 4%	log-odds -1.72, OR 0.18; range across patients -1.74 to -1.70	108, 70	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C12	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (direct contribution only;	20% to 4%	log-odds -1.70, OR 0.18; range across patients -1.70 to -1.70	108, 70	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: accelerating; 75% of patients same; fallback*	15% to 28%	log-odds 0.97, OR 2.63; range across patients 0.75 to 1.15	242	train; test unclear (AUC gain 0.005 [-0.017, 0.031], n=61)	yes	on the training data only
C14	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (direct contribution only, not through any part) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	20% to 33%	log-odds 0.71, OR 2.03; range across patients 0.71 to 0.71	160, 82	train; test agrees (AUC gain 0.012 [0.001, 0.030], n=61)	yes	clearly
C15	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	20% to 31%	log-odds 0.55, OR 1.73; range across patients 0.39 to 0.71	160, 82	train; test unclear (AUC gain 0.008 [0.000, 0.022], n=61)	yes	on the training data only
C16	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: a reversible defect on the thallium scan, thallium stress test result coded 4 (meaning not recorded), a normal thallium scan, a fixed defect on the thallium scan*	20% to 31%	log-odds 0.52, OR 1.69; range across patients -0.22 to 0.85	135, 2 (small n)	train; test unclear (AUC gain 0.014 [-0.009, 0.042], n=61)	no	the model has little data on this; on the training data only
C17	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (direct contribution only, not through any part) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	20% to 9%	log-odds -0.86, OR 0.42; range across patients -0.86 to -0.86	135, 15 (small n)	train; test unclear (AUC gain 0.013 [-0.011, 0.041], n=61)	yes	the model has little data on this; on the training data only
C18	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (direct contribution only, not through any part) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	20% to 10%	log-odds -0.77, OR 0.46; range across patients -0.77 to -0.77	209, 33	train; test disagrees (AUC gain -0.004 [-0.020, 0.009], n=61)	yes	on the training data only
C19	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	26% to 16%	log-odds -0.57, OR 0.56; range across patients -0.57 to -0.57	242	train; test unclear (AUC gain 0.006 [0.000, 0.022], n=61)	yes	on the training data only
C20	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 11%	log-odds -0.72, OR 0.49; range across patients -0.72 to -0.72	108, 42	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C21	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 11%	log-odds -0.72, OR 0.49; range across patients -0.72 to -0.72	108, 42	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C22	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	20% to 11%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	209, 33	train; test disagrees (AUC gain -0.003 [-0.019, 0.009], n=61)	yes	on the training data only
C23	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model)	20% to 11%	log-odds -0.69, OR 0.50; range across patients -0.75 to -0.58	135, 15 (small n)	train; test unclear (AUC gain 0.014 [-0.009, 0.042], n=61)	yes	the model has little data on this; on the

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	direction: resting ECG: a normal resting ECG to signs of left ventricular hypertrophy on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, an ST-T wave abnormality on the resting ECG, a normal resting ECG*	20% to 25%	log-odds 0.21, OR 1.24; range across patients -0.32 to 0.55	121, 119	train; test disagrees (AUC gain -0.003 [-0.023, 0.013], n=61)	no	on the training data only
C26	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	20% to 25%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	121, 121	train; test unclear (AUC gain 0.000 [-0.016, 0.015], n=61)	yes	on the training data only
C27	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	17% to 22%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	242	train; test disagrees (AUC gain -0.005 [-0.023, 0.01], n=61)	yes	on the training data only
C28	direction: resting ECG: a normal resting ECG to an ST-T wave abnormality on the resting ECG (whole model) *shape: categorical; highest to lowest: signs of left ventricular hypertrophy on the resting ECG, an ST-T wave abnormality on the resting ECG, a normal resting ECG*	20% to 21%	log-odds 0.06, OR 1.06; range across patients -0.32 to 0.32	121, 2 (small n)	train; test disagrees (AUC gain -0.003 [-0.023, 0.013], n=61)	no	slightly; the model has little data on this; on the training data only
C29	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	20% to 87%	log-odds 3.30, OR 27.2	242	train; test unclear (AUC gain 0.025 [-0.049, 0.102], n=61)	yes	on the training data only
C30	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	20% to 61%	log-odds 1.84, OR 6.32	115, 110	train; test disagrees (AUC gain -0.017 [-0.061, 0.016], n=61)	yes	on the training data only
C31	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 0.4 mm; 70% of patients same; fallback*	16% to 50%	log-odds 1.64, OR 5.15	242	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C32	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	13% to 34%	log-odds 1.25, OR 3.48	242	train; test disagrees (AUC gain -0.028 [-0.071, 0.002], n=61)	yes	on the training data only
C33	magnitude: chest pain type: no chest pain (asymptomatic) to typical angina, typical patient *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	20% to 2%	log-odds -2.42, OR 0.09	108, 22	train; test agrees (AUC gain 0.032 [0.008, 0.062], n=61)	yes	clearly
C34	structure: made of ischaemia_gate, bp_offset_gate, ischaemia_damping_thal6, lvh_without_exercise_angina, high_fasting_sugar and inputs age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added; ann_33 feeds ann_34; ann_33 feeds ann_35		log-odds n/a, OR n/a	242	train	yes	
C35	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 50%	log-odds 1.22, OR 3.40; range across patients 0.77 to 1.69	242	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C36	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting*	20% to 16%	log-odds -0.26, OR 0.77; range across patients -0.64 to 0.00	121, 121	train; test disagrees (AUC gain -0.002 [-0.016, 0.012], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	direction: thallium stress test result: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) to thallium stress test result coded 4 (meaning not recorded) (through this part) *shape: categorical; highest to lowest: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded), thallium stress test result coded 4 (meaning not recorded)*	20% to 16%	log-odds -0.33, OR 0.72; range across patients -1.07 to 0.00	240, 2 (small n)	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; the model has little data on this; on the training data only
C38	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by thallium stress test result: thallium stress test result coded 4 (meaning not recorded) minus thallium stress test result other than thallium stress test result coded 4 (meaning not recorded); stronger in thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) *shape: hinge at 0.4 mm; 74% of patients same; fallback*	thallium stress test result coded 4 (meaning not recorded): 31% to 31%; thallium stress test result other than thallium stress test result coded 4 (meaning not recorded): 16% to 50%	log-odds difference -1.23, OR 0.29	2, 240 (small n)	train; test unclear (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	the model has little data on this; on the training data only
C39	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.4 mm; 57% of patients same; fallback*	resting ECG other than a normal resting ECG: 25% to 48%; a normal resting ECG: 16% to 50%	log-odds difference -0.68, OR 0.50	121, 121	train; test agrees (AUC gain -0.002 [-0.016, 0.012], n=61); group difference on test -0.65 [-0.72, -0.58]	yes	clearly
C40	interaction: effect of thallium stress test result (thallium stress test result other than thallium stress test result coded 4 (meaning not recorded) to thallium stress test result coded 4 (meaning not recorded)) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in resting ECG other than a normal resting ECG *shape: categorical; highest to lowest: thallium stress test result other than thallium stress test result coded 4 (meaning not recorded), thallium stress test result coded 4 (meaning not recorded)*	resting ECG other than a normal resting ECG: 25% to 44%; a normal resting ECG: 20% to 31%	log-odds difference 0.14, OR 1.16	121, 121, 2 (small n)	train; test unclear (AUC gain -0.002 [-0.016, 0.012], n=61); group difference on test 0.13 [-0.12, 0.35]	yes	the model has little data on this; on the training data only
C41	threshold: ST depression on exercise: fold at about 1.3 mm (starts) *shape: hinge at 1.3 mm; 70% of patients same; fallback; one point with C41 (median fold 1.3 mm, typical patient 0.4 mm, 10th-90th percentile 0.4 mm to 1.3 mm, wide)*	16% to 62% (typical patient with a fold)	fold 1.28 mm (IQR 0.35 to 1.28); past it log-odds 1.70, OR 5.47; range across patients 1.25 to 2.17	242, 109	train; test unclear (AUC gain 0.009 [-0.008, 0.033], n=61)	yes	on the training data only
C42	subgroup: who this part acts on: 109 of 242 patients	76% to 89% (typical patient this part acts on (risk without the part, then with it))	share 0.45; log-odds 0.74, OR 2.09; range across patients 0.02 to 1.62	109, 133	train; test unclear (AUC gain 0.006 [-0.014, 0.032], n=61)	yes	on the training data only
C43	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	242	train	yes	
C44	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (through this part) *shape: hinge at 125 mm Hg; 75% of patients same; fallback; one point with C52 (median fold 125 mm Hg, typical patient 148 mm Hg, 10th-90th percentile 122 mm Hg to 165 mm Hg, wide)*	23% to 17%	log-odds -0.27, OR 0.77; range across patients -0.48 to -0.08	242	train; test disagrees (AUC gain -0.003 [-0.014, 0.004], n=61)	yes	on the training data only
C45	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part)	20% to 18%	log-odds -0.04, OR 0.96; range across patients -0.08 to 0.00	160, 82	train; test disagrees (AUC gain -0.001 [-0.006, 0.000], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C49	interaction: effect of chest pain type (chest pain type other than non-anginal chest pain to non-anginal chest pain) by exercise-induced angina: chest pain brought on by exercise minus no chest pain on exercise; stronger in chest pain brought on by exercise *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	chest pain brought on by exercise: 31% to 7%; no chest pain on exercise: 20% to 4%	log-odds difference -0.001, OR 1.00	82, 160	train; test unclear (AUC gain -0.001 [-0.006, 0.000], n=61); group difference on test -0.008 [-0.02, 0.001]	no	on the training data only
C50	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by exercise-induced angina: chest pain brought on by exercise minus no chest pain on exercise; stronger in no chest pain on exercise *shape: accelerating; 81% of patients same; fallback*	chest pain brought on by exercise: 25% to 45%; no chest pain on exercise: 15% to 28%	log-odds difference -0.05, OR 0.95	82, 160	train; test unclear (AUC gain -0.001 [-0.006, 0.000], n=61); group difference on test -0.08 [-0.17, 0.007]	no	on the training data only
C51	interaction: effect of resting blood pressure (110 mm Hg to 155 mm Hg) by thallium stress test result: a fixed defect on the thallium scan minus thallium stress test result other than a fixed defect on the thallium scan; stronger in thallium stress test result other than a fixed defect on the thallium scan *shape: accelerating; 76% of patients same; fallback*	a fixed defect on the thallium scan: 8% to 17%; thallium stress test result other than a fixed defect on the thallium scan: 15% to 28%	log-odds difference -0.05, OR 0.95	15, 227 (small n)	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61); group difference on test -0.08 [-0.22, 0.12]	no	slightly; the model has little data on this; on the training data only
C52	threshold: resting blood pressure: fold at about 125 mm Hg (stops) for patients without chest pain on exercise *shape: hinge at 125 mm Hg; 92% of patients same; verified; risk saturates near 0%; one point with C52 (median fold 125 mm Hg, typical patient 122 mm Hg, 10th-90th percentile 122 mm Hg to 165 mm Hg, wide)*	4% to 3% (typical patient with a fold (patients without chest pain on exercise))	fold 125 mm Hg (IQR 122 to 132); past it log-odds -0.30, OR 0.74; range across patients -0.63 to -0.17	143, 78	train; test disagrees (AUC gain -0.003 [-0.014, 0.004], n=61)	yes	slightly; on the training data only
C53	subgroup: who this part acts on: 145 of 242 patients	51% to 68% (typical patient this part acts on (risk without the part, then with it))	share 0.60; log-odds 0.52, OR 1.69; range across patients 0.03 to 1.22	145, 97	train; test unclear (AUC gain 0.004 [-0.013, 0.024], n=61)	yes	on the training data only
C54	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	242	train	yes	
C55	direction: thallium stress test result: thallium stress test result other than a fixed defect on the thallium scan to a fixed defect on the thallium scan (through this part) *shape: categorical; highest to lowest: a fixed defect on the thallium scan, thallium stress test result other than a fixed defect on the thallium scan*	20% to 23%	log-odds 0.18, OR 1.20; range across patients 0.12 to 0.29	227, 15 (small n)	train; test unclear (AUC gain 0.002 [0.000, 0.01], n=61)	yes	slightly; the model has little data on this; on the training data only
C56	subgroup: who this part acts on: 234 of 242 patients	24% to 20% (typical patient this part acts on (risk without the part, then with it))	share 0.97; log-odds -0.23, OR 0.79; range across patients -0.48 to -0.12	234, 8	train; test unclear (AUC gain 0.000 [-0.011, 0.01], n=61)	yes	slightly; on the training data only
C57	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C58	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (through this part) *shape: categorical; highest to lowest: no chest pain on exercise, chest pain brought on by exercise*	30% to 25% (typical patient this part acts on)	log-odds -0.12, OR 0.89; range across patients -0.24 to 0.00	160, 82	train; test disagrees (AUC gain -0.003 [-0.013, 0.000], n=61)	yes	slightly; on the training data only
C59	direction: resting ECG: resting ECG other than signs of left ventricular hypertrophy on the resting ECG to signs of left ventricular hypertrophy on the resting ECG (through this part)	20% to 24%	log-odds 0.16, OR 1.17; range across patients 0.00 to 0.24	123, 119	train; test unclear (AUC gain 0.001 [-0.01, 0.012], n=61)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C61	subgroup: who this part acts on: 74 of 242 patients	25% to 30% (typical patient this part acts on (risk without the part, then with it))	share 0.31; log-odds 0.24, OR 1.27; range across patients 0.24 to 0.24	74, 168	train; test disagrees (AUC gain -0.002 [-0.016, 0.006], n=61)	yes	slightly; on the training data only
C62	null: this part barely changes predictions (switched off)		max change 0.059 (max |change in predicted risk|)	242	train	no	
C63	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C64	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (through this part) *shape: categorical; highest to lowest: fasting blood sugar above 120 mg/dL, fasting blood sugar of 120 mg/dL or less*	20% to 21%	log-odds 0.07, OR 1.08; range across patients 0.07 to 0.07	209, 33	train; test unclear (AUC gain 0.000 [-0.005, 0.004], n=61)	yes	slightly; on the training data only
C65	subgroup: who this part acts on: 33 of 242 patients	24% to 25% (typical patient this part acts on (risk without the part, then with it))	share 0.14; log-odds 0.07, OR 1.08; range across patients 0.07 to 0.07	33, 209	train; test unclear (AUC gain 0.000 [-0.005, 0.004], n=61)	yes	slightly; on the training data only
C66	null: this part barely changes predictions (switched off)		max change 0.019 (max |change in predicted risk|)	242	train	yes	
C67	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model.

- C16: direction claim about thallium stress test result in the whole model: not supported by the model (15% of patients have the opposite sign (allowed: 10%)).
- C25: direction claim about resting ECG in the whole model: not supported by the model (34% of patients have the opposite sign (allowed: 10%)).
- C28: direction claim about resting ECG in the whole model: not supported by the model (41% of patients have the opposite sign (allowed: 10%)).
- C48: interaction claim about resting blood pressure and chest pain type in part ann_34: not supported by the model (the group difference is below 0.1 log-odds).
- C49: interaction claim about chest pain type and exercise-induced angina in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C50: interaction claim about resting blood pressure and exercise-induced angina in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C51: interaction claim about resting blood pressure and thallium stress test result in part ann_34: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C62: null claim about resting ECG and exercise-induced angina in part ann_36: not supported by the model (predictions move by up to 0.059).

12.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every

annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (5):** `-13` (restecg:1), `-20` (slope:2), `-26` (thal:7), `-3` (sex:1), `-5` (cp:2).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

12.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 12.6.1)

▷ A2: ischaemia_gate (leaf, section 12.6.2)
▷ A3: bp_offset_gate (leaf, section 12.6.3)
▷ A4: ischaemia_damping_thal6 (leaf, section 12.6.4)
▷ A5: lvh_without_exercise_angina (leaf, section 12.6.5)
▷ A6: high_fasting_sugar (leaf, section 12.6.6)

12.4 Network figures

12.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-12` into `-12_a`, `-12_b`.
- #27: split `-17` into `-17_a`, `-17_b`, `-17_c`.
- #28: split `-25` into `-25_a`, `-25_b`, `-25_c`.
- #29: split `-8` into `-8_a`, `-8_b`.
- #30: split `-6` into `-6_a`, `-6_b`.
- #31: identity node `identity_290` before `290` (1 connection(s)).
- #32: identity node `identity_1416` before `1416` (1 connection(s)).

12.6 Annotations

12.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 12.6.2), A3 (section 12.6.3), A4 (section 12.6.4), A5 (section 12.6.5), A6 (section 12.6.6).

- **Entry nodes:** `-1` (age), `-2` (sex:0), `-3` (sex:1), `-4` (cp:1), `-5` (cp:2), `-6_a` (cp:3_a), `-7` (cp:4), `-8_a` (trestbps_a), `-9` (chol), `-10` (fbs:0), `-12_a` (restecg:0_a), `-13` (restecg:1), `-15` (thalach), `-16` (exang:0), `-17_a` (exang:1_a), `-19` (slope:1), `-20` (slope:2), `-21` (slope:3), `-22` (ca), `-23` (thal:3), `-25_a` (thal:6_a), `-26` (thal:7).
- **Exit nodes:** `0`.
- **Internal nodes:** none.

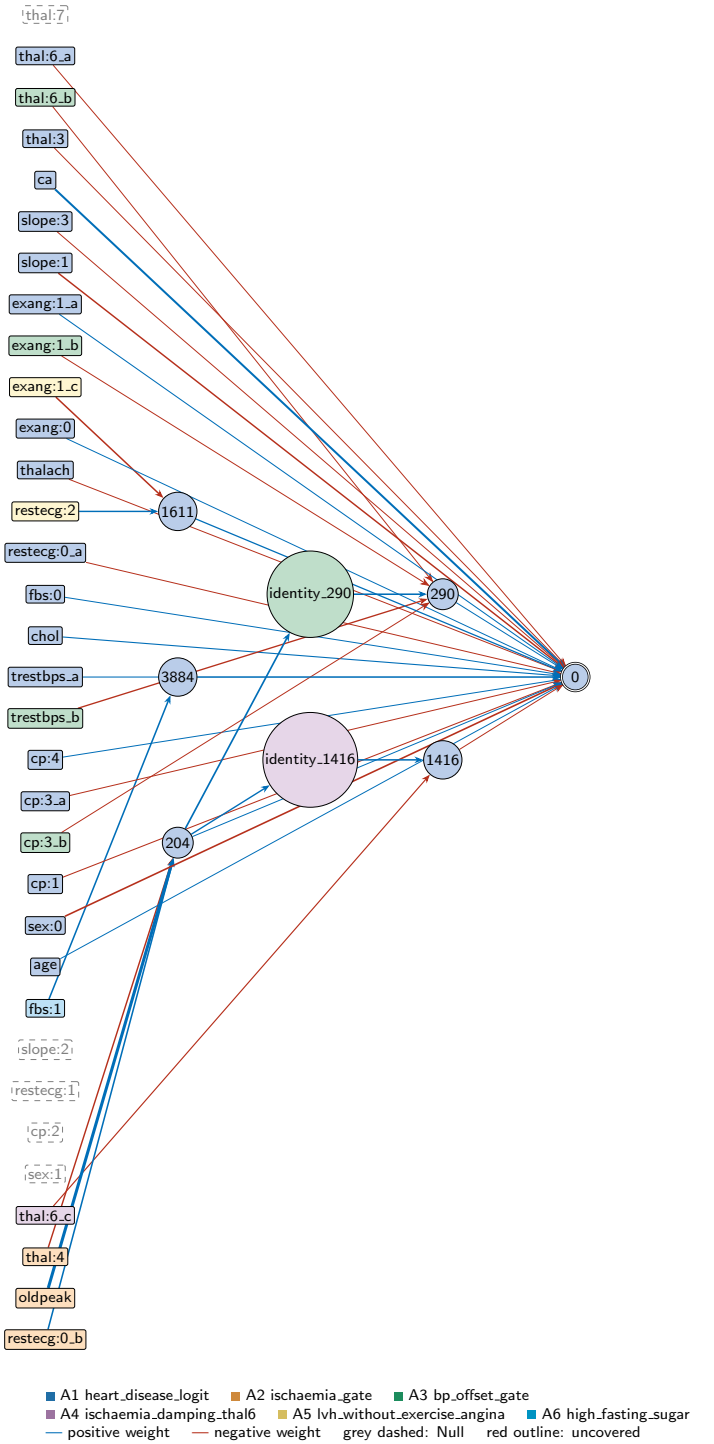


Figure 36: heart-s5-c2: full view of the explained model state.

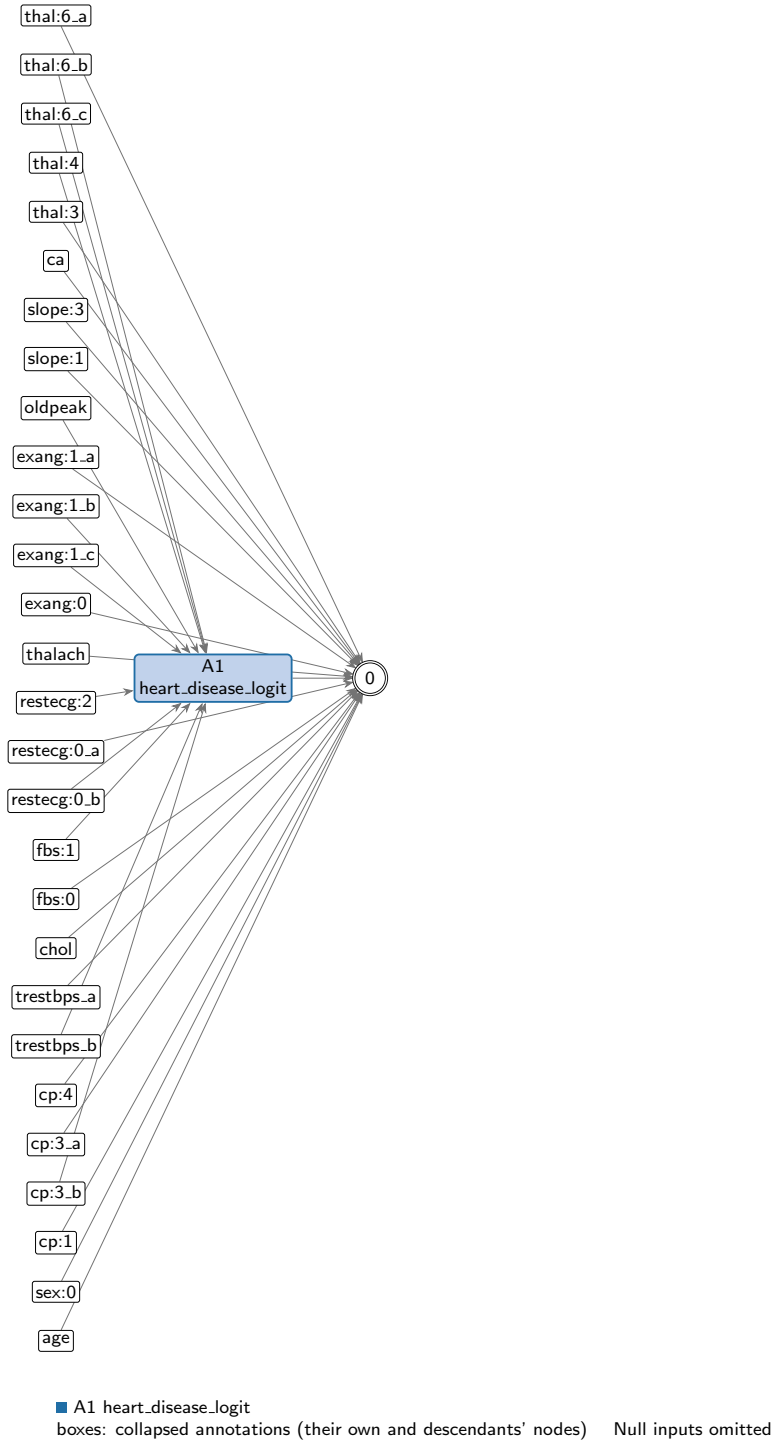


Figure 37: heart-s5-c2: collapsed view at hierarchy level 0 (fully recomposed).

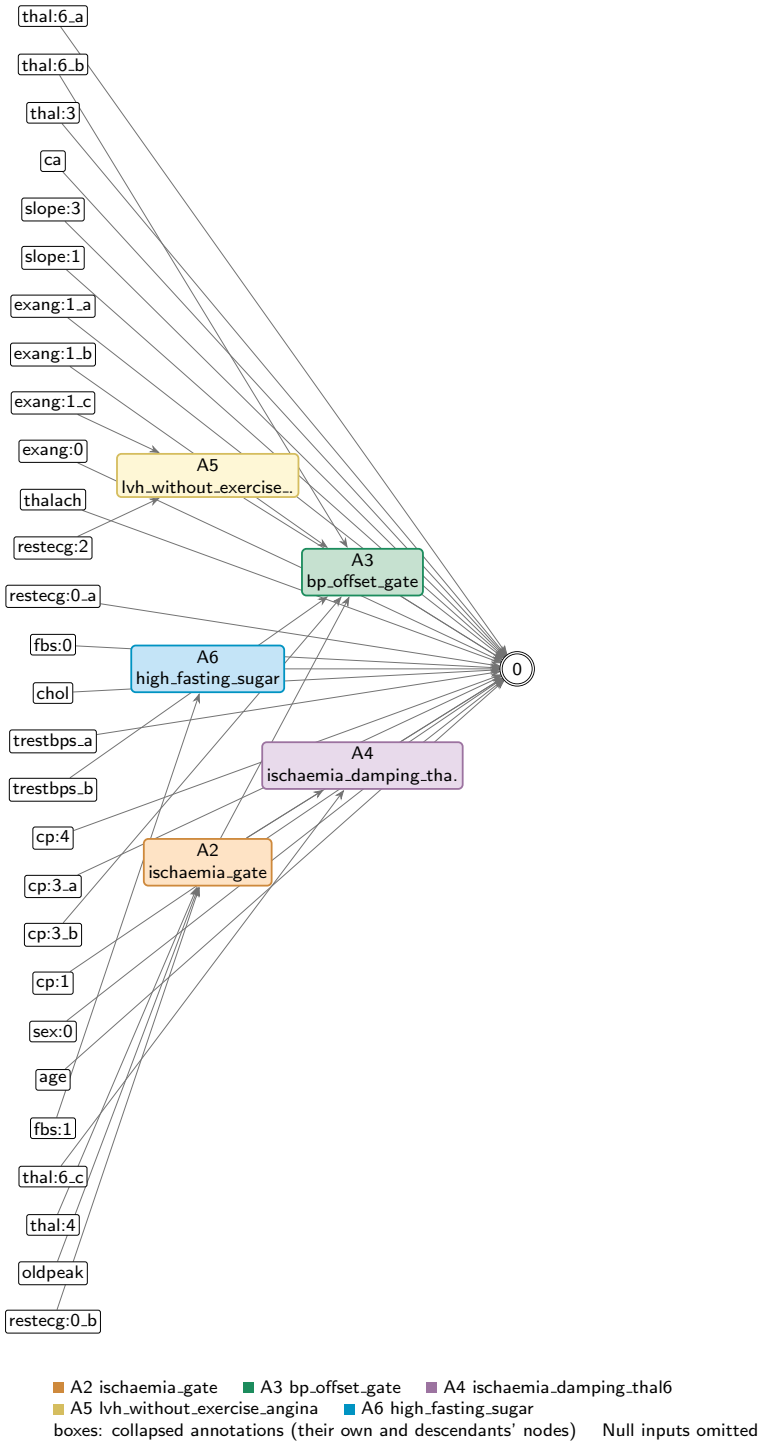


Figure 38: heart-s5-c2: collapsed view at hierarchy level 1.

- **Edges (weight):** -1→0 (0.126), -2→0 (-1.06), -4→0 (-0.489), -7→0 (0.357), -9→0 (0.522), -10→0 (0.264), -15→0 (-0.226), -16→0 (0.136), -19→0 (-0.921), -21→0 (-0.37), -22→0 (1.47), -23→0 (-0.423), 204→0 (0.135), 290→0 (0.252), 1416→0 (-0.118), 1611→0 (0.729), 3884→0 (0.0482), -12_a→0 (-0.158), -17_a→0 (0.471), -25_a→0 (-0.413), -8_a→0 (0.498), -6_a→0 (-0.444), 204→identity_290 (1.14), 204→identity_1416 (0.841).
- **With its descendants:** 41 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 40 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = -0.487 + LINEAR CORE + 0.136*ischaemia_gate + 0.252*bp_offset_gate - 0.118*ischaemia_damping_thal6 + 0.729*lvh_without_exercise_angina + 0.048*high_fasting_sugar. LINEAR CORE (17 direct input->output edges, z-scored inputs): ca +1.473, sex:0 -1.062, slope:1 -0.921, chol +0.522, trestbps +0.498, cp:1 -0.489, exang:1 +0.471, cp:3 -0.444, thal:3 -0.423, thal:6 -0.413, slope:3 -0.370, cp:4 +0.357, fbs:0 +0.264, thalach -0.226, restecg:0 -0.159, exang:0 +0.136, age +0.126. Five inputs are unconnected (sex:1, cp:2, restecg:1, slope:2, thal:7; zero-effect entries of this root). Oldpeak, thal:4, restecg:2 and fbs:1 have no direct edge and act only through the children. COMPOSITION: additive readouts, with a serial fan-out inside. ischaemia_gate (oldpeak above a resting-ECG-dependent threshold) is read out directly and also feeds bp_offset_gate (the ischaemia score minus a resting-BP offset, the largest nonlinear readout) and ischaemia_damping_thal6 (a near-linear term that takes back part of the ischaemia score and adds a fixed-defect term); identity nodes identity_290 and identity_1416 carry those two links so each gate is its own leaf. Two independent indicators add to z: lvh_without_exercise_angina (+0.236 for LVH on the resting ECG without exercise angina) and high_fasting_sugar (+0.074, negligible). EVIDENCE (train, n=242): full model AUC 0.9389, acc 0.880 (compute_performance). Linear core alone AUC 0.933; logit SD 3.19 (core) vs 0.57 (all hidden readouts). Removing the ischaemia gate with its downstream effects gives 0.932; readout-only ablations: ischaemia 0.938, BP 0.933, damping 0.938, LVH 0.939, fbs 0.939. Treating all five ReLUs as identities gives 0.917 (the gating matters); one least-squares linear fit of all hidden readouts on their inputs gives 0.938 (R2 0.84); an L2 logistic regression on all 26 inputs gives 0.936. Mean |SHAP| (train, all 242): ca 0.145, sex:0 0.109, slope:1 0.108, thal:3 0.050, oldpeak 0.047, cp:3 0.045, exang:1 0.044, chol 0.043. Interpretation: close to logistic regression plus one ischaemia correction: ST depression raises risk only above a threshold that is lower with a normal resting ECG, more steeply when resting BP is low; LVH without exercise angina adds a little. Counter-directional signs reported, not explained: restecg:0 positive in the ischaemia gate, trestbps negative in the BP gate, thal:6 (fixed defect) negative directly, fbs:0 +0.26 directly. Same core design as the 10-seed heart-s5 record (key 9330) and the sibling cluster lead heart-s5-c1 (ischaemia gate -> resting-BP gate), with a damping node, an LVH indicator and a fasting-sugar indicator instead of c1's age gate.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -0.487 + 0.136x_{exang:0} + 0.522x_{chol} + 0.471x_{exang:1_a} + 0.126x_{age} \\
&\quad + 0.357x_{cp:4} \\
&\quad + 0.252\,\text{bp_offset_gate}\big(1.14\,\text{ischaemia_gate}\big(x_{restecg:0_b}, \\
&\quad x_{oldpeak}, x_{thal:4}\big), x_{trestbps_b}, x_{cp:3_b}, x_{exang:1_b}, \\
&\quad x_{thal:6_b}\big) + 0.264x_{fbs:0} \\
&\quad + 0.135\,\text{ischaemia_gate}\big(x_{restecg:0_b}, x_{oldpeak}, x_{thal:4}\big) \\
&\quad + 0.729\,\text{lvh_without_exercise_angina}\big(x_{restecg:2}, x_{exang:1_c}\big) \\
&\quad + 1.47x_{ca} + 0.0482\,\text{high_fasting_sugar}\big(x_{fbs:1}\big) \\
&\quad + 0.497x_{trestbps_a} - 1.06x_{sex:0} - 0.423x_{thal:3} - 0.444x_{cp:3_a} \\
&\quad - 0.158x_{restecg:0_a} - 0.37x_{slope:3} - 0.413x_{thal:6_a} - 0.226x_{thalach} \\
&\quad - 0.921x_{slope:1} \\
&\quad - 0.118\,\text{ischaemia_damping_thal6}\big(0.841\,\text{ischaemia_gate}\big(x_{restecg:0_b},
\end{aligned}
$$

$$x_{oldpeak}, x_{thal:4}\big), x_{thal:6_c}\big) - 0.489x_{cp:1}$$

Expanded over the children:

$$\begin{aligned}
y_0 &= \sigma\big(z_0\big) \\
z_0 &= -0.487 + 0.136x_{exang:0} + 0.522x_{chol} + 0.471x_{exang:1_a} + 0.126x_{age} \\
&\quad + 0.357x_{cp:4} + 0.252\,\mathrm{relu}\big(-0.447 + 1.14\,\mathrm{relu}\big(0.385 \\
&\quad + 0.992x_{restecg:0_b} + 2.54x_{oldpeak} - 0.876x_{thal:4}\big) - 0.0688x_{cp:3_b} \\
&\quad - 0.153x_{exang:1_b} - 0.00683x_{thal:6_b} - 0.773x_{trestbps_b}\big) + 0.264x_{fbs:0} \\
&\quad + 0.135\,\mathrm{relu}\big(0.385 + 0.992x_{restecg:0_b} + 2.54x_{oldpeak} \\
&\quad - 0.876x_{thal:4}\big) + 0.729\,\mathrm{relu}\big(-1.26 + 0.887x_{restecg:2} \\
&\quad - 0.953x_{exang:1_c}\big) + 1.47x_{ca} + 0.0482\,\mathrm{relu}\big(-0.98 \\
&\quad + 1.0x_{fbs:1}\big) + 0.497x_{trestbps_a} - 1.06x_{sex:0} - 0.423x_{thal:3} \\
&\quad - 0.444x_{cp:3_a} - 0.158x_{restecg:0_a} - 0.37x_{slope:3} - 0.413x_{thal:6_a} \\
&\quad - 0.226x_{thalach} - 0.921x_{slope:1} - 0.118\,\mathrm{relu}\big(0.853 \\
&\quad + 0.841\,\mathrm{relu}\big(0.385 + 0.992x_{restecg:0_b} + 2.54x_{oldpeak} \\
&\quad - 0.876x_{thal:4}\big) - 0.594x_{thal:6_c}\big) - 0.489x_{cp:1}
\end{aligned}$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00243, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=true); output is sigmoid.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, ablations, linear replacements and SHAP.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.6.2 A2: ischaemia_gate

Kind: Leaf annotation (depth 1); parent A1 (section 12.6.1).

- **Entry nodes:** `-12_b` (restecg:0_b), `-18` (oldpeak), `-24` (thal:4).
- **Exit nodes:** `204`.
- **Internal nodes:** none.
- **Edges (weight):** `-12_b`→`204` (0.992), `-18`→`204` (2.54), `-24`→`204` (-0.876).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. Ischaemia-evidence gate (raises risk). 204 = ReLU(0.385 + 2.536*oldpeak + 0.992*restecg:0 - 0.876*thal:4). Hypothesis: a one-sided ST-depression score that opens above an oldpeak threshold set by the resting ECG (z-oldpeak -0.57 with a normal ECG, +0.21 without) and then rises with oldpeak; it adds 0.136 per unit to the logit directly, a further 0.286 per unit through the BP gate while that gate is open, and -0.099 per unit through the damping node, so about 0.32 logit per unit in all. EVIDENCE (train, n=242): active for 45% (51% with a normal ECG, 40% without); corr with oldpeak 0.92; disease rate 60% when active vs 32% when inactive. Mean |sensitivity| oldpeak 1.14 > restecg:0 0.45 > thal:4 0.39 (thal:4 has 2 patients). This is the main nonlinear effect: removing it with its downstream effects lowers train AUC 0.939 -> 0.932 (below the linear-core level 0.933); an affine replacement recovers 0.938. Supported. Oldpeak and

thal:4 have no direct edge and act only here. The positive restecg:0 weight (a normal ECG raises the score; restecg:0's direct edge is -0.16) is counter-intuitive and is reported, not explained.

Closed form:

$$y_0 = \mathrm{relu}\big(0.385 + 0.992x_{restecg:0_b} + 2.54x_{oldpeak} - 0.876x_{thal:4}\big)$$

where y_0 is node `204`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.0173, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical); thresholds: thresholds = z-oldpeak at which 204 opens: -0.57 with a normal resting E.... Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.6.3 A3: bp_offset_gate

Kind: Leaf annotation (depth 1); parent A1 (section 12.6.1).

- **Entry nodes:** `identity_290`, `-8_b` (trestbps_b), `-6_b` (cp:3_b), `-17_b` (exang:1_b), `-25_b` (thal:6_b).
- **Exit nodes:** `290`.
- **Internal nodes:** none.
- **Edges (weight):** `identity_290`→`290` (1), `-8_b`→`290` (-0.773), `-6_b`→`290` (-0.0688), `-17_b`→`290` (-0.153), `-25_b`→`290` (-0.00683).
- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. Blood-pressure offset gate on the ischaemia score (raises risk). 290 = ReLU(-0.447 + 1.137*ischaemia_gate - 0.773*trestbps - 0.153*exang:1 - 0.069*cp:3 - 0.007*thal:6), read out +0.252. Hypothesis: passes the ischaemia score on, minus an offset that grows with resting blood pressure (and slightly with exercise angina and non-anginal pain). With the ischaemia gate closed it still opens for low resting blood pressure, so within this gate lower resting BP raises risk (0.252*0.773 = 0.195 per SD while open), partly offsetting trestbps's direct +0.50 edge. EVIDENCE (train, n=242): active for 60% (87% with the ischaemia gate open, 38% without); with the ischaemia gate open, active for 100%, 85% and 77% of the low, middle and high resting-BP groups. Disease rate 50% when active vs 35% when inactive. Mean |sensitivity| ischaemia score 0.60 > trestbps 0.46 > exang:1 0.09 > cp:3 0.04 > thal:6 0.004. It is the largest nonlinear readout (logit SD 0.50): readout ablation lowers train AUC 0.939 -> 0.933; an affine replacement fits it well (R2 0.97, AUC 0.939), so its nonlinearity matters little. Supported. The negative BP term inside a risk-raising gate is reported, not explained.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.447 + 1.0x_{identity_290} - 0.0688x_{cp:3_b} - 0.153x_{exang:1_b} - 0.00683x_{thal:6_b} - 0.773x_{trestbps_b}\big)$$

where y_0 is node `290`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00119, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.6.4 A4: ischaemia_damping_thal6

Kind: Leaf annotation (depth 1); parent A1 (section 12.6.1).

- **Entry nodes:** `identity_1416`, `-25_c` (thal:6_c).
- **Exit nodes:** `1416`.
- **Internal nodes:** none.
- **Edges (weight):** `identity_1416`→`1416` (1), `-25_c`→`1416` (-0.594).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Ischaemia damping with a fixed-defect term (near-linear, small). 1416 = ReLU(0.853 + 0.841*ischaemia_gate - 0.594*thal:6), read out -0.118. Hypothesis: almost always open, so it acts as a linear correction: it subtracts 0.099 per unit of the ischaemia score from the logit (damping the ischaemia gate's direct +0.136) and adds 0.070 per SD of thal:6 (about +0.29 logit for a fixed thallium defect, against thal:6's direct -0.41 edge). It closes only for fixed-defect patients whose ischaemia score is below 1.73. EVIDENCE (train, n=242): active for 97% (closed for 8 of the 15 fixed-defect patients). Mean |sensitivity| ischaemia score 0.97, thal:6 0.57. Readout ablation: train AUC 0.939 -> 0.938; affine replacement R2 0.99, AUC 0.939. Supported; effectively linear.

Closed form:

$$y_0 = \mathrm{relu}\big(0.853 + 1.0x_{identity_1416} - 0.594x_{thal:6_c}\big)$$

where y_0 is node `1416`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00134, and the exact function by at most 2.22×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.6.5 A5: lvh_without_exercise_angina

Kind: Leaf annotation (depth 1); parent A1 (section 12.6.1).

- **Entry nodes:** `-14` (restecg:2), `-17_c` (exang:1_c).
- **Exit nodes:** `1611`.
- **Internal nodes:** none.
- **Edges (weight):** `-14`→`1611` (0.887), `-17_c`→`1611` (-0.953).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. LVH without exercise angina (AND indicator, raises risk, small). 1611 = ReLU(-1.261 + 0.887*restecg:2 - 0.953*exang:1), read out +0.729. Hypothesis: a binary AND: open (constant 0.324, +0.236 logit) only for patients with left ventricular hypertrophy on the resting ECG and no exercise-induced angina, zero otherwise. EVIDENCE (train, n=242): active for exactly the 74 patients with LVH and no exercise angina; their disease rate is 41%, against 20% for patients with neither. Mean |sensitivity| exang:1 0.29, restecg:2 0.27. Readout ablation: train AUC 0.939 -> 0.939 (-0.0003). restecg:2 acts only here. Supported; near-negligible for ranking.

Closed form:

$$y_0 = \mathrm{relu}\big(-1.26 + 0.887 x_{restecg:2} - 0.953 x_{exang:1_c}\big)$$

where y_0 is node `1611`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 2.52×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.6.6 A6: high_fasting_sugar

Kind: Leaf annotation (depth 1); parent A1 (section 12.6.1).

- **Entry nodes:** `-11` (fbs:1).
- **Exit nodes:** `3884`.
- **Internal nodes:** none.
- **Edges (weight):** `-11`→`3884` (1).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. High fasting blood sugar indicator (negligible). 3884 = ReLU(-0.980 + fbs:1), read out +0.048: +0.074 logit for the 33 patients with fasting blood sugar > 120 mg/dl. Hypothesis: a small additive term for high fasting sugar (exactly linear on the two values of fbs:1); together with fbs:0's direct +0.26 edge, high fasting sugar overall lowers the logit (fbs:0 is the normal-sugar flag). EVIDENCE (train, n=242): active for exactly the 33 high-sugar patients; disease rate 45% vs 44%. Readout ablation changes train AUC by less than 0.0001; affine replacement R2 1.0. Supported; negligible.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.98 + 1.0 x_{fbs:1}\big)$$

where y_0 is node `3884`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 3.12×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula (force) with readouts and thresholds.
- EMPIRICAL, `empirical_viz` (sensitivity (max_samples 100000)); data split: train. Train-split sensitivity, activity, outcome rates and ablation.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

12.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft

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13. heart-s5-c3: verification checklist

Already recorded: nothing (no verdict in the record).

Recorded vs live. Agent = the drafting agent's report; orchestrator = the record's automated re-check; live = this export (current code, training split).

Check	Agent	Orchestrator	Live	Status
Function preservation, max |change in output|	2.22e-16	2.22e-16	2.22e-16	agree
Closed-form max abs error (all leaves)	0.0205	0.0205	0.0205	agree
Numeric-function max abs error	n/a	1.78e-15	1.78e-15	agree
Structural coverage	1	1	1	agree
Compositional coverage	1	1	1	agree
Hierarchy complete	yes	yes	yes	agree
Root annotation	n/a	heart_disease_logit	heart_disease_logit	agree
Leaf / composition counts	2 / 1	n/a	2 / 1	agree
Operation counts	1 splits, 0 identity	annotate 3, rename node 26, split node 1	annotate 3, rename node 26, split node 1	agree
Full coverage	yes	yes	yes	agree

What to look at.

- Largest closed-form error: `ischaemia_chain` (0.0205). This is the cost of the 3 s.f. display rounding, in the exit's own units; the numeric function is exact. Check that the rounded formula still supports the hypothesis as worded.
- 1 split(s), all of input nodes (an input feeding two annotations). These are routine; check only that each input copy sits in the annotation whose hypothesis uses it.
- Read the root's hypothesis against its closed form: does the stated combination of children (additive, gated, serial) match the formula?

Tool notes (record): get_input_distribution does not accept sample_fraction (called with max_samples only; all 242 rows used). get_viz_summary requires viz_type (sensitivity and line_plot used). add_identity_node cannot intercept a node's only input (by design), which forced the two-exit leaf. compute_shap (kernel, all 242 train rows) gives the seven unconnected inputs mean |SHAP| about 0.001 instead of exactly 0 (sampling noise). The agent sandbox refused compound shell commands, so tools were called through a small Python subprocess wrapper in scratch (same species_leads/xn.py).

Plain-language summary (AI-assisted draft for review (unverified); the full read-down document with its figures is `heart-s5-c3-plain-v5.pdf` in this genome's folder):

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and two parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 105 of 242 training patients, and an age term that acts on 148. [C25, C30, C34] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to about 2%, typical angina to about 2%, non-anginal chest pain to about 4% and atypical angina to about 10%; fasting blood sugar above 120 mg/dL lowers it to about 9%; and a higher maximum heart rate lowers it steadily, from about 23% at 118 to about 15% at 175 beats per minute. [C10, C09, C11, C15, C12, C14] Each extra vessel coloured by fluoroscopy adds about the same, from about 18% at 0 to about 86% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 56%, a reversible defect on the thallium scan to about 38% and chest pain brought on by exercise to about 35%; higher cholesterol raises it steadily, from about 11% at 193 mg/dL to about 33% at 308 mg/dL, and higher resting blood pressure steadily, from about 12% at 110 mm Hg to about 30% at 155 mm Hg. [C01, C04, C06, C08, C05, C07] Through this sum alone, a resting ECG other than normal raises it from about 18% to about 24%; the resting ECG also acts through the ST depression signal (below). [C16] Some rarer results also count (the model has little data on this, n = 2 to 17): a

thallium result coded 4 raises the risk to about 79%, a fixed defect on the thallium scan lowers it to about 9%, and a downsloping ST segment leaves it at about 19%. [C02, C13, C19] This part reads ST depression on exercise and the resting ECG, and acts on 105 of 242 training patients, 66 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 91% with it. [C30] Through it, more ST depression adds nothing until about 0.5 mm, then raises the predicted risk steadily: for a typical patient from about 16% at 0 mm to about 60% at 2.8 mm. [C26] ST depression counts more with a normal resting ECG, nothing until about 0.5 mm and then steadily, from about 16% to about 60%, against about 21% to about 48% with any other resting ECG. [C28] Through it, a resting ECG other than normal lowers a typical patient's risk from about 18% to about 16%. [C27] This part acts on 148 of 242 training patients, 81 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 91% without it and about 92% with it. [C34] Through it, age adds nothing until about 53 years, then raises the risk steadily, for a typical patient from about 18% at 42 to about 19% at 66 years. [C32, C33]

Worth checking: Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 9%). [C12] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C09, C11, C15] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C13] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C02] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C28]

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 86%	log-odds 3.33, OR 28.0; range across patients 3.33 to 3.33	242	train; test unclear (AUC gain 0.029 [-0.039, 0.100], n=61)	yes	on the training data only
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 79%	log-odds 2.85, OR 17.4; range across patients 2.85 to 2.85	135, 2 (small n)	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	the model has little data on this; on the training data only
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 56%	log-odds 1.75, OR 5.75; range across patients 1.75 to 1.75	115, 110	train; test disagrees (AUC gain -0.024 [-0.071, 0.009], n=61)	yes	on the training data only
C05	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	11% to 33%	log-odds 1.32, OR 3.73; range across patients 1.32 to 1.32	242	train; test disagrees (AUC gain -0.019 [-0.058, 0.008], n=61)	yes	on the training data only
C06	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 38%	log-odds 1.00, OR 2.72; range across patients 1.00 to 1.00	135, 90	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	on the training data only
C07	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	12% to 30%	log-odds 1.19, OR 3.30; range across patients 1.19 to 1.19	242	train; test unclear (AUC gain 0.008 [-0.018, 0.037], n=61)	yes	on the training data only
C08	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	18% to 35%	log-odds 0.89, OR 2.44; range across patients 0.89 to 0.89	160, 82	train; test agrees (AUC gain 0.022 [0.003, 0.049], n=61)	yes	clearly
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 2%	log-odds -2.43, OR 0.09; range across patients -2.43 to -2.43	108, 22	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to 2%	log-odds -2.19, OR 0.11; range across patients -2.19 to -2.19	159, 83	train; test unclear (AUC gain 0.025 [-0.007, 0.066], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 4%	log-odds -1.69, OR 0.18; range across patients -1.69 to -1.69	108, 70	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C12	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 9%	log-odds -0.80, OR 0.45; range across patients -0.80 to -0.80	209, 33	train; test unclear (AUC gain 0.000 [-0.013, 0.013], n=61)	yes	on the training data only
C13	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4*	18% to 9%	log-odds -0.80, OR 0.45; range across patients -0.80 to -0.80	135, 15 (small n)	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	the model has little data on this; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C14	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	23% to 15%	log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	242	train; test agrees (AUC gain 0.014 [0.002, 0.035], n=61)	yes	clearly
C15	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 10%	log-odds -0.71, OR 0.49; range across patients -0.71 to -0.71	108, 42	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C16	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	18% to 24%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	121, 121	train; test unclear (AUC gain 0.005 [-0.005, 0.024], n=61)	yes	on the training data only
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 19%	log-odds 0.05, OR 1.05; range across patients 0.05 to 0.05	115, 17 (small n)	train; test disagrees (AUC gain -0.024 [-0.071, 0.009], n=61)	yes	slightly; the model has little data on this; on the training data only
C25	structure: made of ischaemia_chain, age_hinge and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C26	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	log-odds 1.65, OR 5.20; range across patients 1.22 to 2.08	242	train; test unclear (AUC gain 0.023 [-0.001, 0.055], n=61)	yes	on the training data only
C27	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 16%	log-odds -0.36, OR 0.70; range across patients -0.86 to 0.00	121, 121	train; test unclear (AUC gain 0.006 [-0.009, 0.028], n=61)	yes	slightly; on the training data only
C28	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.5 mm; 100% of patients same; verified*	resting ECG other than a normal resting ECG: 21% to 48%; a normal resting ECG: 16% to 60%	log-odds difference -0.86, OR 0.42	121, 121	train; test agrees (AUC gain 0.006 [-0.009, 0.028], n=61); group difference on test -0.86 [-0.86, -0.86]	yes	clearly
C30	subgroup: who this part acts on: 105 of 242 patients	76% to 91% (typical patient this part acts on (risk without the part, then with it))	share 0.43; log-odds 1.03, OR 2.80; range across patients 0.03 to 2.33	105, 137	train; test unclear (AUC gain 0.026 [-0.001, 0.062], n=61)	yes	on the training data only
C32	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 19%	log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10	242	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C33	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 20% (typical patient with a fold)	fold 52.8 years (IQR 52.8 to 52.8); past it log-odds 0.12, OR 1.13; range across patients 0.12 to 0.12	242, 148	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C34	subgroup: who this part acts on: 148 of 242 patients	91% to 92% (typical patient this part acts on (risk without the part, then with it))	share 0.61; log-odds 0.06, OR 1.07; range across patients 0.009 to 0.12	148, 94	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only

Every paragraph and claim: the certificate's "Plain-language summary" subsection. Record a verdict on the plain-language layer with `uv run python -m explaneat export record-plain-verdict heart-s5-c3 --verified --minutes N --notes "..."`.

Verdict.
Verified (correct as it stands, or after my changes): ☐ yes ☐ no

Minutes: _____

Changes: _____

_____

Notes: _____

_____

Record it (writes `mike_verification` into `experiments/10seed/explanations/heart-s5-c3.json`; use `--not-verified` for no; rebuild the CSVs with `uv run python experiments/10seed/build_explanations_`
`uv run python -m explaneat export record-verdict heart-s5-c3 --verified --minutes N --changes "..." --notes "..."`

13 Explanation certificate: heart-s5-c3

This section is the explanation certificate of the heart-s5-c3 network. It is rendered from the canonical JSON record `heart-s5-c3.json` (schema 1.4.0, SHA-256 `644db15ecc0191f8`…), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 13 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s5, seed 5
Genome	`96fb45fb-972d-4aaa-b6d5-01fc63fbc4e6` (NEAT key 9698, generation 39)
Explanation	`1b3f669d-9fb5-4126-b6e4-35535db7b3b1` (30 operations)
Train AUC / accuracy	0.9376 / 0.8884 (n = 242)
Test AUC / accuracy	0.9084 / 0.8197 (n = 61)
Output activation	sigmoid
Phenotype	31 nodes (26 inputs, 4 hidden, 1 output), 26 enabled edges
Explained model state	32 nodes, 26 enabled edges
Refactoring operations	split_node 1
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 2.22 \times 10^{-16}$ on the training split
Annotations	3 (2 leaf, 1 composition)
Structural coverage	1 (31 of 31 candidate nodes; 7 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`644db15ecc0191f8`…

Table 13: Explanation certificate summary: heart-s5-c3.

Drafter's summary. Mostly logistic regression: 17 direct edges (logit SD 3.22; train AUC 0.932 alone) plus an ST-depression chain and an age hinge (logit SD 0.80 together) to 0.938. The chain scores oldpeak (threshold lowered by a normal resting ECG) and adds a second, much steeper stage of the same score, so oldpeak's effect is a hinge (about 0.12 logit per SD, then about 1.10 per SD above z -0.39 with a normal ECG or +0.40 without); age enters only through a small hinge starting

just below the median. Same ischaemia gate as the 10-seed heart-s5 record (key 9330) and its cluster-1 sibling, but with no resting-BP offset and a free-standing age term.

What resisted explanation. Nothing resisted coverage. The chain 204 -> 3487 -> 290 (with 204 -> 290 as well, and both 204 and 290 read out) was kept as one two-exit leaf rather than two leaves: 3487's only input is 204, so it cannot be intercepted by an identity node (add_identity_node must leave one input un-redirected), and splitting 204 would have duplicated the gate. 3487 is a near-copy of 204 (ReLU(0.94*204 - 0.002)), so 290 = ReLU(1.818*204 - 0.470): a second, steeper threshold on the same score. Counter-directional signs reported, not explained: a normal resting ECG (restecg:0) lowers the oldpeak threshold inside the chain (raising risk) while its direct edge is -0.188; fbs:0 +0.27 directly; thal:7 only +0.025. The age hinge barely matters (readout ablation -0.0005 AUC). closed_form_max_abs_err 0.021 comes from 3-significant-figure display rounding in the nested chain formula (numeric error 1.8e-15).

13.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and two parts add to that sum: a signal from ST depression on exercise and the resting ECG that acts on 105 of 242 training patients, and an age term that acts on 148. [C25, C30, C34] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to about 2%, typical angina to about 2%, non-anginal chest pain to about 4% and atypical angina to about 10%; fasting blood sugar above 120 mg/dL lowers it to about 9%; and a higher maximum heart rate lowers it steadily, from about 23% at 118 to about 15% at 175 beats per minute. [C10, C09, C11, C15, C12, C14] Each extra vessel coloured by fluoroscopy adds about the same, from about 18% at 0 to about 86% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher; a flat rather than upsloping ST segment raises it to about 56%, a reversible defect on the thallium scan to about 38% and chest pain brought on by exercise to about 35%; higher cholesterol raises it steadily, from about 11% at 193 mg/dL to about 33% at 308 mg/dL, and higher resting blood pressure steadily, from about 12% at 110 mm Hg to about 30% at 155 mm Hg. [C01, C04, C06, C08, C05, C07] Through this sum alone, a resting ECG other than normal raises it from about 18% to about 24%; the resting ECG also acts through the ST depression signal (below). [C16] Some rarer results also count (the model has little data on this, n = 2 to 17): a thallium result coded 4 raises the risk to about 79%, a fixed defect on the thallium scan lowers it to about 9%, and a downsloping ST segment leaves it at about 19%. [C02, C13, C19] This part reads ST depression on exercise and the resting ECG, and acts on 105 of 242 training patients, 66 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 91% with it. [C30] Through it, more ST depression adds nothing until about 0.5 mm, then raises the predicted risk steadily: for a typical patient from about 16% at 0 mm to about 60% at 2.8 mm. [C26] ST depression counts more with a normal resting ECG, nothing until about 0.5 mm and then steadily, from about 16% to about 60%, against about 21% to about 48% with any other resting ECG. [C28] Through it, a resting ECG other than normal lowers a typical patient's risk from about 18% to about 16%. [C27] This part acts on 148 of 242 training patients, 81 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 91% without it and about 92% with it. [C34] Through it, age adds nothing until about 53 years, then raises the risk steadily, for a typical patient from about 18% at 42 to about 19% at 66 years. [C32, C33]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, sex and chest pain type each move the predicted risk on their own, and two parts add to that sum: a signal from ST depression shaped by the resting ECG, and an age term. [C25, C01, C04, C10, C09] *Evidence.* It is made up of the inputs that feed the score directly and two parts, the ST depression signal and the age term (each explained below), and the score is their sum, which was checked on all 242 training patients. [C25] ST depression on exercise and age do not feed the score directly, so they act only through these parts. [C25] Changing one input at a time across the whole model, more ST depression adds nothing until about 0.5 mm, then raises a typical patient's predicted risk steadily, from about 16% at 0 mm to about 60% at 2.8 mm. [C03] Across the whole model, older age adds nothing until about 53 years, then raises it steadily, from about 18% at 42 to about 19% at 66 years. [C18]

ST depression signal. This part raises the predicted risk with ST depression on exercise, adding nothing until about 0.5 mm and then steadily, and more so with a normal resting ECG. [C26, C28] *Evidence.* It acts on 105 of 242 training patients, 66 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 76% without it and about 91% with it. [C30] Through it, more ST depression adds nothing until about 0.5 mm, then raises the predicted risk steadily: for a typical patient from about 16% at 0 mm to about 60% at 2.8 mm. [C26] ST depression counts more with a normal resting ECG, nothing until about 0.5 mm and then steadily, from about 16% to about 60%, against about 21% to about 48% with any other resting ECG. [C28] Through it, a resting ECG other than normal lowers a typical patient's risk from about 18% to about 16%. [C27] Its formula reproduces the part exactly on all 242 training patients. [C31]

Age term. This part adds a little to the predicted risk for older patients: nothing until about 53 years, then steadily. [C32, C33] *Evidence.* It acts on 148 of 242 training patients, 81 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 91% without it and about 92% with it. [C34] Through it, age adds nothing until about 53 years, then raises the risk steadily, for a typical patient from about 18% at 42 to about 19% at 66 years. [C32, C33] Its formula reproduces the part exactly on all 242 training patients. [C36]

Worth checking against what is known. Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 9%). [C12] Worth checking: patients with typical angina get the lowest predicted risk of all chest pain types, below patients with no chest pain symptoms. [C09, C11, C15] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C13] Worth checking: the model gives a higher predicted risk to the 2 training patients whose thallium result is coded 4, a code the dataset documentation does not define. [C02] Worth checking: ST depression on exercise raises the predicted risk more when the resting ECG is normal than when it is not. [C28]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 86%	log-odds 3.33, OR 28.0; range across patients 3.33 to 3.33	242	train; test unclear (AUC gain 0.029 [-0.039, 0.100], n=61)	yes	on the training data only
C02	direction: thallium stress test result: a normal thallium scan to thallium stress test result coded 4 (meaning not recorded) (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 79%	log-odds 2.85, OR 17.4; range across patients 2.85 to 2.85	135, 2 (small n)	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	the model has little data on this; on the training data only
C03	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	log-odds 1.65, OR 5.20; range across patients 1.22 to 2.08	242	train; test unclear (AUC gain 0.023 [-0.001, 0.055], n=61)	yes	on the training data only
C04	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 56%	log-odds 1.75, OR 5.75; range across patients 1.75 to 1.75	115, 110	train; test disagrees (AUC gain -0.024 [-0.071, 0.009], n=61)	yes	on the training data only
C05	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	11% to 33%	log-odds 1.32, OR 3.73; range across patients 1.32 to 1.32	242	train; test disagrees (AUC gain -0.019 [-0.058, 0.008], n=61)	yes	on the training data only
C06	direction: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 38%	log-odds 1.00, OR 2.72; range across patients 1.00 to 1.00	135, 90	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	on the training data only
C07	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	12% to 30%	log-odds 1.19, OR 3.30; range across patients 1.19 to 1.19	242	train; test unclear (AUC gain 0.008 [-0.018, 0.037], n=61)	yes	on the training data only
C08	direction: exercise-induced angina: no chest pain on exercise to chest pain brought on by exercise (whole model) *shape: categorical; highest to lowest: chest pain brought on by exercise, no chest pain on exercise*	18% to 35%	log-odds 0.89, OR 2.44; range across patients 0.89 to 0.89	160, 82	train; test agrees (AUC gain 0.022 [0.003, 0.049], n=61)	yes	clearly
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 2%	log-odds -2.43, OR 0.09; range across patients -2.43 to -2.43	108, 22	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to 2%	log-odds -2.19, OR 0.11; range across patients -2.19 to -2.19	159, 83	train; test unclear (AUC gain 0.025 [-0.007, 0.066], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 4%	log-odds -1.69, OR 0.18; range across patients -1.69 to -1.69	108, 70	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C12	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 9%	log-odds -0.80, OR 0.45; range across patients -0.80 to -0.80	209, 33	train; test unclear (AUC gain 0.000 [-0.013, 0.013], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 9%	log-odds -0.80, OR 0.45; range across patients -0.80 to -0.80	135, 15 (small n)	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	the model has little data on this; on the training data only
C14	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: steady; 100% of patients same; verified*	23% to 15%	log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54	242	train; test agrees (AUC gain 0.014 [0.002, 0.035], n=61)	yes	clearly
C15	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 10%	log-odds -0.71, OR 0.49; range across patients -0.71 to -0.71	108, 42	train; test agrees (AUC gain 0.033 [0.008, 0.066], n=61)	yes	clearly
C16	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (direct contribution only, not through any part) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	18% to 24%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	121, 121	train; test unclear (AUC gain 0.005 [-0.005, 0.024], n=61)	yes	on the training data only
C17	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: resting ECG other than a normal resting ECG, a normal resting ECG*	18% to 21%	log-odds 0.02, OR 1.02; range across patients -0.49 to 0.38	121, 121	train; test disagrees (AUC gain 0.003 [-0.013, 0.021], n=61)	no	slightly; on the training data only
C18	direction: age: 42 years to 66 years (whole model) *shape: hinge at 53 years; 100% of patients same; verified; matches the fold at 53 years*	18% to 19%	log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10	242	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C19	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 19%	log-odds 0.05, OR 1.05; range across patients 0.05 to 0.05	115, 17 (small n)	train; test disagrees (AUC gain -0.024 [-0.071, 0.009], n=61)	yes	slightly; the model has little data on this; on the training data only
C20	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 86%	log-odds 3.33, OR 28.0	242	train; test unclear (AUC gain 0.029 [-0.039, 0.100], n=61)	yes	on the training data only
C21	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	log-odds 2.08, OR 8.01	242	train; test unclear (AUC gain 0.023 [-0.001, 0.055], n=61)	yes	on the training data only
C22	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 56%	log-odds 1.75, OR 5.75	115, 110	train; test disagrees (AUC gain -0.024 [-0.071, 0.009], n=61)	yes	on the training data only
C23	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	11% to 33%	log-odds 1.32, OR 3.73	242	train; test disagrees (AUC gain -0.019 [-0.058, 0.008], n=61)	yes	on the training data only
C24	magnitude: thallium stress test result: a normal thallium scan to a reversible defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result coded 4 (meaning not recorded), a reversible defect on the*	18% to 38%	log-odds 1.00, OR 2.72	135, 90	train; test unclear (AUC gain 0.020 [-0.012, 0.058], n=61)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	structure: made of ischaemia_chain, age_hinge and inputs sex, cp, trestbps, chol, fbs, restecg, thalach, exang, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C26	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 60%	log-odds 1.65, OR 5.20; range across patients 1.22 to 2.08	242	train; test unclear (AUC gain 0.023 [-0.001, 0.055], n=61)	yes	on the training data only
C27	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 16%	log-odds -0.36, OR 0.70; range across patients -0.86 to 0.00	121, 121	train; test unclear (AUC gain 0.006 [-0.009, 0.028], n=61)	yes	slightly; on the training data only
C28	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 0.5 mm; 100% of patients same; verified*	resting ECG other than a normal resting ECG: 21% to 48%; a normal resting ECG: 16% to 60%	log-odds difference -0.86, OR 0.42	121, 121	train; test agrees (AUC gain 0.006 [-0.009, 0.028], n=61); group difference on test -0.86 [-0.86, -0.86]	yes	clearly
C29	threshold: ST depression on exercise: fold at about 0.9 mm (starts) *shape: hinge at 0.5 mm; 100% of patients same; verified*	16% to 74% (typical patient with a fold)	fold 0.92 mm (IQR 0.45 to 1.38); past it log-odds 2.30, OR 9.93; range across patients 1.86 to 2.73	242, 105	train; test unclear (AUC gain 0.023 [-0.001, 0.055], n=61)	no	on the training data only
C30	subgroup: who this part acts on: 105 of 242 patients	76% to 91% (typical patient this part acts on (risk without the part, then with it))	share 0.43; log-odds 1.03, OR 2.80; range across patients 0.03 to 2.33	105, 137	train; test unclear (AUC gain 0.026 [-0.001, 0.062], n=61)	yes	on the training data only
C31	structure: its formula reproduces it on 242 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	242	train	yes	
C32	direction: age: 42 years to 66 years (through this part) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 19%	log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10	242	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C33	threshold: age: fold at about 53 years (starts) *shape: hinge at 53 years; 100% of patients same; verified; one point with C33 (median fold 53 years, typical patient 53 years, 10th-90th percentile 53 years to 53 years)*	18% to 20% (typical patient with a fold)	fold 52.8 years (IQR 52.8 to 52.8); past it log-odds 0.12, OR 1.13; range across patients 0.12 to 0.12	242, 148	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C34	subgroup: who this part acts on: 148 of 242 patients	91% to 92% (typical patient this part acts on (risk without the part, then with it))	share 0.61; log-odds 0.06, OR 1.07; range across patients 0.009 to 0.12	148, 94	train; test disagrees (AUC gain -0.003 [-0.017, 0.004], n=61)	yes	slightly; on the training data only
C35	null: this part barely changes predictions (switched off)		max change 0.032 (max |change in predicted risk|)	242	train	no	
C36	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model.

- C17: direction claim about resting ECG in the whole model: not supported by the model (44% of patients have the opposite sign (allowed: 10%)).
- C29: threshold claim about ST depression during exercise (compared with rest) in part ann_27: not supported by the model (the fold changes the curve by only 0.095 log-odds compared with a straight line (needed 0.1)).
- C35: null claim about age in part ann_28: not supported by the model (predictions move by up to 0.032).

13.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (7):** `-11` (fbs:1), `-13` (restecg:1), `-14` (restecg:2), `-16` (exang:0), `-20` (slope:2), `-3` (sex:1), `-5` (cp:2).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

13.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 13.6.1)

▷ A2: ischaemia_chain (leaf, section 13.6.2)

▷ A3: age_hinge (leaf, section 13.6.3)

13.4 Network figures

13.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 2.22×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-12` into `-12_a`, `-12_b`.

13.6 Annotations

13.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 13.6.2), A3 (section 13.6.3).

- **Entry nodes:** `-26` (thal:7), `-25` (thal:6), `-24` (thal:4), `-23` (thal:3), `-22` (ca), `-21` (slope:3), `-20` (slope:2), `-19` (slope:1), `-17` (exang:1), `-16` (exang:0), `-15` (thalach), `-14` (restecg:2), `-13` (restecg:1), `-11` (fbs:1), `-10` (fbs:0), `-9` (chol), `-8` (trestbps), `-7` (cp:4), `-6` (cp:3), `-5` (cp:2), `-4` (cp:1), `-3` (sex:1), `-2` (sex:0), `-12_a` (restecg:0_a).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `-2`→`0` (-1.04), `-4`→`0` (-0.494), `-6`→`0` (-0.442), `-7`→`0` (0.354), `-8`→`0` (0.483), `-9`→`0` (0.551), `-10`→`0` (0.274), `-15`→`0` (-0.213), `-17`→`0` (0.423), `-19`→`0` (-0.873), `-21`→`0` (-0.434), `-22`→`0` (1.49), `-23`→`0` (-0.471), `-24`→`0` (0.172), `-25`→`0` (-0.421), `-26`→`0` (0.0253), `204`→`0` (0.0487), `290`→`0` (0.216), `1987`→`0` (0.0824), `-12_a`→`0` (-0.188).

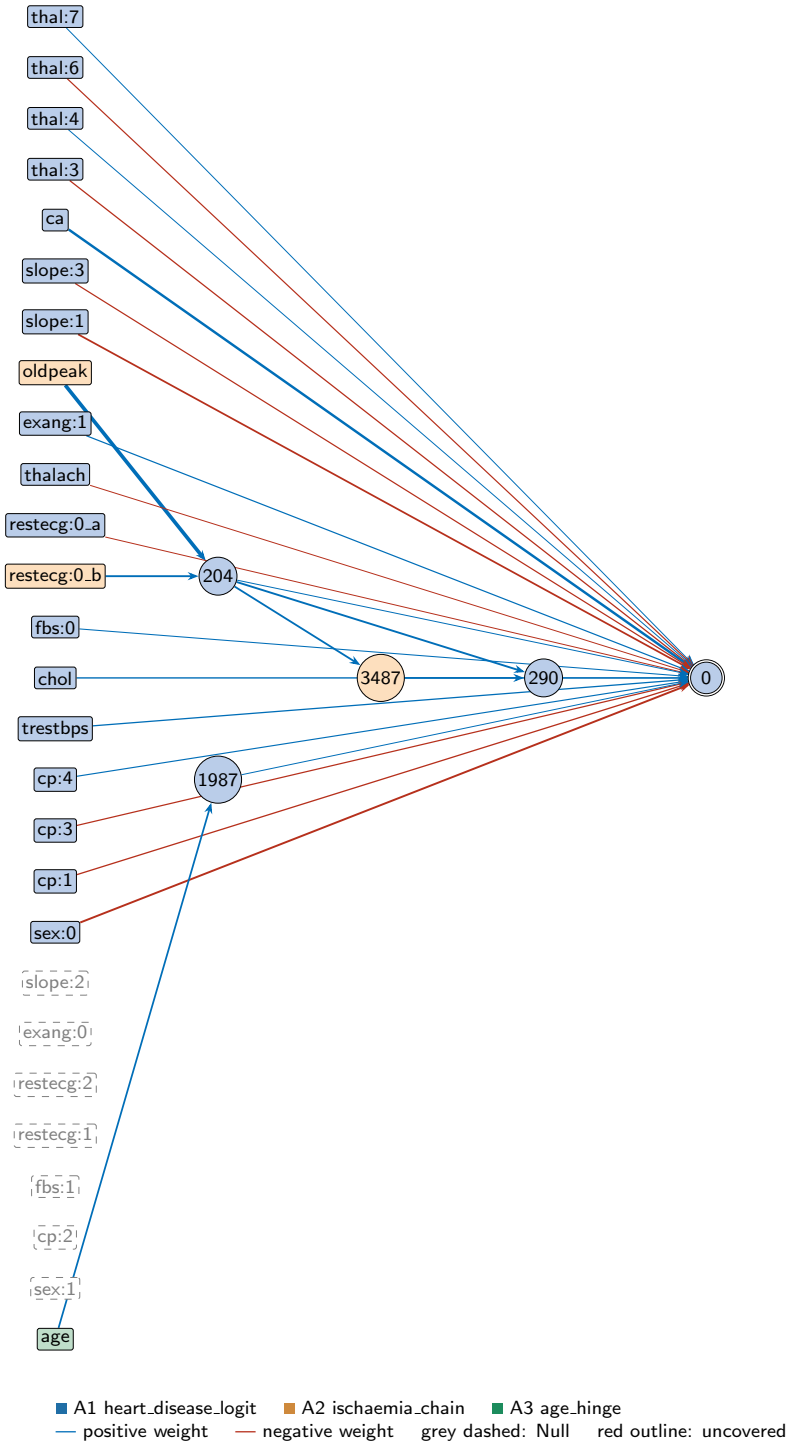


Figure 39: heart-s5-c3: full view of the explained model state.

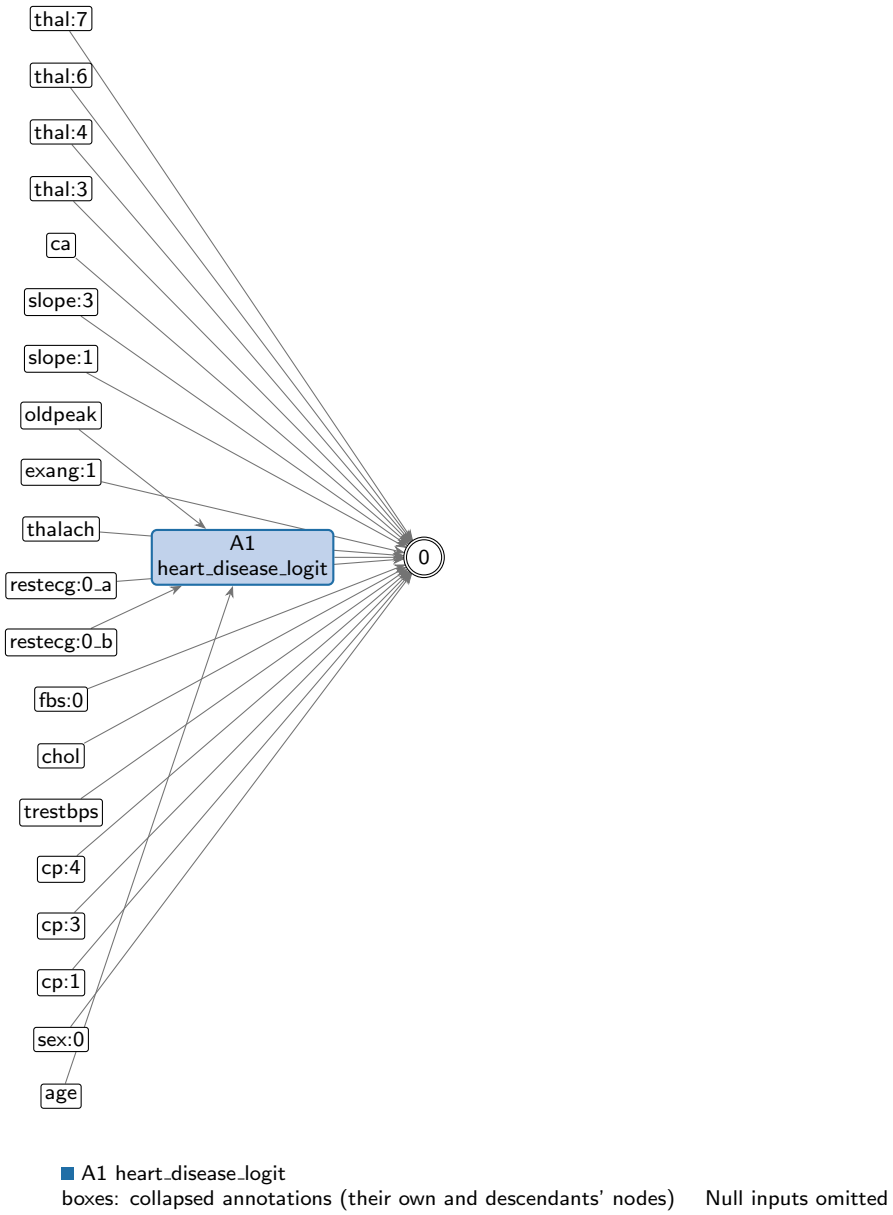


Figure 40: heart-s5-c3: collapsed view at hierarchy level 0 (fully recomposed).

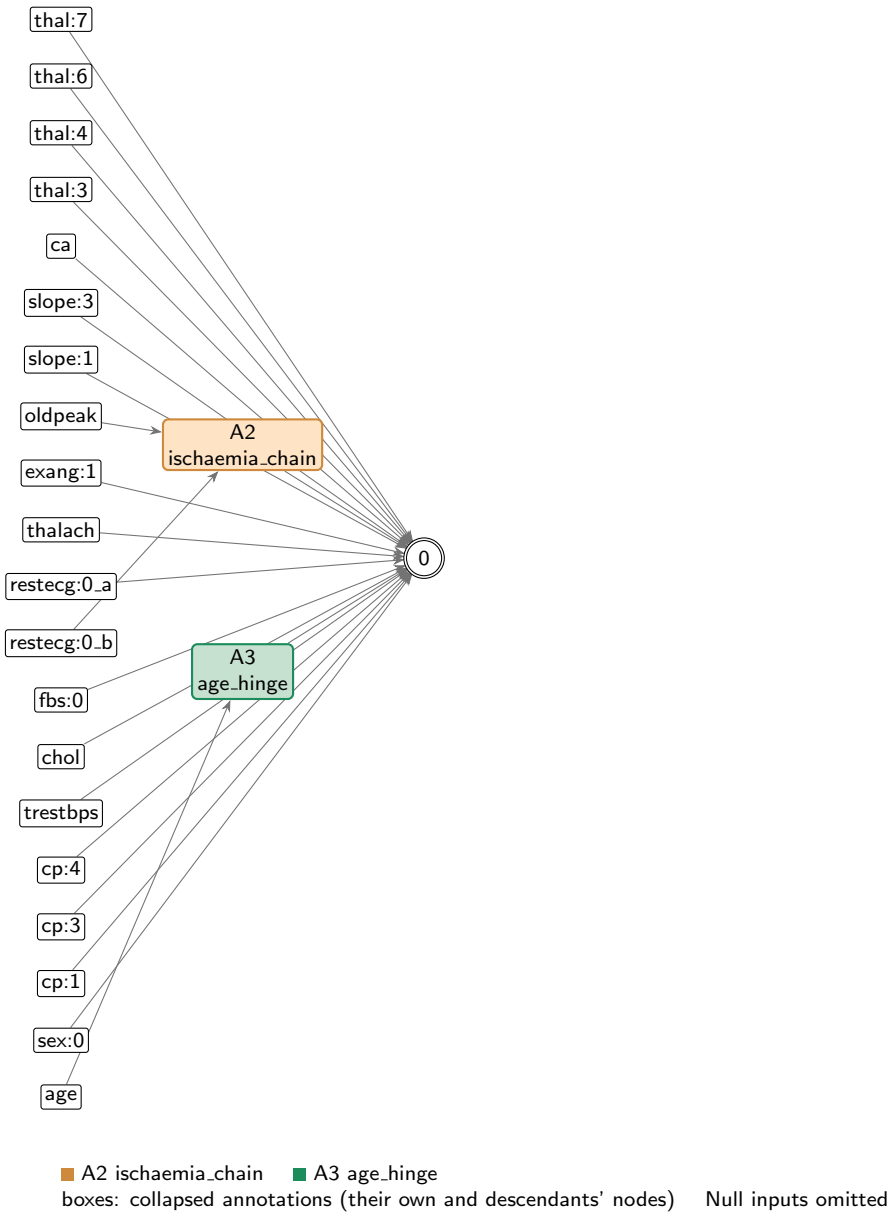


Figure 41: heart-s5-c3: collapsed view at hierarchy level 1.

- **With its descendants:** 32 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 31 candidate nodes.

Hypothesis. Root: p = sigmoid(z), z = -0.536 + LINEAR CORE + 0.049*ischaemia_chain[204] + 0.216*ischaemia_chain[290] + 0.082*age_hinge. LINEAR CORE (17 direct input->output edges, z-scored inputs): ca +1.485, sex:0 -1.042, slope:1 -0.873, chol +0.551, cp:1 -0.494, trestbps +0.483, thal:3 -0.471, cp:3 -0.442, slope:3 -0.434, exang:1 +0.423, thal:6 -0.421, cp:4 +0.354, fbs:0 +0.274, thalach -0.213, restecg:0 -0.188, thal:4 +0.172, thal:7 +0.025. Seven inputs are unconnected (sex:1, cp:2, fbs:1, restecg:1, restecg:2, exang:0, slope:2; zero-effect entries of this root). Oldpeak and age have no direct edge and act only through the children; restecg:0 acts both directly and through the ischaemia chain (split into restecg:0_a direct and restecg:0_b). COMPOSITION: additive. The two children are read out additively into z: ischaemia_chain (two exits: a one-sided ST-depression score opened earlier by a normal resting ECG, and a steeper second stage of the same score) and age_hinge (an age term that starts just below the median age). Together the chain gives oldpeak a hinge-shaped effect: about 0.12 logit per SD between its two thresholds and about 1.10 per SD above the second (z-oldpeak -0.39 with a normal ECG, +0.40 otherwise). EVIDENCE (train, n=242): full model AUC 0.9376, acc 0.888 (compute_performance). Linear core alone AUC 0.9317; logit SD 3.22 (core) vs 0.80 (nonlinear readouts). Removing the whole chain (204 with downstream) gives 0.9317; readout-only ablations: 204 0.9380, 290 0.9341, age 0.9371. All ReLUs as identities 0.9304; one least-squares linear fit of all nonlinear readouts on restecg:0, oldpeak and age gives 0.9355 (R2 0.88); an L2 logistic regression on all 26 inputs gives 0.9355. Mean |SHAP| (kernel, all 242 train rows): ca 0.141, sex:0 0.104, slope:1 0.099, oldpeak 0.060, thal:3 0.054, exang:1 0.045, chol 0.044, cp:3 0.043, cp:4 0.040, trestbps 0.040. Interpretation: close to logistic regression plus one ST-depression correction that switches on above a threshold (lower for patients with a normal resting ECG) and a small age term above median age. Counter-directional signs reported, not explained: a normal resting ECG lowers the oldpeak threshold inside the chain (raising risk), fbs:0 (normal fasting sugar) +0.27 directly, thal:7 (reversible defect) only +0.025. Compared with the 10-seed heart-s5 record (generation 37, key 9330: ischaemia gate -> resting-BP gate -> age term) and its cluster-1 sibling: the same ischaemia gate on oldpeak and restecg:0, but here the second stage is a steeper copy of the score (no resting-BP offset) and the age term is a free-standing hinge (not switched off by the chain).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -0.536 + 0.216\,\text{ischaemia_chain_1}\big(x_{\text{restecg:0_b}}, \\
&\qquad x_{oldpeak}\big) + 0.354x_{\text{cp:4}} + 0.551x_{chol} + 0.172x_{\text{thal:4}} + 1.49x_{ca} \\
&\qquad + 0.0487\,\text{ischaemia_chain_0}\big(x_{\text{restecg:0_b}}, x_{oldpeak}\big) \\
&\qquad + 0.274x_{\text{fbs:0}} + 0.0253x_{\text{thal:7}} + 0.483x_{trestbps} \\
&\qquad + 0.0824\,\text{age_hinge}\big(x_{age}\big) + 0.423x_{\text{exang:1}} - 0.213x_{thalach} \\
&\qquad - 0.494x_{\text{cp:1}} - 1.04x_{\text{sex:0}} - 0.188x_{\text{restecg:0_a}} - 0.421x_{\text{thal:6}} \\
&\qquad - 0.471x_{\text{thal:3}} - 0.434x_{\text{slope:3}} - 0.442x_{\text{cp:3}} - 0.873x_{\text{slope:1}}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -0.536 + 0.216\,\text{relu}\big(-0.468 \\
&\qquad + 0.862\,\text{relu}\big(-0.00219 + 0.94\,\text{relu}\big(0.245 \\
&\qquad + 2.5x_{oldpeak} + 0.98x_{\text{restecg:0_b}}\big)\big) + 1.01\,\text{relu}\big(0.245 \\
&\qquad + 2.5x_{oldpeak} + 0.98x_{\text{restecg:0_b}}\big)\big) + 0.354x_{\text{cp:4}} + 0.551x_{chol} \\
&\qquad + 0.172x_{\text{thal:4}} + 1.49x_{ca} + 0.0487\,\text{relu}\big(0.245 + 2.5x_{oldpeak} \\
&\qquad + 0.98x_{\text{restecg:0_b}}\big) + 0.274x_{\text{fbs:0}} + 0.0253x_{\text{thal:7}} + 0.483x_{trestbps} \\
&\qquad + 0.0824\,\text{relu}\big(0.191 + 0.872x_{age}\big) + 0.423x_{\text{exang:1}}
\end{aligned}
$$

$$- 0.213x_{thalach} - 0.494x_{cp:1} - 1.04x_{sex:0} - 0.188x_{restecg:0_a}$$
$$- 0.421x_{thal:6} - 0.471x_{thal:3} - 0.434x_{slope:3} - 0.442x_{cp:3} - 0.873x_{slope:1}$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00216, and the exact function by at most 2.78×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form of the full model and its collapsed form in terms of the children.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split performance, whole-model SHAP and ablation / linear-replacement analysis.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

13.6.2 A2: ischaemia_chain

Kind: Leaf annotation (depth 1); parent A1 (section 13.6.1).

- **Entry nodes:** `-12_b` (restecg:0_b), `-18` (oldpeak).
- **Exit nodes:** `204`, `290`.
- **Internal nodes:** `3487`.
- **Edges (weight):** `-12_b`→`204` (0.98), `-18`→`204` (2.5), `204`→`3487` (0.94), `204`→`290` (1.01), `3487`→`290` (0.862).
- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. Ischaemia chain (raises risk with ST depression, in two stages). 204 = ReLU(0.245 + 2.501*oldpeak + 0.980*restecg:0); 3487 = ReLU(0.940*204 - 0.002) (a near-copy of 204); 290 = ReLU(-0.468 + 1.008*204 + 0.862*3487) = ReLU(1.818*204 - 0.470) whenever 204 > 0.0023. Readouts: logit += 0.049*204 + 0.216*290. Hypothesis: an ST-depression score that opens above an oldpeak threshold set by the resting ECG (z-oldpeak -0.49 with a normal ECG, +0.29 otherwise) and then a second, much steeper stage once the score passes 0.259 (z-oldpeak -0.39 with a normal ECG, +0.40 otherwise). Net effect on the logit: about 0.12 per SD of oldpeak between the two thresholds and about 1.10 per SD above the second, i.e. a hinge in oldpeak whose position moves about 0.78 SD lower with a normal resting ECG. EVIDENCE (train, n=242): 204 active for 105/242 (58 of 121 with a normal ECG, 47 of 121 without); 290 active for 96/242; disease rate 66% when 290 is active vs 30% when not; corr(204, oldpeak) 0.91. Sensitivities (mean |d exit/d entry|): 290: oldpeak 1.82, restecg:0 0.71; 204: oldpeak 1.09, restecg:0 0.43. Ablation: removing 204 with its downstream (the whole chain) lowers train AUC 0.9376 -> 0.9317, the linear-core level; removing the steepener readout (290) alone 0.9341; removing 204's own readout 0.9380 (no loss). An affine replacement of 290 on 204 fits almost perfectly (R2 0.998, AUC 0.9376): the steepening is a hinge that the data barely probe on its flat side. Supported. Oldpeak has no direct edge and acts only here. Counter-intuitive sign reported, not explained: a normal resting ECG (restecg:0) lowers the oldpeak threshold (raising risk) inside the chain, partly offset by its direct -0.188 edge.

Closed form:

$$\begin{aligned} y_0 &= \mathrm{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big) \\ y_1 &= \mathrm{relu}\big(-0.468 + 0.862\,\mathrm{relu}\big(-0.00219 \\ &\qquad + 0.94\,\mathrm{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big)\big) \\ &\qquad + 1.01\,\mathrm{relu}\big(0.245 + 2.5x_{oldpeak} + 0.98x_{restecg:0_b}\big)\big) \end{aligned}$$

where y_0 is node `204`; y_1 is node `290`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.0205, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split sensitivity, line plot, activity, outcome rates and ablations.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

13.6.3 A3: age_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 13.6.1).

- **Entry nodes:** `-1` (age).
- **Exit nodes:** `1987`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`1987` (0.872).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Age hinge (raises risk with age above a threshold, small). 1987 = ReLU(0.191 + 0.872*age), read out +0.082. Hypothesis: age's only route to the output; nothing below z-age -0.22 (just under the median age), then +0.072 logit per SD of age. EVIDENCE (train, n=242): active for 148/242; disease rate 55% when active vs 28% when not; line plot correlation 0.90; mean |sensitivity| 0.53. Removing it lowers train AUC 0.9376 -> 0.9371; an affine replacement (R2 0.80) gives 0.9376. Supported, near-negligible.

Closed form:

$$y_0 = \mathrm{relu}\big(0.191 + 0.872 x_{age}\big)$$

where y_0 is node `1987`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 2.75×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Closed form from get_formula.
- EMPIRICAL, `empirical_viz`; data split: train. Train-split sensitivity, line plot, activity, outcome rates and ablations.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

13.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft